

FREE AGRICULTURE K.C.S.E 2012 to 2015

K.C.S.E REVISION SERIES 2012 to 2015 AGRICULTURE

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SECTION I : QUESTIONS

INTRODUCTION TO AGRICULTURE

This topic entails the following:-

- Definition of agriculture
- Main branches of agriculture
- Farming systems
- Farming methods
- Role of agriculture to Kenya's economy
- Varied opportunities in agriculture.

The following relevant questions and their answers in this topic will help and motivate the user to comprehend and understand the required concepts and practices:

1. Give **two** factors which characterize intensive farming
2. State **three** reasons why organic farming is encouraged in farming
3. State **two** ways in which agriculture contributes to industrial development
4. State **four** ways by which wind affects the growth of crops
5. State **one** physical characteristic used in classifying soil
6. Outline **four** advantages of organic farming
7. State **two** conditions under which shifting cultivation is practiced
8. Differentiate between the following terms as used in Agriculture:-
 - (a) Oleculture and floriculture
 - (b) Apiculture and aquaculture

FACTORS WHICH INFLUENCE AGRICULTURE

In this topic, the following factors influence agriculture.

- Human factors e.g. -level of education, -Health HIV/AIDS, -Economic status of the farmer e.t.c
- Biotic factors e.g. pests, parasites, decomposers, pathogens, pollinators, predators e.t.c.
- Climatic factors e.g. rainfall, temperature, wind and relative humidity, light

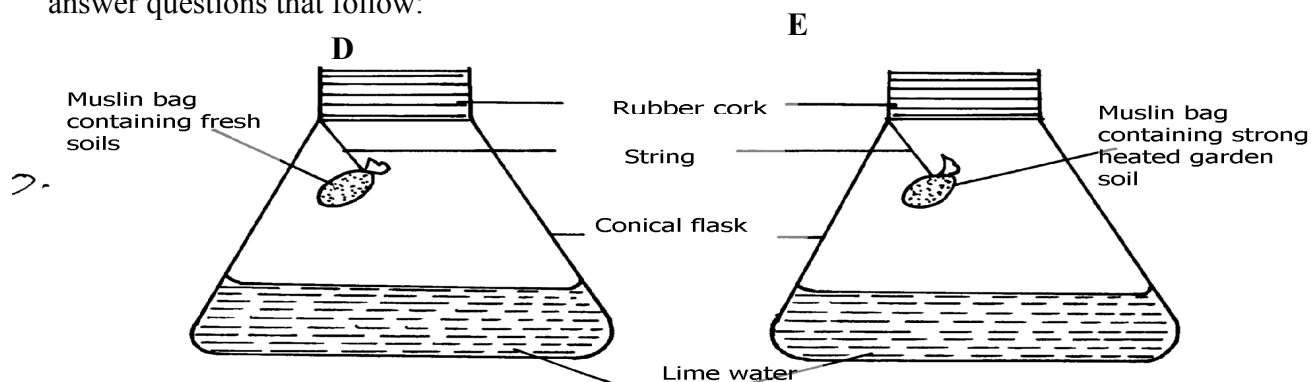
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- Edaphic factors e.g. type of soils, soil profile, soil structure, soil texture, soil chemical properties.

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the required concepts:

1. State **two** roles of humus in the soil that are beneficial to crops
2. a) outline **five** activities that may be undertaken in organic farming
3. List **four** effects of temperature on crop growth
4. State **four** ways by which wind affects the growth of crops
5. Name **two** factors related to light that affect crop production and distribution in Kenya
6. Describe the environmental conditions that may lead to low crop yields
7. List **three** environmental factors that affect crop distribution in Kenya
8. State **one** physical characteristic used in classifying soil
9. Outline **four** advantages of organic farming
10. The diagrams below show an experiment carried out by a form 1 class. Study them carefully and

answer questions that follow:



- (a) What was the aim of the experiment?
- (b) What was the observation that form 1 students made at the end of the experiment in flasks **D** and **E**?

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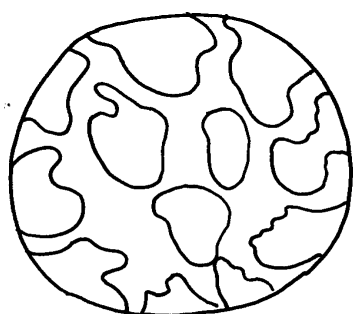
(c) Give the reason for the observation made in flask **D**

12. Briefly explain how sub-soil as a horizon in a soil profile can affect soil productivity

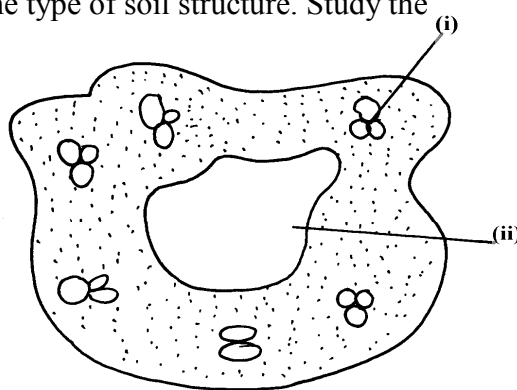
13. (a) What are the **three** aspects of light that are important to a farmer?

(b) Mention **three** ways through which relative humidity affect crop production

14. The diagram labeled **E** and **F** below illustrates some type of soil structure. Study the diagrams carefully and answer the questions that follow:



E



F

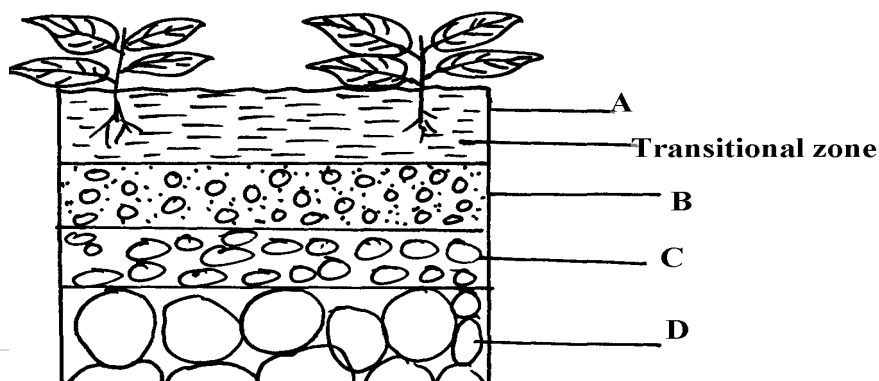
(a) Identify the types of soil structure illustrated in diagrams **E** and **F**

(b) Identify the parts labeled **(i)** and **(ii)** in diagram **F**

(c) Outline the influence of physical characteristics of soil on its properties

15. State **three** physical characteristics of soil

16. Study the diagram below and answer the questions that follow



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- a) State merits of horizon **A**
- b) State distinct features of horizon **B**
- c) What does the term **transition zone** refer to in soil profile
 - i) Name horizon **C** and state its importance
- 17. Outline **two** ways temperature affects crop production
- 18. List **four** ways by which biological agents can enhance the process of soil formation
- 19. List **four** environmental factors that affect crop production in Kenya
- 20. Explain the role played by topography in soil formation
- 22. Mention **two** importance of parent's material in soil profile
- 23. Mention **four** ways of modifying soil temperature in crop production
- 24. a) Mention **two** factors that affect selectivity of herbicides
b) Name **two** farming practice that cause water pollution
- 25. Give **four** factors that influence soil formation
- 26. State **three** properties of soil that is influenced by soil texture
- 27. Name any **three** agents of biological weathering

CROP PRODUCTION I (LAND PREPARATION)

- Land preparation entails the following farming practices.
- Land clearing or bush clearing tools, chemicals and equipment used.
- Primary cultivation, tools and equipment as machines used.
- Primary cultivation, tools and equipment as machines used.
- Secondary cultivation, tools and equipment used.
- Tertiary operations e.g. ridging, rolling and leveling.

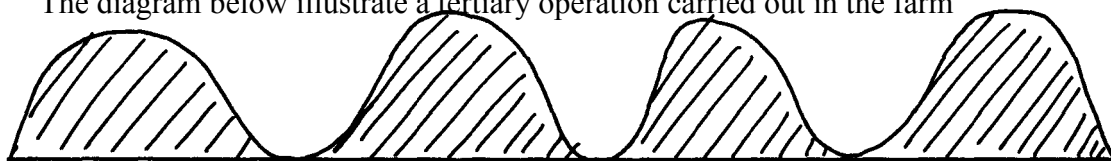
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- Sub-soiling, tools used and reasons for the same.
- Minimum tillage and reasons for the same.

The following relevant questions and their answers in this topic will greatly help and motivate

the user to comprehend and understand the required concepts and farming practices:

1. Give **three** factors that determine depth of ploughing during land preparation
2. List **four** reasons for cultivating land before planting
3. (a) What is minimum tillage?
(b) Give four farming practices that help in achieving minimum tillage.
4. (a) Describe the establishment of grass pasture from the time the land is ploughed using a mould board plough to the time the pasture is ready for grazing
(b) Explain **five** practices that a farmer should carry out to ensure uniform germination of seeds
(c) Describe **five** factors that determine the number of cultivations when preparing a seedbed
5. State **four** physical conditions of the seedbed that need to be changed to facilitate germination
6. State **four** importance of sub soiling as a tertiary operation
7. Outline **four** advantages of rolling in seedbed preparation
8. State **four** disadvantages of minimum tillage
9. The diagram below illustrate a tertiary operation carried out in the farm



- a) Identify the tertiary operation
 - b) (i) State the importance of the tertiary operation identified in **20(a)** above
(ii) Give **two** other tertiary operations carried out in the field besides the one identified above
10. Give **two** reasons why it is advisable to cultivate the field during the dry season

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11. How are hard pans caused by cultivation?
12. Give **four** factors that determine the number of secondary cultivation operations
13. Define the term minimum tillage
14. List four advantages of timely planting
15. State any **two** factors that determine the number of cultivation on a field before it is ready for planting
17. Give **three** benefits of timely planting of annual crops
18. State **four** factors determining the depth of ploughing land

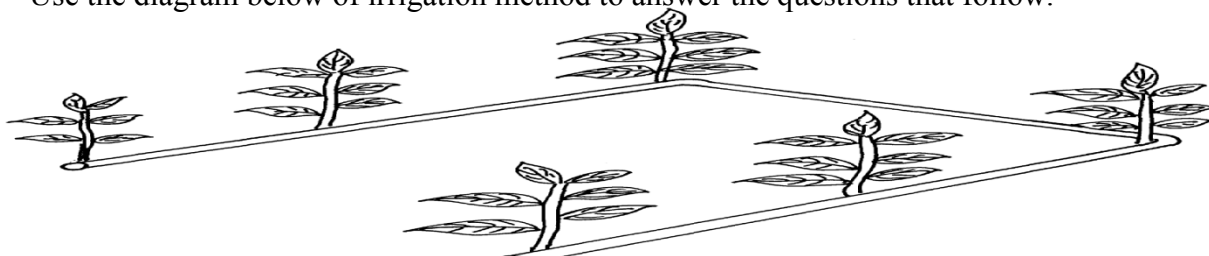
WATER SUPPLY, IRRIGATION AND DRAINAGE

This topic entails the following:

- Hydrological cycle
- Sources of water on the farm
- Water collection and storage
- Pumps and pumping of water
- Types water pipes
- Water treatment
- Uses of water of the farm.
- Types of irrigation advantages and disadvantages.
- Importance and methods of drainages
- Water pollution causes and prevention.

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the required concepts and practices:

1. State **two** reasons for treating water for use on the farm
2. State **three** reasons for draining swampy land before growing crops
3. Use the diagram below of irrigation method to answer the questions that follow.

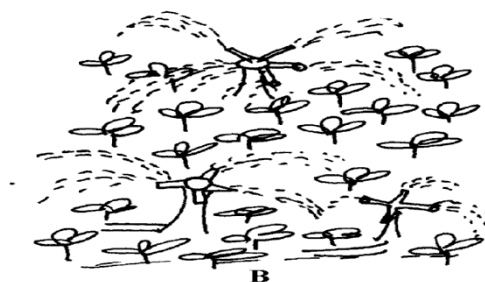
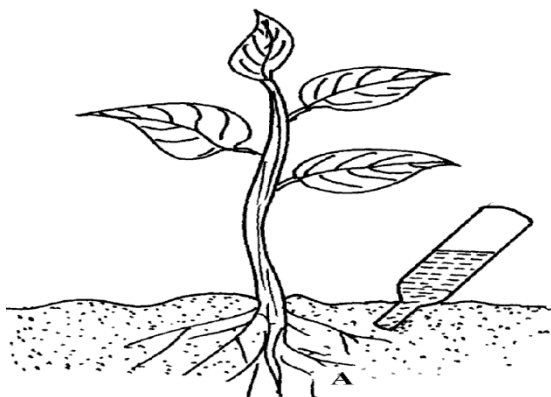


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- a) Identify the method of irrigation
 - b) State **four** advantages of the above irrigation system
 - c) State **three** factors that determine the type of irrigation on the farm
 - d) State **two** disadvantages of the above system of irrigation
4. a) What is **irrigation**
- b) Outline **three** methods of irrigation
5. a) List **four** uses of water on the farm
- b) Give **four** methods of harvesting water on the farm
- c) Outline the stages involved in water treatment process
6. List any **four** uses of water in the farm
7. State **two** types of irrigation used in Kenya
8. Outline **four** disadvantages of cambered beds
- Describe the process of water treatment
9. Give **four** roles of drainage as a method of land reclamation
10. Name **two** types of water pumps which can be used in the farm
11. Name any **four** examples of working capital in maize production
12. List **four** types of water pumps which can be used in the farm
13. State **four** methods of drainage

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14. Distinguish between a **dam** and a **weir**
15. How do the government control prices of essential farm produce
16. What is the difference between pumping and piping of water in the farm?
17. List four reasons of draining water logged soils before planting.
18. Give three Agricultural practices which lead to water pollution
19. The diagrams below illustrate some methods of irrigating crops in the field. Study the diagrams
and answer the questions that follow:

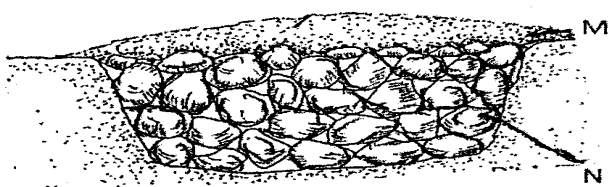


- (a) Identify the methods used ; (i) A (ii) B
 - (b) State **two** advantages of method **A** over method **B**
 - (c) What material should be inserted at point **T**
 - b) Name **two** farming practice that cause water pollution
20. Give **four** reasons for practicing irrigation
 22. a) State **four** importance of water to plants
 - b) State **four** reasons for treating water before use
 - c) Describe water treatment system in a chemical treatment plant
 23. Name **four** diseases caused to man by drinking untreated water
 24. State the functions of the following chemicals as used in water treatment;
(a) Chlorine.

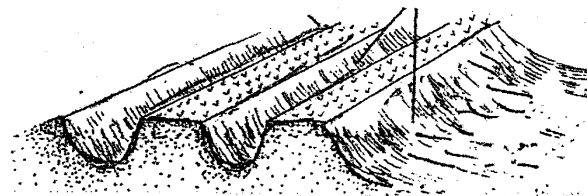
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(b) Aluminum sulphate (Alum)

21. The diagrams labeled **S** and **T** illustrate some methods of draining waterlogged fields; use it to answer the questions that follow:



S



T

(a) Identify the methods illustrated

(b) What are the materials in **S** labeled **M** and **N**

(c) Name **two** types of crops that can be planted in the field instead of carrying out the practice illustrated in **S** and **T**

(d) What is the importance of carrying out land reclamation?

SOIL FERTILITY 1 (ORGANIC MANURE)

This topic entails the following:

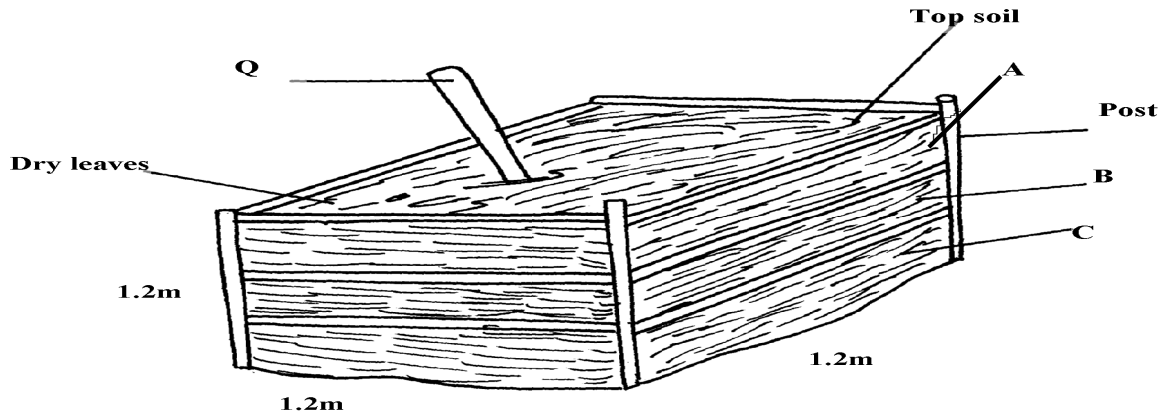
- Characteristics of a fertile soil
- How soil loses soil fertility
- Soil fertility maintenance
- Reasons of adding organic matter to soil
- Disadvantages of organic manure
- Types of organic manure i.e green manure, farm yard manure and compost measure.

The following relevant questions and their answers in this topic will greatly help the user to comprehend and understand the required concepts and practices:

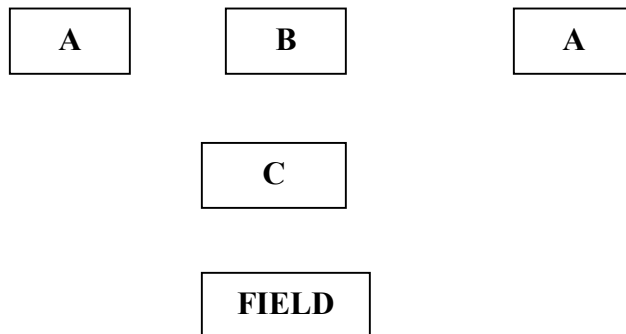
1. State **two** roles of humus in the soil that are beneficial to crops
2. List **four** characteristic of fertile soil

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3. The diagram below illustrates a compost heap. Study it and answer the questions that follow



- a) Name the part labeled **Q** and state its function
- b) What is the function of each of the following components in preparation of compost manure
- i) Top soil
 - ii) Wood ash
 - iii) Rotten manure
4. The illustration below shows a four heap system of making compost manure. Study it and answer the questions that follow.



- (a) By use of arrows indicate on the diagram above how the following material should be transferred from one heap to another till the manure is applied in the field

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- (b) How long does the material take to be ready for application in the field as manure?
- (c) Give a reason for turning the material in the heap regularly
- (d) Give **two** reasons why it is necessary to sprinkle water on the heap
5. Name **four** indicators of well-decomposed manure
6. (a) State **two** factors that should be considered when siting a compost manure heap
- (b) When preparing compost manure, explain the importance of each of the following:-
- (i) Addition of ash
- (ii) Regular turning of the compost manure
7. What is **leaching**?
8. State **four** advantages of adding organic matter to a sandy soil
9. (a) Describe the preparation of the following farm materials:-
- (i) Farm yard manure
- (ii) Hay
- (b) Explain the factors considered in timely planting of annual crops
10. A ration containing 18% protein is to be made from maize and sunflower cake. Given that maize contains 7% protein, and sunflower seed cake 34% protein. Use Pearson square method to calculate the value of feedstuffs to be used to prepare 100kgs of the feed
- ii) A part from Pearson square method, name **two** other methods that can be used to formulate feed ration

AGRICULTURE ECONOMICS (BASIC CONCEPTS AND FARM RECORDS)

This topic entails the following

- Definition of scarcity, preference and classic, opportunity cost as used in agriculture production.
- Uses of farm records

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- Types of farm records i.e production records, filed operation records, breeding records, feeding records, health, labour records and master roll.

The following relevant questions and their answers in this topic will greatly help and motivate

the user comprehend and understand the concepts and practices.

1. (a) What are the uses of farm records to a farmer?
2. Identify the farm record below and the questions that follow:

Date	Disease symptoms	Animals affected	Drug used	Cost of treatment	Remarks

(a) Identity of the record

(b) State **two** different information that should be entered in the remarks column

(c) Give **two** importance of keeping the farm record illustrated above

3. State **four** uses of farm records
4. State **four** uses of farm records
5. Outline **two** ways the level of education and technology influence the efficiency of agricultural production
6. Study the illustration below of farm records:- Use it to answer the questions that follow:
Enterprise
Month

Name of cow	DAYS IN MONTH											
	1		2		3		4		5		6	
	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM

(a) Name the type of the farm record illustrated above

(b) Give **three** reasons for keeping health records in a livestock production

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(c) Give **three** pieces of information a dairy farm manager should collect for planning purposes

7. List down four pieces of information recorded in a field operation record.
8. List **two** events occur during induction stroke in a four stroke engine
9. Give **two** conditions under which a farmer may prefer the use of donkey trailed cart instead of a tractor drawn trailer in his farm

SOIL FERTILITY II (IN ORGANIC FERTILIZERS)

This topic entails the following;

- Essentials elements required by crops
- Classification of essential elements
- Role of micro-nutrients
- Deficiency symptoms of macro-nutrients and micro-nutrients.
- Identification and classification of fertilizers.
- Soil sampling and testing methods of fertilizer application.
- Effect of soil acidity/alkalinity on crops
- Fertilizer rate calculations

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the required concepts and practices:

1. State **four** advantages of applying lime in clay soil
2. a) Give the form in which the following elements are absorbed by crops
 - i) Sulphur
 - ii) Nitrogen
 - iii) Carbon
 - iv) Magnesiumb) List **three** effects of nitrogen on plants
3. Mr. Malombe of Shinyalu village prepared to top dress 10 hectares of napier grass using sulphate of ammonia (21%N). Sulphate of ammonia is applied at rate of 150kg per hectare. Calculate

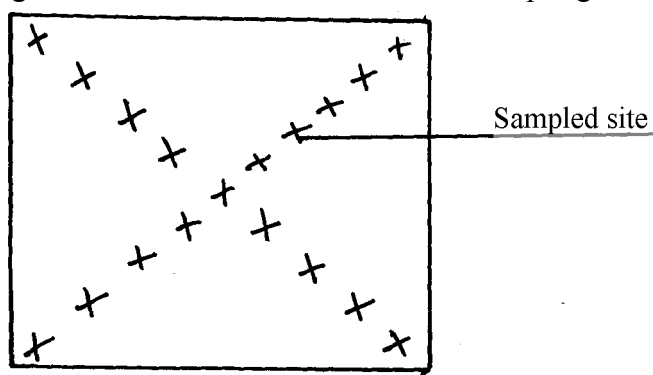
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- a) The quantity of sulphate ammonia fertilizer the farmer will need for 10 hectares
- b) The number of 50kg bags of fertilizer he will purchase
4. Give **two** disadvantages of using farmyard manure
5. State **four** factors which influence the stage at which the crops are harvested
6. A form **four** student was given a sample of a fertilizer with the following characteristics:
- (i) Grey in colour
 - (ii) It is granular
 - (iii) Causes no corrosion
 - (iv) It is highly hygroscopic
 - (v) It is neutral
- (a) Identify the fertilizer
- (b) At what stage of growth of maize should it be applied?
- (c) Calculate the amount of K_2O contained in 400kg of a compound fertilizer 25:10:5
7. State **two** pieces of information that soil sample should have before being taken to the laboratory for testing
8. A compound fertilizer bag has the labels 20-20-0. What do the figures stand for?
9. Give **four** functions of sulphur in crops
10. State **four** advantages of lining as a measure of soil improvement
11. State **two** methods of increasing soil PH
12. (a) State **three** factors that determine the amount of inorganic fertilizers needed to be applied to crops
- (b) What are the necessary precautions observed when carrying out soil sampling?
13. List **three** functions of nitrogen in crops
14. (a) Distinguish between fertilizer grade and fertilizer ratio

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(b) List **four** elements whose deficiency results into chlorosis in plants

15. The diagram below shows a method of soil sampling



(a) Name the method illustrated in the diagram

(b) State **three** precautions taken when collecting the soil for testing using the above method

(c) Give **four** reasons why soil from the farm is tested

16. A farmer was advised to apply compound fertilizer 20-20-10 on an orchard measuring 20m X 10m at the rate of 80kg/ha. Calculate the amount of fertilizer the farmer would require

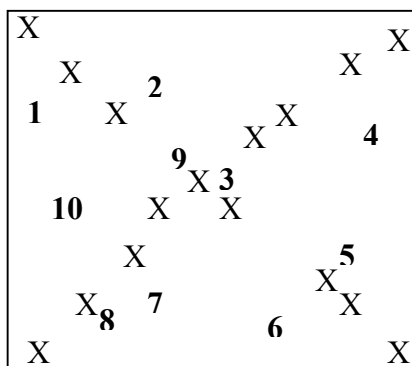
for the orchard.

(Show your

working)

17. a) A compound of fertilizer has a fertilizer grade of 25:10:5. calculate the amount of phosphorus present in 400kg of this fertilizer

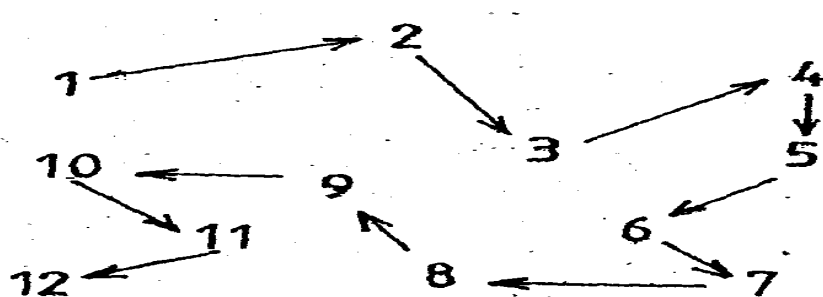
b) The diagram below illustrates methods of collecting soil sample from a field



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- i) Identify the methods illustrated 1-
ii) xx
iii) State **three** importance of carrying out soil sampling and testing
18. (a) What is an incomplete compound fertilizer?

(b) State **four** reasons why a maize crop continued showing deficiency of potassium despite applications recommended amount of potassic fertilizer
19. The diagram below shows a soil sampling method.



- (a) Identify the method illustrated above

(b) Name any **two** spots in a farm that should be avoided during sampling

(c) Describe the steps followed while carrying out the exercise in (a) above

CROP PRODUCTION II (PLANTING)

This topic entails the following:

- Correct planting materials for various crops
- Selection and preparation of planting materials
- Determination of optimum time of planting
- Factors which determine planting depth
- Planting procedure for different crops
- Factors which determine seed rate, spacing and plant population.
- Calculation of plant population
- Economic value of land.

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The following relevant questions and their answers in this topic will greatly help and motivate

the user to comprehend and understand the required concepts and practices:

1. State **two** reasons for seed treatment of tree species before planting
2. Give **three** factors that determine spacing of beans
3. State **four** reasons for using certified seeds for planting
4. Below are diagrams showing vegetative material used for propagation.

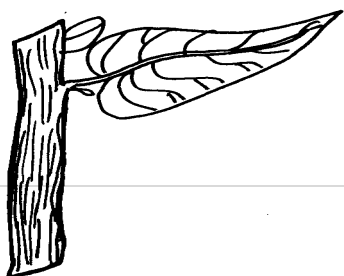


- a) Name the propagation materials A, B, C, D
 - b) What is the term used for inducing **B** to start germinating?
 - c) State **four** advantages of vegetative propagation on crop production
5. Differentiate between hybrid and composite
 6. a) A farmer planted 100 maize seeds and 90 seeds germinated.
Calculate the germination percentage

b) Given that maize is planted at a spacing of 75cm by 25cm, calculate the plant population in a
plot measuring 4m by 3m
 7. Give **four** qualities of a mother plant which should be considered when selecting vegetative material for propagation

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8. Explain **five** practices that a farmer should carry out to ensure uniform germination of seeds
9. State **two** factors which determine the depth of planting
10. State **two** advantages of adding organic matter to sandy soil
11. Calculate the number of tea plants in two hectares (2ha) given that the spacing is 150cm x 75cm
and one seedling is planted per hole
12. Outline **four** reasons why training is important in some crops
13. Give **four** factors that influence the depth of planting
14. Two precautions taken when harvesting cotton
15. Outline **four** reasons why training is important in some crops
16. Give **four** factors that influence the depth of planting
17. Two precautions taken when harvesting cotton
18. State **four** factors that determine the spacing of annual crops
19. Outline **four** advantages of rolling in seedbed preparation
20. List **two** factors that effect rooting of cuttings in crop production
21. Outline **three** ways of preparing materials before sowing
22. Outline **three** ways of preparing materials before sowing
23. Distinguish between over sowing and under sowing
24. Study the illustration below of a tea vegetative material and answer the questions that follow



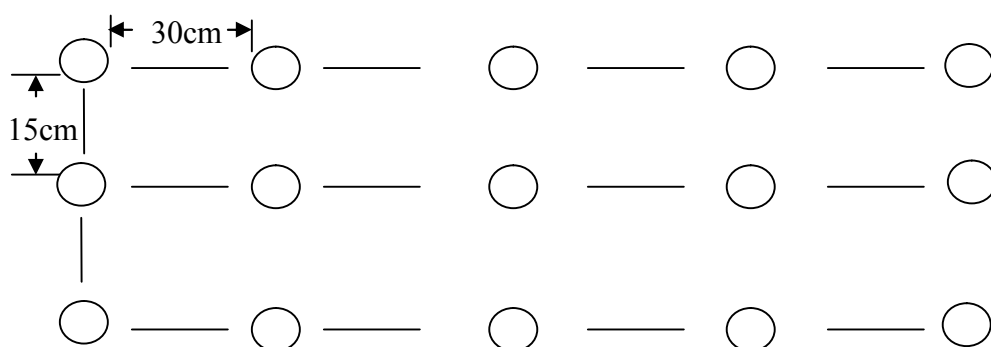
a) What name is given to the vegetative material drawn above for tea propagation

b) State **two** devisable characteristics of the selected plants used to develop the plant shown

c) Give **two** precautions observed during the preparation of the material above before planting

25. The diagram below illustrates the spacing which is used when planting beans. Study the diagram

and answer the questions that follow:



a) State the spacing illustrated above

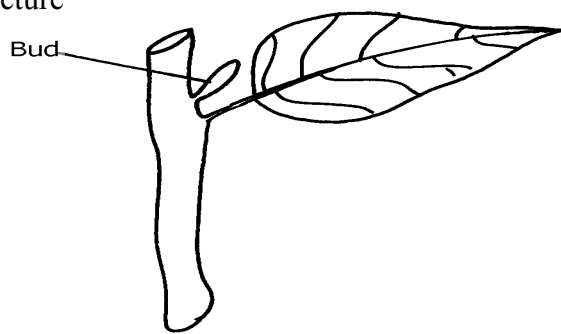
b) Suppose the student is asked to use the illustrated spacing to plant in a plot 4m by 3m leaving

30cm distance from the edge; calculate;

i) The number of rows on the wider side of the plot

ii) Calculate the plant population

26. Using planting material whose diagram is shown below, list **four** factors that would influence the rooting of the structure



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27. Describe the selection, preparation and raising of vegetative tea seedlings in the nursery
28. Explain the factors considered in timely planting of annual crops
29. Give four disadvantages of broadcasting as a method of planting.
30. Define the following terminologies as used in Agriculture
31. Give two advantages of producing crops by use of seeds over vegetative propaganda
32. State **four** ways of preparing planting materials before planting

CROP PRODUCTION III **NURSERY MANAGEMENT PRACTICES**

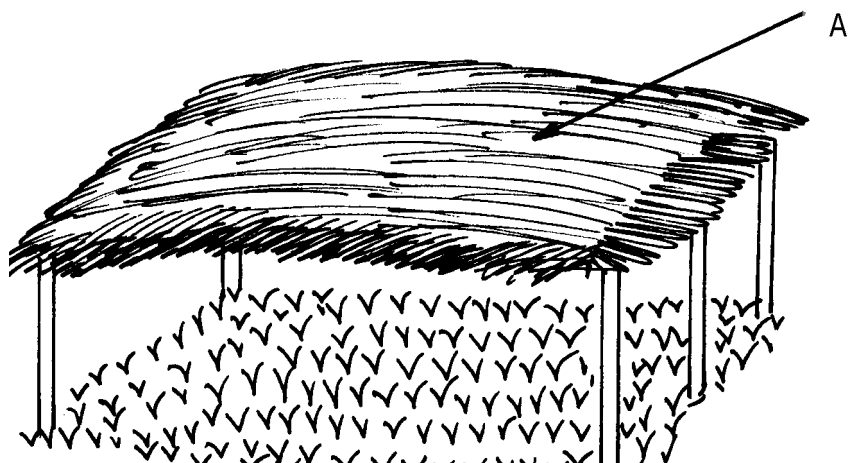
This topic entails the following:

- A nursery bed
- A nursery bed and a seed bed
- Reasons of establishing nursery bed
- Suitable site for nursery bed
- Nursery bed preparation
- Nursery bed management practices
- Transplanting seedling crops from nursery bed
- Budding a seedling
- Grafting a seedling
- Reasons for budding, grafting and layering
- Tissue culture
- Damage caused by animals to a seedling and prevention.

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the required concepts and practices.

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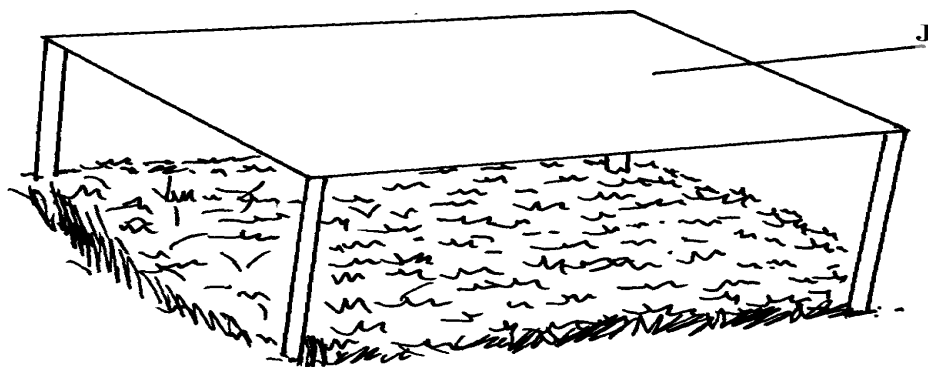
1. Name **three** methods of grafting that are used in propagation of plants
2. State **two** practices done during hardening-off of seedlings in a nursery bed.
3. List **two** methods of budding used in crop propagation
4. List **four** management practices carried out on a nursery bed
5. Outline **two** importance of tissue culture in crop propagation
6. Differentiate between a nursery bed and a seedling bed
7. Give **four** advantages of under sowing in pasture production
8. Give **four** advantages of under sowing in pasture production
9. The diagram below shows a structure used in crop production:



- (a) Identify the structure above
 - (b) Give a reason for carrying out each of the following practices in the structure shown above
 - (i) Pricking out
 - (ii) Hardening off
 - (c) State three importance of the part labeled A in the above structure
10. (a) Describe the siting and establishment of a crop nursery
- (b) Explain management practices in a crop nursery

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11. State **four** importance of thinning seedlings in the nursery bed
12. State the difference between a seedling bed and a seedbed.
13. Below is a diagram of a nursery for raising the seedlings



- (a) State **two** advantages of having the part labeled J
- (b) State any **three** management practices that should be carried out on the nursery from the time seedlings emerge to the stage of transplanting

CROP PRODUCTION IV (FIELD MANAGEMENT PRACTICES)

This topic entails the following:

- Crop rotation
- Reasons for crop rotations
- Crop rotation programme
- Terms used in crop farming
- Mulching
- Reasons for various field management practices
- Correct stage for harvesting crops
- Harvesting practices of various crops.

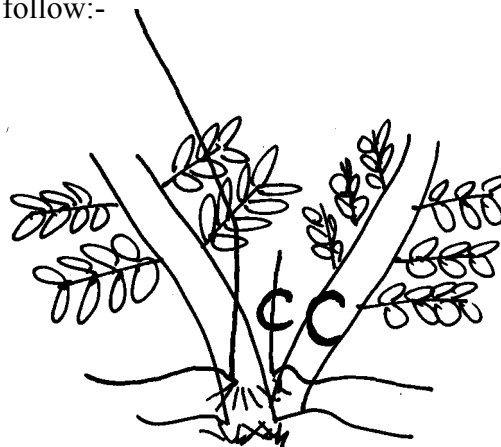
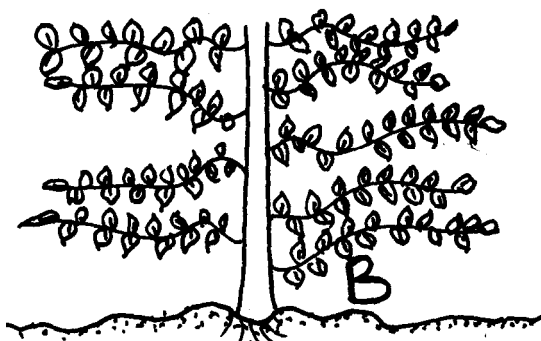
The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Distinguish between **staking** and **propping** as a field management practice on crops
2. Explain five advantages of crop rotation

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3. State **four** factors which influence the stage at which the crops are harvested
4. The diagrams labelled **B** and **C** below are illustrations of coffee plants established using two different formative pruning systems.

Examine the diagrams and answer the questions that follow:-

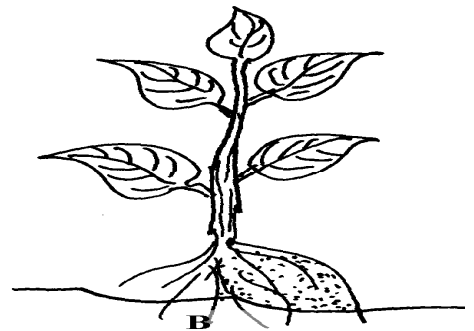
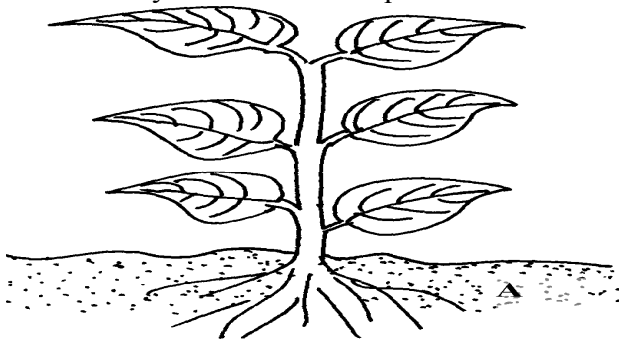


- (a) Identify the system of pruning illustrated in **B**
- (b) Identify the system of pruning in **C**
- (c) Outline the procedure of how pruning in diagram **C** is carried out
5. Give **two** functions of earthing up in crop production
6. Describe the factors which determine the stage of harvesting of crops
7. Give **two** ways in which inorganic much helps to conserve water
8. State **four** factors that determine the spacing of annual crops
9. Explain the importance of each of the following practices: (i) Hardening off
(ii) Pricking out
(iii) Gapping
10. Outline **two** factors that determine the stage of harvesting crops
11. (a) List **four** factors that determine harvesting sage of a crop

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- (b) Give **four** practices that can be used to control storage pests
12. List **four** benefits of pruning in crop production
13. (a) What is winnowing?
- (b) Give **one** importance of the following practices: (i) Mulching
(ii) Threshing
- (c) Which factors are considered when carrying out a crop rotation program?
14. What is frelishing?
15. Briefly explain how each of the factors listed below will determine the stage at which a crop is harvested
- (a) Intended use of the crop
- (b) Market demand
16. What is roguing in crop production?
17. What is meant by the term “changing the cycle” in coffee growing?
18. The diagram below shows a practice carried out on various crops on the farm. Study them

carefully and answer the questions that follow;



- (a) Identify the farm practice represented by **B**
- (b) State the importance of the above practice in the following crops;
- (i) Maize
(ii) Irish potatoes

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(c) At what stage of growth should the above practice be carried out in maize?

19. Mention **four** factors which determine the stage at which crops are harvested
20. State **two** limitation of using polythene sheets as mulching materials in a field of tomatoes
21. Give **two** management practices carried in a banana stool
23. State **two** functions of polythene sheet when used as mulch material
24. Give **four** crops requiring training

CROP PRODUCTION V (VEGETABLES)

This topic entails the following:

- Growing or production of a vegetable crop from nursery establishment to harvesting
- Keep records of crop production.
- Market vegetable crop produce
- Give reasons or importance of growing vegetable crops.
- The vegetable crops include the following: Tomatoes, cabbages, onions, carrots, kales.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. The diagram below is of a tomato plant. Study it and answer the questions that follow:-



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- a) State **three** management practices that have not been carried on the plant above
- b) For each management practice state **one** reason why it should be carried out
- c) Name **two** diseases that attack the crop above in the field

2. Describe the production of tomatoes (*lycopersiconesculentum*) under the following subheadings

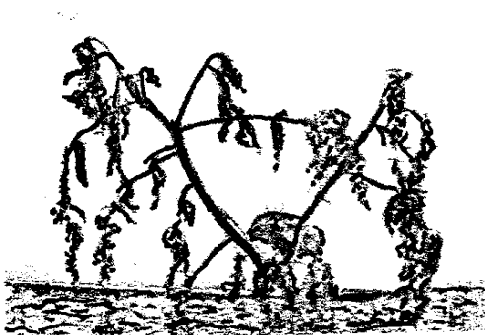
- a) Varieties
- b) Nursery establishment
- c) Field management practices

3. List **four** symptoms of late blight in tomatoes

4. State any **four** factors considered when grading tomatoes for fresh market

5. State **two** ways of controlling purple blotch in onions

6. The following is an illustration of an infected tomato plant. Study it carefully and answer the questions below:-



- (a) Identify the disease which may have caused the condition shown in the illustration
 - (b) Name any other crop which may be affected by the disease identified in (a) above
 - (c) Mention **two** other factors which can lead to the same condition as shown by the illustration
 - (d) State **two** measures that can be used to control the disease named in (a) above
7. Give **two** ways in which pruning helps to control diseases in tomatoes
8. Outline **four** ecological requirements for cabbages

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9. a) Mention **two** pests which attack tomatoes
b) Give **two** causes blossom end rot disease in tomatoes
10. List three ecological requirements of tomatoes.

LIVESTOCK HEALTH **(INTRODUCTION TO LIVESTOCK HEALTH)**

This topic entails the following:

- Definition of Health and disease.
- Signs of sickness in animals livestock diseases
- Categories of livestock diseases
- Reasons for keeping livestock in good health
- Disease control practices
- Appropriate methods of handling livestock.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. Identify **four** physical appearances to be observed in a sick animal
2. State **two** reasons why tsetse fly control is considered to be a land reclamation method
3. a) Explain **five** factors to consider when siting a fish pond
b) Explain the measures used to control livestock diseases
4. a) Name **four** notifiable diseases in livestock
b) Discuss **four** ways in which livestock disease are spread in the farm
c) Describe the methods of controlling livestock disease giving an example of different disease in each case

AGRICULTURAL ECONOMICS II **(LAND TENURE AND LAND REFORM)**

- Definition of land tenure.
- Description of tenure systems.
- Descriptions of land reforms.

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The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. State **four** ways by which Re-afforestation help in land reclamation
2. State **three** objectives of land reforms that are taking place in Kenya
3. State **two** causes of land fragmentation in Kenya since independence
4. Outline the process followed in land adjudication
5. State **four** benefits of a farmer having land title deed
6. State **four** reasons for practicing land consolidation
7. Give **four** advantages of communal land tenure system
8. State **four** advantages of landlordism and tenancy
9. Outline **four** objects of land tenure reform
10. State **three** advantages of communal land tenure system
11. List down four important details in a land title deed.

SOIL AND WATER CONSERVATION

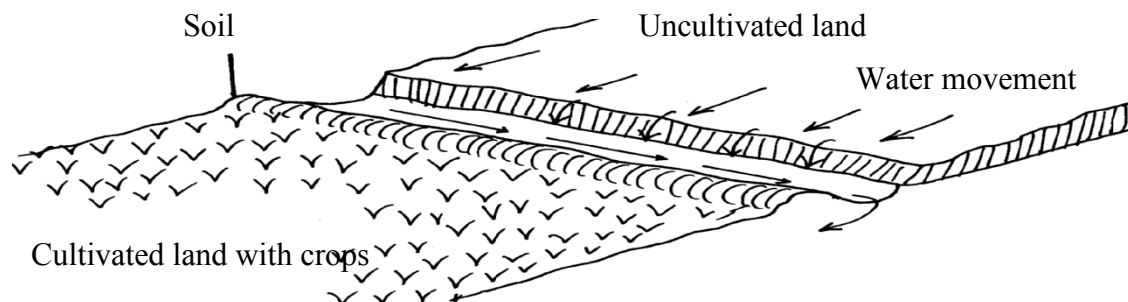
This topic entails the following:

- Definition of soil erosion
- Explanation of various factors which influence erosion.
- Agents of erosion
- Description various methods of erosion
- Description of various methods of erosion control
- Description of micro-catchments and then uses.

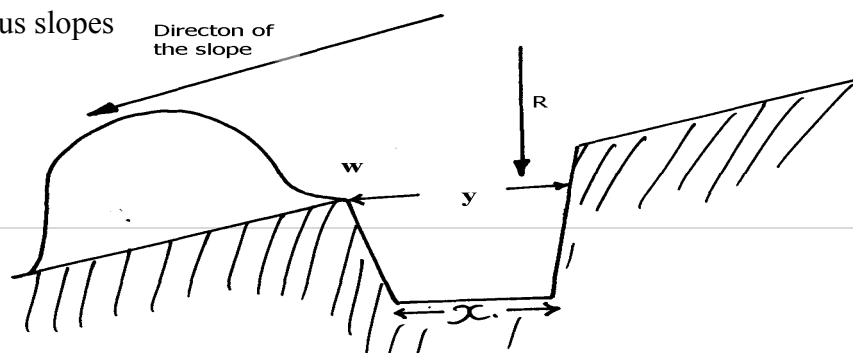
The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Name **three** human activities that may influence soil erosion
2. Below is a diagram showing soil erosion control method

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- a) Identify the structure used to control soil erosion
 - b) What is the function of the structure made
 - c) Why was soil not put on the upper side of the trench made
 - d) State **four** effects if water was allowed into the cultivated land
3. Give **two** roles played by Grassley in soil erosion control
 4. List **three** materials that may be used for constructing a gabion
 5. State **one** factor that would determine the width and depth of a cut off drain
 6. Explain **five** ways by which grass helps to conserve soil
 7. Give **four** farming practices that help in reducing the effects of water shortage in crop production
 8. Name **two** types of terraces
 9. Name **two** forms of gully erosion
 - 9 -V- shaped
 - U – shaped
 10. Explain the cultural methods of soil erosion control
 11. Mention **four** control measures of river bank erosion
 13. The figure below represents a physical soil and water conservation measure used on various slopes



- a) Identify the measure represented above
 - b) Describe the construction of the identified measure above
15. a) What is soil erosion
- b) Give **four** types of water erosion
- c) Explain factors which influence soil erosion
- d) State any **seven** cultural ways of controlling weeds
16. Give **two** ways through which gabions control soil erosion

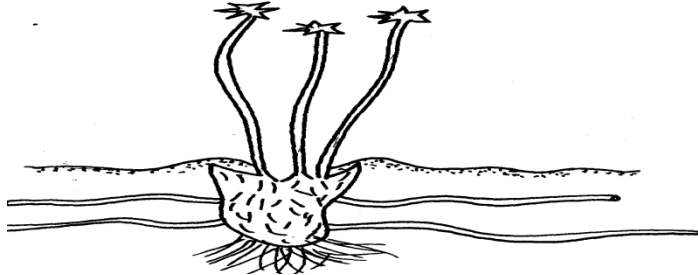
WEEDS AND WEED CONTROL

This topic entails the following:

- Definition of a weed
- Identification of weeds
- Classification of weeds
- Competitive ability of weeds
- Description of weed control methods
- Harmful effects of weeds

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

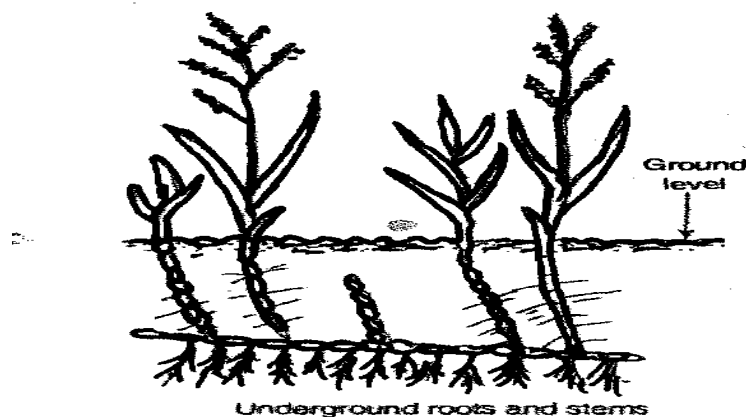
1. Study the diagram below and answer the questions that follow:



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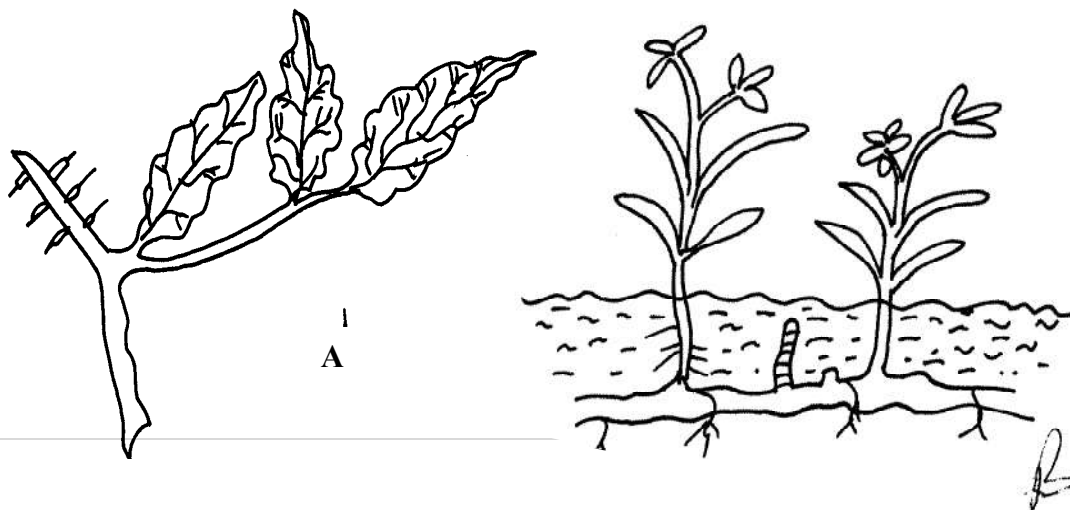
- (a) Identify the weed
- (b) Why is it difficult to control the weed?
- (c) State the economic importance of the weed shown above

2. The following is an illustration of a common weed of arable land



- (a) Identify the weed
- (b) Why is it difficult to control the weed?
- (c) State **one** harmful effect of the weed on crop production
- (d) Give **two** measures used to control the weed

3. Below is a diagram of some common weeds. Study then and answer the questions that follow:-



(i) Identify weed **A** and **B**

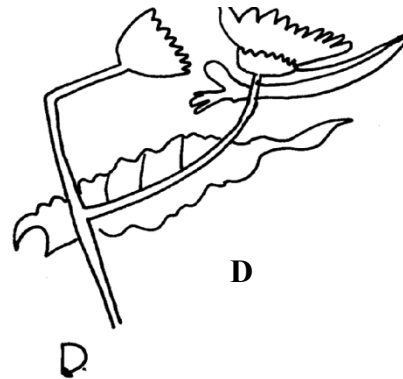
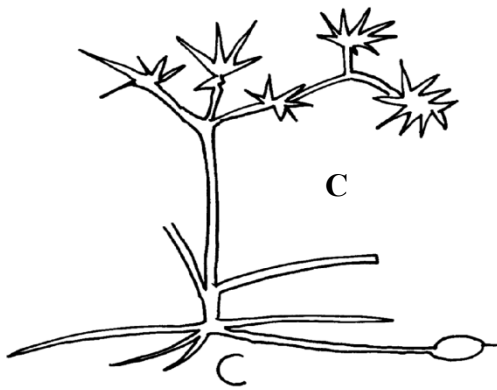
(ii) State **two** effects of weed **B** on crop production

(iii) Give a reason why weed **B** is difficult to control

(iv) State the effective method of controlling weed **B**

4. Give **four** methods of propagation which make weeds have a high competitive ability over crops

5. The diagrams below show weeds:-



Identify the weeds **C** and **D**

(b) State why it is difficult to control weed **C**

(c) State **two** economic importance of weed **D**

6. (a) List any **four** types of weeds known to be poisonous to livestock

(b) At what stage of life cycle is best recommended to control weeds?

7. State **two** disadvantages of using herbicides

8. a) Name **four** herbicides that can be used to control weeds in a field of maize

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- b) At what stage of growth of maize should the weeds be controlled by use of a post emergence herbicides
9. (b) Give an account of economic importance of weeds
10. a) mention **two** factors that affect selectivity of herbicides
11. Listing specific examples of weeds describe their harmful effects in agricultural production

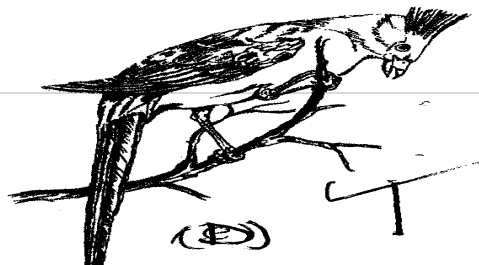
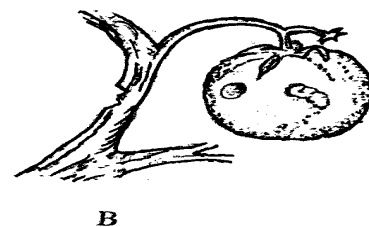
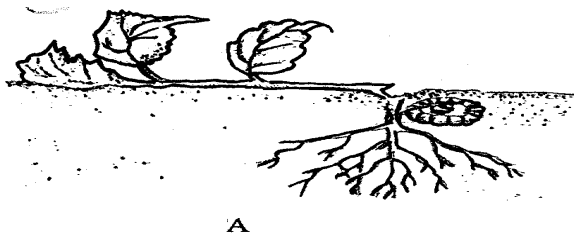
CROP PESTS AND DISEASES

This topic entails the following:

- Definition of a pest and a disease
- Main cause of crop diseases
- Description of harmful effects of crop pests and diseases
- Identification of crop pests and diseases
- Control measures of crops pests and diseases

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. State **two** possible causes of swelling on the roots of legume crops
2. Discuss the various cultural methods of controlling pests in crops
3. What does the term 'close season' mean in crop production?
4. Explain **five** cultural methods of pest control in stored grains
5. Name **four** symptoms of viral infections in plants
6. Study the illustrations given below and then answer the questions that follow:-



C

D

(a) Identify the pests shown by the illustrations **B**, **C** and **D**

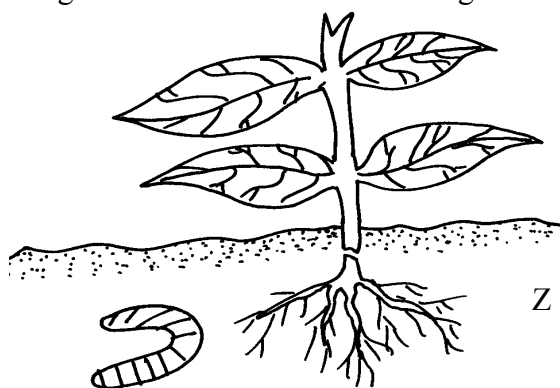
(b) State **one** effective method of controlling the pest labelled

(c) Name the type of crop commonly attacked by the pest labelled ‘**C**’

7. State the various practices carried out in the field to help control crop diseases

8. State **two** ways in which pesticides kill crop pests

9. The diagram below shows kale seedling attacked by a pest



(a) Identify the pest

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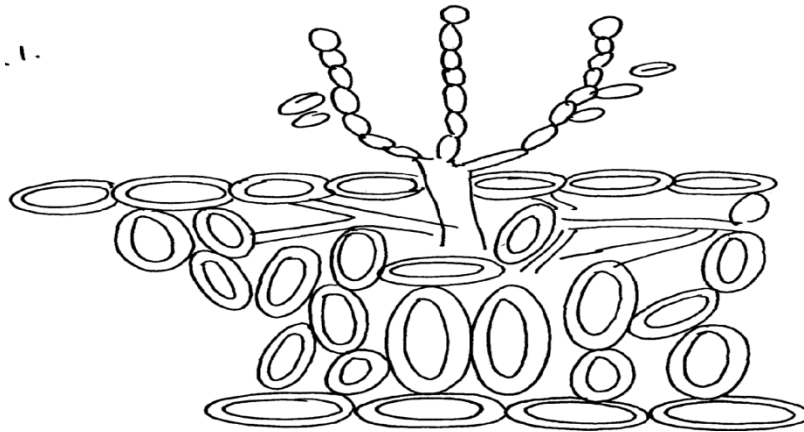
(b) What damage does the pest cause to the crop

(c) State **two** methods of controlling the pest

11. Define the term 'economic injury level of a pest and integrated pest control management

12. List **four** harmful effects of crop pests

13. The diagram below shows a fungal disease in a section of potato crop leaf



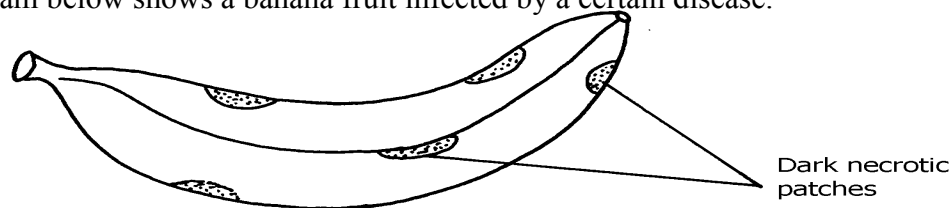
a) Identify the fungal disease

b) (i) State the causal organism of the disease identified above

ii) Give **two** symptoms of the above disease

c) Give **two** control measures of the disease

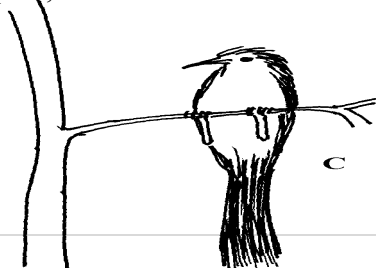
14. The diagram below shows a banana fruit infected by a certain disease.



(a) Identify the disease

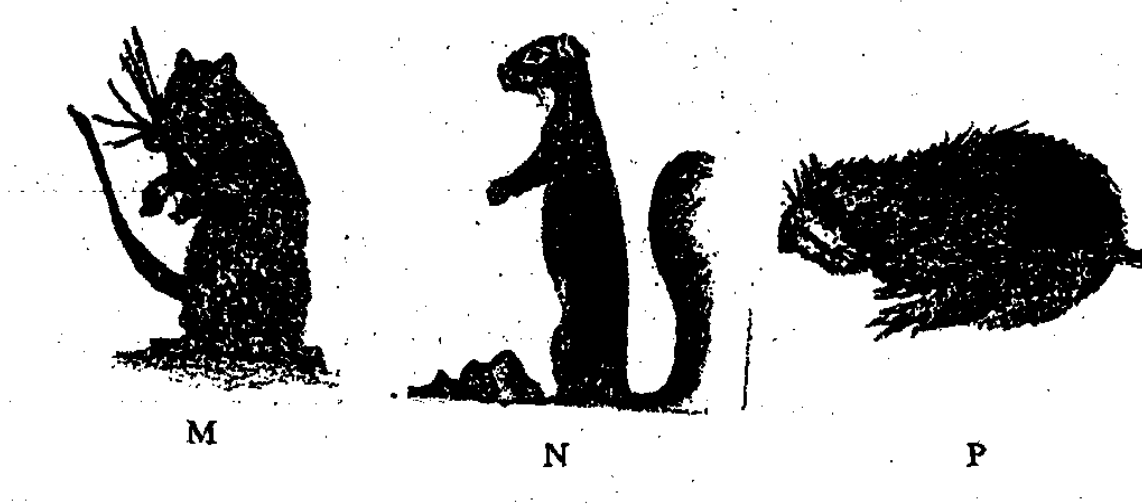
(b) Suggest any **one** control measure for the disease you have named in (a)

15. The diagram below shows a crop pest;



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- (a) Identify the crop pest
- (b) State **two** damages the pest would cause to crops
16. Give two symptoms of nematode attack on bananas.
17. Give **four** cultural practices used in controlling crop pests
18. State **three** symptoms of coffee berry disease.
- 19 (i) Use the diagrams below to answer the question that follow.



- (a) Identify the crop pests labeled M, N and P
- (b) State **one** control measures of crop pests labeled M
- (c) State the damage caused to crops by crop pest labeled N
20. State and explain the cultural methods of pest control

CROP PRODUCTION VI

FIELD PRACTICES FOR MAIZE, MILLET, SORGHUM, BEANS AND RICE: HARVESTING OF COTTON PYRETHRUM, SUGAR CANE COFFE AND TEA

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This topic entails the following:

- Description of management practices of the food crops from planting to harvesting and marketing.
- Economic value of food crops and industrial crops

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. List **four** management practices carried out in maize field at 45cm high
2. Why is it not recommended to use sisal bags in handling cotton
3. Describe the production of rice under the following sub-headings:-
 - (a) Land preparation
 - (b) Water control
 - (c) Fertilizer application
 - (d) Weed control
 - (ii) Describe the environmental conditions that may lead to low crop yields
4. Describe the production of maize for dry grain production under the following sub-headings:-
 - (a) Seedbed preparation
 - (b) Planting
 - (c) Weed control
 - (d) Pest control
 - (e) Disease control
 - (f) Harvesting
5. Two precautions taken when harvesting cotton
6. Describe production of maize under the following sub-headings ;
 - (a) Varieties

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- (b) planting
 - (c) pest and pest control
 - (d) harvesting and storage
7. Describe the field production of maize under the following sub headings
- a) Ecological requirement
 - b) Varieties
 - c) Seedbed preparation
 - d) Pests and diseases
 - e) Harvesting
8. a) Discuss harvesting of cotton
- b) Explain the roles of Agricultural co-operatives in Kenya
9. Describe the production of maize under the following sub-headings:
- (a) Ecological requirements
 - (b) Land preparation
 - (c) Planting and field management
 - (d) Pests and disease control
 - (e) Harvesting and marketing
10. Give **two** precautions measures a farmer should put into consideration when harvesting cotton
11. Describe the production of beans under the following sub headings:
- (a) Ecological requirements
 - (b) Seedbed preparation
 - (c) Planting
 - (d) Pest and pest control

FORAGE CROPS

This topic entails the following:

- Definition and classification of pasture crops.
- Identification of pasture crops
- Description of ecological requirements of forage crops
- Description of the establishment of pasture and fodder crops
- Description of forage utilization and conservation.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. State **three** factors which affects the quality of standing forage given to livestock
2. Explain Napier grass production under the following sub-headings
 - i) Seedbed preparation
 - ii) Planting
 - iii) Fertilizer application
 - iv) Weed control
 - v) Utilization
3. Why is it necessary to allow freshly cut nappier grass to wilt before ensiling
4. Describe the establishment of grass pasture from the time the land is ploughed using a mould board plough to the time the pasture is ready for grazing
5. State **two** causes of failure in pasture establishment
6. State **two** advantages of grass-legume mixture
7. List **four** factors that determine the quality of hay
8.
 - i) Discuss the production of Guatemala grass (*Trypsacum Laxum*) under the following headings
 - a) Ecological required

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b) Land preparation

c) Planting

d) (i) Utilization and defoliation

ii) Discuss **six** effects of late defoliation of fodder

9. Give **four** factors that determine the nutrient content in hay

10. (a) Describe the preparation of the following farm materials:-

(i) Farm yard manure

(ii) Hay

(b) Explain the factors considered in timely planting of annual crops

11. (a) What is topping in pasture management

(b) State two methods used in topping in pasture management

12. Give **two** advantages of grass-legume pasture over pure grass pasture

13. Why are farmers encouraged to conserve excess forage in the farm?

14. Give **two** factors affecting the quality of hay

AGRICULTURAL ECONOMICS III **(PRODUCTION ECONOMICS)**

This topic entails the following:

- Parameter of national development
- Factors of production
- Law of diminishing returns
- Farm planning and budgeting
- Agricultural services
- Risks and uncertainties
- How to adjust to risks and uncertainties.

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The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. State **four** ways of increasing labour efficiency on the farm
2. The table shows egg production from individual birds with varying mounts of layers mash

100 layers (Fixed number)	Layers mash Kgs/week	Total egg production per week	Marginal production per week
100	0	140	0
100	10	155	15
100	20	180	25
100	30	240	60
100	40	340	100
100	50	470	130

- (a) Sketch a graph representing the total egg production per week against amount of feed given
 - (b) Identify the type of production function represented by the graph in (a) above
3. (a) What are the uses of farm records to a farmer?
(b) Explain **four** ways in which a farmer may improve Labour productivity in the farm
(c) Outline the process followed in land adjudication
 4. Name any **three** types of agricultural services available to the farmer
 5. Outline **four** management guideline questions which assist a farm manager in making accurate farm decisions
 6. Give **four** ways of improving labour productivity
 7. List **four** variable inputs in sorghum production
 8. List **four** agricultural support services available to a crop farmer in Kenya
 9. Define the following as used in Agricultural economics:-

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(a) Gross domestic product (GDP)

(b) Per capita income

10. Explain the various ways in which farmers may adjust to risks and uncertainties

11. (a) The table below represents the yield of maize in 90kg bags in response to application of

different quantities of planting fertilizer

(i) Fill in the blank spaces

Input 50kg bag fertilizer	Out put 90kg bag maize	Average product (AP)	Marginal product (MP)
0	6	-	-
1	10	-	-
2	24	-	-
3	31	-	-
4	36	-	-
5	40	-	-
6	43	-	-
7	43	-	-
8	40	-	-

(ii) Suggest the best level of production in relation to the inputs and output

(b) A farmer is considering undertaking the production of either maize or beans. Study the

following information about the two crops then answer the questions that follow:

(i) Maize

Yield per hectare

5,500 kg

Price

15 per kg

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Cost of cultivation / ha	Kshs. 3000
Amount of seeds/ha	25kgs
Cost of DAP fertilizer/bag	Kshs.1,500
Amount of DAP fertilizer/ha	3bags
Cost of seeds/kg	Kshs.100
Labour requirements/ha	50 man days
Cost of labour	Kshs.150 per man day
Amount of CAN fertilizer/bag	3 bags
Cost of CAN fertilizer/bag	Kshs.1000

(ii) Beans

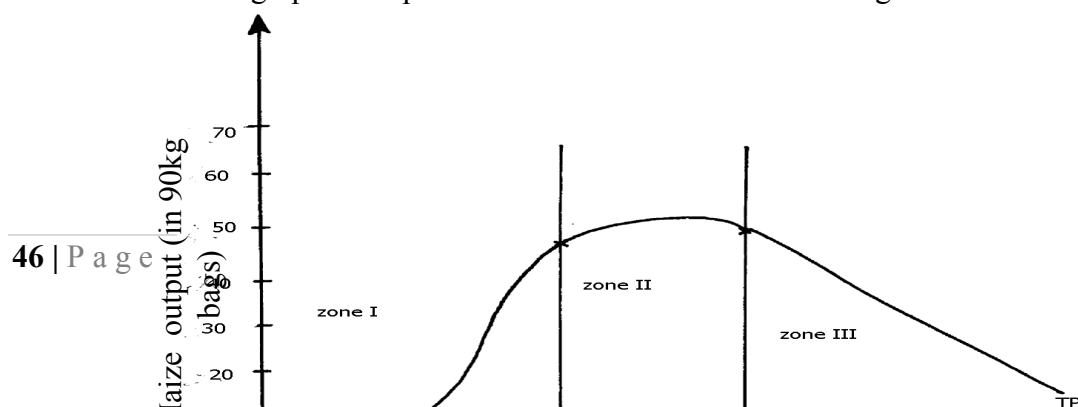
Yield per hectare	5000kg
Price	50per kg
Cost of cultivation / ha	KShs.3600
Labour requirements/ha	75 man days
Cost of labour	Kshs. 200 per man day
Cost of DAP fertilizer/bag	Kshs. 1500
Amount of DAP fertilizer/ha	2bags
Cost of seeds/kg	Kshs.800
Amount of seed/ha	20kg
Amount of CAN fertilizer/bag	1bag
Cost of CAN fertilizer/bag	Kshs.1,000
Cost of sprays	Kshs.3,000

(i) Calculate the gross margins for each crop

(14mks)

(ii) From your calculation, which crop is profitable to grow?

12. Below is a graphical representation of the law of diminishing returns.



NPK fertilizer input (in 50kg bags)

- (a) Explain what happens in each of the three zones marked I and III in relation to the output of maize and the NPK fertilizer input
- (b) Which of the three is a rational zone of production
13. Give **four** variable costs in maize production
14. A farmer has the following yield from a two hectare millet crop enterprise at Oluch irrigation schemes.

Study it and prepare his gross margin. is it profitable to grow millet? He spent the following in his operations

Weed	800/=
Seeds	20kg/ha
Irrigation	600/= /ha
Ploughing	500/= /ha
Clearing the land	1200/=
Cost of seeds	300/= /10kg bag
Planting	400/= /ha
Harvesting	1200/= /ha
Yield	32bags
DAP fertilizer	2 bags at 10 000/= /50kg bags
CAN fertilizer	2 bags at 700/= /50kg bags
Gunny bags	40/= /bag
Transport to market	2000/=

14. A farmer has the following yield from a two hectare millet crop enterprise at Oluch irrigation schemes. Study it and prepare his gross margin. is it profitable to grow millet? He spent the following in his operations

FREE AGRICULTURE K.C.S.E 2012 to 2015

Weed	800/=	
Seeds	20kg/ha	
Irrigation	600/=/ha	
Ploughing	500/=/ha	
Clearing the land	1200/=	
Cost of seeds	300/= /10kg bag	
Planting	400/= /ha	
Harvesting	1200/= /ha	
Yield	32bags	
DAP fertilizer	2 bags at 10 000/= /50kg bags	
CAN fertilizer	2 bags at 700/= /50kg bags	
Gunny bags	40/= /bag	
Transport to market	2000/=	(20mks)

15. What is profit maximization in Agricultural Economics
16. a) A farmer is considering undertaking the production of either maize or beans. Study the following information about the two crops and then answer the questions that follow:

Maize	
Yield per hectare	5500kg
Price	Kshs.15 per kg
Cost of cultivation/ ha	Kshs. 3000/=
Amount of DAP fertilizer/ bag	Kshs.1500/=
Amount of DAP fertilizer/ ha	3 bags
Cost of seed/ Kg	Kshs.100
Labour requirements / ha	50 man days
Cost of labour	Kshs.150 per man day
Amount of CAN fertilizer	3 bags
Cost of CAN fertilizer/ bag	Kshs.1000

Beans	
Yield per hectare	Kshs.5000
Price	Kshs.50 per kg
Cost of cultivation/ ha	Kshs.3600
Labour requirements/ ha	75 man- days
Cost of labour	Kshs.200 per man day
Cost of DAP fertilizer/ bag	Kshs.1500

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Amount of DAP fertilizer/ ha	2 bags
Cost of seed/ kg	Kshs.80
Amount of seed/ ha	20kg
Amount of CAN fertilizer/	1 bag
Cost of CAN fertilizer/ bag	Kshs.1000
Cost of sprays	Kshs.3000

i) Calculate the gross margin for each crop

ii) From your calculation which crop is profitable to grow

b) Discuss **five** factors considered when planning a farm

17. Using the data provided in the table below, make an interpretation and advice the farmer on which crop to grow ;

Type of crop	Gross margin (Ksh)
Cotton	18,400
Ground nuts	20,050

18. Outline **three** advantages of budgeting in farm business

19. A farmer has 1 Ha piece of land on which he grows maize. His farm record on maize production for nine years is as shown in the table below:

Year	Fertilizer applied (bags)	Total output of maize (bags)
1995	0	4
1996	2	10
1997	4	28
1998	6	42
1999	8	52
2000	10	60
2001	12	66
2002	14	66
2003	16	64

- (a) i) Using an appropriate scale, with input on the X-axis draw a graph to show the relationship between inputs and total output

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(ii) From the graph you have drawn, how many bags of maize would the farmer produce if

he applied 9bags of fertilizer?

Calculate the farmers marginal products and average products for the years

(i) From the data given, what rate of fertilizer application would the farmer choose if he

wanted to grow maize in 2004?

(ii) Give an explanation for your choice in (c) (i) above

(b) Assuming that the average price of fertilizer over the years recorded was shs. 1,200/= per bag

and the price of maize was ksh.1000/= per bag :

Calculate the gross income for the years 2002 and 2003

Calculate the net income for the year 1999. (Assume no other costs were incurred)

20. Name five types of costs incurred in a farming business

22. List any four sources of credit to farmers.

23. List **three** ways in which labour peaks can be overcome in the farm (1½ mks)

24. State **four** ways of improving farm labour productivity

25. A farmer had a plot of land measuring 5 hectares in which he intended to plant maize. He was advised to apply 150 kg of P₂O per hectare at planting and 200kg N per hectare during

top dressing. The fertilizer available in the market was Calcium Ammonium Nitrate containing

20% N and Di-ammonium phosphate 46% P₂O₅. Calculate.

(a) (i) The amount of Di—ammonium phosphate required

(ii) The amount of calcium ammonium nitrate required

(b) Baraka farm manager plans to grow Irish potatoes or maize for grains. Study the information

below and answer the questions that follow:

Irish potatoes

Cost of fertilizers/ha _____ Kshs 10,000.

Labour requirements/ha _____ Kshs 50 man - days

Yield /ha _____ 10,000kg

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Seed potato/ha _____ Kshs 20, 000
Cost of labour _____ Kshs 200 per man day
Cost of fungicides _____ Kshs 5000
Cost of ploughing _____ Kshs 4000
Selling price of potatoes per kg _____ Kshs 30.

Maize

Yield per hectare _____ Kshs. 7,500kg
Selling price of maize per kg _____ Kshs 20.
Cost of ploughing /ha _____ Kshs. 4000
Seed maize/ha _____ Kshs. 3000
Labour requirement /ha _____ 200 man days.
Cost of fertilizers /ha _____ Kshs 10,000
Cost of top dressing fertilizers _____ Kshs 4,800
Cost of labour _____ Kshs 150 per man - day

- (i) What is gross margin?
- (ii) Calculate the gross margin of each of the crops
- (iii) From the calculation above which crop should the farm grow?
- (d) Describe the environmental factors that may lead to poor yields in crop production

AGRICULTURAL ECONOMICS IV

This topic entails the following:

- Importance of farm accounts
- Financial documents and their uses

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Analysis of financial statements
- Books of accounts and their uses.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Name **three** methods of grafting that are used in propagation of plants
 2. a) The following transactions were extracted from Mr. Tembo's financial books for the year ending 31st Dec 2003. study and answer the questions that follow:

Particulars	cost (ksh)
Milk sale	8 000
Goat sale	500
Purchase of farm tools	1 000
Construction of zero grazing unit	10 000
Depreciation of machinery	800
Closing stock	16 000
Veterinary bills	400
Interest payable	750
Wages	4 800
Sales of cabbages	750
Sales of tea	4 700
Opening stock	12 000
Sales of heifers	9 400
Purchase of pesticides	300

 - (a) i) Prepare a profit and loss account for Mr. Tembo's farm for the year ending 31st Dec. 2003
 - ii) Calculate the percentage profit or loss made by the farm
 - b) i) Give **five** functions of farmer's cooperative societies
 - ii) Outline **five** common risks and uncertainties in farming
3. State **four** reasons for using certified seeds for planting
4. List any **two** financial statements which may be prepared on a farm
5. The following information was obtained from Lang'at's farm records for the year ending

FREE AGRICULTURE K.C.S.E 2012 to 2015

December, 2004. Study it and answer the questions that follow:-

Goats	4,000
Poultry	15,000
Causal workers	12,000
Opening valuation	150,000

His sales and receipts are as follows:

Mohair	75,000
Rabbits	3,600
Eggs to hotel	15,000
Closing valuation	200,000

(a) Prepare the profit and loss A/C of Lang'at's farm

(b) State the benefit of a profit and loss A/C to Mr. Lang'at

6. (a) List any **four** financial documents used in the farm

(b) Prepare a profit and loss account for Mr. Rob's farm for the year ending 31st Dec. 2009, given

the following information:-

Sale of milk	Kshs.10,000
Sold two heifers	kshs.10,000
Cabbage sold	Kshs. 20,000
Debts payable	Ksh.4,200
Sold tomatoes	Kshs. 3,000
Veterinary bills	Kshs.2,500
Bought livestock feeds	Kshs.2,500
Purchase fertilizers	Kshs.5,000
Bought seeds	Kshs. 4,000
Debts receivable	Kshs.20,000
Opening valuation	Kshs.150,000
Closing valuation	Kshs.200,000

(c) Did the farm make a profit or a loss? Calculate the percentage profit or loss made by the Farm

FREE AGRICULTURE K.C.S.E 2012 to 2015

(d) Explain the various ways in which farmers may adjust to risks and uncertainties

7. a) The following transactions were extracted from Mr. Tembo's financial books for the year

ending 31st Dec 2003. study and answer the questions that follow:

Particulars	cost (ksh)
Milk sale	8 000
Goat sale	500
Purchase of farm tools	1 000
Construction of zero grazing unit	10 000
Depreciation of machinery	800
Closing stock	16 000
Veterinary bills	400
Interest payable	750
Wages	4 800
Sales of cabbages	750
Sales of tea	4 700
Opening stock	12 000
Sales of heifers	9 400
Purchase of pesticides	300

i) Prepare a profit and loss account for Mr. Tembo's farm for the year ending 31st Dec 2003

ii) Calculate the percentage profit or loss made by the farm

b) i) Give **five** functions of farmer's cooperative societies

ii) Outline **five** common risks and uncertainties in farming

8. At the end year ended 31/12/2005 Bidii farm recorded the following:

Perennial crops	250,000
Bank loans	30,000
Cash at hand	5,000
Bank overdrafts	15,000
Land	350,000
Unpaid wages	3,000
Debts receivable	20,000
Stocks in store	25,000
Livestock	200,000
Bank balances	100,000

FREE AGRICULTURE K.C.S.E 2012 to 2015

(a) Prepare a balance sheet as at 31/12/2005

(b) Did Bidii farm qualify for a loan and why?

9. State **one** condition in which each of the following documents is used.

i) Invoice

ii) Delivery note

iii) Receipt

10. Below is a transaction showing Mrs. Okello's financial position in her business for the year 2009

-Purchase of pesticides	3,000 00
-Milk sales	8,000 00
-Sales of goats	5,000 00
-Construction of store	10,000 00
-Closing valuation	16,000 00
-Depreciation of machinery	3,000 00
-Interest payable	1,750 00
-Purchase of farm tools	800 00
-Veterinary bills	1,400 00
-Sales of tomatoes	1,750 00
-Wages	10,000 00
-Sales of heifer	10,000 00
-Opening valuation	12,000 00
-Sales of coffee	5,000 00

i) Prepare a profit and loss account for Mrs. Okello's farm

ii) Calculate the percentage profit or loss that Mrs. Okello made during the year 2009

iii) Explain **six** ways in which farmers adjust to risk and uncertainties in farming

11. Name **two** examples of liabilities in a balance sheet

AGRICULTURE ECONOMICS (V)

This topic entails the following:

- Market and marketing
- Types of markets
- Supply and demand

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Marketing functions
- Problems of marketing
- Agricultural organizations

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. a) Explain the principle that govern the operations of farmers' co-operative societies
b) Explain the role of agricultural cooperatives in Kenya
c) Explain various functions of agricultural marketing
2. a) Give **four** marketing functions
b) Outline **four** problems associated with marketing of agricultural products
3. State **two** roles of agricultural society of Kenya
4. (a) What is elasticity of demand for a commodity

(b) Given that at a price of shs.1000 per bag, 20 bags of maize are demanded but when the price changes to shs.800 per bag, 22 bags are demanded. Calculate the elasticity of demand. Show your working

(c) Outline **six** problems of marketing maize as an agricultural product

(d) Determine **nine** principles governing cooperatives in Kenya
5. Outline **four** reasons why training is important in some crops
6. (a) Explain marketing activities in Agriculture

(b) Discuss problems experienced in marketing of Agricultural products
7. What is elasticity of supply

FREE AGRICULTURE K.C.S.E 2012 to 2015

8. How do the governments control prices of essential farm produce
 - b) Explain the roles of Agricultural co-operatives in Kenya
9. State the law of demand and supply.
10. What do the following initials stand for?
 - (i) K.N.F.U – Kenya National Farmers Union
 - (ii) H.C.D.A – Horticultural Crops Development Authority
11.
 - (a) What is a co-operative society
 - (b) List **two** functions of co-operatives
 - (c) State and explain the nine principles of governing co-operatives
12. Give **four** factors which influenced the demand of tomatoes in the market

AGROFORESTRY

This topic entails the following:

- Definition of agro forestry
- Importance of agro forestry
- Forms of agro forestry
- Importance of trees
- Selection of trees to plant
- Routine management practices on trees
- Methods of tree harvesting.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. State **two** reasons for seed treatment of tree species before planting
2. State **four** ways by which Re-afforestation help in land reclamation
3. List **four** advantages of agro-forestry

FREE AGRICULTURE K.C.S.E 2012 to 2015

4. The illustrations below are techniques of harvesting agroforestry trees. Study them carefully and then answer the questions below:-



A



B



C

- (a) Identify the harvesting techniques represented by techniques **A** and **B**
- (b) Give an example of a tree species suitable for technique **B** and **C** as a method of harvesting
5. State **four** factors considered when choosing trees for Agroforestry
6. (a) Five characteristics of trees used in agroforestry are;
(b) The benefits of agroforestry are:
7. Name **four** forms of agro- forestry
8. Give **four** characteristics that good agro-forestry tree should possess

ANSWERS SECTION 1

INTRODUCTION TO AGRICULTURE

1. two factors which characterize intensive farming
- Small farms
 - Huge capital

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Skilled labour
 - Produce for sale
 - Mechanization done
2. three reasons why organic farming is encouraged in farming
- Cheap
 - Environmental friendly
 - No chemical residues in produce
3. two ways in which Agriculture contributes to industrial development.
- Provide raw materials for industries.
 - Provide market for industrial goods.
 - Is a source of capital for starting industries.
4. four ways by which wind affects the growth of crops.
- Causes physical damage to crops.
 - Cause rapid spread of diseases/ pests/ weeds.
 - Can cause water stress as a result of evaporation.
 - Causes stress of crops due to chilling caused cold winds.
 - Encourage transpiration hence water and mineral uptake.
- Agriculture – rearing of fish in fish ponds
5. One physical characteristic used in classifying soil is:
- Colour,
 - Texture,
 - Structure
6. Four advantages of organic farming
- Environmental friendly
 - Products do not have organic farming
 - Products do not have organic chemical residue
 - Improve soil structure
 - Replenishes nutrients in the soil as it uses organic manure
 - Enhances soil water retention
 - Provides food for soil microbes
 - Enhances soil water infiltration
7. Two conditions under which shifting cultivation is practiced are:
- Can only be practiced where land is abundant
 - Practiced where population is sparse
 - Practiced where number of livestock per units low (2 ½mks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

8. Four factors that determine the time of planting a crop

- (a) Olericulture – production of vegetables
- Floriculture – production of flowers
- (b) Apiculture – bee keeping
- Agriculture – rearing of fish in fish ponds

FACTORS WHICH INFLUENCE AGRICULTURE

1. two roles of humus in the soil that are beneficial to crops

- Provide nutrients
 - Increase water holding capacity
 - Increase soil temperature
- Neutral soil PH

2. a) five activities that may be undertaken in organic farming

- Mulching
- Apply manure
- Use medicinal plants to control parasites and diseases
- Crop rotation
- Rear livestock on natural organically grown pasture
- Physical/ cultural/ biological/ pests, weeds and disease control

3. four effects of temperature on crop growth

Low temp-slow growth rate

- increase incidence of negative infection e.g. CBD
- improve quality of some crop

High temp-cause wilting

- increase growth rate
- improve quality of some crops
- increase pest and disease incidences (1/2x4=2mks)

4. four ways by which wind affects the growth of crops.

- Causes physical damage to crops.
- Cause rapid spread of diseases/ pests/ weeds.
- Can cause water stress as a result of evaporation.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Causes stress of crops due to chilling caused cold winds.
 - Encourage transpiration hence water and mineral uptake.
5. Two factors related to light that affect crop production and distribution in Kenya:-
- Light intensity
 - Light duration
 - Light wavelength
6. The environmental conditions that may lead to low crop yields
- Poor soil fertility /infertile soil
 - Damage by hailstorms
 - Less rainfall/unreliable/drought
 - Poor soil type resulting into leaching or water logging
 - Inappropriate soil PH
 - Inappropriate temperature (too low or high)
 - Excessive wind leading to increase in water loss from the soil
 - Extreme relative humidity
 - Extreme of light intensity
 - Topography / some attitudes e.g. very high may limit crop growth (1mk x any 7pts = 7mks)
7. - Rainfall
- Soil
 - Topography
 - Light
 - Wind
8. One physical characteristic used in classifying soil is:
- Colour,
 - Texture,
 - Structure
9. Four advantages of organic farming
- Environmental friendly
 - Products do not have organic farming
 - Products do not have organic chemical residue
 - Improve soil structure
 - Replenishes nutrients in the soil as it uses organic manure
 - Enhances soil water retention
 - Provides food for soil microbes

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Enhances soil water infiltration (4x ½ = 2mks)
10. (a) The aim of the experiment was:- to show presence of living organisms in the soil
- (b) observations were:
- Flask D - Limewater turns milky/turbid (1mk)
 - Flask E – Lime water remains clear (1mk)
- (c) The reason for the observation in flask D is:-
Carbon dioxide which turns water milky in flask D would have been produced only during the respiration of living organisms present in fresh soil
11. - It may have hard pan which interfere with water infiltration
12. a) - Light duration
- Light intensity
 - Light wave length (½ x3=1 ½ mks)
- b) Evapotranspiration
- Presence of pest
13. a) - E – Single grained structure
- F – Granular structure (1x1=1 mk)
 - b) i) Humus with clay (1x1=1 mk)
 - ii) Air space (1x1=1 mk)
 - c)- Colour affects soil texture and hence micro- organisms in the soil ✓
 - Texture – affects drainage, aeration and capillary
 - Structure – affects aeration and root penetration
14. three physical characteristics of soil (1 1/2mks)
- Soil structure
 - Soil texture
 - Soil colour
15. a) State merits of horizon A
- source of plant nutrients
 - support/anchor the crops
 - store of water for the crops
 - sources of soil micro organism
- b) State distinct features of horizon B
- deficient of humus(nutrients)
 - contain leached nutrients
 - contains more compact soil particles

FREE AGRICULTURE K.C.S.E 2012 to 2015

- presence of hard pans in some soils
- c) Transitional zone-this is a zone bordering two adjacent layer of soil profile
- i) Weathered rock

Importance

- Give rise to sub soil
- Source of minerals
- Determine mineral content of soil and type of soil

16

- Low temperatures encourages crop diseases such as leaf rust
- Low temperatures may increase or lower the quality of farm produce
- High temperatures hastens maturity/ improves the quality/ lower the quality
- Increases the rate of evapo transpiration which may result loss plant moisture/ leading to wilting of crops

17.

- Movement of animals in large numbers
- Decomposition of plant and animal remains by soil micro- organisms
- Physical breaking of rocks by roots of higher plants
- Man's activities e.g. cultivation, mining and road construction
- Mixing up of soil by animals e.g. earth worms and

18. - Temperature/ Altitude

- Soil type;
- Prevailing winds;
- Rainfall; (4x ½ =2mks)

19. - It influences the movement of the weathered materials hence affecting the depth of soil development;

22. two importance of parent's material in soil profile

- Determine soil characteristics
- Determine soil depth
- Determine soil nutrients

23. four ways of modifying soil temperature in crop production

- Mulching
- Pruning
- Shading of crops
- Irrigation (4x ½ mks)

24. a) two factors that affect selectivity of herbicides

- Stage of plants growth

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- Plants morphology and anatomy
 - Mode of action
 - Environmental factors (2x1=2mks)
 - b) Name two farming practice that cause water pollution
25. four factors that influence soil formation
- Parents rock material
 - Climate
 - Topography
 - Biotic/organic/living organism
26. - Drainage
-Aeration
-Water-holding capacity
-capillary
27. -large animals e.g. Buffaloes
-Man activities e.g. farming
-Root pressure of plants
-Burrowing animals e.g moles, termites

CROP PRODUCTION I (LAND PREPARATION)

1. three factors that determine depth of ploughing during land preparation
- Crop to be planted
 - Implement available
- Type of soil
2. Four reasons for cultivating land before planting.
- To improve soil aeration.
 - To improve germination.
 - Destroy weeds.
 - Destroy weeds.
 - Incorporate organic matter in the soil.
 - Increase water infiltration.
3. (a)Is a situation in which least possible cultivation operations are carried out in crop production.
- (b) - Clearing the land / bush clearing.
- Using appropriate chemicals to kill the existing vegetation.
 - Weeding using herbicides.
 - Planting / drilling seeds directly into the stubble of previous crop.
4. (a) - Harrow the land to a fine filth;

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- Harrow during the dry or before the rains;
- Make the seed bed weed – free / ensure clean seed bed;
- Firm the seed bed using rollers after sowing;
- Select a desirable variety of seed for the ecological zone,;
- Sow seeds at the onset rains/ early planting;
- Apply phosphatic fertilizers at appropriate rate of 200 – 300 kgs/ ha at planting time;
- Drill or broadcast the seeds evenly;
- Use a recommended seed rate for the variety / seed rate of 1.5 – 2.0 kg/ha pure seeds;
- Bury seeds at 2 ½ times their diameter;
- Control weeds by uprooting/ apply a suitable herbicide;
- Apply nitrogenous fertilizers about 6 weeks after germination in split application.
- Avoid grazing when the pasture is too young.
- Practice light grazing in the field phase of pasture establishment. (10 x 1 = 10 mk)
- (b) - Select seeds of the same size, variety, age and free from pests and diseases.
- Plant seeds at the same time.
- Prepare the whole field to required uniform tilth.
- Plant at the right moisture content of the soil / irrigation uniformly.
- Treat seeds before planting i.e. break dormancy.
- Plant at the correct depth.

(5 x 1

= 5 mks)

(c) - Soil moisture content.

- Type of soil.
- Cost of operation.
- Size of seed/ type of planting material/ type of crop.
- Type of machinery available / use of tractors.
- Topography / gradient of the land/ liability of soil erosion.
- Skills of the operator.
- Initial conditions of the land/ the cropping history of the land.
- Time available to carry out the operation before planting.

5. Four physical conditions of the seedbed that need to be changed to facilitate Germination

- Size of soil clods (clods made small or medium size)
- Appropriate soil depth
- Soil looseness
- Should be weed free
- Soil moisture content improved

6. Four importance of sub soiling as a tertiary operation

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Brings leached nutrients to the surface
- Breaks hard pans
- Promotes aeration of the soil
- Promotes water infiltration
- Ensures better root penetration

7. Four advantages of rolling in seedbed preparation are:

- Press the seeds against the soil moisture
- Controls soil erosion
- Ensure uniform germination
- Controls removal of small seeds by wind
- Breaks large soil clods

8. four disadvantages of minimum tillage

- The less porous surface increased soil erosion especially in heavy soils
- Difficulty in weed control
- Speed of planting to reduce due to large amount of residues in the soil and big clods
Leads to accumulating of soil borne pests and diseases

9. a) - ridging

b) (i) Encourage tuber expansion

- Allow easy harvesting of crop roots
- (ii) Rolling
 - Leveling

10.

- Leads to timely planting
- Weeds are appropriately controlled especially the perennial such as couch grass
- Farmers take advantage of availability of labour reducing the cost of labour
- Control of soil borne pests
Gives time for better organic decomposition

11. - By repeated cultivation at the same depth;

- Cultivating the soil when wet using heavy machinery;

12. - Type and size of planting material;

- Topography/slope of land;
- Soil moisture content;
- (Initial) condition of land/amount of vegetation on the land;
- Capital available

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Type of implement used;
13. It is the least number of cultivation operations either during preparation of the seed bed or during the management of the crops.
- 14.
- Market demand
 - Type of crop to be planted
 - Moisture condition of the soil and rainfall pattern
 - Prevalence of pests and diseases
 - Prevalence of weeds
15. two factors that determine the number of cultivation on a field before it is ready for planting
- purpose of crop
 - moisture content
 - concentration of desired chemical
 - weather
 - market demand
16. - Enables crop to benefit maximumly from available moisture
-Crops make use of nitrogen flush available at that time
-Crops fetch high market prices
-Crops escape from pests and diseases
-There is high vigour in crops that resist diseases
-Ensures timely harvesting
17. - type of crop to be planted
-Implements available
-Type of soil
-Climatic conditions

WATER SUPPLY, IRRIGATION AND DRAINAGE

1. State two reasons for treating water for use on the farm
- Remove chemical impurities
 - Remove foreign material
 - Remove disease causing organisms
 - Remove bad smell & taste
2. State three reasons for draining swampy land before growing crops
- Increase soil volume
 - Improve aeration
 - Increases activities of micro organisms
 - Control erosion

FREE AGRICULTURE K.C.S.E 2012 to 2015

Reduce toxic substance in soil

- 3 . a)Sprinkle irrigation
b) four advantages of the above irrigation system

- Little water required
- Done on nay topography
- Control weeds between rows
- Water under low pressure
- Prevent fungal diseases

c) three factors that determine the type of irrigation on the farm

- Where tree crops are planted
- Little water supply
- Enough capital for the method is available
- Slope land

$$3 \times \frac{1}{2} = 1 \frac{1}{2}$$

mks

d) two disadvantages o f the above system of irrigation

- Difficult to carry field mechanization
- Require a lot of capital
- Require clean water
- Regular repair of broken pipes and blocked pipes

Applicable where tree plants are grown

4. a) Artificial√ application of water to the soil surface for purpose of supplying enough moisture√ for plants growth(mark whole)

b) Surface,
overhead,
subsurface,
drip/trickle

5. a) four use of water on the farm

- Irrigation
- Domestic use
- Diluting chemicals
- Construction work
- Watering livestock and washing buildings
- Processing farm produce(1/2x4=2mks)

b) four methods of harvesting water on the farm (2mks)

- roof cantonment
- weirs
- rock cantonment

FREE AGRICULTURE K.C.S.E 2012 to 2015

- dams
- ponds

c) the stages involved in water treatment process

- filtration of water intake
- softening
- coagulation and sedimentation
- filtration in tanks
- chlorination storage ($1/2 \times 6 = 3\text{mks}$)

6. four uses of water in the farm.

- For diluting chemicals used to control pests.
- For watering livestock.
- For watering plants e.g. irrigation.
- For washing utensils, calf pen bully sheds.
- For domestic use e.g. drinking, cooking.
- For rearing fish.
- For recreation
- Processing of farm produce.
- In construction of buildings.

7. two types of irrigation used in Kenya.

- Overhead / sprinkler.
- Surface / Flood / furrow/ basin.
- Drip/ trickle.

8. Four disadvantages cambered beds

- High cost of maintenance
- Provides breeding ground for vectors of malaria
- Prevents proper mechanization of the farm
- Labour intensive

(c) Stage I: Filtration of water intake.

- Water from source river is made to pass through a series of sieves.
- Large particles of impurities are trapped by the sieves.
- Water then enters into the large pipe to be directed to the mixing chamber.

Stage II: Softening of the water.

- Water circulates in the mixing chamber and doses of soda ash to soften the water.

Stage III: Coagulation and sedimentation

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Water is passed through coagulation tank where fresh air enters to remove bad smell/ chloride of lime used.
- Water stays for 36 hours thus solid particles settle and bilharzias causing organisms killed.
- Alum added to coagulated solid particles which settle at the bottom.

Stage IV: Filtration

- Water is passed through filtration tank with layers of sand and gravel to filter it.
- Water leaving the filtration tank is clean.

Stage V: Chlorination

- Water is passed through chlorination tank where chlorine is added.
- Micro-organisms in the water are killed by chlorine.

Stage VI: Storage

- The treated water is stored in large overhead tanks before distribution and use.

9. - Improves soil aeration

- Raises soil temperature
- Increases activities of micro- organisms
- Increases soil volume
- Prevent accumulation of poisonous substances in the soil

10. - Semi-rotary

- Hydram
- Piston/ reciprocating
- Centrifugal
- Rotary

11. Four examples of working capital in maize production are;

- Seeds
- fertilizer
- Herbicides
- Pesticides
- Fuel
- Casual labour ($4 \times \frac{1}{2} = 2\text{mks}$)

12. four types of water pumps which can be used in the farm

- Centrifugal/rotadynathic pumps
- Piston/reciprocating pump
- Semi-Rotan pump
- Hydram pump

13. four methods of drainage

FREE AGRICULTURE K.C.S.E 2012 to 2015

Open ditches

- Under ground drain pipes
- French drains
- Cambered beds
- Pumping
- Planting of trees/planting of trees such as Eucalyptus

14. A dam is a barrier constructed across a river or a dry valley to hold water and raise its level

to form a reservoir or lake

- A weir is a barrier constructed across a river to raise the level of water and still allow water to flow over it

15. - Giving subsidies by reducing the cost of production inputs

Fixes prices of the related products

16. – Piping is the conveyance of water through pipes from one place to the other while pumping is the lifting of water from one point to another by use of mechanical force;

17.

- To facilitate the action of soil living organisms
- To check or reduce leaching
- To moderate or increase soil temperature
- To reduce accumulation of dissolved soil salts
- To reduce erosion rate of top soil
- To improve soil structure
- To increase effectiveness of phosphorous fertilizer and conserve soil nitrogen
- As a way of reclaiming areas such as coastal plains and the river belts which may have high water tables
- In rice fields, water should be controlled by draining the water for a different crop cycle.

18.

- Allowing livestock to graze near water sources often results in organic waste products being washed into the water ways.
- Fertilizer application
- Pesticides
- Over grazing
- Irrigation
- Over cultivation
- Use of farm machinery

19. a) i) A – Drop/ trickle irrigation

ii) B – Sprinkler/ overhead irrigation

b) Two advantages of method A over method B

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Conserves water
- Does not damage flowers, leaves
- Does not cause splash/ splatter irrigation
- Does not encourage spread of fungal diseases from crop to crop
- Does not encourage the growth of weeds all over the field
- Agro- chemical can be dissolved in the water and directly applied to the crop

- c) i) Cotton wool
ii) Rough sand

21. four reasons for practicing irrigation

- Increase crop production by applying adequate moisture
- To reclaim dry areas
- To meet moisture requirement of crops
- To produce and benefit from off season crops
- Growing of paddy vice

23. - to prevent rotting
-For processing
- For long storage
- Prevent pest and disease attack

24. (a) Kill germs
(b) For sedimentation

21. S- French drainage T- Vambedred peds M – soil- stones

SOIL FERTILITY 1 (ORGANIC MANURE)

1. two roles of humus in the soil that are beneficial to crops

- Provide nutrients
 - Increase water holding capacity
 - Increase soil temperature
- Neutral soil PH

2. four characteristic of fertile soil (2mks)

- Well drained
- Correct PH
- Good water holding capacity
- Adequate plant nutrients
- Free from pest and diseases
- Correct soil nutrients

3. a) Q-stick✓

Function-checking temperature✓ and other conditions within the heap

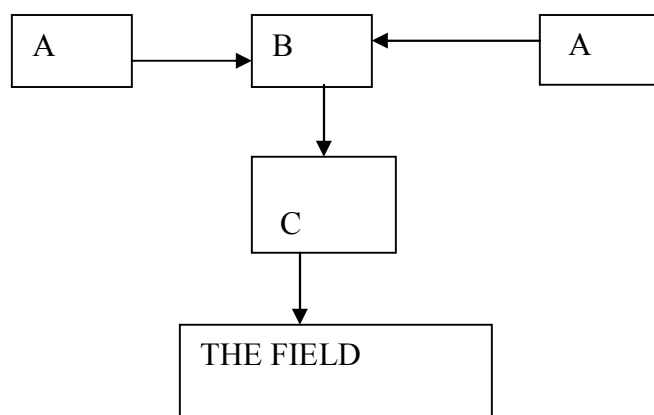
b) i) Top soil-introduces organisms to effect composition✓

ii) Wood ash-increases the level of phosphorus and potassium✓

FREE AGRICULTURE K.C.S.E 2012 to 2015

iii) Rotten manure-provides food for micro-organism✓

4. The illustration below shows a heap system of making compost manure. Study it and answer the questions that follow.



a) use of arrows indicate how the decomposing material should be transferred from one heap

to another till the manure is applied in the field.

b)- 3 - 6 wks

c) one reason for turning the material in the heap regularly.

- Proper decomposition.
- Facilitate air circulation.
- Microbial activities.

d) two reasons why it is necessary to sprinkle water on the heap.

- To regulate the internal temperatures in the heap.
- Create moist environment for microbial activity.

5. Four indicators of well-decomposed manure

- Absence of bad odour and instead the smell of forest soil
- Light weight
- Brown colour
- Moist but not wet
- - Original nature of material not noticeable ($\frac{1}{2} \times 4\text{pts} = 2\text{mks}$)

6. (a) Two factors that should be considered when siting a compost manure heap are:-

- Accessibility
- Drainage
- Direction of prevailing wind

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Size of the farm/proximity
 - (b) Five advantages of rotation grazing are: (5mks)
 - Livestock with maximum use of pastures
 - Reduces build up of parasites and diseases
 - Animal waste evenly distributed
 - Pasture area given time to regenerate
 - Excess pasture conserved
 - Possible to apply fertilizer in the parts of the pasture which are not in use (5x1=5mks)
7. It is movement of dissolved nutrients from p soil to lower horizons of soil becoming Unravel able to crops
- 8.
- Improves soil structure
 - Adds nutrients
 - Increases cation exchange capacity
 - Increases microbial activity in the soil
 - Improves water holding capacity/ reduces leaching
 - Buffers soil PH
 - Moderates soil temperature
9. (a) (i) Preparation of farm yard manure:-
- Collect animal waste/refuse/dung and urine;
 - Collect animal bedding/litter and other rotten plant residues;
 - Store collected materials under roof/shed to prevent leaching and oxidization of nutrients;
 - Turnover the materials regularly;
 - Sprinkle water if dry;
 - leave the material to rote completely before use; (6x1=6mks)
- (ii) Preparation of Hay
- Cut the grass /legume in the field when 50% of it is starting to flower;
 - The cut forage is spread in the field for four continuous days (sunny days)
 - The cut forage is turned daily for even for four uniform drying;
 - Gather the dried material in a central spot;
 - Bale the material;
 - Properly store the baled hay (6x1=6mks)
- (b) Factors to consider in timely planting of annual crops
- Escape from serious weed competition;
 - Utilization of early rainfall;

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Exploitation of Nitrogen flush in the soil that has accumulated during dry season;
 - Escape from serious pest + disease attack e.g. stalk borer in maize;
 - Fetch high market prices when harvested early;
 - Reduce competition for labour during labour peak period;
 - For harvesting season to coincide with dry period to reduce losses e.g. cotton
- Early planting means early farming/calendar for the farmer to enable him /her to finish up other farm activities; (8x1=8mks)

10. i) A ration containing 18% protein is to be made from maize and sunflower cake.

Given

that maize contains 7% protein, and sunflower seed cake 34% protein. Use Pearson

square

methods to calculate the value of feedstuffs to be used to prepare 100kgs of the feed

(3mks)

ii) two other methods that can be used to formulate feed ration (2mks)

- Linear programming
 - Trial and error
- Graphical method.

AGRICULTURE ECONOMICS

(BASIC CONCEPTS AND FARM RECORDS)

1. (a) - Help to determine the value of the farm/ determine assets and liabilities.

- Provide history of the farm.
- Assist in planning and budgeting in various fields.
- Helps to detect losses or theft in the farm.
- Assists when sharing losses or profits (dividends)for communal owned farms/

partnership.

- Help to settle disputes in the farm among heirs.
- Help to support insurance claim e.g. against fire and theft.
- Provide labour information like terminal benefits, NSSF due, Sacco dues for all

employees.

- Help to compare the performance of different enterprises within a farm or other farms.
- Help in the assessment of income tax to avoid over or under taxation.
- Records, helps to show whether the farm business is making profit or losses. This

information

helps in obtaining credit.

(10 x 1

= 10 mks)

2. (a)Health record

FREE AGRICULTURE K.C.S.E 2012 to 2015

(b) Next date of treatment /vaccination

- Occurrence of the disease
- Response to treatment ($\frac{1}{2} \times 2 = 1\text{mk}$)

(c) - Select and cull animals on health ground

- Know the course of action to be taken in the event of a disease and maintenance of good health
- Know the prevalent disease
- Calculate cost of treatment

3 Four uses of farm records are;

- Used to compare the performance of different enterprise
- Shows history of the farm
- Help in planning and budgeting of farm operations
- Assessment of income tax
- Calculation of profits and losses
- Securing loans
- Settling disputes where no will is left

4. four uses of farm records

- Help compare performance of different enterprises within the farm and other farms
- Shows the history of the farm
- Guide a farmer in planning and budgeting of farm operations
- Helps to detect losses or theft on the farm
- Helps to avoid over taxation or under taxation
- Helps to determine the value of the farm in terms of assets and liabilities
- Helps in sharing of profits and losses in partnership
- Helps in settling disputes among heirs in absents of a will
- Shows whether the farm business is making profit or losses
- Helps in supporting insurance claims of farm assets
- Provide labour information

5.

- Understanding the technical language used in agriculture
- Application of the right amounts of inputs
- Correct/ appropriate measurements in farming

FREE AGRICULTURE K.C.S.E 2012 to 2015

Uses of appropriate technology

6. (a) Milk production record (1x1=1mk)
- (b) Determine prevalent diseases;
- Establish treatment of diseases;
- Establish disease control method;
- Determine cost of medication/health care;
- Determine the health status of different animals; (4x1=4mks)
- (c) Price trends/market situation
- Production techniques
- Labour trends
- Breeds of dairy cattle
- Production constraints/risks and certainties
- 7.
- Field
 - Area of the land
 - Season
 - Crop planted
 - Crop variety
 - Land preparation date
 - Type of fertilizer at planting
 - Type of fertilizer at top dressing
 - Seed rate used
 - Type of weed and date of weed control
 - Type of pest and date of pest control
 - Date of harvesting
 - Remarks (Any 4)
8. List two events occur during induction stroke in a four stroke engine. (1mk)
- Piston moves down from TDC
 - Exhaust valve is closed
 - Inlet valve is open
 - Air / fuel mixture get into combustion chamber
 - Piston reaches BDC. (4x ½ = 2mks)
9. – Where the resources are free

FREE AGRICULTURE K.C.S.E 2012 to 2015

- where there is no alternative
- where the alternatives are very many.

SOIL FERTILITY II (IN ORGANIC FERTILIZERS)

- four advantages of applying lime in clay soil
 - Lower soil acidity
 - Increase calcium content
 - Hastens decomposition of organic matter
 - Improve soil structure/ improve drainage
 - Facilitates availability and absorption of nitrogen and phosphorous
 - Improve legume nodulation and nitrogen fixationIncrease multiplication of micro- organisms
- i) Sulphur- SO_4^{2-} , SO_2
 - ii) Nitrogen- NO_3^- , NH_4
 - iii) Carbon- CO_2
 - iv) Magnesium- mg^{2+}
 - three effects of nitrogen to plants (1 1/2mks)
 - delayed maturity
 - excessive succulence
 - excessive vegetative growth
 - weak stems
 - lodging
- The quality of sulphate ammonia fertilizer the farmer will need for 10 hectares (1mk)
100kg SA supplies 21kg N
1ha requires 150kg SA
10ha requires (150x10) kg SA
= 1500kg SA in 10ha
 - The number of 50kg bags of fertilizer he will purchase (1mk)
50kg fill 1 bag
$$\frac{1500}{50} \times 1500 = 30 \text{ bags} \quad (1)$$
- two disadvantages of using farm yard manure
 - It's bulky hence difficult to apply by one laborer.
 - It may spread weeds.
 - It may spread diseases.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- It releases nutrients slowly.
5. four factors which influence the stage at which the crops are harvested.
- Purpose of the crop / maturity.
 - Moisture content.
 - Concentration of certain chemicals.
 - Water condition.
 - Market demand.
6. a) - Calcium Ammonium Nitrate CAN.
b)- Knee high
- 30 – 45 cm height.
c) Calculate the amount of K_2O contained in 400 kg of a compound fertilizer 25 : 10 : 5.

$$\begin{aligned} & 5 \text{ kg of } K_2O \text{ is in } 100 \text{ kg of } 25 : 10 : 5 \\ \therefore & 400 \text{ kg of compound fertilizer} \\ & \frac{400 \times 5}{100} = \underline{20 \text{ kg of } K_2O} \end{aligned}$$

7. Two pieces of information that soil sample should have before being taken to the laboratory for testing are:-

- Name of the farmer
 - Address of the farmer
 - The type of test to be carried out
 - Date of sampling
8. A compound fertilizer bag has the labels 20-20-10. What do the figures stand for
- 20 – 20% Nitrogen (N)
 - 20 – 20% Phosphorous Pentoxide (P_2O_5)
 - 0- 0% Potassium Oxide (K_2O)
9. Four functions of sulphur in crops are:-
- Amino acids/protein synthesis
 - Formulating enzymes and hormones
 - Increase oil content and hormones
 - Needed for formation of chlorophyll
10. - Improves soil PH hence microbial activities
- Allows wide production of different crop varieties
 - Improves soil aeration
 - Improves drainage

FREE AGRICULTURE K.C.S.E 2012 to 2015

Improves soil structure

11. -Liming

- use of alkaline fertilizer ($2 \frac{1}{2} = 1 \text{mk}$)

12. a) -Type of crop

- Fertilizer characters
 - Type of soil
 - Environmental condition
- mks)

($\frac{1}{2} \times 3 = 1 \frac{1}{2}$)

b) - Avoid unique sites e.g. compost, along fence

- Remove all vegetation parts
 - Mix the samples thoroughly
- Make as many samples as possible

13. three functions of nitrogen in crops

(1

1//2mks)

- Protein synthesis/formation
- Forms part of chlorophyll molecule
- Encourages vegetative growth
- Regulate availability of phosphorus and potassium in plant
- Increase to size of grains and protein content increases

14. Fertilizer ratio is the proportion of various nutrients in a fertilizer grade is the percentage of

nutrients in a fertilizer;

(b)- Nitrogen

- Calcium
- Potassium
- Magnesium
- Sulphur

15. (a) Diagonal/transverse method; (1x1=1mk)

(b) – Avoid contamination/use sterilized container;

- Avoid sampling soil from unusual sites e.g ant hills
- Avoid mixing top soil with sub-soil; (3x1=3mks)

(c) – To determine the nutrient status of the soil;

- To determine the soil PH/type of fertilizer to add to the soil/lime to add to the soil;
- To determine the type of crop to grow;
- May help in diagnosing low crop yield/mineral deficiency;

FREE AGRICULTURE K.C.S.E 2012 to 2015

16. Area – $10 \times 20 = 200\text{m}^2$ (1mk)
100,000m² require 80kgs of 20-20-10
200m² will require
 $\frac{200 \times 80}{10,000}$
 $= 1.6\text{kgs}$ (1mk)
17. a) A compound of fertilizer has a fertilizer grade of 25:10:5. calculate the amount of phosphorus present in 400kg of this fertilizer
- N:P:K
25:10:5
If 10kg P₂O₅/1 = 100kg NPK
? = 400kg NPK/1
 $= \frac{400 \times 10}{100}$
 $= 40\text{kg P}_2\text{O}_5/1$ (3 steps x 1=3mks)
- b) i) - zigzag method
ii) xx – traverse/diagonal
iii) State three importance of carrying out soil sampling and testing
- determine the type of crop to grow
 - determine the type of fertilizer to be used
 - determine type of nutrients in the soil (3x1=3mks)
18. (a) Lacks one of the major fertilizer NPK elements
(b) – The soils could be very acidic
- Too much rainfall.
19. (a) Random/zigzag soil sampling
(b) – old manure heaps
- Ant hills
- Dead furrows
- Fence lines
- Cattle bomas
(c) - Clear vegetation
- Mark points using pegs
Collect top soil and sap soil in different paper bag.
- Dry the soil and mix thoroughly
- Sent dispatch to laboratory for testing

CROP PRODUCTION II (PLANTING)

1. two reasons for seed treatment of tree species before planting

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Break dormancy
- Control pests and diseases
- 2. three factors that determine spacing of beans
 - Type of soil
 - Moisture in soil
 - Species/ size of bean plant
 - Machinery to be used
 - Purpose of beans
- Stand in the field
- 3. four reasons for using certified seeds for planting
 - High yielding
 - Quality produce
 - High germination percentage
 - Grow faster
- 4. a) A Banana sucker
B Stem tubes
C Bulb
D Stem cutting
b) Chitting

c) four advantages of vegetative propagation on crop production
 - Grow faster
 - True copy of mother plant
 - Have no dormancy period
 - Easy to obtain

4x ½

5. Differentiate between hybrid and composite

Hybrid	composite
Seeds produced by crossing inbreedlines and controlled pollination✓	Seeds produced by growing different varieties together under uncontrolled pollination✓ 2//

6. a) $\frac{\text{Germinated seed}}{\text{Total seeds planted}} \times 100\%$

$$\frac{90}{100} \times 100 = 90\%$$
(2)
- b) Given that maize is planted at a spacing of 75cm by 25cm, calculate the plant population in a plot measuring 4m by 3m

FREE AGRICULTURE K.C.S.E 2012 to 2015

plant ppl = $\frac{\text{land area}}{\text{Spacing}}$

4mx3m

75x25cm

400cmx300cm

$\frac{75\text{cm} \times 25\text{cm}}{75\text{cm} \times 25\text{cm}} = 64\text{plants}$ ✓ (2)

7. four qualities of a mother plant which should be considered when selecting vegetative material for propagation.

- High quality.
- High yielding.
- Disease resistance / healthy/ disease free.
- Fast growth/ fast maturity.

8. - Select seeds of the same size, variety, age and free from pests and diseases.
- Plant seeds at the same time.
- Prepare the whole field to required uniform tilth.
- Plant at the right moisture content of the soil / irrigation uniformly.
- Treat seeds before planting i.e. break dormancy.
- Plant at the correct depth.

(5 x 1

= 5 mks)

9. two factors which determine the depth of planting

- Soil type
- Size of seed
- Soil moisture content
- The type of germination

10. - Reduces leaching

- Improves water holding capacity
- Improves soil structure
- Suffer soil pH
- Moderate soil temperature
- Increases microbial activities
- Increases cation exchange capacity
- Improve fertility of the soil after decomposition

11. Area

Spacing

Tea population ($\frac{10,000 \times 2}{1.5 \times 0.75}$) ✓

1

20,000m² ✓

FREE AGRICULTURE K.C.S.E 2012 to 2015

$$\frac{1.125\text{m}^2}{1} = \underline{17,777 \text{ plants}}$$

Needed in carbohydrate metabolism

12. Four reasons why training is important in some crops

- Facilitate field practices of spraying and harvesting
- Improves crop quality by preventing solving
- Enable crop grow in the required direction
- Improve yield
- Control pest and diseases

13. Four factors that influence the depth of planting are:

- The size of the seed
- Soil moisture content
- Type of soil /soil texture
- Type of germination

14.

- Avoid mixing with foreign materials
- Harvesting during the dry weather
- During harvesting separate grade A and B

Don't put in gunny/sisal bags

Needed in carbohydrate metabolism

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 - Harvesting during the dry weather
 - During harvesting separate grade A and B
- Don't put in gunny/sisal bags

FREE AGRICULTURE K.C.S.E 2012 to 2015

18. - Moisture content of soil
- Use of which the crop is to be put
 - Number of seeds per hole
 - Prevalence of certain diseases/ pests
 - Machinery to be used in subsequent operations
 - Fertility status of the soil
19. Four advantages of rolling in seedbed preparation are:
- Press the seeds against the soil moisture
 - Controls soil erosion
 - Ensure uniform germination
 - Controls removal of small seeds by wind
 - Breaks large soil clods
20. two factors that effect rooting of cuttings in crop production
- Temperature
 - Relative humidity
 - Light intensity
 - Oxygen supply
 - Chemical treatment
 - Leaf area
- 21.
- Breaking seed dormancy
 - Seed dressing
 - Seed inoculation
- 22.
- Breaking seed dormancy
 - Seed dressing
 - Seed inoculation
23. Under sowing is the establishment of pasture under a cover crop usually maize while over sowing is the establishment of pasture legume in an existing grains pasture
24. a) Stem cutting
- b)
- High yielding
 - High quality
 - Good rooting ability

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Adaptable to the ecological zone

c)

- Make top cut near the auxiliary bud as close as possible and sloping away from it
- Lower cut must be sloping at an angle and be 2.5 – 4cm below the leaf
- Single leaf internodes cuttings must be kept shaded and wet floating in water from the time of cutting to planting

25 a) $(30 \times 15) \text{ cm}^2 / 30\text{cm} \times 15\text{cm}$

b)

$$\frac{4M - 0.6 M}{0.3M} + 1$$

3.4 + 1

0.3 = 12 rows

c) Plant population

$$\frac{3M - 0.6}{0.15} + 1$$

$$= \frac{2.4}{0.15} + 1$$

$$= 16 \text{ plants} \times 12 \text{ rows} = 192 \text{ plants}$$

26. – Temperature;

- Relative humidity;
- Light intensity;
- Oxygen supply;
- Leaf area;
- Chemical treatment;

27. Selection of mother plants (tea);

- Select healthy bushes/free of pests and diseases;
- Select high quality bushes;
- Select those which are high yielding;
- Select those with good rooting ability;
- select those which adapt to a wide range of ecological conditions; (4x1=4mks)

Preparation of planting materials

- Prune the selected tea bushes and leave unchecked for six months;
- Select and cut good branches for making cuttings
- Obtain the cuttings from the middle of the branches/discard the brown and the hard bottom part/ the green soft top part

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Make single leaf internodes cutting carefully 2.5-4cm long;
- Make slant cut with the use of scalpel/sharp knife taking away from the node;
- Make top cutting near the auxiliary bud as much as possible;
- Keep the cuttings wet in the water to avoid dehydration until they are planted;
- Keep the cutting under the nursery;

Raising of tea seedlings in the nursery

- Plant cuttings on rooting medium in polythene sleeves/sleeves measure 25 x 7.5-10cm and sealed cut.
- Rooting medium consists of fertile sub-soil and phosphate fertilizer;
- Plant single leaf internodes per polythene sleeve,
- Place the sleeves in the vegetative propagation units,
- Erect wooden hoops over the sleeves cuttings, then place polythene sheet over it/erect shade over the nursery;
- Water sleeved seedlings every 3 weeks/main high humidity;
- Uproot weeds when they appear;
- Hardening off done 4 months after raising (9x1=9mks)

28. Factors to consider in timely planting of annual crops

- Escape from serious weed competition;
 - Utilization of early rainfall;
 - Exploitation of Nitrogen flush in the soil that has accumulated during dry season;
 - Escape from serious pest + disease attack e.g. stalk borer in maize;
 - Fetch high market prices when harvested early;
 - Reduce competition for labour during labour peak period;
 - For harvesting season to coincide with dry period to reduce losses e.g. cotton
- Early planting means early farming/calendar for the farmer to enable him /her to finish up other farm activities; (8x1=8mks)

29.

- It is wasteful because a higher seed rate is used.
- It is not possible to use machines.
- It is not possible to establish plant population.
- Lack of uniformity in seed establishment.

30. (i) Seed inoculation;

- It is the treatment of legume seeds with Nitro-culture/artificial bacteria to increase their Nitrogen fixation in the soil ,if grown in Nitrogen deficient soils.

(ii) Chitting;- Breaking of dormancy in Irish potatoes before planting

(iii) Tipping;- Removal of three leaves and a bud from each shoot above the required height of the

table in tea during plucking table formation / formation of a uniform and flat plucking table in tea.

31. two advantages of producing crops by use of seeds over vegetative propagation

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Seed treatment is easier
- Seeds can be stored for a long time
- Faster and uniform germination
- Mechanization of farm operation is easy/possible
- Application of fertilizer/manure is easy and can also be mechanized

32. four ways of preparing planting materials before planting

- Breaking seed dormancy
- Seed dressing
- Chitting
- Seed cleaning
- Seed inoculation

Root trimming as in banana or tree seedlings

CROP PRODUCTION III

NURSERY MANAGEMENT PRACTICES

1. three methods of grafting that are used in propagation of plants

- Whip and tongue grafting
- Side grafting
- Approach grafting
- Bark grafting
- Notch grafting

2. two practices done during hardening-off of seedlings in a nursery bed.

- Gradual removal of shade
- Gradual reduce of watering

3. two methods of budding used in crop propagation (1mk)

- T-budding
- Top budding
- Patch budding

4. four management practices carried out on a nursery bed (2mks)

- Watering
- Shading
- Pest and disease control
- Weed control
- Mulching
- Hardening off
- Pricking out (1/2x4=2mks)

5. Two importance of tissue culture in crop propagation

- Propagate pathogen free plants

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- Appropriate soil depth
 - Soil looseness
 - Should be weed free
 - Soil moisture content improved
6. - A nursery bed is a portion of land specially prepared to raise seedlings before transplanting while a seedling bed is a specially prepared portion of land for receiving pricked out seedlings from the nursery bed Mark as a whole (1 mk)
7. Four advantages of under sowing in pasture production
- Amino acids/protein synthesis
 - Formulation of enzymes and hormones
 - Increase oil content and hormones
 - Needed for formation of chlorophyll
 - Aid in nitrogen fixation in legumes
 - Needed in carbohydrate metabolism
8. Four advantages of under sowing in pasture production
- Amino acids/protein synthesis
 - Formulation of enzymes and hormones
 - Increase oil content and hormones
 - Needed for formation of chlorophyll
 - Aid in nitrogen fixation in legumes
 - Needed in carbohydrate metabolism
9. (a) The structure is a nursery
(b) Reason for carrying out each of the following practices in the structure shown above is:-
- Pricking out – to avoid overcrowding /allow seedling to grow strong and healthy transfer seedlings from one nursery to another
 - Hardening off – To prepare seedlings to ecological conditions in the main field/reduce transplanting shock
- (c) Three importance of the part labeled A in the above structure (1/2 x 3= 1 ½ mk)
- To reduce the amount of water through vaporization
 - To modify nursery temperature
 - To reduce the impact of raindrops/hailstones hence minimizing damage on seedlings
 - Reduce splash erosion
 - Reduce the scorching effect in the seedlings
 - Reduce the scorching effect in the seedlings

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10. a) - Sitting crop nursery
- Good soil fertility
 - Security against destruction
 - Accessibility
 - Should be near source of water
 - Topography should discourage water logging (1x5=5 mks)
- b)- Establishment
- Prepare fine filth
 - Add manure or fertilizers to the nursery
 - Sterilize soil against soil borne pests/ diseases
 - Shade the nursery bed
 - Ensure nursery is 1m wide
 - Plant seeds in drills and cover with light soil layer (1x5=5 mks)
- b) Management practices
- Mulch to conserve moisture and suppress weeds
 - Water regularly in the morning and afternoon
 - Pricking – remove excess seedlings and transfer to another nursery or use polythene sleeves
 - Weed control – done by hand uprooting
 - Pest and disease control – use clean seeds and apply chemicals as recommended
 - Hardening off – Done by removal of shade
 - 1 week to transplanting to make seedlings survive after transplanting
11. State four importance of thinning seedlings in the nursery bed
- To control spread of pests and diseases
 - To create space for other seedlings
 - To avoid competition for light, nutrients
 - Allow rapid growth of seedlings/vigorous(1/2x4=2mks)
12. Seedling bed is where overcrowded seedlings from the nursery bed are transferred while

seedbed is the final land where planting materials are raised until they are ready for harvesting.

13. a) two advantages of having the part labeled J
- To reduce the amount of water loss through evapo- transpiration
 - To modify the temperature
 - To reduce the impact of the raindrops thereby minimize the damage of seedlings/ reduce splash
 - Retaining water

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b) Management practices carried out on the nursery from the time the seedlings emerge to stage of transplanting

- Proper watering
- Controlling weeds
- Hardening off
- Pricking out

CROP PRODUCTION IV **(FIELD MANAGEMENT PRACTICES)**

1. Staking is supporting tall varieties of tomatoes using a stick fixed next to the plant and tied with

sisal string while propping is supporting banana plant with sticks Mark as a whole= 1

mk

2. five advantages of crop rotation

- Improves soil fertility: where legumes are included nitrogen is fixed/ added in the soil
- Control pests and diseases: disrupts the life cycle of certain pests and diseases
- Control weeds: control weeds which are specific to certain crops e.g. striga in cereals/ cover crops in a rotation will smother certain weeds
- Better use of the soil nutrients: different crops (due to differing root systems) draw nutrients from varying soils horizons/ different crops require different nutrients
- Control of soil erosion: cover crops included reduce soil erosion
- Improve soil structure: When grass lays are included which during the period organic matter will accumulate to enrich the soil and improve soil structure

3. four factors which influence the stage at which the crops are harvested.

- Purpose of the crop / maturity.
- Moisture content.
- Concentration of certain chemicals.
- Water condition.
- Market demand.

4. a) B. - Single stem pruning system. (1x1=1mk)

b) Identify the system of pruning in C.

- Multiple stem pruning system. (1x1=1mk)

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c) Outline how pruning in diagram C is carried out.

- Main stem of the seedling is capped/ cut stem at 38 – 60 cm high.
- Two or three suckers are selected and allowed to grow while the rest are removed

5. Two functions of earthing up in crop production

- To influence tuber expansion
- To retain water between the ridges which increases water conservation/infiltration
- Reduces soil erosion
- To prevent the greening effect in potatoes

6. (a) The factors which determine the stage of harvesting of crops

- Stage maturity of the crops
- Use of the crop
- Tastes and preferences of consumers
- Weather conditions
- Chemical conditions
- Chemical concentration of the chemical
- Moisture content (1mk x any 6pts = 6mks)

7. - Reduces runoff thus increasing amount of water into the soil

Reduces evaporation thus increasing the amount of water retained

8. - Moisture content of soil

- Use of which the crop is to be put
- Number of seeds per hole
- Prevalence of certain diseases/ pests
- Machinery to be used in subsequent operations
- Fertility status of the soil

9. i) To acclimatized the seedlings to direct light/ conditions in the seedbed

ii) To avoid overcrowding and reduce competition for light.

iii) Produce healthy and strong seedling

iv) To obtain correct plant population

10. Two factors that determine the stage of harvesting crops are:-

- Purpose of crop
- concentration of required chemical

11 a) - Weather condition

- Use/ purpose

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- Stage of growth
 - Concentration of the required chemicals (i.e. tea) ($\frac{1}{2} \times 4 = 2$ mks)
 - b) - Proper drying of produce
 - Keeping storage facility/ structure clean
 - Use of pesticides on storage structure
 - Treating produce with pesticides e.g. cereals
 - Use of rodent guards
12. Enable controlling over bearing by ensuring required leave ratio
- Open up canopy for air and light penetration
 - Reduce chemical waste
 - Control pests and diseases
 - Allow other field operation
 - Give crop desired shape
- (c) Which factors are considered when carrying out a crop rotation program?
13. a) -Removing of chaffs by use of wind
- b)i)- Prevent erosion
- Add organic matter
 - Conserve soil moisture
 - Raise soil temperature ($\frac{1}{2}$ mk)
- ii) - improve quality of grains by separating them from rest of plant ($\frac{1}{2} \times 1 = \frac{1}{2}$ mks)
- c) - Root systems of crops
- Botanical relation of crops
 - Nutrient absorption by plant
 - Susceptibility to pests, diseases or weeds
14. a) This is support given to crops with weak stem. So as to grow in the desired direction
- b) -Makes the plant to get enough sunlight to manufacture food i.e. prevent shading
- leads to more productivity
 - reduced infections of diseases from the grounds
15. (a) Some crops are harvested earlier e.g. maize for silage at silking stage while maize for grains when the grains are dry;
- (b) A crop can be harvested earlier when the market demand is high;
16. It is a farming practice that involves the removal and destruction of crop plants which are heavily infested with pests and diseases from the field.
17. It is the replacement of old bearing stems by suckers. The cycle is usually changed

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after 4 – 6 years.

18. a)- Earthening up
b) Importance of the above practice
i) Maize - provides support to prevent lodging
ii) Irish potatoes – Improves tuber formation
c) During second weeding
19. Four factors which determine the stage at which crops are harvested
- Original conditions of land
 - Soil type
 - Cost
 - Size of planting materials
 - Soil moisture type of implement
20. Two limitations of using polythene sheets as mulching materials in a field of tomatoes
- Expensive
- Requires skilled labour
 - Does not decompose
- May overheat soil around crop roots
22. - suckering
-Pruning of leaves
-Propping
-Mulching
-Earthing up
23. - Smothers weeds
-Regulate soil temperature
-Conserve moisture
25. -passion fruits
-Deep rooted
-Nitrogen fixing
-Good by-products
- Friendly to crops / not an enemy crop

CROP PRODUCTION V **(VEGETABLES)**

1. a) three management practices that have not been carried on the plant above
- Pruning
 - Staking
 - Weed control
- mks

3x ½ = 1 ½

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b) For each management practice state one reason why it should be carried out

- Pruning
- Staking – prevents lying on ground where fruits are soiled and get diseases
Easy movement in field 1x ½

= ½ mk

- Weeding – reduce competition for nutrients, water and space 1x ½
=1/2 mk

c) Name two diseases that attack the crop above in the field

- Tomato blight
- Bacterial wilt
Blossom – end rot

2. the production of tomatoes (*lycopersicon esculentum*) under the following subheadings

a) Varieties

- fresh market e.g. money maker, beef eater, marglobe supermande
- processing varieties-cal -j, marzano, Kenya beauty (1/2x4=2mks)

b) Nursery establishment

- select site and clear
- dig to remove weeds and narrow to fine tithe
- mark out nursery beds 1 wide with convenient length and level it
- make drills 10cm apart and crop seeds singly in furrows
- cover with thin layer of soil, mulching, water (1/2x4=2mks)

c) Field management practices

- gapping-to reduce bad seeds and maintain optimum plant population
- weed control-remove weeds mechanically to prevent competition avoid during flowering
- top dressing-use fertilizers at 20kg/ha when plants are 25-30cm tall
- staking-train plants to grow in desired shape; to produce clean fruits, control pests and disease
- pruning-to remove unwanted branches to input micro climate facilitate spraying
- pest control-use pesticides and other appropriate methods to control pests e.g. aneucal ball worm, leaf hopper
- disease control-use fungicides, legislative methods ,etc to control early blight, damping off, bacterial wilt (7x2=14mks)

3. two symptoms of late blight in tomatoes.

- Rapid drying of leaves.

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- Brownish dry rots of fruits.
- Destruction / drying of the whole plant.

4. four factors to consider when grading tomatoes for fresh market.

- Size i.e. large, medium, small.
- Degree of ripeness of fruit.
- Damage of tomatoes e.g. bruises on skin.
- Shape of the fruit.

5. Two ways of controlling purple blotch in onions

- Crop rotation
- Application of fungicides

6. (a) The disease which may have caused the condition shown in the illustration

Bacterial with (*Pseudomonas solana cerum*). ($\frac{1}{2} \times 2 = 1\text{mk}$)

(b) Any other crop which may be affected by the disease identified in (a) above

- Irish potatoes($\frac{1}{2} \times 2 = 1\text{mk}$)

(c) Two other factors which can lead to the same condition as shown by the illustration

- Nematode attack
- Lack of water
- Physical damage on the roots/male attack ($\frac{1}{2} \times 2 = 1\text{mk}$)

(d) Two measures that can be sued to control the disease named in (a) above

- Crop rotation
- Regueing /field hygiene
- Use of certified seeds

7. -Enables efficient coverage of plant with chemicals

- Creates unfavorable micro climate for disease causing organism
 - Diseased branches are removed hence reduced incidences of disease spread
- Remove branches touching the ground to avoid infection

8. - Altitude – 900-2900 meters above sea level

- Rainfall - well distributed throughout the growing period
 - 750mm – 2000 mm per annum
 - practice irrigation

- Soils – well drained

- Deep rich

- Slightly acidic PH of 6.5

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9. a)

- American boll worm
- Cut worm
- Red spider mite
- Nematodes

b) Lack of calcium

- Irregular watering
- Excessive application of Nitrogen in early stages of the plants growth

10

- Altitude 0 – 2100m above sea level.
- Temperature range 18 - 29° C
- Sunlight is important during ripening to give the tomatoes a bright red or yellow colour depending on the variety
- Rainfall – well distributed during the growing season. (760 - 1300mm pa)
- Soils - warm and well drained soils with a PH of 5.5 - 7.0
-

LIVESTOCK HEALTH

(INTRODUCTION TO LIVESTOCK HEALTH)

1. four physical appearances to be observed in a sick animal
 - Behaviour of animal-over excitement, aggression, abnormal sores
 - General appearance-dull eyes, restlessness
 - Movement of animals -limping/straining when walking
 - Posture (4x ½ =2mks)
2. two reasons why tsetse fly control is considered to be a land reclamation method
 - Allows livestock rearing
 - Enables human settlement
3. a) Siting a fish pond
 - i) Soil type- clay soil is the best
 - ii) Topography – requires gentle slope not Lilly and flat
 - iii) Source of water – near reliable source
 - iv) Marketing centre should be close

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- v) Accessibility from the homestead
- vi) Security – protected against predators
- vii) far from natural source of fish

b)

- General farm hygiene, cleanliness of houses, equipment proper carcass disposal by burning/ burying/
- Disinfection to destroy pathogens e.g. Anthrax and calf diseases
- Isolation of sick animals – separated from healthy ones to avoid spread of diseases e.g. foot mouth
- Drenching/ deworming to control internal parasites e.g. tapeworms and roundworms
- Treatment of the sick animal – to prevent spread of diseases
- Vaccination to create resistance to diseases on regular basis e.g. foot and mouth, anthrax, new castle
- Control vectors – to avoid disease transmission e.g. ECF, nagana/ specific method
- Prophylactic approach/ use of drugs to avoid injection e.g dry cow therapy against mastitis
- Trypanocidal drugs to control trypanosomiasis
- Proper breeding to control breeding diseases e.g. brucellosis
- Proper feeding to prevent nutritional disorders e.g. milk fever, anaemia
- Slaughtering/ killing – to prevent spread of contagious diseases e.g. anthrax
- Quarantine – to avoid spread of diseases
 - prevent introduction of diseases
- Proper housing to avoid predisposing the animal to diseases e.g. ventilation, spacing
- Foot trimming to minimize occurrence foot rot

4. a) four notifiable diseases in livestock (4mks)

- Lumpy skin disease
- Newcastle
- Anthrax
- African swine fever
- Rinderpest
- Rabbits (4x1 = 4 mks)

b) Discuss four ways in which livestock disease are spread in the farm (4mks)

- Ingestion of contaminated feed and water.
- By carrying agents /vectors
- Through wounds
- Through inhalation of pathogens
- By abrasion in the body

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- Through contact with the disease causing organisms. (4x1 = 4mks)

c) the methods of controlling livestock disease giving an example of different disease in each case

- Use of prophylactic drugs e.g. coccidiostat to control coccidiosis
- Use of antiseptics / disinfectants to maintain farm hygiene to control calf scours
- Quarantine to control foot and mouth disease
- Isolation to control infectious diseases e.g. fowl pox
- Mass slaughter to control zoonotic diseases e.g. anthrax
- Vaccination to control black quarter
- Control vectors like tsetse flies to control nagana.
- Use of healthy breeding stock / to prevent breeding diseases
- Proper nutrition to control bloat and milk fever.
- Treat sick animals to prevent spread e.g. mastitis
- Drenching / control of internal parasites like fascioliasis , ascariasis
- Keep resistant breeds e.g. zebu to control ECF
- Foot trimming to reduce occurrence of foot rot
- Proper housing to control pneumonia. (12x1 =12mks)

AGRICULTURAL ECONOMICS II **(LAND TENURE AND LAND REFORM)**

1. Four ways by which Re-afforestation help in land reclamation
 - Add organic matter from falling leaves
 - Recycles soil erosion
 - Control soil erosion
 - Improve drainage of swampy areasPlay part in hydrological cycle
2. three objectives of land reforms that are taking place in Kenya
 - Increase output from land
 - To meet changing national and market demands
 - To achieve increasing productivity of both land and labour
 - Proper utilization of national land resources and arid lands
 - Increases commercial farmingIncrease land conservation and improvement
3. two causes of land fragmentation in Kenya since independence.
 - People buying pieces of land elsewhere.
 - Compensation when the government takes part of one's land for public use.

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- Inheritance of land.

4 - Establishment of land ownership.

- Measurement of land size.
- Description of the land.
- Recording and mapping of the surveyed land.
- Solving objections if any.
- Submission of the maps and records to the district land office registration

5. Four benefits of a farmer having land title deed

- As security to get a loan
- As security of land ownership
- Minimizes land disputes
- Acts as an encouragement to the farmer to carry out long term investment on the land

6. Four reasons for practicing land consolidation are

- To save time and transport
- Make supervision of land easier and more effective
- To facilitate mechanization of farm
- To make it easy for long term investments in the farm
- To have sound farm planning

7. four advantages of communal land tenure system

- Problems of landlessness does not exist
- Land cannot be easily fragmented
- The system allows free movement of livestock
- The land is left rest for a while so as to allow pasture regeneration
- No land dispute

8. four advantages of landlordism and tenancy

- Enables landlords who cannot use land to get income from tenants
- Idle land can be put into good agriculture use increasing agricultural production
- The landless can rent land to earn a living
- Ensure equitable distribution of land as a natural resource
- It reduces land dispute since the land lord or the state control its allocation

9.

- To achieve flexibility in farming patterns to meet changing national and market demands
- Achieve effective utilization of National land and introduction of irrigation schemes
- Encourages measures on the land and general improvement of land
- To achieve increasing productivity of both land and labour

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- Encourage commercial instead of subsistence production in order to ensure meaningful self employment in rural areas
 - Encourages farmers to invest more through offering security of tenure
10. – Freedom to use land by all members
- Promotes unity among members
 - Promotes use of land according to general requirements of the community
- 11.
- The number of the title deed
 - Size of the land
 - The name and identity of the owner
 - Date of registration
 - Type of land ownership
 - Seal of the government
 - Issuing officer's signature

SOIL AND WATER CONSERVATION

1. Three human activities that may influence soil erosion
- Ploughing up and down slope
 - Over cultivation
 - Planting annual crops on slopes
 - Overstocking/ burning vegetation/ clean weeding
 - Over irrigation
2. a) Cut off drain 1x1=1
- mk
- b) Collect water from uncultivated land and drain it into a water bay 1x1=1
- mk
- c) Will be washed by water back to the trench, filling it then to cultivated land
- d) four effects if water was allowed into the cultivated land
- Will wash top fertile soil away
 - Will uproot planted crops
 - Expose unproductive soil
- Soil washed will cover crops on the lower end of the farm
3. two roles played by Grassley in soil erosion control (1mk)
- Improves soil structure by holding loose particles together

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- Provides ground cover to prevent exposure of soil to agents of soil erosion
 - Reduce movement of surface run off
4. three materials that may be used for constructing a gabion.
- Wires.
 - Stones.
 - Concrete/ sand/ cement/ water/ ballast.
 - Wood/ poles/ metal pegs/ rods.
5. one factor that would determine the width and depth of a cut off drain.
- Expected volume of run – off.
 - Bed rock / soil type.
6. (a) - Reducing the speed of surface run-off – hence reducing the runoffs water erosive power.
- Trap soil from surface run-off/ filter out soil.
 - Reduce the impact of rain drops on the soil thus reducing splash erosion.
 - Grass holds soil particles together hence reducing soil erosion.
 - Soil structure is improved by organic matter from grass thus rate of water infiltration increases.
 - Water stays for 36 hours thus solid particles settle and bilharzias causing organisms killed.
 - Alum added to coagulated solid particles which settle at the bottom.

Stage IV: Filtration

- Water is passed through filtration tank with layers of sand and gravel to filter it.
- Water leaving the filtration tank is clean.

Stage V: Chlorination

- Water is passed through chlorination tank where chlorine is added.
- Micro-organisms in the water are killed by chlorine.

Stage VI: Storage

- The treated water is stored in large overhead tanks before distribution and use.

7. Four farming practices that help in reducing the effects of water shortage in crop production are:-
- Mulching
 - Early planting
 - Planting early maturing crops
 - Practice land fallowing
 - Contour cropping/Contour farming
8. - Fanya juu terraces
- Broad base terraces
 - Bench terraces

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- Narrow based terraces
- Fanya chini terraces

10. The cultural methods of soil erosion control are:

- Planting cover crops – The more the soil is covered by a crop or grass, the less erosion will occur
- Early planting- Established an early ground cover by crops thus reducing the risk of soil erosion
- Inter-cropping – Increases the ground cover protecting the soil from erosion
- Crop rotation – improves soil structure where the rotation includes a grass
- Strip cropping/contour/field strip cropping – This is the growing of alternate strips of different crops in the same field with the purpose of interrupting the continuous flow of water or wind
- Weed or disease and pest control – This ensures a good crop stand that covers the soil more
- Harvesting procedures that leave crop residues on the field
- Mulching- The covering of the soil with organic or inorganic materials
- Contour farming follows the contours during ploughing, ridging and planting which reduces surface run off
- Grass strips formed by either leaving narrow strips of land un ploughed or planting grass on strips along the contour
- Afforestation and reafforestation
- Fallowing – leave the land uncultivated for same time (any 10x2=20mks)

12.

- Construction of dams
- Construction of dykes
- Planting trees along river bank to hold soil together

Observing government regulation on leaving a sizeable strip of an uncultivated land along the river bank

13. a) Bund

b) It is constructed along the contours

- A channel is dug with the upper width (y) 1.5cm and bottom width (x) 90cm
- Excavated soil is put on the lower part of the channel leaving the part (W) the ledge
- The steeper the slope the closer the bunds

14. a) It is the process by which top soil is detached, removed and carried away from one place to another place where it is not useful

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b) four types of water erosion

- rain drop/splash erosion
- sheet erosion
- rill erosion
- gully erosion

c) factors which influence soil erosion

- Amount and intensity of the rainfall
- Excess water run off take with it loose soil articles
- Slope of the land (topography)
- Sped of water as it flows to determine by the slope of the land .the steeper
- The slope the higher the rate of erosion
- Type of soil
- Some soils drain water faster than the other as sandy soil is easily eroded than sandy soil
- Soil depth
- Shallow soil become saturated with water quickly than deep soils
- Vegetation cover
- Forests protect soil against erosion than bare soil
- Overstocking
- Overstocking increase soil erosion
- Deforestation
- Cutting down of trees expose soil to agents of erosion leading tom soil erosion
- Planting annual crops in steep slope
- It leads to frequent cultivation hence exposure soil to erosion
- Indiscriminate burning of vegetation before cultivation
- The land is exposed to erosive forest of rain and wind
- Clear weeding
- This leaves the soil less protected against water erosion
- Pruning up and down the slope
- Increase soil erosion

d) seven cultural ways of controlling weeds

i)matching-matching smother weeds

ii)cover cropping-cover crop smother weeds

iii) Crop rotation-some weed only grow well when in association with certain crops

e.g. Striga grow only where some cereal crops and sugar cane are growing .when

these crops are rotated with dicots, striga does not germinate

iv) Timely planting-crops establish early before weeds thus smothering them

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v) use of clean seed/planting material-prevents the introduction of weeds to the farm land

vi) Proper spacing-helps to create little space for weed growth

vii) Clean seedbed-this starts off the crops on a clean bed so that they effectively compete

with weeds

viii) flooding-mainly practiced in rice fields

15. -Slow down surface run – off

-Filter soil particles from surface run off

WEEDS AND WEED CONTROL

1. Oxalis (sorrel)

-(oxalis latifolia)

b)- The weed contain bulbs i.e Elaborate & extensive root system that support the plant.

- Because it has rhizomes.

c) State the economic importance of the weed shown.

- Reduces yields of crops.

- Increases cost of production.

- It's a livestock feed.

- Fixation of nitrogen.

2. (a) Couch grass Digitaria Scalarum (½mk)

(b) Why is it difficult to control the weed?

- It has got underground rhizomes which grow deep in the soil (½ x 1pt = 1mk)

(c) One harmful effect of the weed on crop production

- Competes with crops for nutrients/soil moisture and space resulting to low yields

- Increases the cost of production when controlling it

(d) Two measures used to control the weed

- Use of appropriate herbicides

- Physical removal of rhizomes

3. i)- A-Double thorn (oxygoniumsinuatum)

- B- Coach grass (Digiteriascalarum)

ii) - Lower the quality of produce

- Lower yields

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- Compete with intended plants for nutrients and water
- iii) Weed B is difficult to control as it has underground rhizomes

iv) Can be effectively controlled by use of chemicals

4. Four methods of propagation which make weeds to have a high competitive ability over crops

- Availability to produce many viable seeds
 - Ability to propagate vegetative –with bulbs, rhizomes
 - Ability to regenerate woody stems-quickly
 - Efficient means of propagation
- Ability to remain viable in the soil for a long period of time

5. Weed C – Nutgrass (*Cyperus rotundus*)

Weed D – Sow thistle (*sonchus oleraceus*)

6. a) - Thorn apple

- Sodom apple
 - Oxalis
 - Tick berry
- mks)

($\frac{1}{2} \times 4 = 2$

b) - Before flowering to avoid spread through seeds

- Early stage before spreading underground organs

7.

- Requires skilled labour
 - Have long residual effect which interferes with future crops
- It is not environmental friendly/ pollutes the environment

8. a)

- MCPA
- 2 - 4 - D
- Bentazon
- Bromoxynil
- Linuron
- Loxyyril
- Atrazine
- Metribuzin

b) - 10 – 15cm high

- 2 – 4 weeks after emergence

9.

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- Complete for nutrients/ light/ space
- ACD as alternate host of insect pests
- Some produce poisonous substances
- Blocks water cords
- Lowers the quality of pasture
- Poisonous to man and livestock
- Parasites of desired crops
- Aquatic weeds affect navigation and water animals
- Increase the cost of production
- Cause irritation to workers
- Some have medicinal value
- Eaten by man and livestock
- Acts as soil cover
- Add organic matter in the soil
- Some are legumes

10. a) two factors that affect selectivity of herbicides

- Stage of plants growth
- Plants morphology and anatomy
- Mode of action
- Environmental factors (2x1=2mks)

11. specific examples of weeds describe their harmful effects in agricultural production

- Compete with crops for nutrients spacing, light, moisture lowering yield e.g. MacDonald's eye etc
- Some are parasitic e.g. wihiweed
- Low quality of produce e.g. Mexican marigold lowering quality of milk/pigweed seeds in finger millet
- Poisonous to both man and livestock e.g. Dahira stramonium, Bracken fern
- Alternate hosts for pests and diseases e.g. mallow weed –for cotton strainer
- Some are allelopathic/hinder germination e.g. Mexican marigold
- Block irrigation channel e.g. salvinia/water hyacinth
- Affect fishing and navigation-salvinia and water hyacinth
- Lower quality of pasture e.g. manyatta grass
- Reduce workers efficiency/irritate e.g. double thorn, shnging nelthe, devil's horse whip

CROP PESTS AND DISEASES

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1. two possible causes of swelling on the roots of legume crops
 - Infection by nematodes
 - Nodulation/ nitrogen fixing bacteria
2. the various cultural methods of controlling pests in crops
 - Timely planting-done early e.g. maize escape stalk borer
 - proper tillage-to expose soil boring pest i.e. white grubs
 - Timely harvesting-enables crops i.e. maize to escape weevil attacks
 - close season-avoids growing susceptible crops for some period to control pink worms in cotton
 - trap cropping-plant crops together with main crop to trap/attract pests before they attack e.g. sorghum round the maize plantation
 - trap rotation-rotate crops preferred by certain pests with those which are not e.g. groundnuts and potatoes rotated with maize and beans to starve pests
 - establishing resistance crop varieties-this enables crops to resist pest using natural mechanism e.g. goose neck sorghum against birds, tilting sorghum against shoot fly
 - field/farm hygiene-keep the field free from any plant material harboring pests by rouging, removal of crop residues
 - Alteration of environmental conditions by creating microclimates that are not conducive to some pests e.g. open pruning, mulching for traps
 - Destruction of alternative host especially weeds that host pests e.g. Removal of mallow weeds help control cotton strainers. Use of clean planting material e.g. seeds, suckers, crown bananas weevils are controlled
 - Proper spacing-makes it difficult for pests to move from one plant to another through close spacing in ground nuts controls aphids
 - Using organic manure which discourages eelworms
 - Irrigation-overhead irrigation in cabbages controls aphids(any 10x2 must be discussed)
3. What does the term close season mean in crop production?
 - A period during when a particular crop is not supposed to grown in a given areas so as to control diseases and pests built up.
4.
 - Use of clean planting materials.
 - Timely planting.
 - Proper seed bed preparation.
 - Use of resistant crop varieties.
 - Proper weed control/ destruction of alternate host.
 - Observing field hygiene.
 - Mulching.
 - Use of close season.

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- Use of trap crops.
- Proper spacing.
- Timely harvesting.
- Use of crop rotation.

6. Four symptoms of viral infections in plants

- Leaf curling
- Mosaics
- Malformation/distortions
- resetting (short internodes)
- Leaf chlorosis

7. (a) B - American bollworm $\frac{1}{2}$ mk
C - Weaver bird $\frac{1}{2}$ mk
D – Mongoose bird $\frac{1}{2}$ mk

(b)- Flooding with water

- Fumigation of the soil with furadan
- Physical killing ($\frac{1}{2} \times 1 = \frac{1}{2}$ mk)
- (c) Fruits ($\frac{1}{2} \times 1 = \frac{1}{2}$ mk)

8. (b) The various practices carried out in the field to help control crop diseases

- Crop rotation
- Rugueing/destroy infected plants
- Plant disease-free plant/use certified seeds
- Closed season
- Early planting /timely planting
- Proper spacing
- Timely weed control
- Use of resistant varieties
- Application of appropriate chemicals
- Use of clean equipment
- Quarantine
- Heat treatment to kill pathogens
- Pruning to create unfavorable micro-climate for diseases
- Proper nutrition to prevent deficiency

9. - Some pesticides cause suffocation of pests by blocking respiratory surfaces

- Some pesticides are stomach poisons that kill pests by damaging the cells/ tissues
- Some pesticides damage the pests nervous system
- Some pesticides kill pests by destroying digestive system

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10. a) Cutworm
- b) - Cuts the stem causing lodging
- Reduce plant population
- c) - Use of appropriate insecticides
Removing and killing it
11. i) This is a situation in which pest population caused damage beyond tolerance
ii) This is the use of combination of both chemical and cultural pest control methods
12. four harmful effects of crop pests
- Some e.g. nematodes damage crop roots causing wilting and death of the plant
 - Some like squeals unearth planted seeds leading to low plant population some destroy crop leaves lowering photosynthetic area-result to reduced yield
 - Sucking pest deprive plants of food by sucking plants sap
 - Some pests attack fruits berries and flowers lowering their quality and quantity
 - Some pests destroy embryo seeds lowering their germination potential
 - Some transmit crop diseases
 - Some e.g. stalk borer eat the growing points causing retarded growth
 - They lower marketability of crops produce by lowering quality
 - Where the leaf is the major product pest damage lower the quality and quantity through defoliation
13. a) Potato blight
- b) (i) *Pythophthora infestans*
ii) dry patches i.e. necrotic lesions on leaves and fruits
- affected fruits appear rotten and fall off prematurely
- c) spraying with copper fungicides
- roguing the affected crop
14. (a) – Anthracnose of Bananas / Banane anthracnose; (1x1=1mk)
(b) – Spray with (appropriate) fungicide
- Plant resistant varieties
15. a)- Mouse bird
- b) - Destroys grains in records
- Destroys fruits e.g. tomatoes
- 16.
- Cause swellings called galls on the roots.
 - Leads to blockage of the vascular vessels which transport materials within plants leading to wilting and stunting growth of the crops.
17. four cultural practices used in controlling crop pests
- Timely planting

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- Proper tillage
 - Close season
 - Trap cropping
 - Timely harvesting
 - Crop rotation
 - Planting resistant varieties
 - Field hygiene
18. three symptoms of coffee berry disease.
- Fungal disease
 - Favoured by high rainfall
 - Flowers have dark brown blotch /stred on brown petals
 - Green servier have small dark sunken parches/lessions
 - Barry (dip in the ground/dry up on the in the black mummified condition and when squeezed they are empty
19. (a) M – hedgehog N- Squirrel P – Rat
(b) M – Use dogs
N- eat germinating maize/bean seedling
21. - timely planting
- Early planting
 - Timely harvesting
 - Early harvesting
 - Proper tillage
 - Close season- period when that crop is not grown anywhere trap cropping
 - Crop rotation
 - Planting resistant carieties
 - Field hygiene
 - Alteration of environmental
 - Crop nutrition
 - Destroying alternative host
 - Use of clean planting material
 - Proper spacing
 - Lose of organic manure
 -
 - Irrigation.

CROP PRODUCTION VI

FIELD PRACTICES FOR MAIZE, MILLET, SORGHUM, BEANS AND RICE: **HARVESTING OF COTTON PYRETHRUM, SUGAR CANE COFFE AND TEA**

1. four management practices carried out in maize field at 45cm high

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- Weed control
 - Thinning
 - Farthing up
 - Top dressing
 - Pest and disease control
 - Rouging (1/2x4=2mks)
2. To prevent contamination of the cotton by the sisal strings
3. (i) (a) Land preparation
- Land is ploughed /dug
 - Ploughs/jembes used for primary cultivation
 - The land is leveled
 - Bunds are constructed around the plots to control water
 - The land is flooded up to a depth of 5cm
 - The soil-water mixture should be worked on until a fine mud is produced
- (b) Water control
- Bunds are constructed around the plots to control the water level
 - The land is flooded with water to a depth of 5cm before transplanting
 - The level of water is gradually increased to a height of 15cm by the time the rice crop is fully grown
 - Water should be allowed to flow slowly through the field
 - Old water should be drained and fresh one added where the flow of water is not possible
 - Old water should be drained every 2-3weeks
 - The field should be drained off 3weeks before harvesting (1mk x any 4pts = 4mks)
- (c) Fertilizer application
- Sulphate of Ammonia is applied in the nursery before sowing
- Sulphate of Ammonia is applied at the rate of 25kg for each nursery unit of 18.5m x 18.5m
 - Sulphate of Ammonia is applied in two splits before transplanting and 40days after transplanting
 - Sulphate of ammonia is applied at the rate of 125kg/ha before transplanting and 125kg/ha about 40days after transplanting
 - Double super phosphate is broadcasted in the field before transplanting
 - DSP is applied at the rate of 120kg/ha (1mk x any3pts = 3mks)

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(d) Weed control

- Flooding
- Uprooting
- Use of herbicides such as propanil against aquatic weeds (1 x any 3pts = 3mks)

(ii) The environmental conditions that may lead to low crop yields

- Poor soil fertility /infertile soil
- Damage by hailstorms
- Less rainfall/unreliable/drought
- Poor soil type resulting into leaching or water logging
- Inappropriate soil PH
- Inappropriate temperature (too low or high)
- Excessive wind leading to increase in water loss from the soil
- Extreme relative humidity
- Extreme of light intensity
- Topography / some attitudes e.g. very high may limit crop growth (1mk x any 7pts = 7mks)

Seedbed preparation

- Prepare land in dry period/ early/ before onset of rains
- Clear the land
- Remove stumps/ perennial weeds
- Plough/ primary cultivation
- Harrow/ carry out secondary cultivation

(1x5=5 mks)

Planting - Early planting/ plant at onset of rains

- Select suitable variety/ certified seed
- Depth of planting 2.5 cm-10 cm
- Plant with 1-2 seeds per hole
- Plant with DAP/ SSP/ DSP at rate of 120kg/ ha DAP/100-150kg/ha/DSP
- Plant 1-2 seeds per hole
- Spacing 75-90cmx23-30 cm

Weeding- Uproot weeds

- Tillage
- Use herbicides

Pest control

- Use chemicals/ pesticides
- Early planting

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- Planting certify seeds
- Rogueing/ field hygiene

Disease control- Uprooting and burning affected crop/rogueing

- Use appropriate chemicals
- Crop rotation
- Field hygiene

Harvesting —Harvested after 4-6 months depending on variety and ecological time

- Harvested when dry 14-20 % moisture content
- Stalk, are cut and stoked in the field
- Cobs are removed by hand

4. Two precautions taken when harvesting cotton

- Avoid mixing with foreign materials
 - Harvesting during the dry weather
 - During harvesting separate grade A and B
- Don't put in gunny/sisal bags

5. a) - Maize varieties

- Different varieties are developed for different ecological zones
 - Example: Hybrids and composites available
 - Kitale hybrids e.g. 612, 622 for high and medium altitudes
 - Embu hybrids 511, 513 e.t.c. for medium altitudes
 - Composites for lower altitudes like katumani composite, coast composite e.t.c.
- (1x5=5 mks)

b) Planting

- Plant early at the onset of rains
 - Dry planting is encouraged in low rainfall areas
 - Depth of planting 3 – 10 cm
 - One – two (1-2) seeds per hole
 - Spacing vary with variety (i.e. 20-30cm x 75-90cm)
 - Plant either manually or use planters
- (1x5=5 mks)

c) Pest and pest control

- Maize stalk borer – early planting, rogueing, destroy crop remains, apply appropriate pesticides (placed in cone)
- Army worm – use of recommended pesticides
- Aphid – spray with appropriate pesticides

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- Maize weevil – proper drying and dusting with pesticides
- Red flour beetle – good storage
- Rats – use rat proof stores, cats, traps or poison

Pests $5 \times \frac{1}{2} = 2 \frac{1}{2}$

Control $5 \times \frac{1}{2} = 2 \frac{1}{2}$

d) - Harvesting and storage

- Storing in cool areas can be practiced
- Carry out direct delusking in other warm areas
- Store in bulk (grains)
- Stored on cobs

Can be stored in bags

6. a) Ecological requirement (3mks)
- i) Altitude 0-2200m above sea level
 - ii) SOU- fertile alluvial or loam soil well drained
 - iii) temperature-moderate
 - iv) rainfall-moderate
 - v) PH-neutral or alkaline
- b) Varieties (3mks)
- Kenya flat complex
 - Double comb variety
 - Kitale hybrids
 - Embu hybrids
 - Coast composites
 - Katumani composite
- c) Seedbed preparation (5mks)

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PEST	CONTROL
Maize stalk borer	• Early planting • rogueing • Burning infected maize crops • Use of pesticides
Army worm	• Dusting with appropriate chemicals

- early

land

preparation to allow rotting of vegetation

- clearing of land using appropriate tools
- Ploughing done using appropriate implementing e.g. disc or mould board plough
- harrowing done where the seedbed is rough to a medium tilth
- does not require a very fine tilth
- eradicate perennial weeds

d) Pests and diseases (3mks)

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aphids	• Spraying using suitable insecticides
birds	• Scared away
Maize weevil	• Dusting maize comb or shelled maize with appropriate chemical • Proper storage hygiene
rats	• Use of rat proof stores, cats, traps • Bush clearing around stores
(1x4=4mks)	
DISEASES	CONTROL
White leaf blight	• Planting resistant variety
Maize streak	• Early planting • Use of resistant varieties • rogueing
rust	Planting resistant variety
smut	Crop rotation
(2x1=2mks)	

e) Harvesting

- Period varies from one variety to another
- In some cases stalks are cut and stocked in the field to allow combs to dry
Properly followed by removal of the combs which are stocked in the store
- De-husking directly in the field
- Use of the combined harvesters

7. a) Harvesting of cotton

- Harvesting is the picking of the cotton lint
- Harvesting is done 4 – 5 months after planting
- Cotton picking is done at weekly interval
- Picked lint is placed clean containers/ never use sisal bags whose fibres may mix with the lint
- Never pick wet lint
- Grade lint as it is picked in the field
- Place clean lint AR (safi) in one container and BR(fifi) in another container
- Pick lint which is exposed/ fully opened and the lint dry
- Harvesting is done in dry season

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- Avoid picking lint with contamination such as twigs, dry leaves or soil

b) The role of Agricultural Co-operatives in Kenya

- Co-operators pool their resources together to buy expensive machinery e.g. tractor for use by the members
- Provide education/ technical information to members
- Provide loans to members in form of inputs and cash
- Negotiate for higher prices for members
- Reduce overhead costs e.g. transportation, storage and use of machinery
- Bargain with supplier to give discount on seed, fertilizers and other farm inputs/ provide inputs at lower prices
- Provide employment for their members
- Benefit members from lower taxes charged
- Provide strong bargaining power for members on policy issues
- Market farmers produce
- Invest and pay out returns to members in form of dividends
- Help to negotiate for loans for members without security
- Some provide banking services to members

8. Field production of maize under the following sub-headings

a) Ecological requirements

- Altitude (0 – 2200) m above sea level
- Temperatures 23 – 27c
- Rainfall 750mm-1250mm
- Soils, fertile, well drained PH 7 -8

b) Field preparations

- Done during the dry season
- Disc, mold board ploughing to a depth of 20cm
- Disk harrowing to break the soil clods to a medium tilth
- Ridging done at spacing of 75cm apart

c) Planting and field management

- Seeds placed in the ferrous at a spacing of 30cm and covered with soil mixed with

DAP

- Gapping, thinning done depending on germination percentage
- Clean weeding done after every 4 weeks interval.
- Top dressing done at interval i.e. 1st done with CAN when the crop is knee high, 2nd

when the

crop is tussling

d) Pests and disease control

- Spraying the crop with fungicides (head smut control)
- Uprooting fully infected crops and burning them

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- Spraying the crop with insecticides and dusty the base of the leaves (control stalk borer)
- Field hygiene
- e) Harvesting and marketing
 - Hand harvested by plucking the cobs/ or machine harvested
 - Shelled, dusted for storage pests and packed
 - Delivered to millers consumers
 - Delivered to national cereals and produce board
- 9. Give two precautions measures a farmer should put into consideration when harvesting
RCH
 - Lint should not e mixed with foreign matter
 - Use different containers for different cotton grades
 - Avoid picking during wet weatherAvoid using gunning bags
- 10. (a) - Rainfall that is well distributed
 - Well drained fertile soils
 - Neutral soils
 - Warm temperatures(b) Clear the land/vegetation ploughed to appropriate tilth
Levelise for uniform planting
- (c) – Make holes 45 x 15cm
 - put in 1 teas spoonful of DAP and cover with soils lightly
 - Put 1 seed per hole and cover with soil when soils are moist(d)pests
 - Been aphids
 - Been brachids
 - American ball worm
 - Golden ring moth
 - Hens at flowering stageControl – spray using appropriate pesticide e.g diazinol

FORAGE CROPS

1. three factors which affects the quality of standing forage given to livestock
 - Forage species
 - Stage of harvesting
 - Mode of feeding
2. i) Seedbed preparation

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- Done during dry period/ done early
- Clear vegetation/ remove stumps
- Carry primary cultivation/ harrowing to
- Make furrows/ holes
- Spacing 90cm x 50cm for cutting and 90cm x 50cm for splits

ii) Planting

- At on set of rains/ early planting/ irrigate if necessary
- Select variety for ecological condition of the area
- Use health planting material
- Place the planting material in holes/ furrows
- Cover the planting material with soil to an appropriate depth
- Use cuttings or splits
- Select cutting from mature cane/ stems
- Cutting should have 3-5 nodes

iii) Fertilizer application

- Apply phosphatic fertilizer at planting 200kg/ha
- Apply manure before planting, 7-10 tonnes/ha
- Top dress phosphorous fertilizer after 8 weeks after planting
- Apply manure after harvesting and dig it

iv) Weed control

- Cultivation/ tillage/ mechanical
- Uprooting
- Slashing
- Suitable herbicide application
- Interplanting legumes as cover crops e.g. Desmodium,

v) Utilization

- Cut and take to animals when proportion of leaf is higher than the stem/ 3-5 months after planting/ frequency 8 weeks
- Cut down excess foliage to conserve as silage or hay
- Cut and sold
- Cut when mature to get stem cuttings for planting
- Cut stems at 2.5 – 5cm above the ground surface
- Use a sharp panga for harvesting
- Chop forage into small pieces before feeding
- Napier grass is cut, dried and used as mulching material

3. to reduce moisture content ✓ which can lead to rotting instead of formation during ensiling

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4. (a) - Harrow the land to a fine filth;
- Harrow during the dry or before the rains;
 - Make the seed bed weed – free / ensure clean seed bed;
 - Firm the seed bed using rollers after sowing;
 - Select a desirable variety of seed for the ecological zone,;
 - Sow seeds at the onset rains/ early planting;
 - Apply phosphatic fertilizers at appropriate rate of 200 – 300 kgs/ ha at planting time;
 - Drill or broadcast the seeds evenly;
 - Use a recommended seed rate for the variety / seed rate of 1.5 – 2.0 kg/ha pure seeds;
 - Bury seeds at 2 ½ times their diameter;
 - Control weeds by uprooting/ apply a suitable herbicide;
 - Apply nitrogenous fertilizers about 6 weeks after germination in split application.
 - Avoid grazing when the pasture is too young.
 - Practice light grazing in the field phase of pasture establishment. (10 x 1 = 10 mk)
5. Two causes of failure in pasture establishment
- Poor seed germination due to wrong placement of seeds
 - Poor inoculation of legume seeds
 - Lack of nutrients in the soil
 - Unfavourable chemical conditions in the soil
 - Poor drainage
 - Pest and disease attacks
14. - High nutritive value
- relieve bloat
 - Higher yields of forage per unit area
 - Improve soil fertility due to nitrogen fixation
 - Economy in use of nitrogen fertilizers
6. four factors that determine the quality of hay
- Forage species used
 - Stage of harvesting i.e. leaf-stem ratio
 - Length of drying period
 - Weather condition during process
 - Conditions of storage structure
7. a) Altitude - High altitude
- 2000 m above sea level
- Soils - well drained
- deep fertile
- Rainfall High rainfall above 900mm per annum

Well distributed throughout the growing period

b)

- Prepare land early enough/ during dry season/ before onset of rains
- Carry out primary cultivation appropriately
- Harrow to a medium tilth
- Remove all perennial weeds

c) - Establishment from stem cutting or splits or seeds makes furrows at a spacing 1m apart. Plant the grass 0.5m apart within the rows. Holes may also be used
Planting is done on the onset of long rains

Phosphate fertilizers are used as planting fertilizers

DAP fertilizer is applied at the rate of 100 – 150kg/ ha

Organic manure is applied at the rate 10 tones/ ha

d)

Utilization – chopped and fed to livestock as green fodder

Detoliation – can be harvested when it is over 8 – 12 weeks

ii)

- Forage has high DM content hence high DM yield
- High cellulose content hence it is woody and fibrous
- High lignin cuten lannin and silia content which are all indigestible
- It has low crude protein content
- It has low leaf stem – ratio
- It has low dry matter digestibility

8. – Weather conditions during dry process;

-Length of the drying period

- Stage of growth at harvesting time/leaf-stem ratio of the plant species;

- Species of the hay crop;

- Storage facilities/method of storage;

- Period of storage

- Disease and pest attack on the crop;

- Fertility of the soil;

9. (a) (i) Preparation of farm yard manure:-

- Collect animal waste/refuse/dung and urine;
- Collect animal bedding/litter and other rotten plant residues;
- Store collected materials under roof/shed to prevent leaching and oxidization of nutrients;
- Turnover the materials regularly;

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- Sprinkle water if dry;
- leave the material to rot completely before use; (6x1=6mks)

(ii) Preparation of Hay

- Cut the grass /legume in the field when 50% of it is starting to flower;
- The cut forage is spread in the field for four continuous days (sunny days)
- The cut forage is turned daily for even for four uniform drying;
- Gather the dried material in a central spot;
- Bale the material;
- Properly store the baled hay (6x1=6mks)

(b) Factors to consider in timely planting of annual crops

- Escape from serious weed competition;
 - Utilization of early rainfall;
 - Exploitation of Nitrogen flush in the soil that has accumulated during dry season;
 - Escape from serious pest + disease attack e.g. stalk borer in maize;
 - Fetch high market prices when harvested early;
 - Reduce competition for labour during labour peak period;
 - For harvesting season to coincide with dry period to reduce losses e.g. cotton
- Early planting means early farming/calendar for the farmer to enable him /her to finish up other farm activities; (8x1=8mks)

10. (a) It is the constant removal of the sterility fibrous material left behind after continuous grazing

(b) -Done by slashing the whole pasture to the base, leaving only the maintenance forage.

- Burning is also done.

11. two advantages of grass-legume pasture over pure grass pasture

- More nutritious to livestock
- Improves soil fertility through nitrogen fixation
- More total yield per unit area
- Security against total pasture loss (2x1=2mks)
 - to feed animals during the dry seasons
 - forage species used
 - Stage of harvesting / leaf to stem ratio
 - Length of drying period
 - Weather conditions
 - Storage conditions

AGRICULTURAL ECONOMICS III **(PRODUCTION ECONOMICS)**

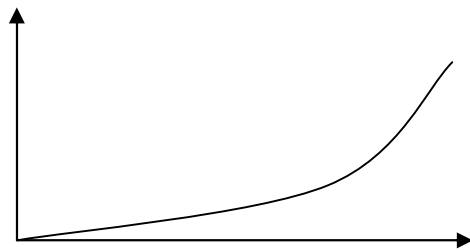
1. four ways of increasing labour efficiency on the farm

- Training them

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- Giving incentives
- Supervision
- Good operator – worker relationship
- Farm mechanization
- Assigning tasks according to skills & specialization
- Proper remuneration : Attractive salaries

2. a) graph representing the total egg production per week.



b) - Increasing returns production function.

3. (a) - Help to determine the value of the farm/ determine assets and liabilities.

- Provide history of the farm.

- Assist in planning and budgeting in various fields.

- Helps to detect losses or theft in the farm.

- Assists when sharing losses or profits (dividends) for communal owned farms/ partnership.

- Help to settle disputes in the farm among heirs.

- Help to support insurance claim e.g. against fire and theft.

- Provide labour information like terminal benefits, NSSF due, Sacco dues for all employees.

- Help to compare the performance of different enterprises within a farm or other farms.

- Help in the assessment of income tax to avoid over or under taxation.

- Records, helps to show whether the farm business is making profit or losses. This information

helps in obtaining credit.

(10 x 1

= 10 mks)

(b) - Training worker e.g. in F.T.C's, during field days, Agricultural shows, through demonstrations and workshops.

- Measuring farm operations to supplement the labour force.

- Providing incentives to workers such as attractive wages, free protective wear, housing, medical facilities, proper feeding, rewarding good workers. Et.c

- Supervising and counseling workers.

- Creating good operator - worker relationships.

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- Assigning specific tasks to the labor force.
- (c) - Establishment of land ownership.
 - Measurement of land size.
 - Description of the land.
 - Recording and mapping of the surveyed land.
 - Solving objections if any.
 - Submission of the maps and records to the district land office registration
- 4. Three types of agricultural services available to the farmer
 - Credit
 - Extension and training
 - Agricultural research
 - Banking
 - Artificial insemination
 - Veterinary
- 5. Four management guideline questions which assist a farm manager in making accurate farm decisions
 - What product to produce?
 - How much to produce?
 - What to produce?
 - For whom to produce?
- 6. - Training
 - Farm mechanization
 - Labor supervision
 - Giving incentives and improving terms and conditions of serviceAssign specific tasks
- 7. - Fertilizers - Seeds
 - Pesticides -Casual labor
- 8. - Banking
 - Extension and training
 - Credit facility
 - Agricultural research
 - Marketing
 - Farm input suppliesTractor hire service
- 9. (a) Is the sum total of goods and services produced by a country within a period of one year
(b) -Per capital income: Is the gross national income divided by the number of people living

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in a country

10. -Diversification- Setting up several and different enterprises on the farm. If one fails the

farmer cannot incur total loss.

- Contracting- farmers can enter into contract with consumers. It guarantees a constant fixed market for goods/services
- Insurance- Taking an insurance cover to compensate them in case of loss
- Input rationing- Farmers can control the quantities of inputs used in various enterprises to reduce losses
- Flexibility in production methods- Ability to change from one enterprise to another in response to demand changes
- Adopting modern methods of production e.g. disease control, irrigation, mechanization e.t.c.

11. Application of fertilizer

Input 50kg bag fertilizer	Out put 90kg bag maize	Average product (AP)	Marginal product (MP)
0	6	-	0
1	10	10	4
2	24	12	14
3	31	10.33	7
4	36	9	5
5	40	8	4
6	43	7.18	3
7	43	6.14	0
8	40	5	-3

(ii) The best level of production in relation to the inputs and out put is level 3

(b) (i) Gross margins for the crops

(i) Maize

Value of maize/income	$5,500 \times 15 = 82,500/=$ (1mk)
Cost of labour	$50 \times 150 = 7,500/=$ (1mk)
Cost of cultivation /ha	$1 \times 3,000 = 3,000/=$ (1mk)
Cost of seed	$25 \times 100 = 2,500/=$ (1mk)
Cost of DAP fertilizer	$3 \times 1,500 = 4,500/=$ (1mk)
Cost of C.AN fertilizer	$3 \times 1000 = 3,000/=$ (1mk)
Total variable costs	$20,500/=$ (1mk)
GM for maize	$82,500 - 20,500 = 62,000$ (1mk)

(ii) Beans

Value of beans/income	$5,000 \times 500 = 250,000/=$ (1mk)
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FREE AGRICULTURE K.C.S.E 2012 to 2015

Cost of labour	$75 \times 200 = 15,000/=$ (1mk)
Cost of cultivation /ha	$1 \times 3,600 = 3,600/=$ (1mk)
Cost of seed	$20 \times 80 = 1,600/=$ (1mk)
Cost of DAP fertilizer	$2 \times 1,500 = 3,000/=$ (1mk)
Cost of C.AN fertilizer	$1 \times 1000 = 1,000/=$ (1mk)
Total variable costs	$27,200/=$ (1mk)
GM for beans	$250,000 - 27,200 = 222,800$ (1mk)

(b) (ii) The crop which is profitable from the calculation is that : (1mk)

- It is more profitable to grow beans than maize

12. a) ZONE I

- For each additional unit of input applied the output of maize increased at an increasing rate because the fertilizer resources are underutilized ✓1 (1x2=2 mks)

ZONE II

- For each additional unit of input applied the output of maize increased at a decreasing rate because the resources are used to the maximum ✓1 (1x2=2 mks)

ZONE III

- For each additional unit of input applied the output of maize decreases because the fertilizer/ resources are excessively applied ✓1 (1x2=2 mks)

b) ZONE II

13. Give four variable costs in maize production

- Cost of fertilizer
- Cost of seeds
- Cost of pesticide
- Cost of weeding
- Cost of harvesting
- Cost of casual labour
- Cost of fuel

14.

ITEM	QUAN TITY	NO. OF UNITS	COST PER UNIT	TOTAL COST	VARIABLES
Weeding	-	-	-	800	00
Seeds	20k	2	300	600	00
Irrigation	g	2ha	600	1200	00

FREE AGRICULTURE K.C.S.E 2012 to 2015

Ploughing	-	2ha	500	1000	00
Clearing land	-	-	-	1200	00
Planting	-	2ha	400	800	00
Harvesting	-	2ha	1 200	2400	00
DAP fertilizer	-	2	10 000	20 000	00
DAN fertilizer	2ba	2	700	1400	00
Gunning bags	gs	32	40	1280	00
transport	2ba	-	-	2000	00
	gs				
	32				
	-				
Total variable cost				32 680	00
income		32bags	1200	38 400	00

$$\begin{aligned}\text{Gross margin} &= \text{total revenue} - \text{total variable cost} \\ &= 38400 - 32680 = 5720.00\end{aligned}$$

15.

- Profit maximization is the profit in a production process where the highest net returns (Net revenue) on invested capital is realized/ when the difference between total revenue (TR) and total cost (TC) is the highest point in a production process/ where profit is highest
- Is where marginal revenue (MR) is equal to or almost equal to marginal costs

16. a) i) Gross margins for the crops

Value of maize/ income	55000 X 15 = 82500	1 mark
Cost of labour	50 X 150 = 7500	1 mark
Cost of cultivation/ ha	1 X 3000 = 3000	1 mark
Cost of seed	25 X 100 = 2500	1 mark
Cost of DAP fertilizer	3 X 1500 = 4500	1 mark
Cost of CAN fertilizer	3 X 1000 = 3000	1 mark
Total variable costs	20500	1 mark
GM for maize	82500 - 20500 = 62000	1 mark

ii) Beans

FREE AGRICULTURE K.C.S.E 2012 to 2015

Value of beans/ income	$5000 \times 500 = 250000$	1 mark
Cost of labour	$75 \times 200 = 15000$	1 mark
Cost of cultivation/ ha	$1 \times 3600 = 3600$	1 mark
Cost of seed	$20 \times 80 = 1600$	1 mark
Cost of DAP fertilizer	$2 \times 1500 = 3000$	1 mark
Cost of CAN fertilizer	$1 \times 1000 = 1000$	1 mark
Cost of sprays	3000	1 mark
Total variable costs	27200	1 mark
GM for beans	$250000 - 27200 = 222800$	1 mark

iii) It is more profitable to grow beans than maize 1 mark

b)

- Size of the farm
- Climatic conditions
- Fairness objectives and preferences
- Existing market conditions
- Available resources
- Expected returns

17. - The farmer should grow groundnuts;

- The crop has a higher gross margin than cotton;

18. – The farmer may be able to estimate the required production resource e.g labour capital
e.t.c

- Assists farmer when e.g. labour capital etc
- Assists farmer in making management decisions;
- Helps to reduce uncertainties in the production process;
- Shows progress or lack of progress in farm business;

19. (a) (i) See the graph paper

(ii) 56 bags; (1x1=1mk)

(b) Table – (16x ½ =8mks)

Year	Fertilizer applied (bags)	Maize output (bgs)	Marginal product	Average product
1995	0	4	-	-
1996	2	10	6	5
1997	4	28	18	7
1998	6	42	14	7
1999	8	52	10	6.5
2000	10	60	8	6
2001	12	66	6	5.5
2002	14	66	0	4.7

FREE AGRICULTURE K.C.S.E 2012 to 2015

2003	16	64	-2	4
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(c) (i) 4 bags (1x1=1mk)

(ii) MP is maximum; AP is maximum;(2x1=2mks)

(d) Gross income = Total output x price per unit

in 2002 $66 \times 1000 = 66000/=$ (1x1=1mk)

in 2003: $64 \times 1000 = 64000/=$ (1x1=1mk)

(ii) Net income = Total income – Total cost

in 1999: Total income was $52 \times 1000 = 52000/=$

total cost was $8 \times 1200 = 9,600/=$

Hence $52000 - 9600 = 42,400$; (1mk)

= Shs. 42,400/= (1mk)

21.

- Flood costs (F.C)
- Variable costs (V.C)
- Total costs (T.C)
- Average costs (A.C)
- Marginal costs (M.C)

22.

- Co-operative societies
- Crop boards
- Commercial banks
- Agricultural finance corporation (A.F.C)
- Settlement fund trustees
- Hire purchase companies
- Insurance companies. (Any 4)

23. three ways in which labour peaks can be overcome in the farm

- Overtime working for casual labourers
 - Greater use of casual workers
 - Mechanization
 - Use of contractors who may be engaged to do some work at a fee
 - Cropping system devised such that ripening of crops could be at different times
- Work study to devise new techniques of doing work more quickly and efficient

24

- training
- giving incentives/motivation
- farm mechanization
- labour

25. (a) 1000kg of NAP con 46kg P₂O₅S

$$\frac{150}{100} \times 100 = 150$$

FREE AGRICULTURE K.C.S.E 2012 to 2015

50

300kg of DAP per hectar

1ha = 300kg of DAP

5ha x 300

1 = 1500g of DAP

1 bag = 50kg

1500 x 1 = 1500kg

50 = 30bags pf DAP

N/B Approximation = 3obags

CAN

100kg contain 20kg of price N

200x 100 = 200kg

30 = 1000kg

1 ha = 1000kg

5ha = 1000 x 5 = 5000kg

1bag = 50kg

5000 x 1 = 5000kg

50

= 100bags of CAN

(b) (i) – cross margin is variable cost – total revenue

Gross margin of irish potatoes

Cost of fert = shs 10000 x 5 = 50000

Cost labour requirement = 50 x 200 x 5 = 50,000

Cost of seed potatoes 20,000 x 5 = 100000

Cost of fungicides 5000 x 5 = 25000

Cost of ploughing 400 x 5 = 50,000

Total variable cost shs.145,000

Total revenue = shs.50,000 x 50 = shs.1,500,00

Gross margin shs. 1,500,000 – shs.145,000 =shs.1,255,100

(i) Maize

Cost of fert. shs 10000 x 5 = shs.50000

Cost of fert. shs. 4800 x 5 = shs.24000

Cost of maize seed shs.3000 x 5 = shs.15000

Cost f labour shs.200 x 150x 5 = shs.150000

Cost of ploughing shs.4000 x 5 = shs.20000

Total cost = shs.259000

FREE AGRICULTURE K.C.S.E 2012 to 2015

Revenue 750000 X 5 X 20= Shs.750000

Gross margin = 750000

- 259000

Shs.481000

(ii)He should grow potatoes

- pests

Diseases

Unreliable rainfall

Change in temperature

Strong wind

Light aspect

Infertile soils

AGRICULTURAL ECONOMICS IV

1. three methods of grafting that are used in propagation of plants

- Whip are tongue grafting
- Side grafting
- Approach grafting
- Bark grafting

Notch grafting

2. a) i) Prepare a profit and loss account for Mr. Tembo's farm for the year ending

31st December 2003

(9mks)

PROFIT AND LOSS ACCOUNT FOR MR. TEMBO✓

FARM FOR THE YEAR ENDING 31ST DEC 2003

		Sales and receipts	
	sh cts		
Opening stock✓	12000 00	Milk sale✓	8000 00
Purchase of farm tools✓	1000 00	Sale of goats✓	500 00
Zero grazing unit	10000 00	Cabbage sale✓	750 00
construction✓	800 00	Sale of heifers✓	9400 00
Machinery depreciation✓	750 00	Sale of tea✓	4700 00
Interest payable ✓	300 00	Closing	16000 00
Pesticide purchase✓	400 00	valuation✓	
Veterinary bills ✓	4800 00		
wages ✓			
	30050 00		39350 00
TOTAL	9300 00		
net profits ✓			

FREE AGRICULTURE K.C.S.E 2012 to 2015

	39 350 00		39 350√ 00

ii) Calculate the percentage profit or loss made by the farm (1mk)

$$\% \text{profit} = \frac{\text{profit}}{\text{Total income}} \times 100$$

$$= \frac{9300}{39350} \times 100 = 23.6\%$$

b) five functions of farmer's cooperative societies

- function of farmers cooperative societies
- marketing farmers produce
- negotiating fair prices for produce and input
- keeping records of the cooperative activities and in forming the members accordingly
- paying dividends to members
- giving loans in kind to members
- educating members on matters relevant to cooperative (5x1=5mks)

ii) Outline five common risks and uncertainties in farming

- risks and uncertainties
- pest and diseases outbreak
- price fluctuation
- sickness and injury
- natural catastrophes e.g. floods, earth quakes, storm, strong wind
- new technologies of production
- ownership uncertainty
- physical yield on what is expected

3. four reasons for using certified seeds for planting

- High yielding
- Quality produce
- High germination percentage
- Grow faster

4. two financial statements which may be prepared on a farm.

- Balance sheet.
- Cash analysis.
- Profit and Loss Account.

5. (a) Profit and loss A/C for Langat's farm for the year ending 31st December, 2004

Purchases & Expenses	Shs.	Cts	Sales & receipts	Shs.	Cts
Opening valuation	150000	00	Mohair	75000	00
Goats	4000	00	Rabbits	3600	00

FREE AGRICULTURE K.C.S.E 2012 to 2015

Poultry	15000	00	Eggs to hotel	15000	00
Casual worker	12000	00	Closing valuation	200000	00
Subtotal	181000	00			
Net profit	112600	00			
Total	293,600			293,600	

Awarding:-

- Title (½mk)
- (Purchases & expenses and sales & receipts) ½mk
- Entries each ½ x 10 (5mks)

(b) State the benefit of a profit and loss A/C to Mr. Lang'at

- Helps the farmer to detect whether he has loss or profit
- Helps in tax assessment to avoid over taxation
- Acts as evidence when a farmer requires a loan

6. a) - Invoice

- Receipt
 - Delivery note
 - Purchase order
 - Statement of account
- (4x1=4 mks)

b)

MR. ROBS, PROFIT AND LOSS ACCOUNT AS AT 31ST DEC, 2009

Purchases and expenses√ ½ mk		Sales and receipt√ ½ mk	
Opening stock	150,000	Sale of milk	10,000
Veternary bills	2500	Sale of cabbages	20,000
Livestock feeds	2500	Sale of two heifers	10,000
Fertilizer	5000	Sale of tomatoes	3000
Seeds	4000	Debts available	20,000
Debts payable	<u>4200</u>	Closing valuation	200,000
TOTAL √ ½ mk	<u>168000</u>	√ ½ mk	263,000
Profit √ 1 mk	<u>94,800</u>	√ 1 mk	263,000

FREE AGRICULTURE K.C.S.E 2012 to 2015

263000	(½ mk each entry 6 mks) Total 10 mks
--------	--

c) It made profit

Profit ksh. 94,800✓ 1 mk

% profit= $\frac{\text{profit}}{\text{Opening Valuation}} \times 100$

Opening Valuation

94,800x100

150,000

= 63.2%✓ 1 mk

d) -Diversification- Setting up several and different enterprises on the farm. If one fails the

farmer cannot incur total loss.

- Contracting- farmers can enter into contract with consumers. It guarantees a constant fixed market for goods/services
- Insurance- Taking an insurance cover to compensate them incase of loss
- Input rationing- Farmers can control the quantities of inputs used in various enterprises to reduce losses
- Flexibility in production methods- Ability to change from one enterprise to another in response to demand changes
- Adopting modern methods of production e.g. disease control, irrigation, mechanization e.t.c.

FREE AGRICULTURE K.C.S.E 2012 to 2015

7. a) i) profit and loss account for Mr. Tembo's farm for the year ending 31st Dec. 2003

PROFIT AND LOSS ACCOUNT FOR MR. TEMBO✓ FARM FOR THE YEAR ENDING 31ST DEC 2003

Purchases and expenses		Sales and receipts	
	sh cts		
Opening stock✓	12000 00	Milk sale✓	8001 00
Purchase of farm tools✓	1001 00	Sale of goats✓	501 00
Zero grazing unit construction✓	10000 00	Cabbage sale✓	751 00
Machinery depreciation✓	801 00	Sale of heifers✓	9401 00
Interest payable ✓	751 00	Sale of tea✓	4701 00
Pesticide purchase✓	301 00	Closing valuation✓	16000 00
Veterinary bills ✓	401 00		
wages ✓	4801 00		
	30050 00		39350 00
TOTAL net profits ✓	9300 00		
	39 350 00		39 350✓ 00

ii) Calculate the percentage profit or loss made by the farm (1mk)

$$\% \text{profit} = \frac{\text{profit}}{\text{Total income}} \times 100$$

$$= \frac{9300}{39350} \times 100$$

$$= 23.6\%$$

b) five functions of farmer's cooperative societies

- Function of farmers cooperative societies
- marketing farmers produce
- negotiating fair prices for produce and input
- keeping records of the cooperative activities and in forming the members accordingly

FREE AGRICULTURE K.C.S.E 2012 to 2015

- paying dividends to members
- giving loans in kind to members
- educating members on matters relevant to cooperative(5x1=5mks)

ii) five common risks and uncertainties in farming

- Risks and uncertainties
- pest and diseases outbreak
- price fluctuation
- sickness and injury
- natural catastrophes e.g. Floods, earth quakes ,storm ,strong wind
- new technologies of production
- ownership uncertainty
- physical yield on what is expected

8. a) Prepare a balance sheet s at 31.12.2005

BIDII FARM BALANCE SHEET AS AT 31.12.2005

LIABILITIES	SHS	ASSETS	SHS
CTS		CTS	
Current liabilities		Current assets	
Overdraft	15000	Debt receivable	20000
00		0	
Unpaid wage	3000	Stocks	25000
00		00	
Long term liabilities		Bank balance	100000
Bank l	30000	00	
00		Fixed assets	
Net worth	<u>897000</u>	Perennial crops	250000
00		00	
<u>945000</u>	<u>00</u>	Land	350000
		00	
		Livestock	200000
		00	
		<u>945000</u>	<u>00</u>

FREE AGRICULTURE K.C.S.E 2012 to 2015

- b) Yes it qualified for a loan because it was solvent i.e. has more assets than liabilities
9. one condition in which each of the following documents is used.
- i) Invoice –when goods //services are sold /bought on credit
 - ii) Delivery note-when goods are physically delivered to the buyer
 - iii) Receipt-when goods/services are bought or rendered on cash
10. i) Prepare a profit and loss account for Mrs. Okello's farm

profit and loss A/C for Mrs.Okello's farm for the year ending 31/12/2009

Purchase and expenses		Sales and receipts
Opening valuation	12000 00	Milk sales 8000
Pesticides	3000 00	Sales of goats 5000
Construction of store	10000 00	Sales of tomatoes 1750
Depreciation of machines	3000 00	Sales of heifer 10 000
Interest payable	1750 00	Sales of coffee 5000
Purchase of tools	800 00	Closing valuation 16000
Veterinary bills 1	400	
Wages 10,000		
Net profit	3800	
	45750	45750

Award of marks Title -1mk
 Purchases and expenses side -1mk
 Sales and receipt sales-1mk
 Net profit – 1mk
 Both totals-1mk (5x1=5mks)

Each of the correct entries in purchase and expenses and sales and receipt sides (14x ½ =7mks)

- ii) Calculate the percentage profit or loss that Mrs. Okello made during the year 2009

$$\frac{3800 \times 100}{45750} = 8.3\%$$

- iii) six ways in which farmers adjust to risk and uncertainties in farming

- diversification-production of services products at the same time to avoid risks due to weather, fluctuation in price and disease
- contracting-make contracts with dealers to supply or buy certain commodities at fixed prices thus transfer the risk of drop in demand and supply

FREE AGRICULTURE K.C.S.E 2012 to 2015

- insurance-purchase security by payment of small sum of money for compensation in case of failure
- input rationing-use of inputs sparingly to avoid wastage
- flexibility in production-combination and substitution of inputs and techniques of products for each other use the cheapest
- use of government price stabilization policies
- adapting modern methods of farming-use of researched varieties, breeds better adapted to local conditions

selecting more certain enterprises-engage in enterprises with more surety of success i.e. artificial insemination as opposed to natural insemination (any 6x1=6mks)

11. - Bank overdraft
 -Bank loans
 -Debts payable
 -Tax payable
 -rent

AGRICULTURE ECONOMICS (V)

1. a) the principle that govern the operations of farmers' co-operative societies
- Open membership
 - Equal rights
 - Share limit
 - Neutrality
 - Non- profit motive
 - Loyalty
 - Withdrawal of membership
- b) the role of agricultural cooperatives in Kenya
- Provide education/ technical information to members
 - Negotiate for higher prices for members products
 - Market farmers produce
 - Help to negotiate for loans for members without security
 - Provide inputs to members at lower prices
 - Invest and pay out returns to members in form of dividends
 - Provide transportation, storage and use of machinery to farmers
- c) various functions of agricultural marketing
- Advertising: They advertise farm products in order to increase demand
 - Financing: Provide capital to carry out agricultural activities
 - Transportation: Provide transport to farm produce to the areas of consumption
 - Storage: store farm produce after harvest in order to minimize losses

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- Selling: Sell on behalf of the farmer
- Packing: pack the farm produce to reduce storage space and make transportation easy
- Processing: process the farm produce in order to provide a variety, increase value and prolongs shelf life
- Grading: putting into grades to provide uniform standards
- Assembling: gathering the farm produce for bulking and transportation
- Insurance: Bearing risks by protecting farm damage
- d) Explain five problems farmers face in marketing agricultural products
- Perishability: detoxation of quality
- Seasonability: affect price and storage problems
- Bulkiness: occupy large space hence problems of handling and storage
- Storage: lack of storage facilities on farms
- Poor transport system: Lead to spoilage of farm produce
- Change in market demand: time between planning and actual production create lack of market
- Lack of market information: concerning prices, how much to produce, where to sell goods; farmers end up disposing of the surplus at throw away prices
- Fluctuation in market price due to seasonality in nature of agricultural products
- Poor marketing systems for some produce: Result in lack of market

2. a) four marketing functions (4mks)

- Buying and assembling
- Transportation and distributing
- Storage
- Packing
- Processing
- Grading
- Marketing research
- Selling
- Financing
- Bearing risk (4x1=4mks)

b) Outline four problems associated with marketing of agricultural products. (4mks)

- perishability
- seasoning
- bulkiness
- poor storage facilities
- poor transport system
- lack of market information
- limited elasticity of demand

FREE AGRICULTURE K.C.S.E 2012 to 2015

3. Two roles of agricultural society of Kenya

- Promotes the agricultural industry
- Organizes national ploughing competitions
- Publish the Kenya farmer magazine
- Improve useful indigenous animals
- Publish a 'stud' book
- Hold competitive shows
- Hold trade fairs on livestock
- Demonstration on how to use agricultural machinery
- Encourage breeding and importation of pure breed stock

4. (a) Elasticity of demand for a commodity is the degree of responsiveness of demand to price

OR - The sensitivity of demand to change in price (1mk) (mark as a whole)

(b) Calculation of the elasticity of demand:- ✓mk
(4mks)

$$\text{Elasticity of demand} = \frac{\% \Delta \text{Quantity}}{\% \Delta \text{price}}$$

$$\begin{aligned} \% \Delta \text{Quantity} &= \frac{\text{Change in quantity}}{\text{Quantity}} \times 100 \\ &= \frac{(1000 - 800)}{800} \times 100 = 25\% \end{aligned} \quad \checkmark \text{mk}$$

$$\% \Delta \text{Price} = \frac{\text{Change in price}}{\text{price}} \times 100$$

$$= \frac{(20 - 22)}{22} \times 100 = 9\% \quad \checkmark \text{mk}$$

$$Ed = \frac{25}{9} = 2.8 \quad \checkmark \text{mk}$$

(c) Six problems of marketing maize as an agricultural product

- Bulkiness
- Poor storage
- Seasonality of crops hence demand
- Perishability of farm produce
- Poor transport system
- Delayed payments
- Competition from cheap imports (1mk x any 6pts = 6mks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

(d) Nine principles governing cooperatives in Kenya

- Open membership- voluntarily joining on payment of membership fees
- Equal rights – One man one vote run democratically
- Share limit – A member buys shares up to a specific maximum limit.
- Interest on shares – Any money distributed according to shares
- Withdrawal from ownership – voluntarily
- Loyalty – Members to be faithful and loyal
- Education – continuously educating its members
- Co-operative principle- cooperative members joint cooperative movement
- Non-profit motive – cooperatives are non-profit making organizations

5. Four reasons why training is important in some crops

- Facilitate field practices of spraying and harvesting
- Improves crop quality by preventing solving
- Enable crop grow in the required direction
- Improve yield
- Control pest and diseases

6. a) - Buying and assembling – acquisitions from small scale farmers and accumulate

- Transporting and distribution – from area of production to areas of consumption and retailers
- Storage – kept to accumulate in amount before selling off or to wait for time of scarcity
- Packing- to protect agent damage, theft
- Processing – involve preparation of produce for consumption e.g. pasteurizing milk
- Grading and sorting – placing in groups according to size, colour e.t.c.
- Each group attracts different prices
- Packaging – presentation of produce in a way to attract consumers e.g. labeling, good packing materials
- Collecting marketing information – through media, on prices and demands of goods on markets
- Selling – Final presentation of produce to consumer's e.g. advertising, display e.t.c.
- Financing – availing capital for marketing activities
- Bearing risks – at each stage of marketing, apart is to bear the cost of uncertainties

b) - Marketing problems of agri products

- Perishability – short life cycle
- Seasonality – supply depends on the season/ vary with season
- Bulkiness – demand for more space Vs value
- Storage – need costly storage e.g. cold rooms, refrigeration
- Poor transport system especially in rural areas

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Changes in market demand – take long to produce hence cannot meet changes in market
- Limited elasticity of demand – restrict supply
- Lack of market information

7. Is the degree of responsiveness of supply to change in price

8. - Giving subsidies by reducing the cost of production inputs

Fixes prices of the related products

b) The role of Agricultural Co-operatives in Kenya

- Co-operators pool their resources together to buy expensive machinery e.g. tractor for use by the members
- Provide education/ technical information to members
- Provide loans to members in form of inputs and cash
- Negotiate for higher prices for members
- Reduce overhead costs e.g. transportation, storage and use of machinery
- Bargain with supplier to give discount on seed, fertilizers and other farm inputs/ provide inputs at lower prices
- Provide employment for their members
- Benefit members from lower taxes charged
- Provide strong bargaining power for members on policy issues
- Market farmers produce
- Invest and pay out returns to members in form of dividends
- Help to negotiate for loans for members without security
- Some provide banking services to members

9. Law of demand – when the price of a good or service is low, many consumers are able and willing to

buy it and vice versa

- Law of supply – when the price of commodity is high, many sellers are able to provide the commodity in market for sale. (mark as a whole)

10. (i) K.N.F.U – Kenya National Farmers Union

(ii) H.C.D.A – Horticultural Crops Development Authority

11. a)- It is an organization of people who have joined together voluntarily with a common purpose

for a mutual economic benefit

b) Two functions of co-operatives

- Marketing farmers produce
- Negotiating for fair prices for farmers produce and also for Inputs
- Keeping records of the co-operative activities and informing the members accordingly
- Paying dividends to the members
- Giving loans in hand to the members
- Educating the members on matters relevant to their co-operative through field days,

FREE AGRICULTURE K.C.S.E 2012 to 2015

seminars, workshops and demonstrations

c) - Open and voluntary membership: Rights of anybody who fulfills the conditions set out by the

laws of co-operative society

- Democratic control: All members have equal rights to say on the affairs of the co-operative
- Each member is only allowed one vote regardless of shares possessed
- Share limit: All members should have equal chances of buying shares
- Distribution of dividends: Any profits should be distributed to members as dividends depending on their share contributions
- Withdrawal of membership: Should be voluntary
- Selling of produce: Only members can sell their produce through the co-operatives
- Loyalty: members are bound to be loyal to their co-operative society
- Non- profit motive: Co-operatives are not supposed to be profit motivated. They should sell their products to members at seasonable prices
- Co-operative organization: The co-operative should join the co-operative movement from primary level to national or international level
- Co-operative should sell on cash – goods sold on credit are sometimes not paid for
- Continuous expansion: Co-operatives should aim at continuous expansion in terms of membership and physical facilities
- Neutrality: Co-operatives should be neutral in terms of religion, politics or language

12. four factors which influenced the demand of tomatoes in the market

- Quality of tomatoes
- Price of tomatoes
- Presence of other substitutes
- Price of the substitute

AGROFORESTRY

1. two reasons for seed treatment of tree species before planting

- Break dormancy
- Control pests and diseases

2. four ways by which Re-afforestation help in land reclamation

- Add organic matter from falling leaves
- Recycles soil erosion
- Control soil erosion
- Improve drainage of swampy areas
- Play part in hydrological cycle

3. four advantages of agro forestry

- Source of wood fuel (energy)

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Source of income
 - Conserve environment/reduce soil erosion/improve water catmint
 - Source of food /fruits
 - Medicine/value
 - Aesthetic value
 - Feed for livestock
 - Labour saving
5. (a) A – Pollarding technique ($\frac{1}{2}$ mk)
B – Coppicing ($\frac{1}{2}$ mk)
- (b) An example of a tree species suitable for technique B and C as a method of harvesting B
- (i) Calliandra calothyrsus
- (ii) Markahmia lutea
- (iii) Psidium guajava
- (iv) Croton macrostachyus ($\frac{1}{2} \times 1 = \frac{1}{2}$ mk)
- C (i) - Grevillea robusta
- Calliandra calothyrsus
- Croton cacrostachyus ($\frac{1}{2} \times 1 = \frac{1}{2}$ mk)
6. -Deep rooted
- Nitrogen fixing/ leguminous
- Fast growing
- Good in by product production
7. (a) Five characteristics of trees used in agroforestry are;
- Fast growth rate
 - Nitrogen fixing/leguminous
 - Good in by product use timber, fruits etc
 - Deep rooted
 - Nutritious and palatable
 - easily coppiced
 - Non-competitive ability with main crop ($5 \times 1 = 5$ mks)
- (b) The benefits of agroforestry are:
- Provides food
 - Source of energy/fuel
 - Fodder for livestock
 - Aesthetic value
 - Provides shelter
 - Income generation
 - Soil and water conservation

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Balancing of atmospheric gases (8x1=8mks)
- 8. four forms of agro- forestry
 - Alley cropping
 - Woodlots
 - Multi-storey
 - Source of wood fuel
 - Source of income
 - Labour saving
 - Aesthetic value
 - Reduces the carbon dioxide reducing the depletion of Ozone layer
Used as a method of soil and water conservation
- 9. - Border//edge grow
 - Source of wood fuel
 - Source of income
 - Labour saving
 - Aesthetic value
 - Reduces the carbon dioxide reducing the depletion of Ozone layer
 - Used as a method of soil and water conservation
- 10. - fast growth
 - Deep rooted
 - Nitrogen fixing
 - Good by-products
 - Friendly to crops/not affent crop

SECTION II QUESTIONS

FARM TOOLS AND EQUIPMENT

There are five categories of farm tools and equipment namely:

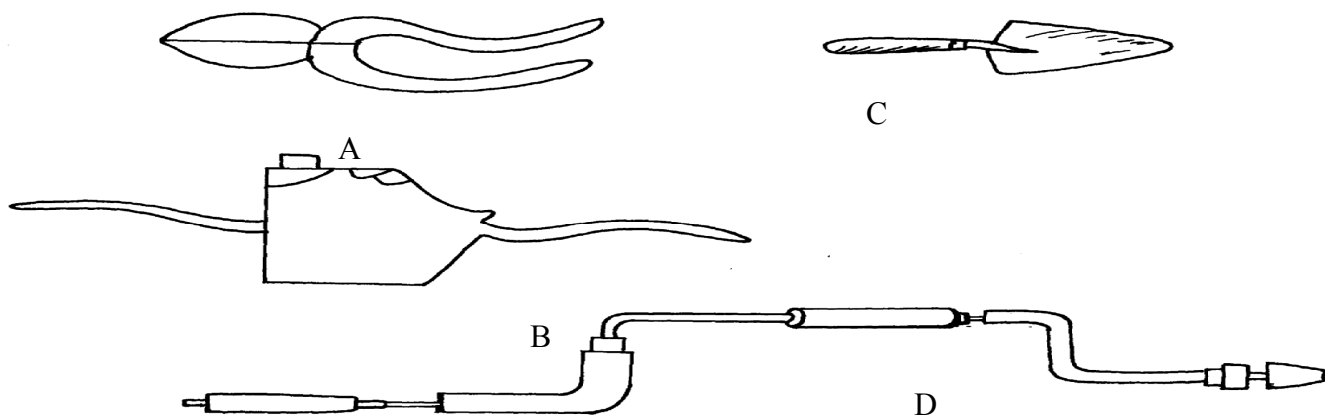
- Garden tools and equipment e.g. pangas, jembe, pick axe, spade e.t.c.
- Workshop tools and equipment e.g. saws, hammers, planes, chisels e.t.c
- Livestock production tools and equipment e.g. milking stool, strip cup, milk churn etc.
- Masonry tools and equipment e.g. wood float, spirit level, plumb bob e.t.c
- Plumbing tools e.g. pipe wrench spanner, stock and die e.t.c

It is very important to identify the farm tools and equipment, give the correct users and maintenance practices.

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the require concepts and practices:

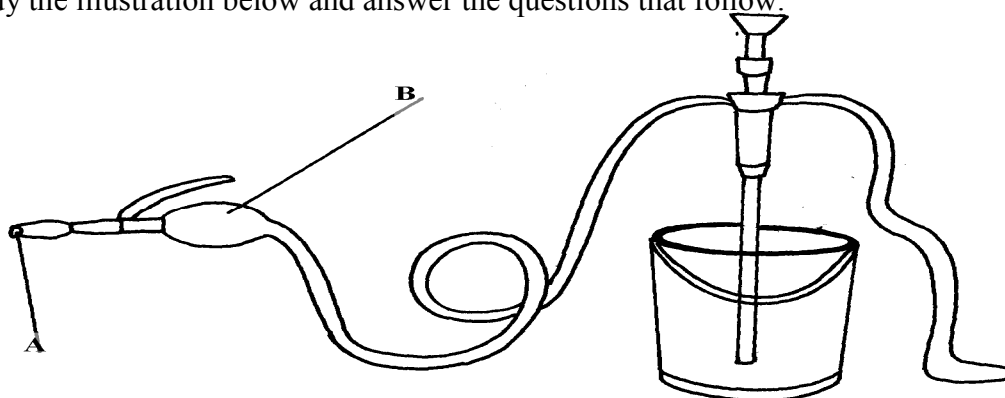
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1. List **four** maintenance practice carried out on a cross-cut saw
2. Identify the following tools and state their functions



Identify A, B, C, D

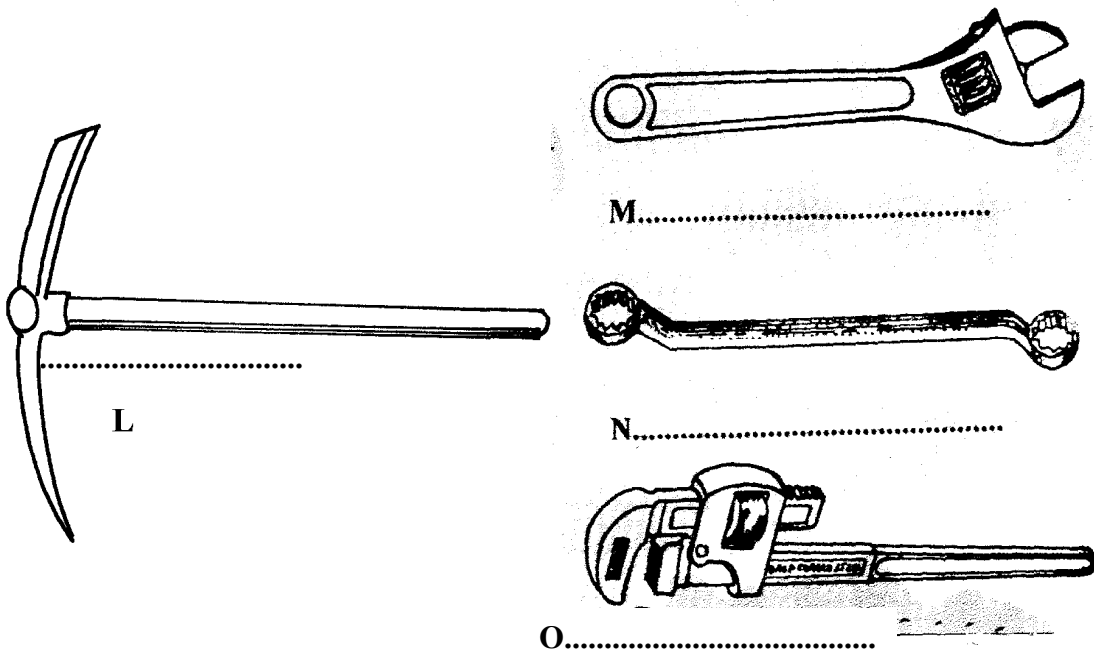
3. Study the illustration below and answer the questions that follow.



- i) Identify the equipment represented by the illustration
- ii) What is the use of the equipment?
- iii) Name the parts labeled **A** and **B**
- iv) What is the function of the part labeled **A** and **B**

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4. What is the use of a garden fork?
5. Name the tool that a builder would use to check the vertical straightness of a wall during construction
6. Name the farm tool that can be used when removing nails from timber
7. Give **two** examples of equipment that a livestock farmer can use in administering oral anti helminthes
8. Below are farm tools, study them and answer the questions that follow:-



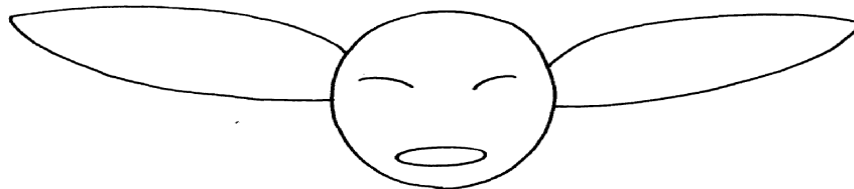
- (a) Identify the tools L, M, N, O
- (b) **Give one** functional advantage of tool **M** over tool **N**
9. Name a tool used to perform the following functions on the farm;
 - (i) Drilling of small holes on metal
 - (ii) Bore holes on wood
10. State the common faults in the operation of Knapsack spray
11. (a) Name the **three** tools in castration of livestock

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12. Name the most appropriate set of animal handling tools that a farmer uses for the following operations:-
- (i) Restraining a large bull when taking it around the show ring
 - (ii) Cutting tail in sheep
13. What is flushing in sheep production?
14. a) The illustration below shows a method of identifying pigs, study the diagram and answer the questions that follow:-



- i) On the diagram A provided below, draw the mark to indicate a pig number 147, using the procedure of ear-notching in diagram

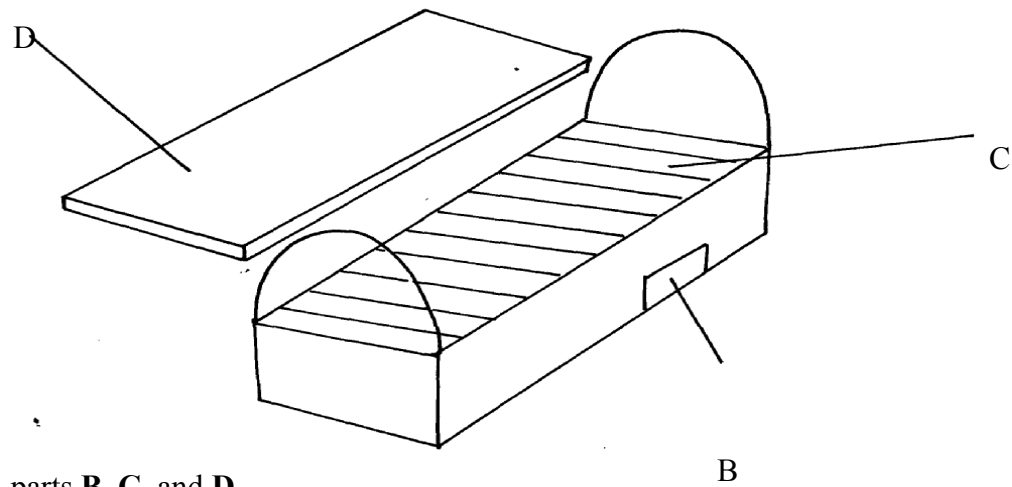


A

- ii) What is the recommended stage of growth in pigs at which the ear-notching should be carried out?
- iii) State any **three** reasons why weight is an important routine management practice in pig production
- b) Below are illustrations showing the various parts of the Kenya top bar hive. Use the illustration

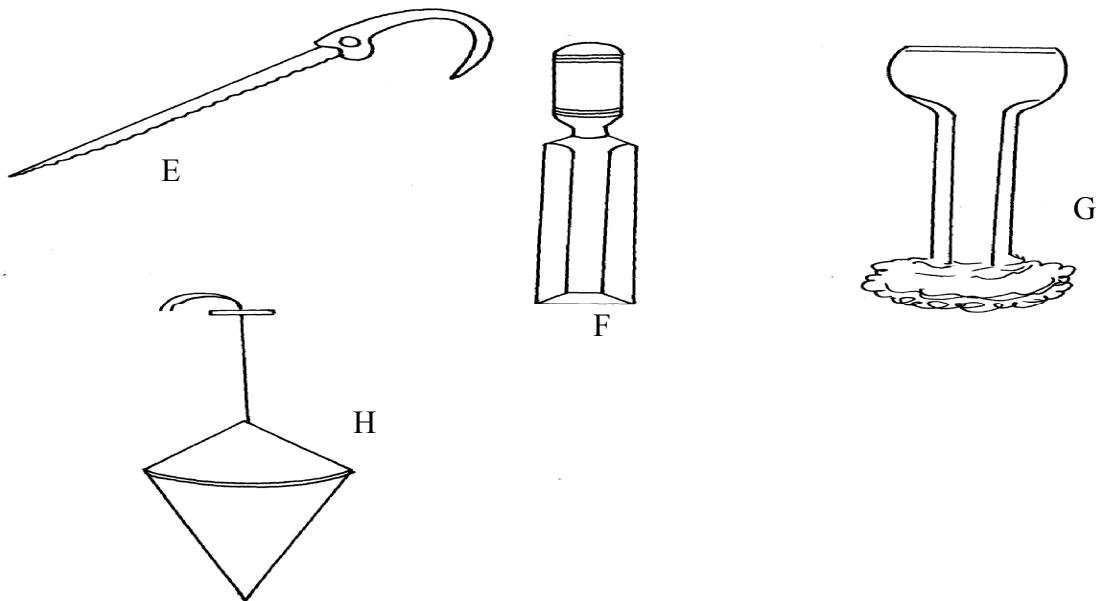
FREE AGRICULTURE K.C.S.E 2012 to 2015

to answer the question that follow



- label the parts **B**, **C** and **D**
- How can a farmer attract bees to colonize a new hive?
- Outline the procedure of opening the hive to harvest honey

15. a) Below are illustrations of farm tools

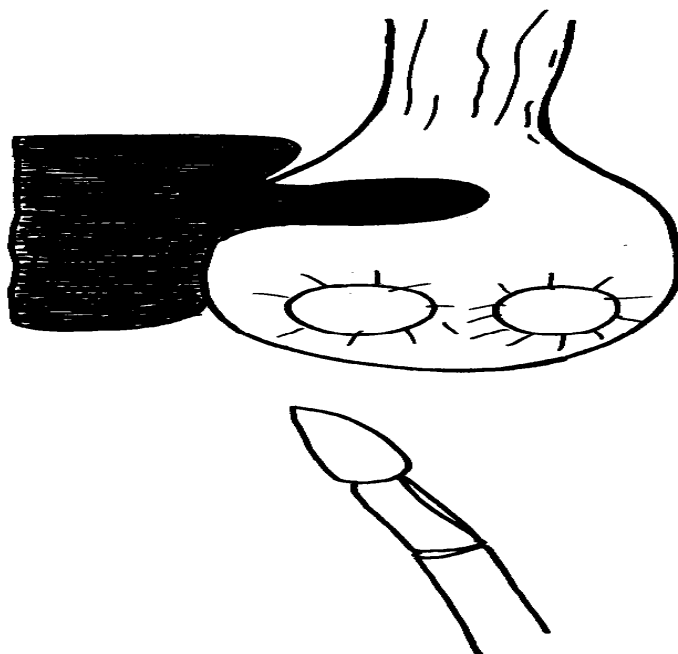


- State the use of the tools
- Name the type of hammer that may be used for driving tool **K** during work

b) Given below is an illustration of one of the routine management practices in livestock

Production.

Study the diagram and answer the following questions



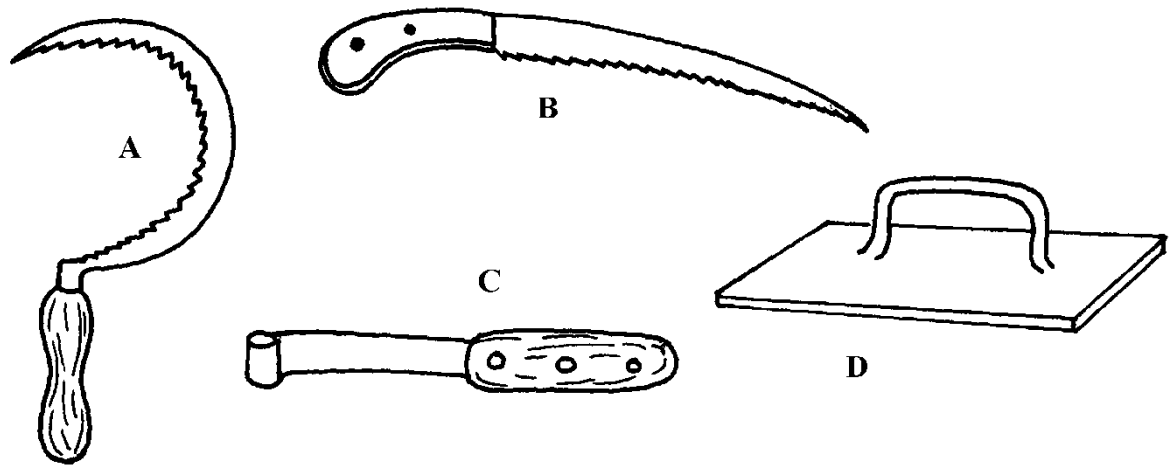
i) Name the practice indicated in the diagram above

ii) Describe the procedure you would follow when carrying out the practice named

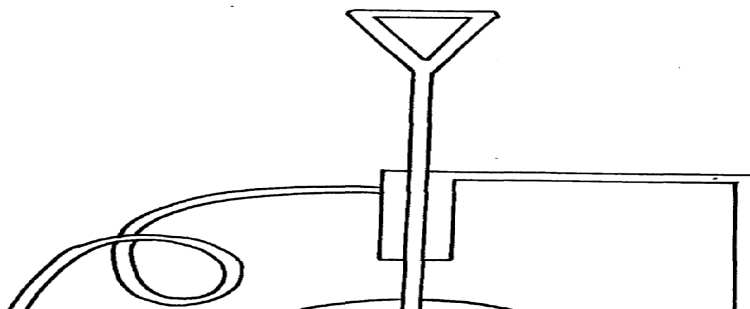
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in (i) above in piglets

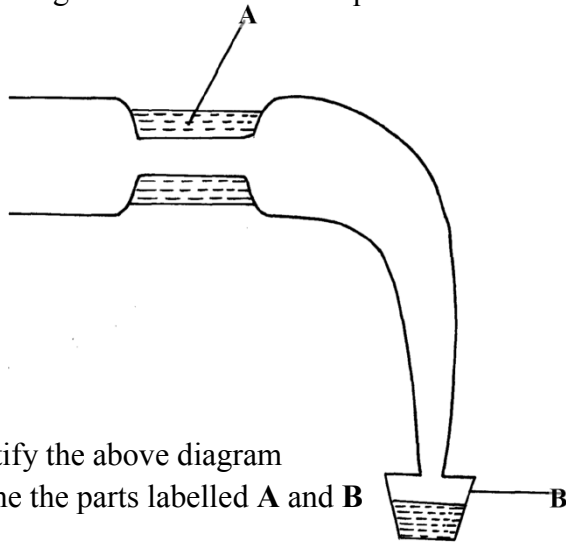
16. Study the diagram below and answer the questions that follow;



- i) Identify the tools
 - ii) Give the use of each of the tools named above
 - iii) State **two** maintenance practices that should be carried out on tool **D**
17. List **two** equipments used in handling cattle during an Agricultural exhibition
18. Mention the use of the following tools.
- i) Dibber
 - ii) Spokeshaves
 - iii) Tinsnips
 - iv) Burdizzo
19. List **four** precautions that should be taken when using workshop tools and equipment.
20. i) Below is an illustration of a farm equipment. Study it and answer questions that follow



- a) Identify the farm equipment illustrated above
b) What is the use of the equipment
c) Name the parts labelled **W**, **X** and
d) What is the functions of **Y** on the equipment
21. Use the diagram below to answer questions which follow



- i) Identify the above diagram
ii) Name the parts labelled **A** and **B**

LIVESTOCK PRODUCTION I
COMMON LIVESTOCK BREEDS

This topic entails the following:

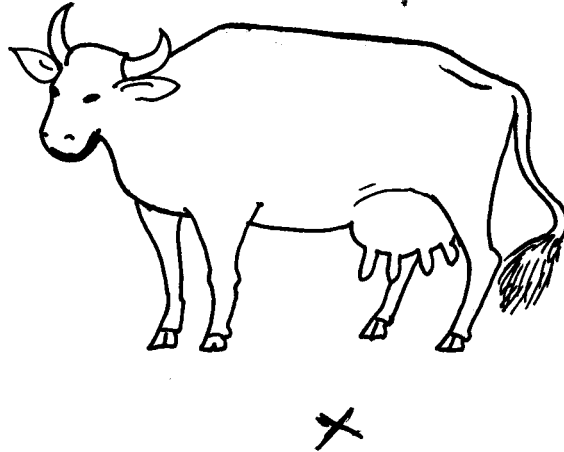
- Reasons of keeping livestock
- Parts of a cow

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Characteristics of indigenous and exotic cattle breeds
- Dairy cattle breeds
- Beef cattle breeds
- Dual purpose cattle breeds
- Pig breeds
- Sheep breeds
- Goat breeds
- Rabbit breeds
- Camel breeds

The following relevant questions and their answers in this topic will greatly help and motivate the user to comprehend and understand the required concepts and practices:

1. State **two** reasons for treating water for use on the farm
2. State **four** advantages of applying lime in clay soil
3. State **four** ways by which Re-afforestation helps in land reclamation
4. Give **two** distinguishing features between the following breeds of rabbits; Kenya white and California white
5. Give **four** reasons why most farmers keep livestock in Kenya
6. Give **three** ideal conformation features of beef cattle
7. (i) Name a dual purpose cattle breed reared in Kenya
ii) State **three** uses of a rotavator
8. Name a pig breed with the following features:
White body colour, erect ears, dished snout, big in body size
9. What does the term '**epistasis**' mean in livestock improvement?
10. (a) Explain the role of livestock industry in Kenya's economy
(b) Outline the general characteristics of indigenous cattle
11. Below is a diagram of a cattle. Study it and answer the questions that follow:-



- (i) What type of animal is represented above?
- (ii) If you stand at a point marked **X**, state **five** characteristics that tells you that the animal belong
to type name in **(i)** above?
- (iii) State **three** areas on the body of a cow where ticks are commonly found
12. Name **four** breeds of dairy goats
13. List **two** distinguishing characteristics of Californian breed of rabbit
14. Name the common milk breed of goats reared in Kenya
15. Name **four** dairy cattle breeds reared in Kenya.
16. Differentiate between 'breed' of animal and 'type' of animal.
- 17 Name the camel breed that is adapted to cooler regions and has a woolly body covering
18. State any **two** channels through which beef is marketed in Kenya.

(LIVESTOCK HEALTH II
(LIVESTOCK PARASITES)

FREE AGRICULTURE K.C.S.E 2012 to 2015

This topic entails the following:

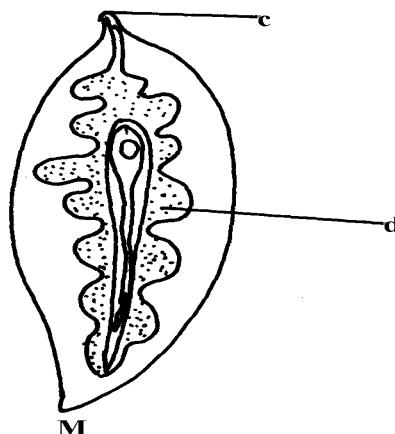
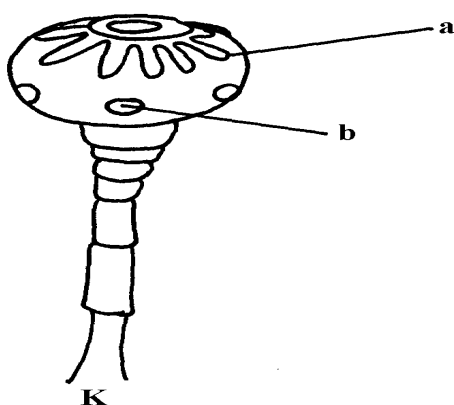
- Host-parasite relationships
- Effects of parasites on livestock
- Life cycle of parasites
- Methods of parasite control in livestock
- Identify different parasites

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Name **two** chemical methods used in deworming cattle
2. a) state **six** effects of parasites

b) Describe the life cycle of *Taenia solium* species of tapeworm

c) State **four** control measures of the tapeworm
3. Give **two** functions of calcium in dairy cows
4. Give **two** control measures of fleas in a flock of sheep
5. Give **two** measures a poultry farmer can use to control fleas in flock
6. State **two** reasons why drenching alone is not an effective method of controlling internal parasites
7. Give **two** forms in which a tape worm is found in livestock
8. Below are diagrams showing different types of internal parasites. Study them carefully and answer the questions that follow:-



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- a) Identify the parasites **K & M**
 - b) Identify the parts labelled
 - c) Name the organs where each parasites is found
 - d) Give the intermediate host of parasite **M**
9. Give any **two** effects of external parasites that are harmful to livestock
10. Outline the procedure followed when hand-spraying cattle to ensure effective use of acaricides
to control ticks
- 11 a) A boar gained 90Kg live weight after eating 360Kg pig finisher meal over a period of time. Calculate the feed conversion ratio
- b) Describe digestion in the four stomachs of the ruminant animal
- c) Give the significance of lubrication system
12. State **four** ways of controlling tsetseflies
13. Name **two** types of roughages
14. Name the common milk breed of goats reared in Kenya
15. Why are the element calcium and phosphorus important in the diet of young livestock?
- 16 Give **two** parasites of cattle which are also disease vectors
- 17 Give **three** control measures of fleas in a flock of layers

LIVESTOCK PRODUCTION II (NUTRITION)

This topic entails the following:

- Identification and classification of livestock feeds.
- Digestion and digestive systems of cattle, pigs and poultry
- Definition of terms used to express feed values
- Preparation of balanced ration for various livestock
- Functions and deficiency symptoms of various nutritional elements.

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The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices;

1. Give **two** ways by which production ration may be utilized by dairy goats
2. Given that the livestock reared require a ration of 18% DCP and the farmer has maize bran of 10% DCP and sunflower meal of 30% DCP. Use Pearson's square method to calculate how much of the feedstuffs a 150kg rations
3. State the importance of the following in livestock nutrition: i) Water ii) Vitamin A
4. Name the compartment of the ruminant stomach where microbial digestion takes place.
5. i) A ration containing 18% protein is to be made from maize and sunflower cake. Given that maize contains 7% protein, and sunflower seed cake 34% protein. Use Pearson square method to calculate the value of feedstuffs to be used to prepare 100kgs of the feed
ii) A part from Pearson square method, name **two** other methods that can be used to formulate feed ration
6. (a) Define the term ration as used in livestock nutrition
(b) A farmer wanted to prepare a 200kg of pig's ration containing 16% D.C.P. Using the square method, calculate the amount of maize containing 10% D.C.P and cotton seed containing 28% D.C.P the farmer would need to prepare the ration (*show your work*)
7. Give **two** livestock feed additives
8. State **three** factors that would determine the amount of concentrate fed to dairy cattle
9. State **three** factors that would determine the amount of concentrate fed to dairy cattle

LIVESTOCK PRODUCTION III (SELECTION AND BREEDING)

This topic entails the following:

- Description of reproduction
- Description of reproduction systems
- Selection of breeding stock
- Description of breeding system.
- Identification of livestock on heat
- Description of methods used in serving livestock.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. State **four** ways of minimizing disowning of lambs by ewes
2. State the gestation period of the following farm animals
 - i) Pigs
 - ii) Rabbits
3. i) What is selection in live stock production
 - ii) State **three** methods used in selection of livestock
4. Give **four** signs of heat observed on female rabbit
5. Name **four** meat breeds of rabbits
6. State **four** reasons for cutting a breeding boar
7. List the methods of selection in livestock
8. Define the following terms as used in livestock breeding
 - i) Heterosis
 - ii) Epistasis
9. State **three** signs of heat in Does (Female Rabbits)

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10. Give the gestation period of the following:
 - i) Cow
 - ii) Sow
11. State **three** disadvantages of natural methods of mating in cattle breeding
12.
 - a) Explain the advantage of battery cage system of rearing layers
 - b) Outline **ten** factors considered when selecting dairy cattle for breeding
13. What is upgrading as used in livestock production?
14. Define the word breech of birth as used in livestock production
15. What does the term “**drift lambing**” mean in livestock production?

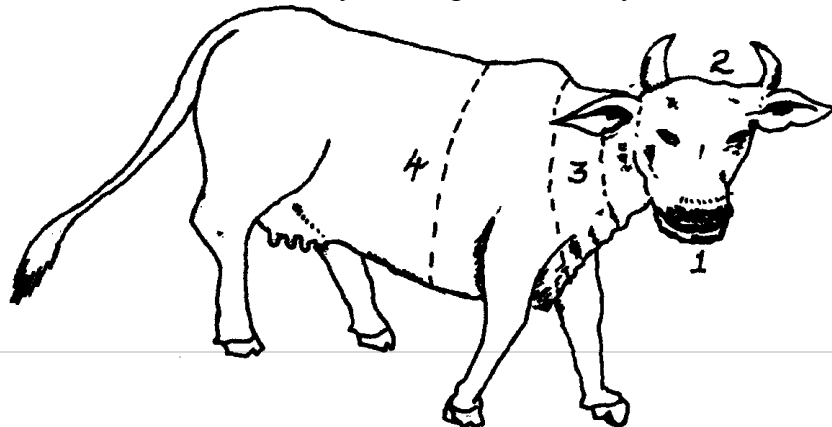
LIVESTOCK PRODUCTION IV (LIVESTOCK MANAGEMENT PRACTICES)

This topic entails the following:

- Description of livestock rearing practices
- Carrying out livestock rearing practices
- Livestock routine management practices i.e. feeding, de-beaking e.t.c.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Give **four** reasons of carrying out crutching sheep management
2. Name **two** ways a farmer can perform closed methods of castration on his male livestock
3. Below is a diagram of a farm animal. Study the diagram carefully and then answer the questions that follow



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a) On the drawing, mark the letters indicated in brackets the part of the animal where:

i) Branding should take place (B)

ii) Vaccination should be carried out (V)

iii) Body temperature of the animal should be taken (BT)

iv) Mastitis infection may occur (M)

b) Name **three** areas of the animal body where ticks are likely to be found

c) Name the parts of the animal numbered 1-4

4. a) What is castration as used in livestock production?

b) State **four** reasons why castration is done in livestock

5. (a) Describe the management of a gilt from weaning to farrowing

(b) Discuss the preparation a poultry farmer should make before the arrival of day old chicks

6. Outline **two** reasons for raddling in sheep management

7. State any **four** reasons for castrating male piglets

8. What are the methods of stocking bees? Give **two**.

9. State **two** ways that show how good feeding help to control livestock diseases

6. Name any **two** recommended methods of docking lambs

7. State three disadvantages of inbreeding

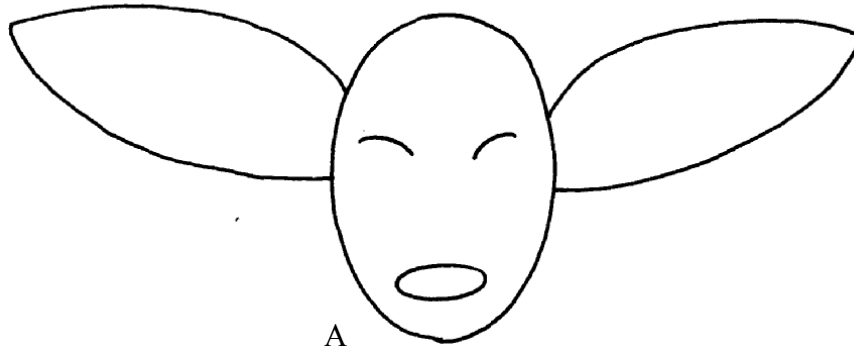
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8. State **four** routine management practices that should be carried out on a lactating ewe
9. (a) Describe the procedure which should be followed to castrate a three weeks old piglet using surgical method
(b) (i) State **five** factors that should be considered when sitting a bee hive in a farm
(ii) Describe the management practices that would ensure maximum harvest of fish from a fish pond
10. Give **three** types of bees found in a bee colony
11. List **three** methods of castrating farm
12. Outline the routine management practices of piglets from the 1st day to the 8th week
13. List **three** types of calf pens
14. State **two** factors that could lead to failure to conceive in sows after service
15. List **three** advantages of hoof trimming in sheep production
16. State **four** factors considered when citing an apiary in the farm
17. Name **three** methods of stocking a beehive with honey bees
18. List **three** common methods of extracting honey from the combs
19. a)The illustration below shows a method of identifying pigs, study the diagram and answer the questions that follow:-



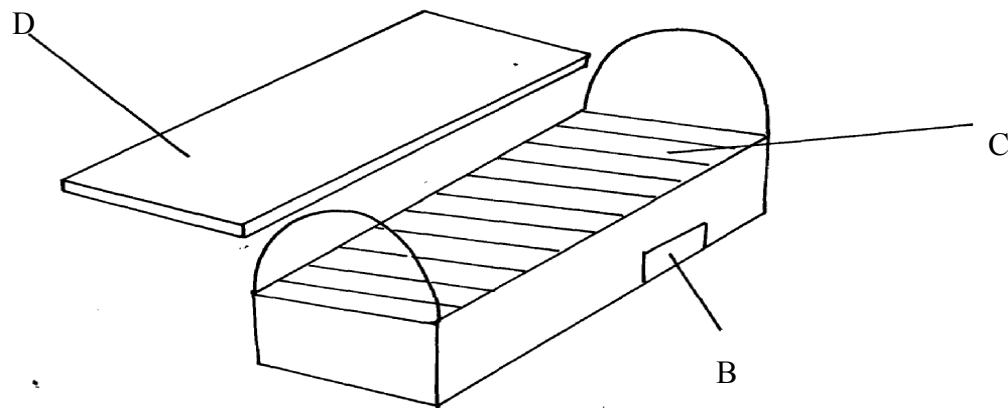
FREE AGRICULTURE K.C.S.E 2012 to 2015

- i) On the diagram **A** provided below, draw the mark to indicate a pig number 147, using the procedure of ear-notching in diagram



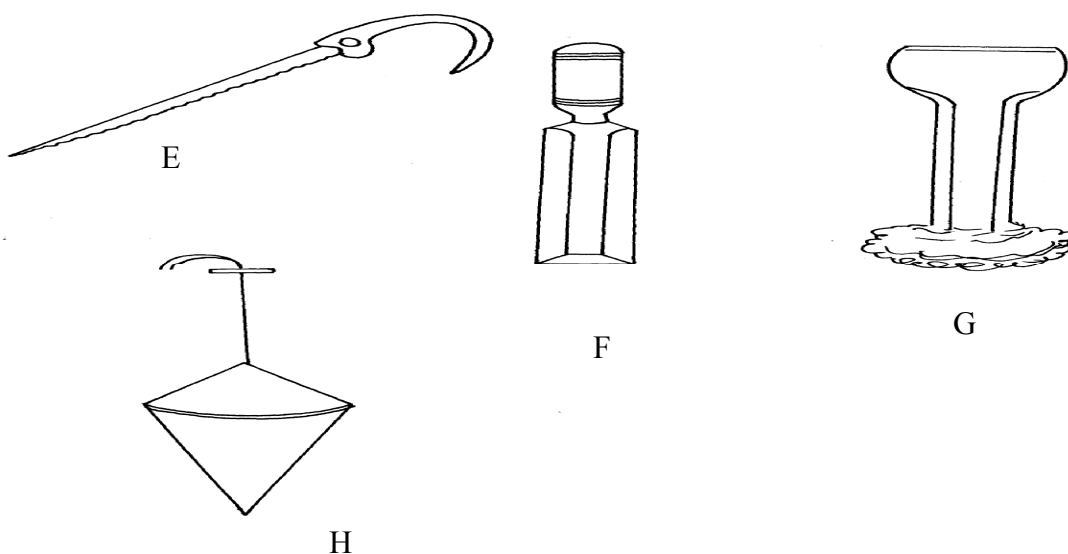
- ii) What is the recommended stage of growth in pigs at which the ear-notching should be carried out?
iii) State any **three** reasons why weight is an important routine management practice in pig production

- b) Below are illustrations showing the various parts of the Kenya top bar hive. Use the illustration to answer the question that follow



- i) label the parts **B**, **C** and **D**
ii) How can a farmer attract bees to colonize a new hive?
iii) Outline the procedure of opening the hive to harvest honey

20. a) Below are illustrations of farm tools



ii) State the use of the tools

iii) Name the type of hammer that may be used for driving tool **K** during work

b) Given below is an illustration of one of the routine management practices in livestock production.

Study the diagram and answer the following questions



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- i) Name the practice indicated in the diagram above
- ii) Describe the procedure you would follow when carrying out the practice named in (i) above in piglets

- 21. a) State and explain **four** advantages of age grouping farm animals as a management practice
 - b) Explain **four** major causes of lamb mortality from birth to weaning
 - c) Describe brucellosis under the following sub headings
 - i) Cause
 - ii) Transmission
 - iii) Symptoms
 - iv) Control measures
- 22. a) Explain the advantage of battery cage system of rearing layers
- 23. Give **two** reasons why it is important to castrate animals when they are still young
- 24. (a) Give **two** reasons why dehorning is carried out in farm animals
 - (b) State **four** methods of dehorning livestock
- 25. (a) What is **steaming up** in livestock production?
 - (b) Give **two** reasons why the practice is important in pig rearing
- 26. State **four** management practices that should be carried on a fish pond in order to obtain maximum fish production.
- 27. List four factors considered when formulating livestock ration. (2mk)
- 28. Study the diagram of a cow below and answer the subsequent questions



- i) Identify the livestock equipment marked **E** above
- ii) State **two** uses of the equipment in **17(i)** above
- iii) Name the **two** types of identification marks applied on the animal above
- iv) Show with an arrow and mark with letter **P** where pye-grease acaricide should be applied on the animal
- v) Give **one** disadvantage for each of the identification marks made on the animal

Identification marks	Disadvantage
A	
B	

- 29. a) Describe the management practices of a gilt from weaning to the time of farrowing
- b) State **five** factors to consider in selecting a gilt for breeding stock.
- 28. State **four** conditions that necessitate the handling of farm animals .
- 29. Give any **two** reasons why docking is an important practice in sheep management.
- 30. Why is crutching a very important management practice in sheep breeding.

FARM STRUCTURES

This topic entails the following:

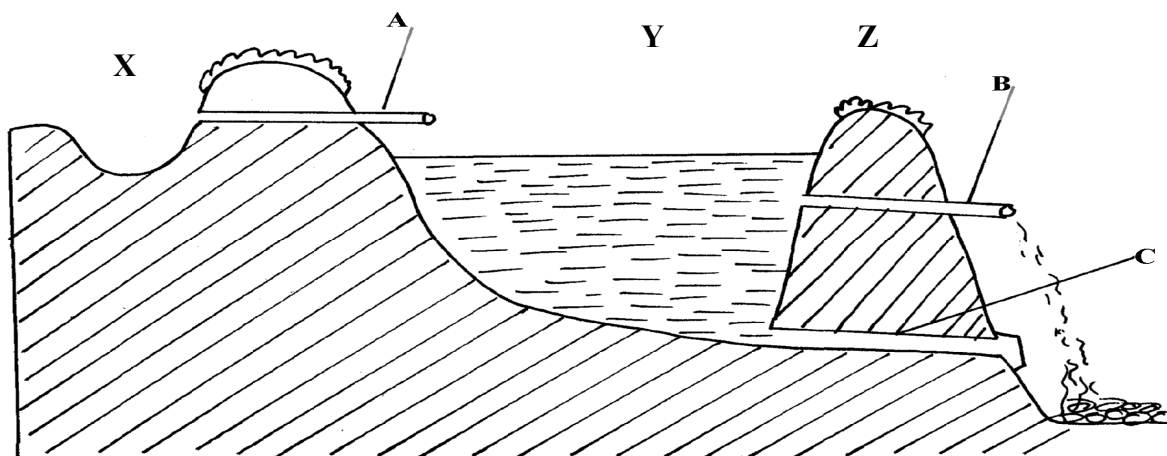
- Description of parts of a building
- Identification of materials for construction
- Description of various farm structures and their uses
- Construction and maintenance of farm structures.
-

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

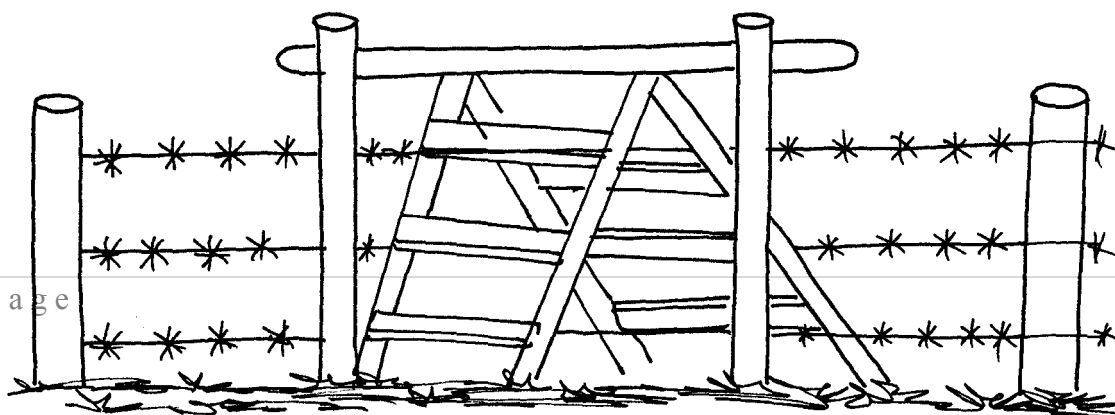
1. Mention **four** disadvantages of using steel materials for construction of farm structures
2. Give **two** reasons for placing a polythene sheet on a foundation of farm buildings
3. What is “**calf crop**” in beef production?
4. a) Outline the procedure in construction of a barbed wire fence
b) List any **four** wood preservatives
c) Give **four** factors that will determine the choice of farm building materials
5. i) Give **four** factors considered when citing a fish pond
ii) Give **four** features of a good laying nest
6. State **two** uses of a footbath in cattle dip
7. a) What is a green house
b) Name **four** materials used in green house construction
c) Give **four** maintenance practices carried out on a green house
d) Explain the importance of maintaining farm structures

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- 8 Give **two** disadvantages of a barbed wire fence when used in paddocking
- 9 The diagram below illustrates a cross-section of a fish pond. Study it carefully and answer the questions that follow:-



- (a) Identify the parts marked **A**, **B**, **C** and **C**
- (b) On the diagram the points marked **X**, **Y** and **Z** are possible sites where fish feeding can be done. Which is the most appropriate point for feeding?
- (c) Give **two** reasons why the floor of the pond should be covered with lime 14days before filling it with water
- (d) Why should part marked **B** be screened?
- (e) State **three** maintenance practices carried out on the structure
10. Study the farm structure below illustrating a pass:-



(a) Identify the type of pass illustrated above

(b) Distinguish between a pass and a gate

(c) Name **one** type of live fence

11 State **four** requirements of a good maize store

12 State **three** factors that determine the depth and size of foundation in a farm building

13 (a) Explain the uses of various hand tools in the construction of a poultry house

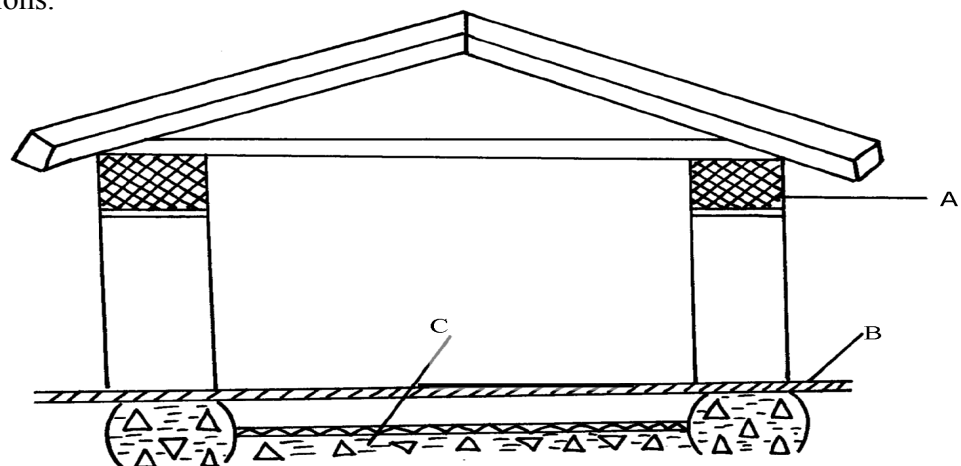
(b) Describe the procedure of erecting wooden rail fence

(c) Explain various factors considered when choosing the construction materials for farm building

14 State **four** features of a good calf pen

15 State **four** uses of a fence

16 Below is a diagram of a cross section of a farm structure. Study it carefully and answer the following questions.



(i) Identify the labeled parts **A**, **B**, and **C**

(ii) What is the importance of the part labeled **B** to the structure

(b) (i) When making concrete blocks the ratio 1:3:5 may be used. What do these figures represent?

(ii) In a concrete mixture 1:3:5, twenty four cubic metres of sand were recommended to be used in putting up of a foundation of a building. Find the volume of the other

two

17 (a) Explain the maintenance practices of a fish pond

(b) Give **four** ways of controlling fish predators in a fish pond

18 Mention **four** reasons of treating timber before roofing farm buildings

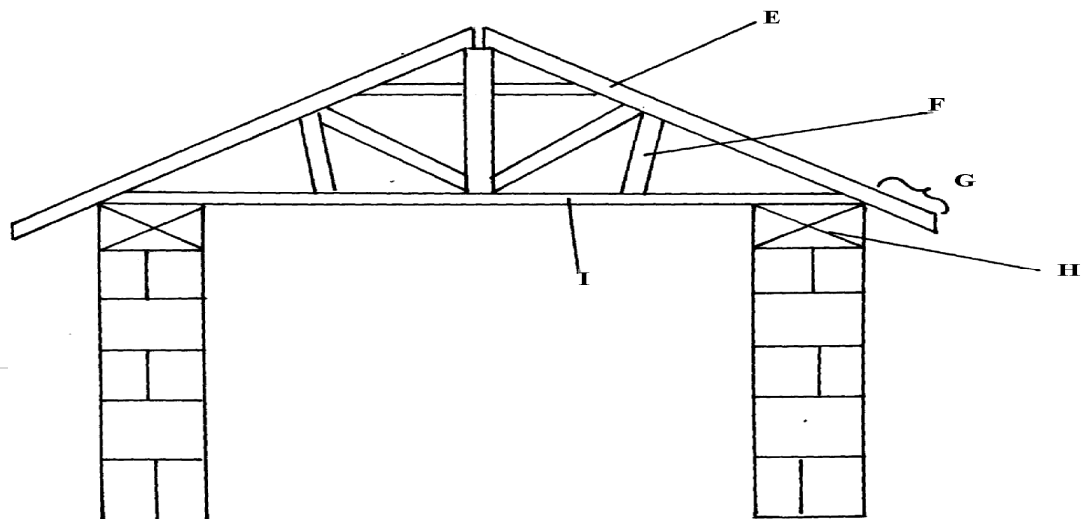
19 List **four** uses of crushing in the farm

20 a) Define the term **fence**

b) List various types of fences

c) Describe advantages of fences

21 Study the illustration of a farm structure below and answer the questions that follow

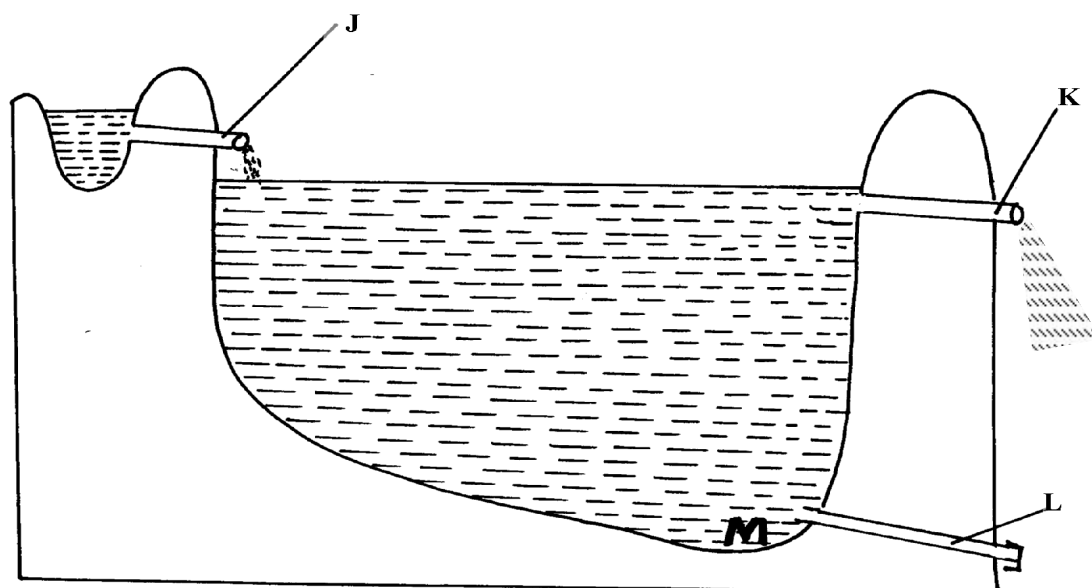


i) Name the parts labeled

ii) State the function of the parts labeled

iii) Name **two** chemicals preservatives used to treat the wooden parts of the structure against insects and fungal damage

21. Below is a diagram of a fish pond, study it and answer the questions that follow



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- i) Label the parts marked **K** and **L**
- ii) Give a reason why part **M** is usually deeper than the rest of the pond
- iii) State **three** maintenance practices carried out in the pond

22. a) Explain **five** factors to consider when siting a fish pond

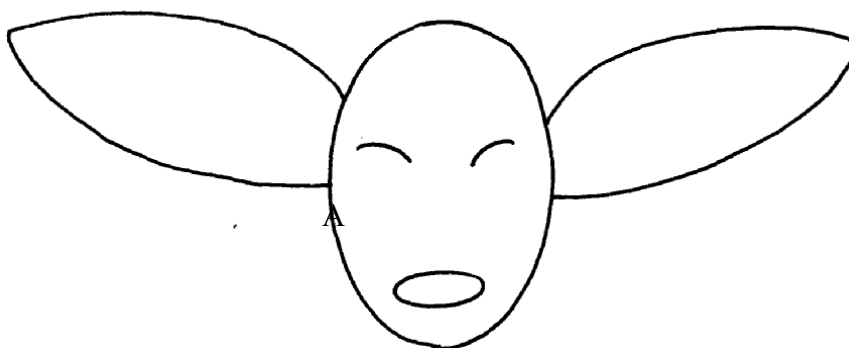
b) Explain the measures used to control livestock diseases

23. State four factors which influence the selection of materials for constructing a dairy shed

24. a) The illustration below shows a method of identifying pigs, study the diagram and answer the questions that follow:-



i) On the diagram **A** provided below, draw the mark to indicate a pig number 147, using the procedure of ear-notching in diagram



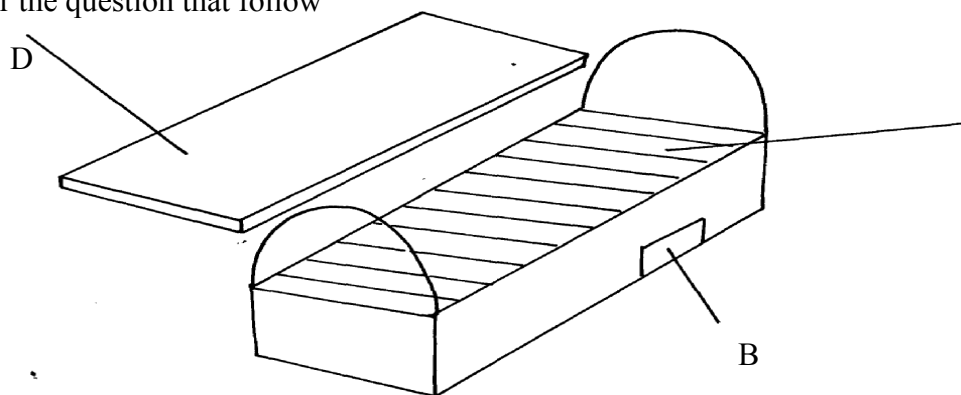
ii) What is the recommended stage of growth in pigs at which the ear-notching should be carried out?

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iii) State any **three** reasons why weight is an important routine management practice in pig production

b) Below are illustrations showing the various parts of the Kenya top bar hive. Use the illustration

to answer the question that follow



i) label the parts **B**, **C** and **D**

ii) How can a farmer attract bees to colonize a new hive?

iii) Outline the procedure of opening the hive to harvest honey

25. a) State **five** maintenance practices of a mould board plough

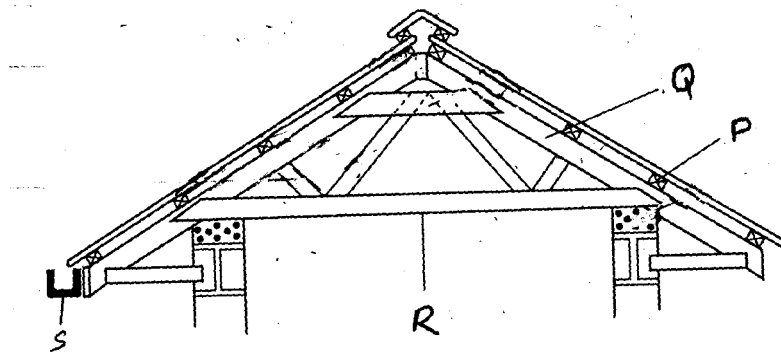
b) Explain **five** structural and functional differences between the petrol and diesel engines

c) List **five** uses of farm fences

26. State **four** uses of farm buildings

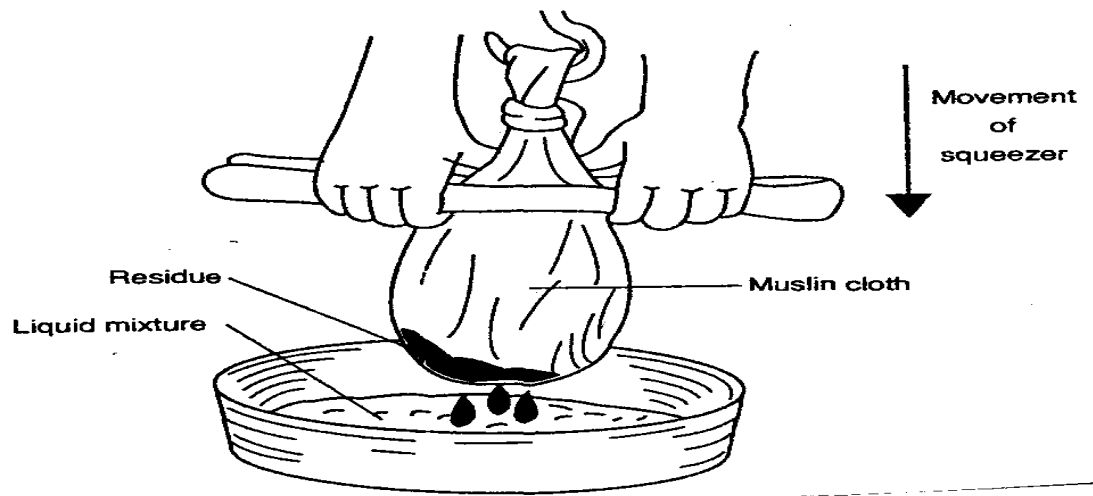
27. List **four** structural requirements for proper housing of farm animal to maintain good health

28. The diagram below is part of a farm structure. Study it and then answer the questions that follow:



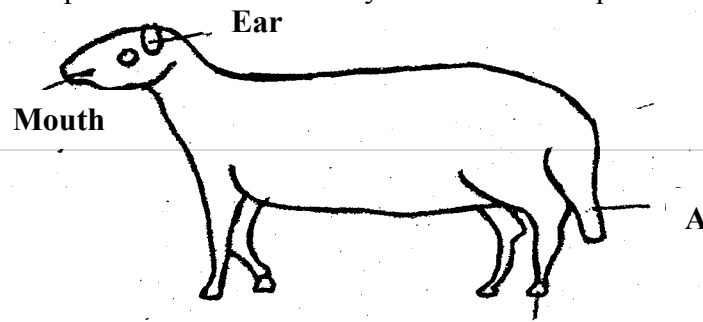
- (a) Name part of structure represented above
- (b) Identify parts labelled **P, Q, R,** and **S**
- (c) Give the functions of the part labeled **P** and **S**

29. The diagram below shows an activity of processing a farm product. Study it and answer the questions that follow:



- (a) Name the product being processed
- (b) Identify the method used in processing the product names in the diagram
- (c) Outline the procedure used when harvesting the product named in the diagram
- (d) List **two** factors that affect quality of the product processed in the diagram

30. Study the diagram of a sheep shown below carefully and answer the questions that follow:



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- (i) What operation is usually carried out on the part labelled **A**?
 - (ii) Give **two** reasons for carrying out the operation in **(i)** above
 - (iii) At what age should the above operation be carried out?
 - (v) Name routine management practice carried out on the part labelled **B**
- 31
- (a) Describe **five** parts of a plunge dip
 - (b) Outline **six** uses of live fences on the farm
 - (c) Outline **four** factors that influence power output of draught animals
- 32.
- a) Explain the factors a farmer needs to consider before siting a farm structure on his farm
 - b) Outline the desirable features of a good grain store
 - c) Describe the uses of farm fences in promotion of agriculture
- 33.
- Name any **two** pests that attack timber used for construction in the farms
- 34.
- State **three** factors that may lead to dip wash being exhausted or weakened while in the dip tank

LIVESTOCK HEALTH III (LIVESTOCK DISEASES)

This topic entails the following:

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- Description of causes and vectors of the main livestock diseases.
- Signs of each stated livestock disease
- Stating predisposing factors where applicable
- Control measures of livestock disease

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. Name **two** diseases that affect female animals only
2. Give **four** pre disposing factors of foot rot disease in sheep
3.
 - i) Name the causative organism of contagious abortion in cattle
 - ii) Give **four** symptoms of contagious abortion in cattle
 - iii) State **three** methods of controlling contagious abortion in cattle
4. What is a vaccine?
5. (a) Discuss black quarter under the following sub-headings:-
 - (i) Animal affected
 - (ii) Casual organism
 - (iii)Symptoms of disease
 - (iv) Control measures
 - (b) (i) Explain **four** measures used to control liver flukes
 - (ii) Name an intermediate host of liver fluke
 - (c) Explain the following terms as used in livestock production
 - (i) Embryo transfer
 - (ii) Artificial insemination
 - (iii) Line breeding
 - (iv) Cross breeding

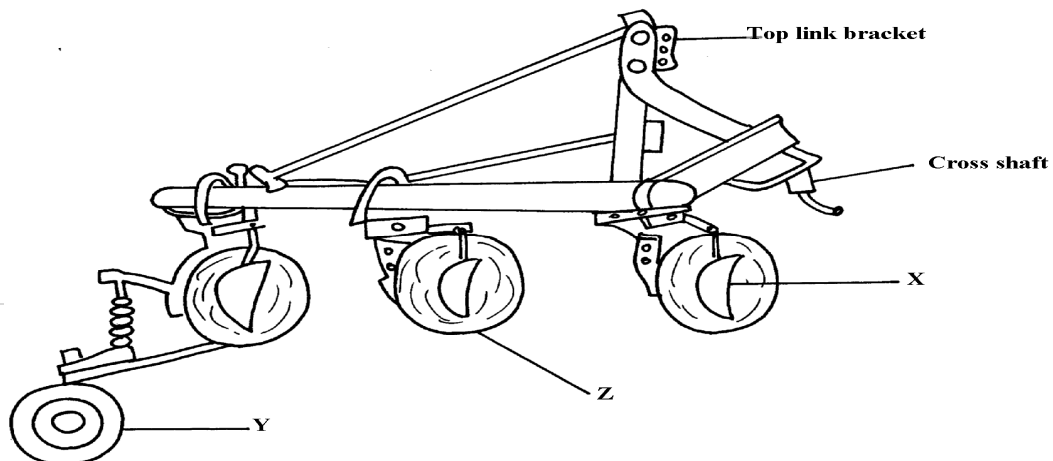
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- (v) Up-grading
6. Give **four** predisposing factors of foot rot disease in sheep
7. (a) Discuss **coccidiosis** disease under the following headings:-
(i) Causal organism
(ii) Livestock species attacked

(iii) Symptoms of attack

(iv) Control measures
- (b) (i) What are the characteristics of an effective acaricide?

(ii) Explain **three** methods of acaricide application
8. Name **four** ways of controlling coccidiosis in the farm
9. State **four** predisposing factors of scour in calves
10. Name **four** notifiable diseases of livestock
11. Name **two** diseases of poultry that are controlled by vaccination
12. State **three** signs of anthrax injection disease observed in the carcass of a cattle
13. Differentiate between active immunity and acquired passive immunity
14. Name **four** systems of a tractor engine
15. State **two** adjustments that should be carried out on a tractor – mounted mould board plough in preparation for ploughing
16. The diagram below illustrates a farm implement. Study it and answer the questions that follow



- a) Identify the implement
 - b) Name the parts labeled **X**, **Y** and **Z**
 - c) State **three** maintenance practices that are carried out on a disc plough
17. List **three** causes of ruminal tympany (Bloat) in ruminant animals
18. (a) Give **four** symptoms of milk fever
- (b) State **two** methods of controlling milk fever
19. (a) Describe East Coast Fever (E.C.F) under the following sub-topics:
- (i) Animal attacked
 - (ii) Causal organisms
 - (iii) Symptoms of attack
 - (iv) Control and treatment
- (b) Explain **ten** measures used to control livestock diseases
20. Name **two** livestock diseases that are spread through natural mating.
21. a) Give the method used in introducing a vaccine to poultry against fowl typhoid
- b) State other methods introducing vaccines to livestock

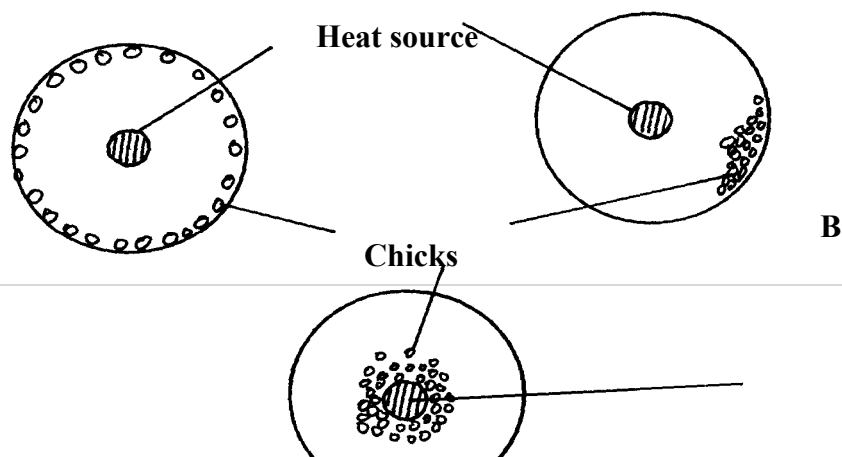
LIVESTOCK PRODUCTION V (POULTRY)

This topic entails the following:

- Identification of parts of an egg.
- Selection of eggs for incubation
- Identification of suitable sources for chicks.
- Descriptions of broodiness
- Description of condition for incubation
- Description of rearing systems
- Categories of poultry feeds according to age-of birds
- Stating causes of stress and vices in poultry and control measures.
- Marketing of eggs and poultry meat.
- Selection, sorting and grading of eggs.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. A deep little poultry house measures 9mx3m. Suppose the amount of space allowed for one bird is 0.27m². Calculate the number of birds that can be kept comfortably in the house. Show your working
2. Give **two** functions of isthmus in female bird
3. Give **four** features of a good laying nest
4. State **four** qualities of marketable eggs
5. Study the diagram showing the behaviour of chicks in a brooder and answer the questions that follow:-



A

Heat source

C

Brooder guard

(a) State the behaviour of chicks in A, B and C

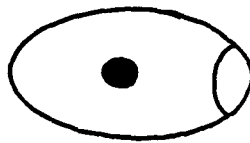
(b) Explain why the brooder guard is rounded as shown in the diagram

6. Mention **six** characteristics of an egg selected for incubation

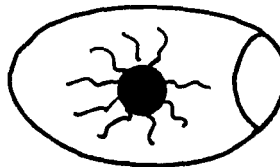
7. Describe the management of layers in deep litter system

8. State **four** reasons for egg breaking and drinking by layers in a deep litter rearing system

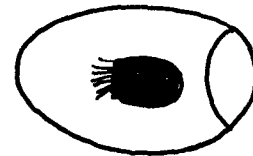
9. Below are diagram showing condition of eggs seven days after incubation study them and answer the questions



(a)



(b)



(c)

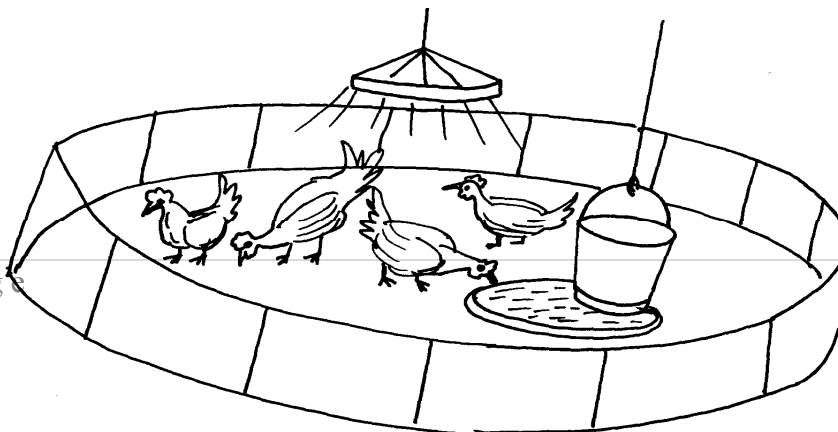
a) Identify the conditions of eggs

b) Identify the egg which suitable for incubation and give a reasons for your answer

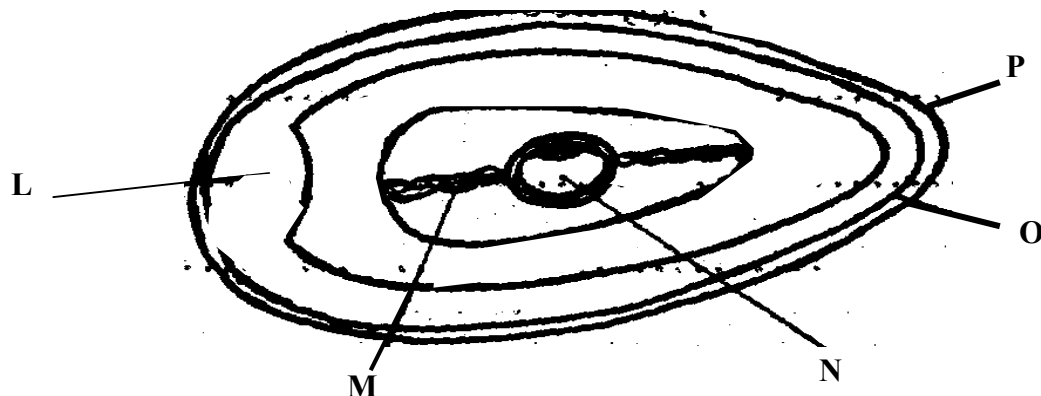
c) Name the practice which used to determine the state of eggs above

10. The diagram U below illustrates an activity carried by a poultry farmer keeping layers. Study the

diagram carefully and answer the questions that follow



- a) Identify the activity carried out using the set up illustrated in diagram U
 - b) List down **four** preparations that should be carried out structure U before arrival of day old chicks
 - c) List down **one** behaviouristic activity which would indicate that the chicks are under stress
11. Give **two** reasons for using litter in a poultry house
12. Give **two** reasons why it is important to castrate animals when they are still young
13. (a) Give **two** reasons why dehorning is carried out in farm animals
- (b) State **four** methods of dehorning livestock
14. State **four** abnormalities of eggs that can be detected during egg candling.
15. Describe the management of day old chicks in a deep litter system from preparation of brooder up to eight (8) weeks old
16. a) Describe the management practices of a gilt from weaning to the time of furrowing
- b) State **five** factors to consider in selecting a gilt for breeding stock.
17. Study the diagram of an egg below and answer the questions that follow:



- i) Name the parts labeled **N**, **O** and **P**
 - ii) State the functions of the parts **M** and **L**
 - iii) Why should the egg be turned during incubation
18. State **three** reactions of chicks in a brooder which has higher temperature than normal.
19. Give **three** types of bedding material a poultry farmer may use in deep litter rearing of layers
20. Give **two** properties of good eggs for incubation

LIVESTOCK PRODUCTION III (LIVESTOCK REARING PRACTICES)

This topic entails the following:

- Raising young stock
- Milk and milk components
- Milk secretion and milk let-down
- Correct milking techniques
- Marketing of milk and beef.

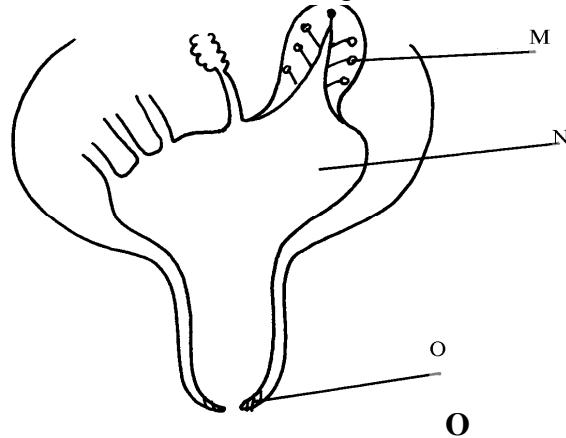
The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. What is “**calf crop**” in beef production?
2. a) Describe the procedure of hand milking in a dairy cow

b) Explain the practices observed in clean milk production

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3. Give **two** reasons for washing a cow's udder with warm water before milking
4. Give **two** roles of uterus in egg formation process
5. The diagram below is a structure of part of a cow's udder



- (i) Name the parts labeled **M**, **N**, and **O** on the diagram above
- (ii) State the functional difference between the following hormones which influence milk let-down;
 - (a) (i) Oxytocin
 - (ii) Adrenalin
 - (b) Mention **three** qualities of clean milk
6. Give **four** characteristics of clean milk
7. State **three** maintenance practices carried out on a milking machine
8. State **four** reasons for feeding Colostrums to calves immediately after calving
9. Give **three** ways of stimulating milk let down in a dairy cow
10.
 - a) Describe the operational differences of a disc plough and mould board plough
 - b) Explain **six** marketing problems affecting dairy farming in Kenya
 - c) State **four** reasons for culling a boar

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11. List **three** advantage of artificial method of calf rearing
12. State **three** methods that may be used to improve milk production in a breed of indigenous goats
13. (a) Outline **ten** physical characteristics between a good layer and a poor layer in a deep litter house

(b) Describe **five** factors that influence milk production in a dairy herd
14. a) State **two** reasons for washing the udder of a cow with warm water before milking.

b) Name the hormone that causes each of the following in dairy cows:.
i) milk letdown.
ii) lactogenesis
15. State **four** methods of increasing the depth of penetration of a disc harrow.
16. List **four** farm machines implements that obtain power from P.T.O shaft of a tractor
17. List **two** tractor drawn implements used for breaking hardpan in a crop field
18. State any **three** machines which are used for harvesting crops
19. a) describe the daily maintance and servicing of a tractor before use

b) State **one** function of each of the following parts of a tractor engine.

i) Fly wheel
ii) Ignition coil
iii) Thermostat
iv) Injector
v) Piston

FARM POWER AND MACHINERY

This topic entails the following:

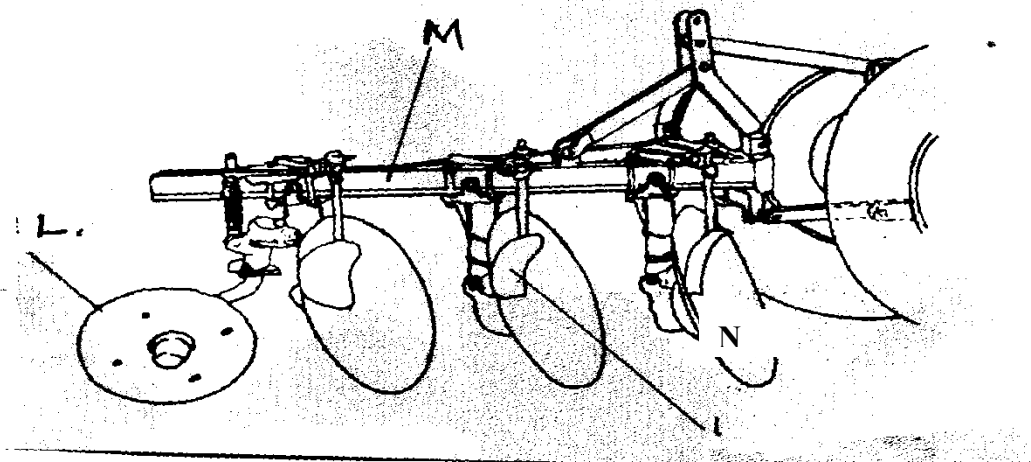
- Sources of farm power
- Systems of a tractor

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- Tractor implements, uses and maintenance
- Animal drawn implements uses and maintenance
- Tractor servicing and maintenance practices
-

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices:

1. Give **four** farm operations powered by engines
2. a) Describe the maintenance practices required on a tractor before it is put to daily use
b) Outline the factors that influence the power output by a draught animal
3. State **two** uses of gear box in a tractor
4. State **two** uses for which wind power is harnessed
5. Name **three** implements that are connected to the power take-off shaft
6. Below is a farm implement, study it keenly and answer the questions that follow:-



- (a) Name the farm implement drawn above
- (b) Identify the parts labelled **L** and **M** above

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(c) Give the function of the part labeled M

(d) State the field condition under which the implement can work better than the others
($\frac{1}{2}$ mk)

7. (a) Explain the factors that a farmer should consider in ensuring fast and efficient cultivation

by oxen

(b) Outline the importance of lubrication system in a tractor

(c) State the daily maintenance and servicing of a tractor

8. State the functions of the following parts of power transmission in a tractor:

(i) Hydraulic system

(ii) Draw bar

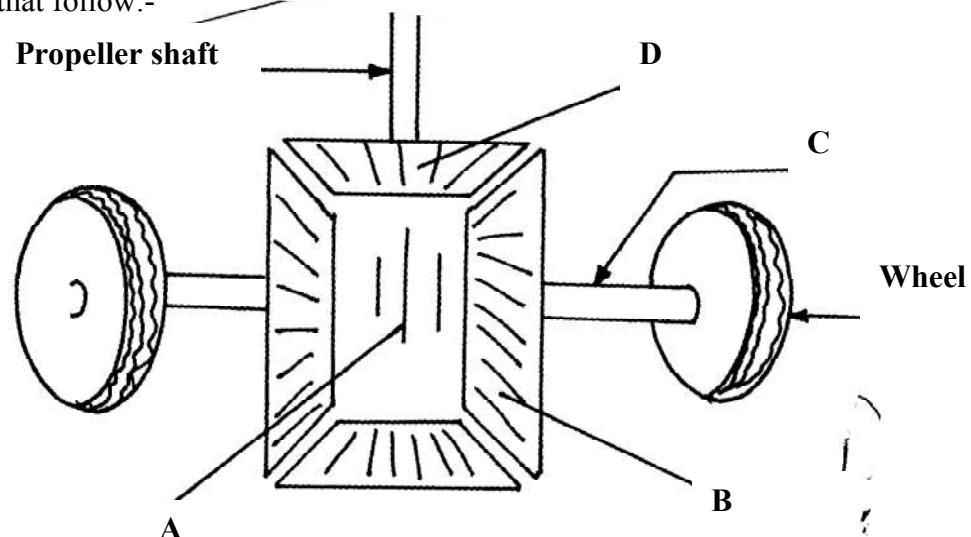
(iii) Propeller shaft

(ii) State **three** sources of tractor hire service

9. (i) What is a tractor hires services (1mk)

(ii) State three sources of tractor hire service ($1\frac{1}{2}$ mks)

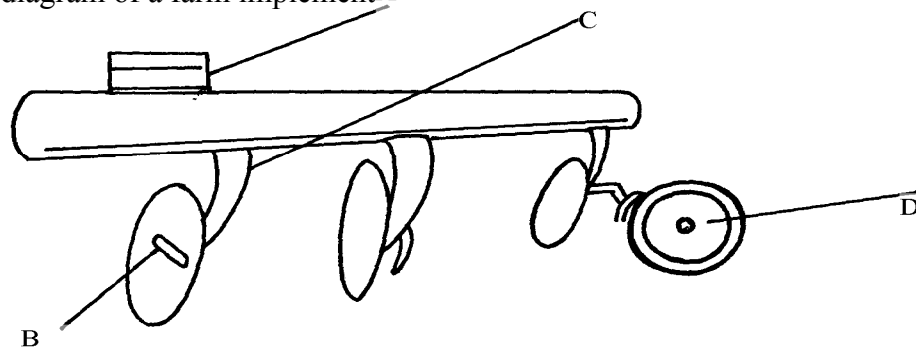
10. The diagram below represents an assembled differential of a tractor. Use it to answer the questions that follow:-



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- (a) Name the parts labeled **A**, **B**, **C** and **D**
- (b) State **two** functions of differential system of a tractor
- (c) Give **two** reasons why wheel skidding of a tractor is not allowed

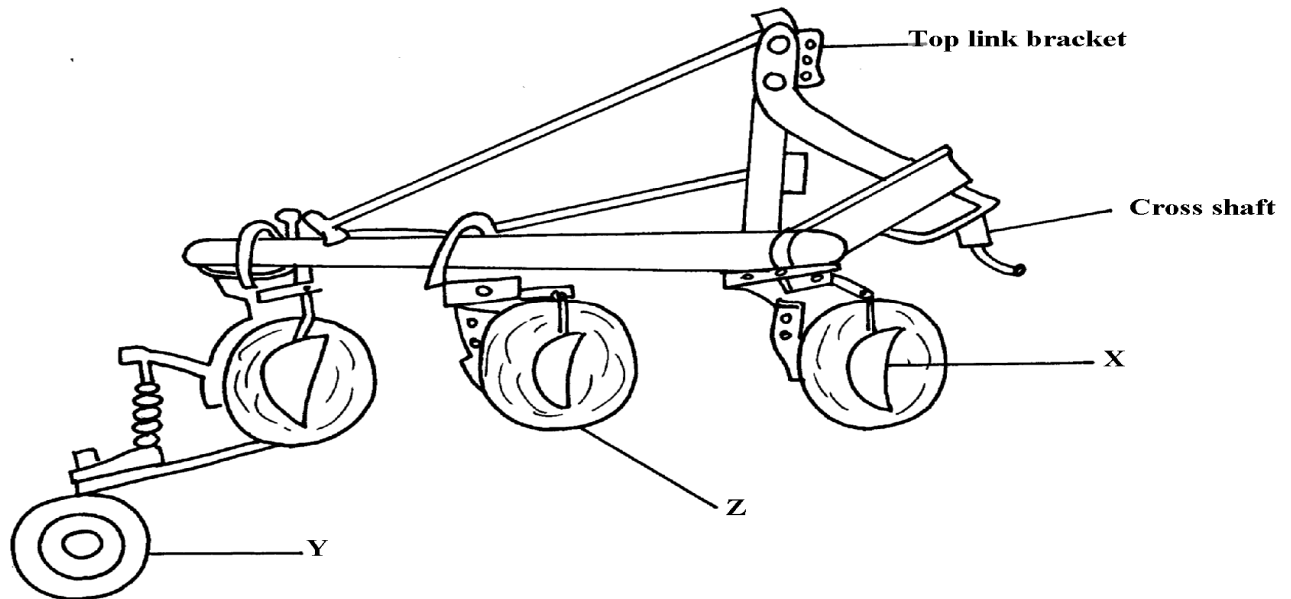
- 11. State **four** sources of power in the farm
- 12. Give the **four** strokes of a four stroke cycle tractor engine
- 13. State **four** factors which ensure efficient working by oxen in the farm
- 14. Mention **two** sources from which farmers can hire tractors
- 15. (a) Below is a diagram of a farm implementA



- (i) State the use of the implement shown above
- (ii) Name the parts labeled **A**, **B**, **C**, and **D**
- (iii) State **two** methods of increasing the depth of penetration of the implement

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16. State **four** ways through which a farmer would ensure maximum power output from ploughing animals
17. State **three** advantages of a disc plough over mould board plough
18. a) Explain the differences between petrol and chisel engine
b) Describe components of transmission system of a tractor
19. Name **four** systems of a tractor engine
20. Give **one** function of the clutch
21. State **two** adjustments that should be carried out on a tractor – mounted mould board plough in preparation for ploughing
22. The diagram below illustrates a farm implement. Study it and answer the questions that follow

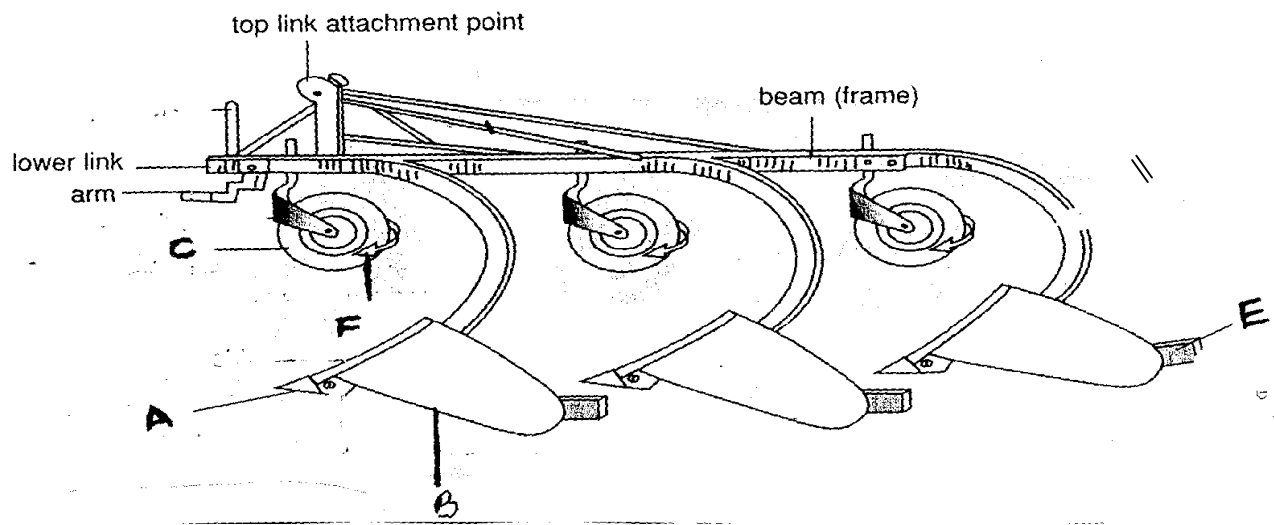


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- a) Identify the implement
 - b) Name the parts labeled **X**, **Y** and **Z**
 - c) State **three** maintenance practices that are carried out on a disc plough
23. a) Describe the operational differences of a disc plough and mould board plough
- b) Explain **six** marketing problems affecting dairy farming in Kenya
- c) State **four** reasons for culling a boar
24. Name the role of the following parts of a mould board plough
- a) Share .
 - b) Mould board
 - c) Land side....
25. a) State **five** maintenance practices of a mould board plough
- b) Explain **five** structural and functional differences between the petrol and diesel engines
- c) List **five** uses of farm fences
26. Give **two** uses of ox-drawn fine harrow
27. List **four** care and maintenance of a tractor battery

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28. Study the diagram of a farm implement shown below and answer the questions that follow:



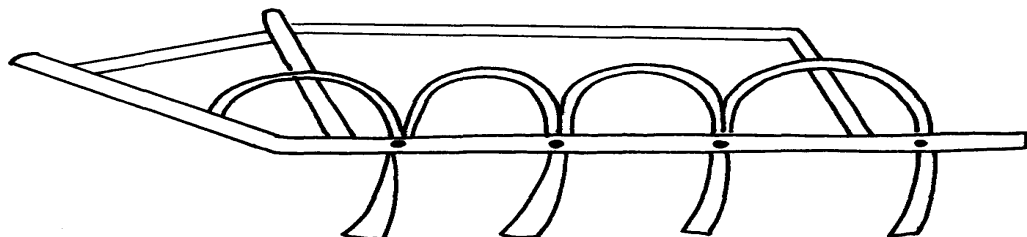
- Identify the farm implement illustrated above
- Label parts **A**, **B** and **C**
- Outline the functions of the parts labeled **E** and **F**
- Give **two** care and maintenance of the above implement

29. Outline **six** uses of live fences on the farm

30. List **two** possible causes of over heating in a tractor engine

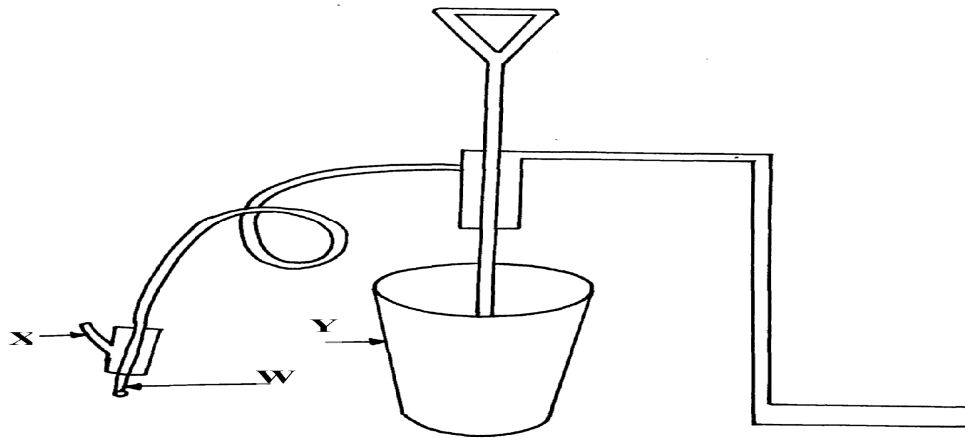
31. List **two** events occur during induction stroke in a four stroke engine

32. i) the diagram below shows a tractor drawn implement.



- a) Name the implement
- b) Give **two** uses of the implement above
- c) State **three** maintenance practices carried out on the above implement.
- ii) Below is an illustration of a farm equipment. Study it and answer

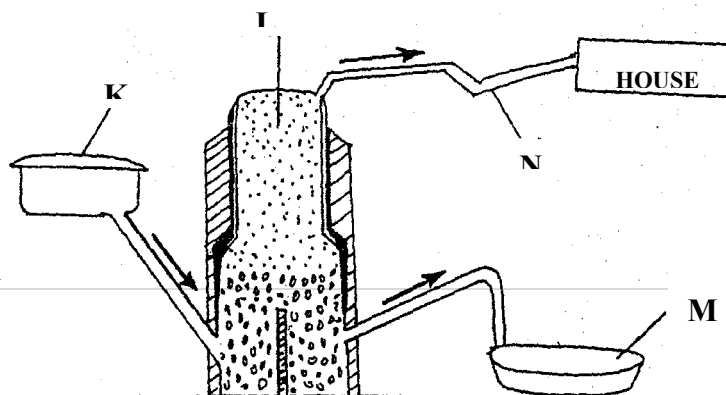
questions that follow



- a) Identify the farm equipment illustrated above
- b) What is the use of the equipment
- c) Name the parts labelled **W**, **X** and **Y**
- d) What is the functions of **Y** on the

equipment

- 33.
 - a) Explain the factors that influence the power output of farm animals
 - b) State the importance of farm fences
- 34. Study the illustration of a biogas digester plant and answer the questions that follow.



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- a) Name the major component of biogas that is trapped in part **L** above
- b) Give the name of the material deposited in part labeled **M** and its use
- c) What is the component of **K** in the biogas production
- d) Give **three** disadvantages of biogas as a source of farm power
35. Other than hydro-electricity mention **two** sources of electrical energy which can be available for use in the farm
36. a) Describe the maintenance practices required on a tractor before it is put to daily use
- b) Discuss the factors that influence the power output by a draught animal

SECTION II ANSWERS FARM TOOLS AND EQUIPMENT

- 1.
- Store the saw properly after use
 - Oil the blade for long storage to avoid rust
 - Sharpen the teeth properly
 - Tighten the handle screw if loose
2. A-Tin snip – cutting thin metal sheets
B-Spokes have – smoothing curved and circular surfaces
C- Ceardeners trowel – uprooting seedlings during transplanting
D-Brace drill – boring/ drilling holes in wood

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3. i) Stir up pump/ bucket pump
ii) Spraying livestock with acaricide
iii) A – Nozzle B – lance handle

iv) A – Atomises the acaricide into spray B – Direct the nozzle to the parts to be sprayed
4. Garden fork is a tool used in weeding in the nursery beds/and in carrot fields
5. The tool for checking the vertical straightness of a wall is Plumb bob and plumbline
6. A tool for removing nails from timber is a claw harmer (1mk)
7. Two examples of equipment that a livestock farmer can use in administering oral anti-helminthes
(1mk)
- Narrow necked bottle
- Drenching gun
- Dosing gun
8. (a) L...- Pickaxe M- Adjustable spanner
N..- Ring spanner O..- Pipe wrench
(b) Give one functional advantage of tool M over tool N (½
9. Tools used to perform functions on the farm
- Hand drill
- Bit brace
10. - Blockage of nozzles and filters leading to uneven or no release of spray.
- Control pump may slacken or loosen, making it difficult to regulate the rate of spraying.
- Inadequate operating pressure due to leakages in the compressor pump, hoses or control valves.
11. (a) - Elastrator and rubber ring.
- The burdizzo.
- Sharp knife / scalpel. O.....

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(iii) Extracting blood samples for laboratory analysis

12. most appropriate set of animal hand tools that a farmer uses for the following operations:-

- (i) Restraining large full when taking it around the show ring-bull ring and lead stick (1mk)
- (ii) Cutting tail in sheep-rubber ring and elastrator (1mk)
- (iii) Extracting blood samples for laboratory analysis-hypodermic needle and syringe

13. Provision of extra and quality feeds to sheep, two or three weeks before mating

14. ai) On the diagram a provided below, draw the mark to indicate a pig number 147, using the procedure of ear-notching in diagram above
ii) the recommended stage of growth in pigs at which the ear-notching should be carried out?

- before weaning/3-7 weeks of age/21-56 days/1 month-22months)
- iii) three reasons why weight is an important routine management practice in pig production
 - to determine growth rate i.e. weight gain
 - facilitate administration of drugs e.g. drenching
 - for feeding i.e. to know the amount of feed to give
 - to determine the service/breeding time (1x3=3mks)

b) i) B-entrance

C-top bar/bar

D-top cover/lid(1x3=3mks)

ii) by applying bees wax/honey/molasses on the sides or top of the hive/jaggery/sheep sorrel/saliva /sugar syrup (Accept concentrated sugar solution-reject-sugar solution)

(1x1=1mk)

iii) Outline the procedure of opening the hive to harvest honey
smoke the hive through the entrance using a smoker then light the lid to remove the top bar(the order must be considered) (1x2=2mks)

15. a) i) E-key hole saw/compass saw
F-wood chisel
G-cold/metal chisel
H-plumb bob (1/2x4=2mks)

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- ii) E-to cut or make key holes F-cutting timber
 G-cutting metal H-checking whether a tall wall is vertical
- iii) wooden hammer/mallet(1mk)

b)i) open castration/surgical castration (1mk)

ii) procedure you would follow when carrying out the practice named in(i) above in piglets

- restrain the piglets
- sterilize the blade
- disinfect the secretal sac
- slit the secretal sac to expose the testicle
- locate and hold the sperm duct
- cut the sperm duct by scrapping with the slide
- sew up the wound
- sterilize/disinfect the wound
- release the animal(piglet)

16. i) A – Sickle

- B- Pruning saw
C- Hoof cutter
D- Metal float

ii) A sickle is used in harvesting grass, rice, wheat by cutting

- B Pruning saw is used for pruning tree crops, cutting hard stems in coffee
C Hoof cutter – Trimming hoofs in animals e.g. cattle, goats, sheep
D Float (wood or metal) smoothing concrete during plastering

iii) Maintenance on B

- Sharpen and reset the teeth
- Replace broken handle
- Oil the blade for long storage

17.

- Halter
- Rope
- Nose ring and leading stick

18. Mention the use of the following tools.

- Dibber.-making holes for transplanting

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- Spoke shaves.-planning curved surface
 - Tin snip.-cutting metal/iron sheets
 - Burdizzo-castration (4x ½ =2mks)
19. four precautions that should be taken when using workshop tools and equipment.
- Use tools for correct purpose.
 - Maintain them in good working conditions.
 - Keep them safely after use.
 - Handle tools correctly during use.
- Use of safety devices / protective clothes
20. a) Bucket pump /stir - up pump (1x1 = 1mk)
b) Spraying acaricide on livestock (1x1 =1mk)
c) W - Nozzle
- X - Trigger
 - Y - Pail /bucket
- d)For holding acaricide solution during spraying. (1x1 = 1)
21. i) Identify the above diagram (1mk)
- Artificial vagina.
 - ii) Name the parts labelled A and B (2mks)
 - A – warm water
 - B – collecting cap

LIVESTOCK PRODUCTION I **COMMON LIVESTOCK BREEDS**

1. two reasons for treating water for us on the farm
- Remove chemical impurities
 - Remove foreign material
 - Remove disease earning organisms
- Remove bad smell & taste
2. four advantages of applying lime in clay soil
- Lower soil acidity
 - Increase calcium content
 - Hastens decomposition of organic matter

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- Improve soil structure/ improve drainage
 - Facilitates availability and absorption of nitrogen and phosphorous
 - Improve legume nodulation and nitrogen fixation
- Increase multiplication of micro- organisms

3. four ways by which Re-afforestation help in land reclamation

- Add organic matter from falling leaves
 - Recycles soil erosion
 - Control soil erosion
 - Improve drainage of swampy areas
- Play part in hydrological cycle

4. Distinguishing feature between Kenya white and California Kenya white is white all over the body, has pink eyes California white is white with black nose and ears; has black/ brown eyes

5. Reasons why farmers keep livestock in Kenya

- Source of food
- Source of income
- Cultural uses e.g. paying dowry, recreation
- Provide source of power e.g. oxen
- Provides raw material to the industries

6. Ideal conformation features of beef cattle

- Blocky/ square/ rectangular
- Deep well fleshed bodies
- Short strong legs to support their heavy bodies
- Compact body

7. (i) Dual purpose cattle in Kenya.

- Sahiwal;
- Red Poll;
- Simmental;

(ii) Uses of Rotavator.

- Mixes trash and the soil;
- Cuts the furrow slices and harrows them; (2 in one pass).
- Break up large soil clods into smaller ones / performs 2⁰ tillage;

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- Cut the furrows slices / primary tillage;

8. Large white is the pig with such characteristics

9. Epistasis is a combination of genes which on their own could have been inferior or undesirable (1mk)

10. (a) The role of livestock industry in Kenya's economy

- Provision of food; various livestock products such as milk, eggs, meat and honey are used as food
- Source of labour: some livestock can be used to provide power for carrying out various activities e.g. oxen, donkeys and camels
- Provides employment; Several people are employed in the livestock industry either directly or indirectly e.g. those employed by KMC
- Social cultural uses: depending on the culture of the society e.g. dowry, status, symbol etc
- Source of income when livestock and livestock products are sold either locally or internationally: income is earned which is used for national development which is used for national development
- Industrial development, various products are processed in industries which are taxable to raise income or national development

(stating any 5pts = 1mk x 5; Explanation each 1mk x 5pts = 5mks (10mks)

(b) The general characteristics of indigenous cattle

- they have large thoracic humps for storing fat
- they are tolerant to high temperatures
- They are able to resist tropical diseases e.g. East Coast fever
- They have relatively long calving interval (beyond one year)
- They have a slow growth rate and mature rate
- They have smooth and short coat of hair
- They have long and narrow head
- they have a relatively short lactation period
- they can walk for long distances without serious loss in condition
- They are relatively small in size even when mature (1mk each for any 10pts = 10mks)

11. (i) Dairy cattle

(ii) five characteristics that tell you that the animal belongs to the type named in (i) above?

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- Straight top line
- Large udder/well develop udder
- Prominent milk vein
- Large swell develop head quarter
- Triangular shape
- Large stomach capacity (any 5x1=5mks)
- (iii) three areas on the body of a cow where ticks are commonly found (1½mk)
 - Base of ear/inside the ears
 - Neck
 - Flanks
 - Tail switch/tail
 - Belly
 - Brisket/dew lap (any 3x ½ =1 ½ mks)

12. 4 dairy goats

- Saanen
- Jamnapari
- Anglo- Nubian
- Toggenburg
- British Alpine

13. two distinguishing characteristics of Californian breed of rabbit

- White body
- Black ears /nose/paws/tail (2x ½ =1mk)

14. two common milk breed of goats reared in Kenya

- British Saaren
- The Toggenburg
- Anglo-Nubian
- Jamnapari
- The British alpine (½ x2=1mks)

15. four dairy cattle breeds reared goats in Kenya.

- Ayrshire
- Friesian
- Guernsey
- Jersey (4x ½ =2mks)

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16. Differentiate between breed of animal and type of animal. (2mk)

- Breed-group of animals with similar characteristics and common origin.
- Type –the purpose for which the animal is kept. (2x1mk = 2mks) (mark as whole)
 - smother weeds
 - Regulate soil temperature
 - Conserve moisture

(LIVESTOCK HEALTH II) **(LIVESTOCK PARASITES)**

1.
 - Use of caustic potash stick (potassium hydroxide)
 - Use of dehorning collodion

2. a)

- Cause anaemia
 - Deprive the hoof animal of food
 - Cause injury and damage to animal tissue and organs
 - Transmit diseases
 - Cause irritation
 - Cause obstruction to internal organs
- b)

- Human beings drop tapeworm segments/ proglottides together with their faeces
- Eggs are released from the segment. Once outside the human body
- Eggs are picked by pigs when feeding
- Eggs hatch into embryos in the intestine of pigs
- The embryo penetrate the intestinal wall and enter into the blood stream
- Embryo localize in the liver
- Embryos are disturbed throughout the muscle where they become cyst/ bladder worms
- Bladder worms get into human beings through eating under cooked pork/ bacon
- Once inside the human intestines the cyst wall dissolves and the bladder worm attach themselves to the wall of intestines
- Bladder worm develop into adult tape worm
- Adult tape worm releases segments/ proglottides containing fertilized eggs with human faeces

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c)

- Use prophylactic drugs/ deworms to kill the internal parasites
- Keep animal houses clean and disinfected
- Practice rotational grazing
- Use of clean feeding and watery equipment
- Use of latrines/ proper disposal of human faeces

Proper cooking of meat

3. Functions of calcium in dairy cows.

- Milk and egg formation / production;
- Bones / skeleton / teeth formation;
- Blood clotting;

4. Control measures of fleas.

- Keep clean animals sleeping places;
- Dust animal surroundings with appropriate insecticides;
- Cover with petroleum jelly to suffocate stick fast fleas;

5. Control measures of fleas

NYR

- Dusting with appropriate insecticide in the pen
- Ensuring cleanliness in poultry house
- Dusting of the birds with correct insecticide
- Applying petroleum jelly on infected parts (1mk each for any 2 pts = 2mks)

6. two reasons why drenching alone is not an effective method of controlling internal parasites

- Cannot kill all stages of parasites
- Cannot kill the eggs (2x ½ = 1mk)

7. -Embryo

-Cyst/bladder worm (2x ½ = 1mk)

8. a) K-tapeworm M-liver fluke

b) a-hooks b-suckers

c-mouth d-digestive glands

c) i) Small intestine

ii) the liver

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d) Water snail (1x1=1mk)

9. - External parasites effects

- Transmit diseases
- Causes anemia/ sucks blood
- Causes irritation/ discomfort
- Causes wounds on the skin that may predispose animal to secondary infection
- Loss of hair

10. - Read the manufacturers instructions carefully

- Mix the acaricide appropriately
- Pour the chemical solution into the knapsack sprayer through the sieve/ stir up pump container
- Restore in the animal
- Spray along the back to loin
- Spray the sides
- Spray under the belly including the udder/ scrotum
- Spray the rear/ hind quarters
- Spray fore limbs
- Spray the face, the ears last
- Allow the animal to drain the chemical
- Release the animal

11. a) Feed conversion is $\frac{90\text{Kg}}{360\text{Kg}} = \frac{1}{4} = 1 : 4$
MMS

b) Digestion in ruminant animal – 4 stomachs

i) Rumen (A pauch)

- Stores food temporarily
- Fermentation of food
- contains bacteria, fungi, protozoa which breaks down cellulose
- Synthesis of amino acids from ammonia gas
- Synthesis of vitamin B complex

ii) Reticulum (Honey comb)

- Sieves and separates fine from coarse food particles
- Retains foreign and undigestible materials

iii) Omasum (Many plies or book)

- Stores food temporarily
- Grinds and sieve food particles

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- Absorption of water
- iv) Abomasum (True stomach)
- Enzymetic digestion takes place

c)

- Prevents rusting of surfaces
- Reduces the rate of wear and tear of moving parts
- Minimizes power loss due to friction
- Acts as a cleaning agent
- Reduces the heat created by the rubbing surfaces and acts as a seal between them

12. Bush clearing to destroy breeding places

- Spraying breeding places with insecticides
- use fly traps with impregnated nets
- use stressing agents e.g. radio isotopes on male file sand then releasing them

13. - Succulent

- Dry

14. two common milk breed of goats reared in Kenya

- British Saaren
- The Toggenburg
- Anglo-Nubian
- Jamnapari
- The British alpine ($\frac{1}{2} \times 2 = 1$ mks)

15. the element calcium and phosphorus important in the diet of young livestock?

- For bone formation and development
- For proper teeth development
- For increased conversion of feed
- Increase livestock appetite ($\frac{1}{2} \times 1 = 1$ mk)

16 two parasites of cattle which are also disease vectors

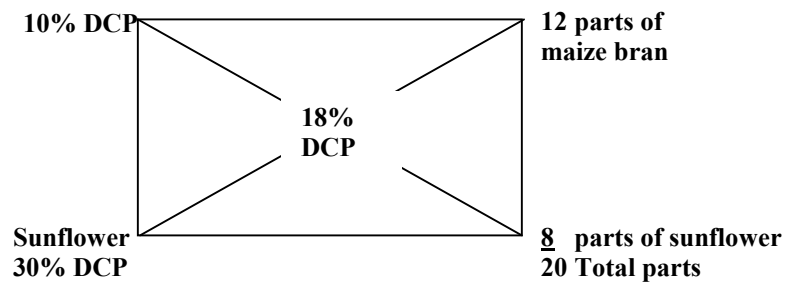
- Ticks
 - Tsetse flies
- Mosquitoes

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- 17.
- training
 - Giving incentives/motivation
 - Farm mechanization
 - Labour supervision

LIVESTOCK PRODUCTION II (NUTRITION)

1. Production ration may be utilized by dairy goats in:
 - Milk formation
 - Growth
 - Foetal/embryo development
2. Use Pearson's square method to calculate how much of the feedstuffs a 150kg rations (5mks)



$$\text{Maize bran} = \frac{12}{20} \times 150 = 90\text{kg maize bran}$$

$$\text{Sunflower} = \frac{8}{20} \times 150 = 60\text{kg sunflower}$$

3.
 - i) Water (1mrk)
 - Transport medium
 - For metabolic processes
 - Regulates body temperature
 - Maintains shell shape

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- Component of livestock product
- Lubricant of body joints ($2 \times \frac{1}{2} = 1\text{mk}$)
 - ii) Vitamin A (1mrk)
- Bone formation
- Prevents diseases / Increases disease resistance
- Improves vision
- Improves vigour/ for proper growth

4. Rumen.

5. i) A ration containing 18% protein is to be made from maize and sunflower cake.

Given

that maize contains 7% protein, and sunflower seed cake 34% protein. Use Pearson

square

methods to calculate the value of feedstuffs to be used to prepare 100kgs of the feed

(3mks)

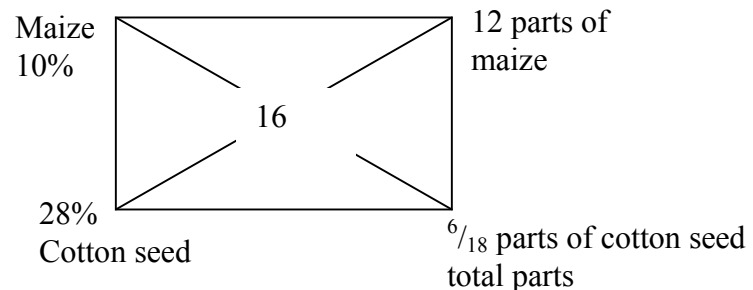
ii) two other methods that can be used to formulate feed ration

- Linear programming
- Trial and error
- Graphical method.

6. (a) The daily amount of food given to an animal/ the amount of food given to an animal per day

$$(b) \text{ Quantity of maize} = \frac{12}{18} \times 200\text{Kg} = 133.33\text{kg}$$

$$\text{Quantity of cotton seed} = \frac{6}{18} \times 200 = 66.67\text{kg}$$



7. Hormones
Antibiotics
Medicants

8.

- Qualities of roughages
- Availability of the concentrates
- Level of production
- Physiological states of the animals
- Quality of concentrates
- Economic factors

9.

- Qualities of roughages
- Availability of the concentrates
- Level of production
- Physiological states of the animals
- Quality of concentrates
- Economic factors

LIVESTOCK PRODUCTION III **(SELECTION AND BREEDING)**

1.

- Steaming up of ewes to have milk to be suckled
- Use of lambing pens
- Ewes as made to recognize lambs after lambs
- Blind folding ewes to activate maternal instinct
- Attend/ treat inflamed udders/ painful

2.

- Pig 112 – 15 days
Rabbits 28 – 32 days

3

(i) Selection.

- It is the process of allowing certain animals to be the parents of the future generations

while

culling others;

(ii) Methods used in selection.

- Mass selection;
- Progeny testing;
- Contemporary comparison;

4.

- Restless.

- Frequent urination.

- Swollen vulva.

- The doe throws itself on its side.

- She (does) rubs herself against the wall or any other solid object.

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- The doe tries to contact other rabbits in the next hutch by peeping through the cage walls.

5. four meat breeds of rabbits

- Flemish giant
- California white
- New Zealand white
- Ear lop
- chinchilla

6. four reasons for cutting a breeding boar

- Bareness/loss of Libido
- Loss of sight, limb, cannot mate
- Old age
- Perpetual sickness/contract
- Reproductive diseases
- Aggressiveness/wildness

7.

- Mass
- Progeria testing
- Comparison

8. (i) Heterosis- Increased vigour/ performance resulting from mating two superior unrelated Breeds

(ii) Epitasis- Is a combination of genes that individually could have been both undesirable
or inferior

9.

- Heat signs in rabbits (doe)
- The doe throws itself on its sides
- She rubs herself against walls or solid objects
- Tries to contact other rabbits in the next hutch (peeping)
- Restlessness
- Swollen vulva
- Frequent urination

10. Gestation periods of :-Cow 270 – 285 days

Sow – 113 117 days

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11. Disadvantages of natural mating

- A lot of semen is wasted
- High chances of inbreeding
- Large males can injure small females
- Expensive to transport bull over long distance
- May need extra pasture, extra cost

12. a) the advantage of battery cage system of rearing layers

- Higher egg production due to less energy wastage by birds
- Accurate egg production records re kept
- Cannibalism and egg eating are controlled
- Eggs are clean
- Allows for mechanization
- Birds do not contaminate food and water
- Makes handling easy as birds are restricted within small areas
- Discourages broodiness
- Increases the sticking rate
- Sick birds can easily be isolated
- The wire floor prevails re-infection on the parasites, worms and coccidiosi
- There is no bully during feedings
- There is low labour requirement (1x10=10mks)

b) Outline ten factors considered when selecting dairy cattle for breeding

- age-young animals have longer productive live and are more productive and Economical to keep than old animals
- level of performance-animals with highest production level be kept
- health-consider animals which are less susceptible to disease
- body confirmation go for ones well people dairy characteristics
- temperature/behaviour select only decline animals which are easy to handle mothering ability
- adaptable under suitability to the environment
- prolycacy ability to give built to many off springs at a time
- physical defects
- quality of products (1x10=10mks)

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13. A type of mating where the female of low grade is mated to a pure bred sire of superior quality (1mk)

LIVESTOCK PRODUCTION IV **(LIVESTOCK MANAGEMENT PRACTICES)**

1. Reasons of carrying out crutching in sheep.
 - Facilitates easy mating;
 - Minimizes blowfly infestation;
 - Provides hygienic conditions at lambing time;
 - Allows easy access to the teats by the lambs;
 - Provides hygienic suckling conditions;
2. Ways of performing closed castration.
 - Use of elastrator and rubber ring;
 - Use of the burdizzo;
3. (a)
 - (i) B – Should take place – Hooks, cheek, rump
 - (ii) V – Is done - Thigh muscles and the shoulders
 - (iii) BT – is taken - Anus
 - (iv) M – May occur - Udder and teatsb) -- Ears (ear lobes)
 - Tail switch..
 - Under the tail head.
 - Both fore and head flanks.
 - Between the hooves.c)
 - 1 – Muzzle
 - 2 – Poll
 - 3 – Shoulder
 - 4 – Heart girth
4. (a) - The removal of the testicles in male livestock or stoppage of production of spermatozoa or semen.
 - It is the rendering of male reproductive organs non- functional.(b) – To control breeding and inbreeding.
 - To make animals, especially the bulls docile.

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- To improve quality of meat by removing unpleasant smell especially in goats.
- To encourage faster growth rates of the castrated male animals.
- To control breeding diseases such as brucellosis, vaginitis trichonomiasis and others

that area

transmitted through mating.

5. (a) The management of a gilt from weaning to furrowing

- Feed gilt on at least 3kg of sow and weaner meal daily/balanced diet
- Provide clean drinking water
- Control any external parasites by spraying with pesticides or washing
- Treat the gilt next to the boar at the age 12months ready to be served/serve gilt at the right age.
- Keep the pen clean by maintaining clean
- Flush the gilt 3-4weeks before service by feeding high quality diet
- Take the gilt to the boar's pen for service and let it stay there for at least 12hours
- Observe the return to heat, if any after three weeks, and repeat the services if necessary
- Steaming up should start 1 ½ months before furrowing by giving 3-4kg of feed
- 7-10days furrowing, the gilt should be washed
- The gilt should be moved to disinfected pen/clean pen
- provide guard rails or furrowing crate in a furrowing pen
- Provide a source of heat
- Sow and weaner meal should be reduced three days before furrowing and fed with bran
- Observe the signs furrowing and supervise the process of furrowing
- Deworm the gilt 7-10days before furrowing
- Weigh the gilt regularly
- Keep proper records

(b) The preparation a poultry farmer should make before the arrival of day old chicks

- Ensure the brooder is ready 2-3days before arrival of chicks
- Brooder should be cleaned and disinfected to reduce risk of disease infection
- Spreading litter on the floor to provide warmth
- Provide a functional heat source e.g. electric bulb, lantern e.t.c
- Provide adequate feeding equipment
- Ensure adequate floor space considering the number of chicks
- Ensure the chick mash is ready before chicks arrive
- Provide wire guard
- Make holes on the walls of brooder for ventilation

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6.

- To identify mated ewes
 - To indicate active rams hence help in culling
- Identify the sire of each lamb

7.

- Promote docility
 - Improve meat quality
 - Control breeding diseases
 - Control interbreeding/ control hereditary defects
- Improve growth rate

8.

- Use of swarm net
 - Use of catcher box
- Placing the hive in a strategic position for bees to occupy

9.

- Control deficiency diseases
 - Impact resistance to diseases
- Good physical appearance/ good coat cover

6. two recommended methods of docking lambs

- Use elastrator and rubber ring
- Sharp knife/scalpel (2x ½ = 1mk)

7. three disadvantages of inbreeding

- Loss of hybrid vigour
- May lead to decline fertility leading to species extinction
- May bring about reduction in performance
- Leads to high rate of prenatal mortality (any 3x ½ = 1 ½ mks)

8. four routine management practices that should be carried out on a lactating ewe

- Inadequate feeding /balance diet
- Spraying/dipping to control external parasite
- Drenching/deworming to control internal parasite
- Provision of clean water ad-lib

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- Tugging
- Avoid extensive movement
- Provide mineral licks (any 4x ½ =2mks)

9. the procedure which should be followed to castrate a three weeks old piglet using surgical Method

- Assemble equipment and sterilize
- Restrain the animal to be castrated
- Thoroughly wash hands before opening up animals skin
- Artery of forces is used to close up the open blood vessel to stop excess bleeding
- Cut the skin of scrotum
- Remove the two tests completely, leaving on empty scrotal sac
- Disinfect the wound
- Animals heals faster since its castrated when young (7x1@=7mks)

(b) (i) State five factors that should be considered when sitting a bee hive in a farm

- Away from homestead, pastures and road
- Sheltered/quiet place
- Near source of water
- Nearest to flowers producing ants
- Safe from predators (5x1=5mks)

(ii) Describe the management practices that would ensure maximum harvest of fish from a fish pond

- Control stocking rate
- Control water pollution
- Supply enough food to fish
- Aerate the water/constant in flow and out flow of water
- Maintain appropriate depth of water
- Control predators
- Harvest fish at correct maturity stage
- Fertilize the pond/adequate water plants

10. three types of bees found in a bee colony

- The queen
- The drone
- The workers

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11. List three methods of castrating farm
- Closed method-use buidizzer
 - Open method-use surgical method
 - Caponization-use hormones
12. the routine management practices of piglets from the 1st day to the 8th week
- Placenta disposal
 - Umbilical cord cutting with sharp and sterilized scalpel
 - Disinfecting the umbilical cord with iodine solution
 - Extracting needles teeth/teeth clipping/di-tusking
 - Keeping piglets in warm creep area
 - Weighing the piglets 24hrs after birth
 - Feeding the piglets on colostrums
 - Iron supplementation through the intra-muscular injection/paste
 - Vaccination against diseases
 - Feed the piglets with creep feeds
 - Water provision at adlibitum
 - Nose ringing
 - Putting if identification marks e.g. ear notching
 - Deworming /drenching with antihelminthes
 - Tail cutting of the piglet
 - Castration of the male piglet
 - Ensure that they are breathing
 - Assist the weak piglets to suckle
 - Changing the beddings regularly
 - Removing dead piglets from the pen
 - Providing furrowing crate to avoid crushing of piglets
- 13.
- Raised pens with slatted floor
 - Permanent calf pen with concrete floor
 - Movable calf pen
 - Temporary calf pen
14. Poor nutrition – infertility
- Poor timing of service
15. three advantages of hoof trimming in sheep production
- Facilitate easy movement

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- Control foot rot disease
 - Prevents the ram from injuring the ewe during mating ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)
16. State four factors considered when citing an apiary in the farm
- Availability of water
 - Availability of flowers
 - A sheltered place
 - An area free from noise/disturbance
 - Away from lime stead and grazing grounds
 - Well drained area ($2 \times 2 = 4$ mks)
17. three methods of stocking a beehive with honey bee
- Use swarm nets
 - Use of a catcher box
 - Use of an empty hive ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)
18. three common methods of extracting honey from the combs
- Use of heat
 - Crushing and straining
 - Centrifugal extractor
19. a) i)
- ii) before wearing/3-7 weeks of age/21-56 days/1 month-22months)
 - iii) -to determine growth rate i.e. weight gain
- -facilitate administration of drugs e.g. drenching
 - -for feeding i.e. to know the amount of feed to give
 - -to determine the service/breeding time ($1 \times 3 = 3$ mks)
- b) i) B-entrance
C-top bar/bar
D-top cover/lid($1 \times 3 = 3$ mks)
- ii) by applying bees wax/honey/molasses on the sides or top of the hive/
jaggery/sheep
sorrel/salvial/sugar syrup
(Accept concentrated sugar solution-reject-sugar solution) ($1 \times 1 = 1$ mk)
- iii) Outline the procedure of opening the hive to harvest honey
smoke the hive through the entrance using a smoker then light the hid to remove
the top bar(the order must be considered) ($1 \times 2 = 2$ mks)
20. a) i) E-key hole saw/compass saw

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F-wood chisel

G-cold/metal chisel

H-plumb bob (1/2x4=2mks)

ii) E-to cut or make key holes

F-cutting timber

G-cutting metal

H-checking whether a tall wall is vertical (1/2x4=2mks)

iii) wooden hammer/mallet(1mk)

b))i) open castration/surgical castration (1mk)

ii) procedure you would follow when carrying out the practice named in(i) above in piglets

- restrain the piglets
- sterilize the blade
- disinfect the secretal sac
- slit the secretal sac to expose the testicle
- locate and hold the sperm duct
- cut the sperm duct by scrapping with the slide
- sew up the wound
- sterilize/disinfect the wound
- release the animal(piglet) (1/2x6=3mks)

a) four advantages of age grouping farm animals as a management practice

- Avoids bullying among the animals
- Facilitate feeding /adequate and economic use of feed
- Facilitate the administration of drugs e.g. drenching
- Easy keeping of management records
- Facilitate breeding/cutting of livestock (2x4=8mks)

b) four major cause of lamb mortality from birth to weaning

- chilling
- scours
- internal parasitic infestation
- loss of mother/lack of foster parents
- inadequate mothers milk/malnutrition
- crushing by the mother (1x4=4mks)

c) i)cause-bacteria/brucella abortus brucells

ii) Transmission-sexually transmitted/it is a breeding diseases

iiI) Symptoms

- abortion/premature birth
- yellowish slimy and odourless discharge through the vulva

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- retained afterbirth/placenta
- the cow may become barren (1x4=4mks)
- iv) Control measures
- vaccination
- use of healthy semen/bull/AI
- cull/destroy affected cattle
- proper disposal of foetus and carcass (1x2=2mks)

23. Reasons for castrating animals when young

- Less pain
- Quick healing
- Little loss of blood

24. a) Reasons for dehorning farm animals

- Reduce space occupied by animal
 - Making handling easier
 - To reduce destruction of farm structures
 - To make them
 - To reduce risk, injury to farmer and other animals
- = 1 mark

2x ½

b) Methods of dehorning livestock

- Use of caustic potash stick (Potassium hydroxide)
- Use of dehorning iron
- Use of dehorning saw or wire
- Use of rubber ring and elastrator
- Use of dehorning collation

25. a) Is the giving of high quality seeds to a gestating animal towards end gestation period

b) Reasons for steaming up

- Increase milk yield after farming
- Help build up body reserves for lactation
- Ensure rapid growth and development foetus
- Ensure healthy and strong young at birth

26. four management practices that should be carried on a fish pond in order to obtain maximum

Fish production.(2mrk)

- Control predators.

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- Control Water pollution.
 - Maintain appropriate water level.
 - Maintain correct stocking rate.
 - Supply adequate food. (4x ½ =2mks)
27. four factors considered when formulating livestock ration. (2mk)
- Body weight / size
 - Available feeds
 - Cost of feeds
 - Nutrient composition of feeds available.
 - Ingredients required in the ratio.
 - Animals level of production.
 - Age / stage of growth.
 - Type of production.
28. four conditions that necessitate the handling of farm animals. (2mk)
- During treatment
 - When spraying or hand dressing
 - When milking
 - When performing some management practices e.g. dehorning
 - When inspecting animals for any signs of a disease
28. - Large animals e.g. buffaloes
- man activities e.g. farming
- root pressure of plants
- burrowing animals e.g. moles, termites
29. (a) – Random/zigzag soil sampling-Arrow roots
(b) -Old manure heaps
- Ant hills
 - Dead furrows
 - Rice
 - Fence lines
 - Cattle bomas

FARM STRUCTURES

1.

- Steel is expensive

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- Require high skilled labour
 - Heavy and difficult to transport
 - Rusts easily
- Low workability

2.

- To prevent termites from rising up to the wall
- To reduce moisture rising up the wall

3. A group of calves kept according to age

4. a) Procedure in construction of a barbed wire fence

- Slash/ clear vegetation around fence line 2 m wide
- Measure and mark spots for holes using pegs
- Dig holes 60cm – 90cm deep depending areas where the poles are to be placed
- Assemble poles and other requirement materials
- Drop pole and struts at respective points
- Prepare concrete mixture
- Erect poles in pole holes
- Align the poles and put concrete using spade or soil
- Compact the concrete in holes
- Allow to settle for a few days while curing
- Put barbed wire around using appropriate tools
- Tighten the wire using wire strainer
- Mail barbed wire using fencing staple at required distance
- Put droppers along the fence as required

12x1=12 mks

b) Wood preservatives

- Creosote
- Old engine oil
- Paint/ far/ tanesc
- Copper sulphate
- Sodium dichromate
- Arsenic pentoxide
- Pentachloroplenol
- Triputyl tin oxide

4x1=4

mks

c) Choice of farm building materials

- Cost of materials

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- Availability/ strength of the material
 - Workability
- Type of enterprise

5. (i) Factors considered when sitting a fish pond.

- Reliable source of water/ water source;
- Soil type / poorly drained clay soil the best;
- Topography / gently sloping;
- Security/ be secure from thieves / predators;
- Water quality / free of pollutants;
- Machine milking;

(ii) Features of a laying nest.

- Dimly lit; dark;
- Spacious / large enough to accommodate bird comfortably;
- Dry clean beddings;
- Have lockable doors;
- Kept in secluded parts of the house;
- Have slanting roofs to prevent birds from perching on;

5. Uses of footbath in cattle dip.

- To wash the foot off mud;
- Contains chemicals for controlling foot rot; CUSO_4 (blue vitriol/ formalin solution;)

6. (a) – A green house is a farm structure made up of glass or translucent material as wall and

roof to enhance and achieve optimum condition for valuable horticultural crop production.

(b) Material used in green house construction.

- Galvanizing iron.
- Aluminium or wooden frame.
- Glass or clear polythene sheet.
- Fibre glass or reinforced panels.

(c) Maintenance practices on green house.

- Dirty polythene sheet should be clear.
- Blocked systems should be repaired and cleared.
- Torn polythene material should be replaced.
- Should be fenced for security.

(d) Importance of maintaining farm structures.

- Last longer/ enhance durability.
- Reduce replacement cost.

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- Protect livestock from predator.
- Prevent straying animals.
- To prevent diseases brought by cold winds.
- Make them effective in their use.

7. Disadvantages of barbed wire fence in paddocking:-

- Can remove wool from sheep
- Barbs can injure the animals
- Smaller animals can pass through if the wire strands are widely spread

8. (a) A .Inlet

B – Spillway/ overflow.

C - Drain pipe/outlet (1 ½ mk each = 1 ½ mks)

(b) The most appropriate part for feeding is part X (1mk)

(c) Two reasons why the floor of the pond should be covered with lime

- Facilitate the work of fertilizer in the pond
- Maintain PH of pond water(½ mk each 2pts = 1mk)

(d) Why should part marked B be screened?

- Prevent escape of fish
- Prevent entry of foreign /unwanted organisms (½ x 1pt = ½mk)

(e) three maintenance practices carried out on the structure

- Removing weeds
- Unblocking inlet and outlets
- Maintaining same water level
- repairing leakages on walls, floor e.t.c
- Draining of water during harvesting

9. (a)Stille (½ mk x 1pt = ½ mk)

(b) A pass allows only human passage while gate allows for both human and livestock in and

out of the farm (½mk mark as a whole)

(c) One type of live fence

- Electric
- Hedges

10. four requirements of a good maize store

- Leak proof
- Rat proof

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- Properly ventilated
 - Easy to clean
 - Raised off the ground properly drained
 - easy to load and unload/spacious (any 4x ½ =2mks)
11. three factors that determine the depth and size of foundation in a farm building
- Function of building
 - Soil type
 - Soil depth
 - Drainage of area (any 3x ½ =1 ½ mks)
12. (a) the uses of various hand tools in the construction of a poultry house
- Jembe- levelling the ground
 - Spade-scooping soil
 - Tape-measuring distance
 - Wheel barrow-carrying small load
 - Spirit level-checking whether surface is vertical or horizontal
 - Rip-saw/tenor saw-cutting timber
 - Hand drill/bit brace-boring in wood
 - Claw hammer-driving in/hitting and removing nails
 - G-clamp-holding objects /wood when joining
 - Tin-snip-cutting iron sheet
 - Chisel
 - Mallet (any 10x1=10mks)
- (b) the procedure of erecting wooden rail fence(7mks)
- Locate the area to be fenced off
 - Determine the amount of material needed
 - Treat the post
 - Clear the area
 - Measure the distance 3-4m apart and place pegs
 - Dig holes up to 60cm deep
 - Put fencing post in hole and reinforce with concrete
 - Place 3-4 horizontal rails
 - Space at about 125mm,175mm, 225mm and 275mm from ground
 - Fixed them onto post using nail (10x1pt=10mks)
- (c) factors considered when choosing the construction materials for farm building
- Purpose of building determine strength and durability of material
 - Availability of capital; depends on ability to purchase
 - Aesthetic aspect:-determine by economic status of farmers

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- Availability of material: easily obtained
- Durability: good quality .not be repaired often
- Resistant to extreme weather condition
- Safely of farm animals and farmer:- not have side effects/workability
- Suitability of the material.(any 5 points x 2 mks = 10 mks)

13. - Have adequate space
 - single housing (1 calf per pen)
 - Properly lit
 - Have proper drainage
 - Well ventilated
 - Drought free
14. - Demarcates boundaries of farms
- Prevents intruders, wild animals, thieves in the farms
 - Facilitate mixed farming
 - Enhance paddocking of farm for effective rotational grazing
 - Control unnecessary movement in the farm
 - Control inbreeding
 - Isolate sick animal
15. (a) (i) A – Wall plate/team beam/lintel
- B – Damp proof course
 C- Hard core
- (ii) - Prevents termite invasion
 - Prevents water capillarity /dampness
- (b) (i) 1 bag of cement
 3 parts/wheel barrows of sand
 5 parts /wheel barrows of ballasts or gravel
- (ii) Sand
 3 parts of sand = 24m^3
 1 bag = $\frac{1 \times 24}{3} = 8\text{m}^3$
 ballast
 parts = 24m^3
 5 parts = $\frac{5 \times 24}{3} = 40\text{m}^3$ ($\frac{1}{2}\text{mk}$)

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16. (a) Maintenance of the fish pond

- Protection of the pond – regularly check pond walls, plant grass on the walls to help control soil erosion
- Pond bottom repair- check water seepage problems regularly, It can be done introducing an even layer of clay to seal off the bottom of pond properly
- Removal of weeds; -regularly remove all weeds that grow on the walls and around the ponds
- Maintenance of appropriate water level: Maintain the same level of water in the pond by use of inlet and outlets
- Inspection of pond: - Regularly check for cracks in the walls and seal immediately
- Cleaning the pond- once n a while to drain out the pond water, remove all stones, silt or roots that may have settled at the pond bottom, lime the water before refilling it with water
- removal of organic materials – any vegetative matter or food remains should be removed as soon as they are noticed to ensure they do not start decomposing
- Repair fence around the pond- In case of worn out posts, repair/replace immediately

(b) Ways of controlling of fish predators in a fish pond

- Put a strong wire fence around the pond
- Provide a wire screen above the pond to guard against prevatory birds
- A sire screen is put in the inlet, outlet and in the spillway
- Scare away in the binds as necessary
- occasionally drain the ponds to kill all unwanted predatora in the pond bottom

17. four reasons of treating timber before roofing farm buildings

- Prevent attack from insects
- Prevent attack from fungi (rotting)
- Resist weather condition:-extreme temperature
- Resist water penetration
- To harden woo-make it durable and more strong
- To avoid warping

18. four uses of crushing in the farm

- Spraying livestock against external parasites
- Identifying animals by use of such methods as branding ,ear-tagging and ear notching
- Vaccination
- Administering prophylactic drugs to the animals
- Treating sick animals
- Dehorning

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- Pregnancy test
- Artificial insemination
- Taking body temperature
- Hoof trimming
- Milking

19. a) A fence is a structure that encloses a designated area and forms a physical barrier for animals and human
b) List various types of fences

- live fence
- electric fence
- barbed wire fence
- chicken wire fence
- wooden fence
- pole and rail fence
- plain wire fence
- trench fence
- wall fence

c) Describe advantages of fences

- keep off intruders/thieves
- prevent damage of crops by animals
- control grazing in paddocks
- control breeding by separating male and female
- acts as wind break
- control pests and disease by controlling wild animals
- add aesthetic value
- provide livestock feed or human fruits or firewood
- add value to the farm
- provide security to the house stead and farm animals
- they form perimeter fence along the boundary to demarcate farm land from the neighbours
- used to isolate sick animals from the rest of the herd to prevent spread of diseases
- separate crop field from pasture facilitating mixed farming

20. i) E – rafter F- Struct
G- Eaves H- Wall plate
ii) E (Rafter) – To provide support for the roofing materials

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F (Strut) – To support the rafter/ holding the weight of the roof

G (Eaves) – Prevent rain from falling on the wall

H- (wall plate)- To support the roof

iii) Chemicals for treating timber

- Tar
- Sodium dichromate
- Copper sulphate
- Arsenic pentoxide
- Old engine oil
- Pentachlorophenox
- Tributyl tin oxide
- Creosote
- Paint

21. i) K- Spill way (reject over flow pipe)

L- Drainage channel

ii) M is deeper to provide breeding place for the fish

iii) Maintenance of the pond

- Cleaning the pond by removing all foreign materials
- Repairing the dyke (bunds)
- Maintain good level of water
- Control predators
- Weed control around the pond
- Plant grass on dykes to prevent erosion
- Remove the silt if accumulated
- Regular pond fertilization
- Apply lime before refilling

22. a) Siting a fish pond

i) Soil type- clay soil is the best

ii) Topography – requires gentle slope not Lilly and flat

iii) Source of water – near reliable source

iv) Marketing centre should be close

v) Accessibility from the homestead

vi) Security – protected against predators

vii) far from natural source of fish

b)

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- General farm hygiene, cleanliness of houses, equipment proper carcass disposal by burning/ burying/
- Disinfection to destroy pathogens e.g. Anthrax and calf diseases
- Isolation of sick animals – separated from healthy ones to avoid spread of diseases e.g. foot mouth
- Drenching/ deworming to control internal parasites e.g. tapeworms and roundworms
- Treatment of the sick animal – to prevent spread of diseases
- Vaccination to create resistance to diseases on regular basis e.g. foot and mouth, anthrax, new castle
- Control vectors – to avoid disease transmission e.g. ECF, nagana/ specific method
- Prophylactic approach/ use of drugs to avoid injection e.g dry cow therapy against mastitis
- Trypanocidal drugs to control trypanosomiasis
- Proper breeding to control breeding diseases e.g. brucellosis
- Proper feeding to prevent nutritional disorders e.g. milk fever, anaemia
- Slaughtering/ killing – to prevent spread of contagious diseases e.g. anthrax
- Quarantine – to avoid spread of diseases
 - prevent introduction of diseases
- Proper housing to avoid predisposing the animal to diseases e.g. ventilation, spacing
- Foot trimming to minimize occurrence foot rot

23. four factors which influence the selection of materials for constructing a dairy shed

- Kind of dairy shed i.e. permanent or temporary
- Availability of materials
- Cost of materials
- Environment conditions of climate and soil type
- Durability of materials a
- Availability of skilled labour for construction
- Capital available (1/2x4=2mks)

24. a)i) On the diagram a provided below, draw the mark to indicate a pig number 147, using the procedure of ear-notching in diagram above 2

ii) the recommended stage of growth in pigs at which the ear-notching should be carried out?

- before weaning/3-7 weeks of age/21-56 days/1 month-22months)

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iii) State any three reasons why weight is an important routine management practice in pig production

- to determine growth rate i.e. weight gain
- facilitate administration of drugs e.g. drenching
- for feeding i.e. to know the amount of feed to give
- to determine the service/breeding time (1x3=3mks)

b) i) B-entrance

C-top bar/bar

D-top cover/lid(1x3=3mks)

ii) by applying bees wax/honey/molasses on the sides or top of the hive/
jaggery/sheep

sorrel/salvia/sugar syrup

(Accept concentrated sugar solution-reject-sugar solution) (1x1=1mk)

iii) Outline the procedure of opening the hive to harvest honey

smoke the hive through the entrance using a smoker then light the lid to remove the top bar(the order must be considered) (1x2=2mks)

25. a) State five maintenance practice of a mould board plough

- Lubricate the moving parts
- Sharpen blunt share
- Tighten bolts and nuts
- Clean the plough after use
- Coat the unpainted parts with old engine oil before any storage
- Replace worn out parts (1x5=5mks)

b) Explain five structural and functional differences between the petrol and diesel

engines

Petrol engine	Diesel engine
i)has a carburetor	i)Has an injector pump
ii)fuel and air mixed in the carburetor	ii)Fuel and air mixed within the cylinder
iii)fuel ignited by an electric spark	iii)fuel ignited by compression of air and fuel mixture in the cylinder
iv)produces little smoke	iv)produces a lot of smoke
v)is light in weight	v)relatively heavy

c) List five uses of farm fences

- keep of wild life ,predation and intruders
- demarcates boundaries

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- separate crop field from pasture land
- divide pasture land into paddock
- control movement of animals and people within the farm and prevent formation of unnecessary pests
- control disease and parasites helps in isolate sick animals
- helps in controlling breeding
- provide security
- act as wind break (1x5=5mks)

26. Uses of farm buildings

- Protect the farm animals from predators
- Provide shelter to the farmer and livestock
- Used to store farm produce and valuable inputs
- Controls livestock diseases and parasites
- Enhances efficiencies in farm planning, budgeting and production

27. Structural requirements for proper housing

- Well ventilation
- Free from cold/ draught
- Adequate space
- Proper drainage
- Leak proof roof
- Well lighting
- Easy to clean/ concrete floor

28. a) - roof
- b) P- purklin Q- Rafter
R- Cross tie S- Gutter
- c) P- Support roofing material/ iron sheet
Q – Collect water and safely directs it away from building

29. a) Honey

b) Crushing and straining

c) Procedure of harvesting the named product

- Wear protective clothes
- Approach hive quietly from behind
- Blow smoke around hive then through entrance

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- Lower hive
- Remove lid/ cover
- Lift top bars and brush off with bees
- Cut honey combs with honey leaving 3cm of wax
- Put combs in container
- Place back bars
- Put lid
- Return hive in position

d) Factors that affect quality of product harvested

- Type of plant from which nectar is obtained
- Maturity stage
- Method of harvesting
- Method of processing

28. i) Docking

ii) Reasons for carrying out the operation

- Avoid incidences of blowfly
- Make mating easy
- Even distribution of fat in body
- Avoid dirtifying wool

iii) Age of operation

- within two weeks from lambing

iv) Methods used for operation

- use of rubber ring and elastrator
- cutting with sterilized docking knife
- use of burdizzo
- use of hot iron bar

v) Routine management practice carried out on part B

- hoof trimming

29. a) Five parts of plunge dip

- i) Holding yard- Hold animal before dipping
- ii) Foot bath- Wash animal feet off dung, mud
 - Prevent foot rot/ contain copper sulphur,
- iii) Jump- Narrow entrance allow single animal easily to jump in dip wash
- iv) Draining race- Animal held while dip wash drain back in dip tank
- v) Drying yard- Animals need to dry before allowed to pasture, avoid contamination
- vi) Silt trap-raps mad, dung before dip wash flow back to dip tank, prevent siltation of dip tank
- vii) Shelter- Prevent evaporation

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-Prevent dilution of dip wash with rains

b) Six uses of live fences

- Thorn species prevent wild animals and other invaders into the farm
- Tall varieties act as wind breakers
- Add aesthetic value to the homestead
- Roots holds soil firmly controlling soil erosion
- Species such as lantana canara can be used to feed livestock
- Provide shade to livestock and man
- Trimmed branches can be used as organic manure, wood fuel
- Some species have medicinal value

1x6=6

marks

c) Four factors that influence power output of draught animal

i) Training- Proper training of oxen will plough better and faster than untrained animal

ii) Feeding- Well fed animals work better than poorly fed animals

iii) Rest-Animals given enough rest work better than those that are not

iv) Housing-Animals housed are protected from harsh condition e.g. cold thus work better

v) Disease control- Animals treated when sick, vaccinated, sprayed/ dipped against

external parasites/ dewormed against internal parasites are more efficient

vi) Age of animals- Young and very old animals give low output than averagely aged

LIVESTOCK HEALTH III **(LIVESTOCK DISEASES)**

1. -Mastitis
- Milk fever
2. Pre-disposing factors of foot rot.

- Overgrown / untrimmed hooves;
- Tick infestation between hooves;
- Muddy / filthy living / grazing areas;
- Presence of sharp objects e.g. stones;
3. (i) Causative agents of Brucellosis.

-Brucella abort – Cattle

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- Brucella suis - Pigs

- Brucella malitensis – sheep and goats;

($\frac{1}{2} \times 1$)

= $\frac{1}{2}$ mk)

(ii) Symptoms of contagious abortions.

- Retained after birth;
- Sterility in cows;
- Spontaneous abortion;
- Yellowish – brown, slimy discharge, odourless discharge from the vulva after abortion;

(iii) Methods of controlling contagious abortion.

- Use of artificial insemination (A.I).
- Vaccination against the disease in young animals;
- Avoid contact with the aborted fetus;
- Blood tests of all breeding animals before mating;
- Cull, slaughter infected animals;

4. A vaccine is an active disease pathogen reduced in strength /virulence or killed and is introduced into an animals body to induce immunity

5. (a) Discuss black quarter under control predators under the following sub-headings:-

(i) animal affected-cattle, sheep, goats

(ii) casual organism-clostridium chauvei

(iii) Symptoms of disease (5mks)

- Rise in body temperature
- Lameness and swelling of upper part of limbs, making animals lie on side
- Swollen shoulders on either side of body, chest or back, formation of gas under skin
- Difficulties in breathing
- Muscle appear black and spongy
- Grunting and grinding of teeth
- Failure to chew cud
- On exposure to air, muscle of body rapidly darken

(iv) control measures (3mks)

- Annual vaccination using black quarter vaccine
- Burning of carcass
- Carcass should not be skinned or opened
- Cleaning and treating of all wounds with antiseptics (any 3x1=3mks)

(b) (i) Explain four measures used to control liver flukes (4mks)

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- Controlling liver flukes
- Draining swampy areas
- Avoid grazing animals in swamps area
- Burning heavily infested pastures
- Apply chemicals (copper sulphate) to kill snails
- Remove and kill snails
- Fencing off swampy area (any 4x1=4mks)
 - (ii) Name an intermediate host of liver fluke
Water snail (limnea sp.)
 - (c) Explain the following terms as used in livestock production
 - (i) Embryo transfer
- Method of breeding which involve removal of ova from a superior animal, fertilize externally in a test tube then transfer to a foster mother which carry pregnancy to term
 - (ii) Artificial insemination
- Introduction of semen in to the females reproductive system by use of a tube
 - (iii) Line breeding
- Is mating of distantly related animals but within the same breed
 - (iv) Cross breeding
- Mating of animals belonging to different breeds e.g. fresian bull and jersey cow
 - (v) Up-grading
- Is the mating of high grade bull/sire to a low grade cow/dam (05x1=5mks)

6. - Overgrown hooves

- Presence of sharp objects/stores
- Muddy living/grazing areas
- Living infestation between hooves

7. (a) (i) Coccidia

(ii) Poultry, calves, young rabbits, kids, lambs

(iii) -Diarrhea

- Dysentery in the dung
- Euraciation
- Ruffled feathers
- Birds become dull with drooping wings
- sudden death in birds, rabbits and kids
- (iv) Drugs such as coccidiostats mixed with food or water
- Isolation of infected animals
- Avoid filthy, unhygienic animal surroundings

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- Avoid common dunking points for livestock from different farms
- Avoid overcrowding in poultry houses
- (b) (i) – Able to kill ticks
- Harmless to both human beings and livestock
- stable – remains effective even after contamination by dung, mud or hair
- (ii) – Spraying animals
- dipping – involves immersing the animal into the acaricide or wetting the animal by the acaricide
- Hand dressing – involves smearing pyegrene on areas not likely to be reached by the acaricide
- e.g. in the ears
- 8. four ways of controlling coccidiosis in the farm
 - Avoid overcrowding in poultry
 - Provision of coccidiostat in feeds and water
 - Use of portable calf pen
 - Practicing proper hygiene
- 9. four predisposing factors of scour in calves
 - Unhygienic condition in the house of the young ones
 - Feeding the calves on cold milk
 - Lack of colostrums and
 - Feeding at irregular intervals
- 10. four notifiable diseases of livestock
 - Rinderpest
 - Foot and mouth
 - Anthrax
 - Rift valley fever
 - New castle disease: mud cow disease
- 11. New castle, fowl pox, fowl typhoid, gumboro, marek – injections, pursaa disease
- 12.
 - Signs of anthrax in carcasses
 - Blood does not clot
 - No rigormatics after death
 - Stomach swells/ bloat
 - Darkened blood oozes out through the natural opening
- 13. Differentiate
 - Active immunity – animal producing antibodies
 - Acquired/ passive immunity – external source of immunity

14.

- Fuel system
- Electrical system
- Cooling
- Lubrication
- Transmission
- Ignition
- Hydraulic

15. 2 adjustment on mould board plough

- Adjust the plough depth
- Front furrow depth
- Lowering/ raising ploughing pitch
- Front furrow width

16. a) Disc plough reject disc alone

- b)
- X – Disc scrapper
 - Y- Rear wheel/ furrow wheel
 - Z - Disc

- c)
- Replace broken discs
 - Clean plough after use reject wash plough
 - Lubricate hubs and furrow wheel bearing/ moving parts reject movable parts
 - Lighten loose nuts and bolts
 - Store in a cool dry place
 - Apply old engine oil to prevent rusting during long storage/ paint implement

17. Causes of ruminal tympany (Bloat)

- Obstruction of esophagus due to bulky food e.g. potatoes
 - Abnormal pressure exerted on esophagus by swelling in wall of chest
- Indigestion due to eating poisonous herbs, soft young green foliage

18. a) Symptoms of milk fever

- Muscular twitching
- Staggering
- Animal lies down on its side most of the time
- Animals lies on sternum with rock twisted on one side
- General paralysis

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- Breathing becomes slow and weak 4x ½
=2 marks

b) Control measures of milk fever

- Feed animal o diet rich in calcium
- Give intramuscular injection of calcium 2-3 days before cavity
- Partial milking
- Cull susceptible animal

19. East coast fever (ECF)

- i) cattle 1x1=1
mark

ii) Theirelia parva Reject if not underlined, spellings are wrong

iii)

- Fever/ high temperature
 - Salivation
 - Lachrimentim/ tears from eyes
 - Difficult in breathing
 - Haemorrhages in vulva/ mouth
 - Coughing
 - Sight impairment
- 6x1=6
- marks

iv)

- Regular spraying/ dipping/ hand dressing with acaricide
 - Fencing/ rotational grazing
 - Treatment using appropriate drugs
- 2x1=2
- marks

-

b) Ten measures used to control livestock diseases

i) Proper breeding and selection

- Animals fed on balanced ration adequate in quantity and quality are strong and able to resist diseases

ii) Proper breeding and selection

- Animals selected that are free from diseases or resistant to diseases will prevent transmission of diseases

iii) Proper housing and hygiene's

- Animals houses should be built to meet construction requirement e.g. ventilation, space, drainage, leak proof, lighting and will prevent and protect animals from contracting diseases

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iv) Isolation of sick animals

- Is separation and confinement of animals from health ones while undergoing treatment, this prevents spread of the diseases

v) Imposition of quarantine

- Is restriction of movement of animals and their products from and into affected areas, thus prevents spreads of diseases

vi) Prophylactic measure and treatment

- Involves use of drugs before disease attack to prevent occurrence e.g. use of coccidiostat to control coccidiosis, drenching to prevent anti helminthes

- Also involve vaccination, spraying with appropriate acaricide and treatment to restore good health

vii) Slaughtering affected animals

- Highly infectious and contagious diseases e.g. rinderpest, new cattle foot and mouth animals should be slaughtered and carcasses will dispose to prevent spread

viii) Use of antiseptics and disinfectants

- Applied on skin to kill germs or clean livestock to maintain hygiene

- Vaginitis.
- Brucellosis.
- Trichomoniasis.

20.

- Vaginitis.
- Brucellosis.
- Trichomoniasis.
- Leptospirosis. ($2 \times \frac{1}{2} = 1 \text{ mk}$)

21. (a) injection (b) oral

LIVESTOCK PRODUCTION V **(POULTRY)**

1. One bird occupies 0.27m^2

Area available $9\text{m} \times 3\text{m} = 27\text{m}^2$

1 bird 0.27m^2

? 27m^2

$= \frac{27\text{m}^2}{0.27\text{m}^2} = 100 \text{ birds}$ $1 \times 2 = 2 \text{ mis}$

0.27m^2

2. Functions of isthmus.

- Shell membranes formed/ determines shape of egg;

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- Water mineral salts and vitamins added;

3. Features of a laying nest.

- Dimly lit; dark;
 - Spacious / large enough to accommodate bird comfortably;
 - Dry clean beddings;
 - Have lockable doors;
 - Kept in secluded parts of the house;
 - Have slanting roofs to prevent birds from perching on;
- Have an appropriate height from floor;

4. Qualities of marketable eggs.

- Smooth texture;
 - Right shape, colour, size and weight;
 - Right shell hardness;
- Clean and fresh;

5. (a) A - Too high temperature

B- draught C chilliness from left side

C – Inadequate heat supply /low temperatures ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mk)

(b) Explain why the brooder guard is rounded as shown in the diagram

To avoid overcrowding in the corners which can lead the suffocation and death

6. - Fertilized

- Medium size (55-60gm)
- oval shaped
- free from abnormalities e.g. double yolk/ blood spots
- Be freshly collected (not more than 1 week)
- Have smooth shells
- Be free from cracks in the shells
- Be clean

7. Adequate space

- Litter to be kept dry and free from dust
- Turn the litter frequently
- Perches and rosters should be adequate and well spaced

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Adequate waterers
- Well distributed waterers
- Provide clean and adequate water
- equipment to be kept clean
- Replenish soft litter in the nest
- ensure the nest is dark
- Collect eggs regularly
- Feed the birds well
- Supply gut to assist in digestion
- Keep the birds busy by hanging greens
- Curl the birds and pool layers
- De-beak birds to prevent cannibalism
- Vaccinate birds regularly
- Check birds for disease symptoms
- control parasites
- Avoid stress factors
- Discourage broodiness
- Maintain and repair the houses
- Provide enough feed troughs
- Keep proper records
- Dispose off dead birds

8. four reasons for egg breaking and drinking by layers in a deep litter rearing System
Bright light in the laying boxes/over crowding/few laying boxes
- Poor feeding without mineral rich feeds
 - Undebeaked birds
 - Irregular egg collection
9. a) a-infertile (clear)
b-fertile egg
c-damage yolk
b) Identify the egg which suitable for incubation and give a reasons for your answer
Egg b-it is fertile and will develop into a chick
c) Name the practice which used to determine the state of eggs above
Candling
10. a) Brooding of chicks (1x1=1mk)
b) four preparations that should be carried out structure U before arrival of day old chicks

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cleaning the house and brooder with disinfectant

- dusting with chemical to kill ectoparasites
- placing the polythene paper on fresh and clean saw dust
- fixing and setting of sources of heat and light
- Fixing the feed and water troughs and putting fresh feed and water.

c) List down one behaviouristic activity which would indicate that the chicks are under stress
Moving away or close to the heat source

- Fighting
- Cannibalism

11. - Litter gives comfort and warmth to the birds

- Helps in drying dropping
- keeps birds bust

12. Reasons for castrating animals when young

- Less pain
- Quick healing
- Little loss of blood

13. a) Reasons for dehorning farm animals

- Reduce space occupied by animal
 - Making handling easier
 - To reduce destruction of farm structures
 - To make them
 - To reduce risk, injury to farmer and other animals
- = 1 mark

2x ½

b) Methods of dehorning livestock

- Use of caustic potash stick (Potassium hydroxide)
- Use of dehorning iron
- Use of dehorning saw or wire
- Use of rubber ring and elastrator
- Use of dehorning collation

14. four abnormalities of eggs that can be detected during egg candling. (2mk)

- Absence of yolk.
- Double / triple yolk.
- Air space in wrong position.
- Excessively large air space.
- Cracks on egg shell.
- Blood / meat spots.

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- Deformed / broken yolk.
15. The management of day old chicks in a deep litter system from preparation of brooder up to eight (8) weeks old (20mks)
- ensure brooder is working well 2-3 days before arrival of chicks
 - provide brooder with litter for warmth and moisture absorption
 - provide heat source
 - put wire gauze around the heat source
 - make holes on the brooder to provide fresh air
 - provide dim light to prevent cannibalism
 - cover litter with polythene sheet or newspaper and place feeds on them to discourage chicks from eating litter
 - check the temperature at above 15cm above the floor to ensure that the temperature is appropriate
 - from 4th -6th week withdraw the heat source gradually
 - feed chicks on chicks mash
 - provide plenty of clean water
 - vaccinate against Newcastle disease after 2-3 weeks
 - keep proper records
 - dust birds with insecticide to control external parasites
 - at 6 weeks introduce growers mash
 - isolate sick birds from healthy ewes
 - remove and treat sick birds
 - clean and disinfect the house
 - provide greens
 - remove dead chicks from the house
 - provide foot bath with disinfectants
16. (a)- Piglets are weaned at the age of 8 weeks and fed on sow and weaner meal.
- Deworming should be done to control internal parasites.
 - Vaccinate to control diseases e.g African swine Fever.
 - Spray to control external parasites
 - Identification is done by ear notching
 - Take it to pork when it shows signs of being heat
 - Trim over grown hooves.
 - Gestation period is 4 months
 - Prepare farrowing pen by disinfecting the walls
 - Sow brought to farrowing pen 3 days to.
- (b) -Age

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- Mothering ability be good
- Physical fitness
- Health of gut
- Body confirmation
- Temperament of behaviour
- Adaptability
- prolificacy

17. (i) M- Chalaza L- Space
 N- Yolk O- Inner shell membrane P- Shell
- (ii) M- Hold the yolk in position at centre of the egg.
 L- Air trapped in this space used by developing embryo
- (iii) – To avoid germinal disc sticking on the egg shell
18. - Move away from the source of heat to the periphery
19. -saw dust
 Wood shavings
 Crushed maize cobs
 Coffee husks
 Rice husks
- 20.
- Be fertilized
 - Should be medium in size i.e 50-65 gm in weight
 - Have smooth shells
 - Be oval in shape
 - Be free of any cracks in shells
 - Be clean to ensure that pores are clean
 - Not have any fresh i.e collected within one week
 - Should be fresh i.e collected within one week

LIVESTOCK PRODUCTION III **(LIVESTOCK REARING PRACTICES)**

1. A group of calves kept according to age
2. (a) - Assemble all milking equipments such as buckets, milking can and towels.

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- Put animals in milking shed and restrain appropriately.
- Wash udder and teat using warm water mixed with an appropriate sanitizing agent.
- Dry the udder using a towel.
- Use trip cup to test the first few drops of milk for mastitis.
- Carry out milking by squeezing out the milk / teats.
- Strip the udder dry.
- Dip the teats in ant-mastitis solution after milking.
- Apply milking jelly (milk salve) on the teats.
- Release the cow.
- Weigh and record the milk.
- Strain the milk into the milking can to cover immediately.
- Cool the milk rapidly to a temperature of 4°C .

(b) - Keep cow healthy/ free from diseases.

- Wash cow flanks, udder and region around the udder using clean water then dry using clean towels.

- Milking shed should be clean, wash after every milking and disinfect.
 - Clean and sterilize milking utensils.
 - Keep milk in a dust free environment.
 - Deliver milk to collecting centres.
 - Don't feed cows on feeds which may taint milk a few hours to milking e.g. Mexican marigold, silage, garlic e.t.c.
 - Do not expose milk to direct sun.
 - Milk should be carried in aluminium container.
- Copper and iron containers may cause oxidation of milk fats.

3. Reasons for washing a cow's udder with warm water

- To remove dirt
- To stimulate milk let down ($\frac{1}{2}$ mk each = 1mk)

4. two roles of uterus in egg formation process

- Additional of calcium which harden egg shell
- Additional of egg pigmentation ($2 \times \frac{1}{2} = 1\text{mk}$)

5. (a) M – Alveolus N – Gland cistem O – Teat ($\frac{1}{2} \times 3 = 1\frac{1}{2}\text{mks}$)

(ii) Oxytocin – Controls the muscle fibres surrounding alveoli to allow milk secretion

adrenalin – A hormone that relaxes the udder muscles to allow milk let down

($1 \times 2 = 2\text{mks}$)

(b) Free from disease causing organisms

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- Has no hair/dirt dust
- Its of high keeping / lasting quality
- Chemical composition is within the expected standards ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)

6. four characteristics of clean milk

- Has normal taste
- Free from physical materials
- Free from pathogens
- Free from foul smell
- It is of high keeping quality
- Is chemical composition is within the expected standards

7. three maintenance practices carried out on a milking machine

- Flushing the tubes under high pressure to deblock it
- Greasing/orling the rotating parts in the pump
- Storing it to dry upside down after through washing

8. four reasons for feeding Colostrums to calves immediately after calving

- Easily digested
- Has high nutritive value
- Contains antibodies which protect the calf from diseases
- Has laxative effect

9.

- Presence of milk man/ milky parlour
- Washing/ massaging udder
- Feeding
- Sounds associated with milking
- Maintain regular milking time

10. a) Differences operational

Disc plough	Mould board plough
Can be used in fields with obstacles	Cannot be used in fields with obstacles
Ploughs/ cuts at varying depths	Ploughs/ cuts at constant depths or confirm depth
Requires less skills to operate	Requires more skills to operate
Works well in sticky soils	Does not work well in sticky soils
Rotates and not easily broken since rolls over	Easily broken by obstacles

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obstacles	
Requires more harrowing	Requires fewer harrowing
Poor furrow slice inversion	Proper furrow slice inversion
Does not require constant replacement of parts	More power to pull

b)

- Poor communication network/ poor infrastructures
- Lack of cooling/ handling facilities/ processing facilities
- Competition with non- dairy products/ cheap imported dairy products
- Prevalence of Zoonotic diseases
- Inefficient/ poor management of marketing society/ dairy boards
- Late/ non- payment by marketing agents/ exploitation by marketing agents/ middle men
- Lack of capital to finance marketing activities
- Price fluctuation due to changes in supply
- Lack of market information

c) Reasons for culling livestock

- Old age
- Poor health
- Low libido/ infertile
- Physical deformities
- Hereditary defects
- To avoid inbreeding

11. three advantage of artificial method of calf rearing

- Accurate records of milk yield may be kept
- It is easy to regulate the amount of milk taken by the calf
- Cows produce milk even in the absence of the calf
- It is easy to maintain high standard of cleanliness/sanitations
- The farmer is likely to sell more milk hence maximizing profit ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)

12. three methods that may be used to improve milk production in a breed of indigenous goats

- Proper selection/culling
- Proper breeding/upgrading/ cross breeding
- Maintaining good health
- Proper feedings
- Proper milking methods
- Proper housing ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

13. a) Physical characteristics between good layer and poor layer

Part/feature	Good	Poor
Comb/wattle	Large warm,wavy	Small, shrunken, dry, scaly pace, cold
Eyes	Bright, orange, alert race	Dark, pace, yellow
Beak	Oval, moist, reddish, active	Yellowish
Vent	Soft, pliable, wide	Round, dry, less active
abdomen	Soft, pliable,wide	Hard, full
Space between keel and petric bone	Wide fits 3-4 fingers	Small fits 1-2 fingers
Temperate	Alert- active	Dull, less active
Moulting	Start late	Start early
Plumage	Dry,rugged, rough	Preened, glossy, smooth
Shanks	Pace	Yellowish
broodiness	rare	common

- b) i) - Young animals produce with high butter fat content than older animals
 ii) Pregnant, emaculated animals have lower butter fat content than normal animals
 iii) early and late stage of lactation has lower butter fat content while middle phase has higher butter fat content
 iv) last drawn milk from udder has more butter fat
 v) - Different breeds of animals produce milk with different % composition e.g. Jersey produces milk with high butter fat content than fresian
 vi) Season of the year
 - Fat % increases during cold season of year but decreases during dry season
 vii) Animals fed roughages produce milk with high fats, protein and lactase than those fed on grains
 viii) mastitis reduces lactose composition in milk
 ix) Certain drugs are known to lower milk composition if animal is under treatment

14. a) - Remove dirt.
 - Stimulate milk letdown ($2 \times \frac{1}{2} = 1 \text{mk}$)
 b) i) Milk letdown – oxytocin
 ii) Lacto genesis – Prolactin ($2 \times \frac{1}{2} = 1 \text{mk}$)

15. four methods of increasing the depth of penetration of a disc harrow
- Exert more hydraulic force.

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- Use fewer discs.
 - Increase space between discs.
 - Add weights.
- Increase cutting angle of discs

16.

- Boom sprayer
 - Spray race
 - Rotavator
 - Maize Sheller
- Mowers

17.

- Chisel plough
- Sub soiler

18.

- Combine harvester
 - Forage harvester
 - Potato lifter
- Mowers

19. a)

- Oil bath air cleaner – check for oil level and add more if low
- Check for cleanliness in oil bath and wash bowl – replace oil if dirty
- Battery – check for electrolyte level and top up if low
- Fuel – check and fill if low
- Radiation- check for water level and top up with clean water if low
- Check for trash in tins and remove if any
- Fan belt- checks for tension and tighten if loose
- Engine oils – check oil level using a dip stick and add more if level is low
- Tyres- check for tyre pressure and add if low
- Bolts, nuts and pins- check for tightness and tighten if loose
- Grease all the moving parts
- Check for any physical abnormalities and rectify accordingly
- Check sediment bowl and drain if dirty

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Use of right type of oil
15x1=15 mks
- b) i) Flywheel – maintain the rotational motion of the crankshaft
ii) Ignition coil – steps up the voltage from the battery
iii) thermostat – controls engine temperature
iv) Injector – Atomises the fuel into very fine spray/ injects fuel into cylinder
v) Piston – compresses air/ fuel mixture in the cylinder/ expels exhaust gases/ transmits power

FARM POWER AND MACHINERY

1. Farm operations powered by Engines.
 - Ploughing and harrowing / land preparations;
 - Transporting farm produce;
 - Spraying of herbicides/ pesticides.
 - Mowing the grass;
 - Lighting of homes;
 - Pumping water for irrigation.
 - Harvesting farm produce;
 - Machine milking.
2. (a) Maintenance practices required on a tractor before setting out to work.
 - Check the engine oil daily using dip stick.
 - Check fuel and add if necessary.
 - Nuts and bolts are tightened whenever they loosen.
 - Water level in the radiator be checked and added if necessary.
 - Battery electrolyte be checked daily and if below level, top up be done using distilled water.
 - Greasing be done on the bearings.
 - Tyre pressure be checked and if low, should be added.
 - Fan belt tension be checked.
 - Break shaft bearing should be greased.
 - Ensure break fluid and clutch fluid levels are maintained.
 - Sediments from the sediment bowls should be removed.
 - Check battery terminals and grease.
 - Oil cleaner be cleaned.

(b) Factors that influence power output by a draught animal.

 - Age – Mature animals produce more power output than young ones.
 - Breed & Type – Indigenous animals are more hardy than exotic.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Training Level – better trained animals have better work output.
- Body Weight – A draught animal can pull 10 – 20% of its body not for 6 – 8 hours.
(The bigger the animal, the more output).
- Harnessing of the animal – well harnesses animal is more efficient at work than poorly harnessed work.
- Condition of working equipment on well maintained equipment have higher work output with the draught animal than poorly maintained ones.
- Environmental/ Ambient temperature – Cool temperatures lead to higher work output with a draught animal than high temperature.
- Health Status – A healthy draught animal has higher workout put than a sick animal.

3. Uses of a gearbox:-

- Stops the tractor without switching off engine
- Provides different forward speeds
- Enables reversing ($\frac{1}{2} \times 2\text{pts} = 1\text{mk}$)

4. Two uses for which wind power is harnessed

- To pump water
- To generate electricity
- For processing /winnowing of grains e.g. millet, rice e.t.c

5. Name three implements that are connected to the power take-off shaft

- Sprayers
- Rotarators
- Reciprocating (1mk each = 3mks)

6. (a) Impelement identity – Disc plough

(b) L – Furrow wheel M - Beam

(c) - Adds weight

- Forms attachment of all the other parts ($\frac{1}{2} \times 1\text{pt} = \frac{1}{2} \text{mk}$)

(d) Where there are hidden obstacles e.g. stumps , rocks e.t.c

- Heavy soils

7. (a) Factors that a farmer should consider in ensuring fast and efficient cultivation by oxen

- A well trained personnel
- Using well trained animals

FREE AGRICULTURE K.C.S.E 2012 to 2015

- use of efficient implement
- Avoiding overworking the animals (allow them to rest
- Good working environment should be created for the animals
- Animals should be handled well
- Feeding the animals properly
- Using of males animals to do the work
- Using healthy animals only to do the work (1mk x 8pts = 8mks)

(b) The importance of lubrication system in a tractor

- Prevents rusting i.e when oiling is done
- It acts as a cleaning agent as it washes off the dirt, dust
- Reduces the heat/cooling
- Increases efficiency of the machine and reduces the rate of wear and tear on moving parts

(c) The daily maintenance and servicing of a tractor

- Checking of maintenance and servicing of a tractor
- Checking of engine oil by use of dip stick and add if low
- The level of electrolyte in battery should be checked daily and add if low
- Inspect the H₂O level in radiator and add if low
- Loose nuts and bolts should be tightened
- Tyre pressure should be checked and added if low
- Level of fuel should be checked and added if low
- Removal of large sediments from the sediment bowl
- Greasing should be done by use of grease on nipples of bearings
- Fan belt tension should be checked to ensure that it deflects between 1.9–2.5cm when pushed
- Break shaft should be greased
- Maintain break fluid level (1mk each for any 9pts= 9mks)

8. (i) Hydraulic system-raise and lower mounted implements like plough
(ii) Draw bar-attachment of trail implement
(iii) Propeller shaft-connect gearbox to differential which has axle to drive wheel making tractors to move backwards or forward. (@ 1mkx3=3mks)
9. (i) Hiring of tractors and implements by farmers who do not have them (correct definition=
(ii) three sources of tractor hire service (1½mks)

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- Government tractors hire service
 - Private contractors
 - Individual farmers
 - Cooperative societies (any 3x ½ = 1 ½ mks)
10. (a) A-ring pinion gear
B-bevel side gear
C-wheel axle
D-drive pinion gear
- (b) State two functions of differential system of a tractor (2mks)
- Change direction of drive to right angle for power to be transmitted to rear wheel
 - Enable rear wheel to travel faster/slower than other when negotiation corner (1x2=2mks)
- (c) Give two reasons why wheel skidding of a tractor is not allowed (1mk)
- To make tyres last longer
 - To make it easy to control the tractor
11. Wind, water, human, animal, biogas, wood fuel, charcoal, kerosene, fossil fuel, petroleum, ethane (natural gas), hydroelectric power, nuclear, Geothermal, storage battery
12. Induction, compression, power exhaust
13. - Keeping them healthy
- Proper feeding
 - Proper handling e.g not over working /not beating them
 - Proper training
 - Not overloading them
14. Individual owners
- Government tractor hire service
 - Co-operative societies
 - Companies
15. (i) Primary cultivation
- A- To link bracket
 - B _scraper
 - C- Standard/disc hanger
 - D – Rear finow wheel / thrust wheel 2mks
- (ii) Adding weight on beam
- exerting more hydraulic force

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- Sharpen the disc blade
- Increased space between the disc
- Loosen the area of disc contact with the soil
- Increase the cutting angle

16. four ways through which a farmer would ensure maximum power output from Ploughing animals

- Feeding and watering animals well before working
- Training animals on draft techniques
- Allowing animals to rest well after a day work/avoid overworking the animals
- Keeping the animals in good health while working

17. three advantages of a disc plough over mould board plough

- Tears and wear is less
- Needs less power to pull
- It can ride over obstacle

18. a) the differences between petrol and diesel engine

Diesel	Petrol
i)use diesel as fuel	Use petrol as fuel
ii)has injector pump	Has carburetor
iii)has no spark plug	Has spark plug for ignition
iv)fuel is ignites by compression	Fuel ignites by spark plug
v)air and fuel first meet in cylinder before ignition	Air and fuel meet in carburetor before ignition
vi)specific fuel consumption is low	Specific fuel consumption is high
vii)higher air compression ratio	Lower air compression ratio
viii)air/fuel ratio is not constant	Air/fuel ratio is constant
ix)has sediment bowls	No sediments bowls
x)operation cost is lower	Operation cost is high
xi)it is heavy in weight and suited to heavy machines	Operation cost is high light in weight and suited to light machines
xii)produces a lot of smoke	Produce minimal smoke

b) Describe components of transmission system of a tractor

i) Clutch- It disconnects the engine from the rest of the transmission system. It is mounted on the flywheel and made up of pressure plates and clutch plate in the middle. The clutch allows the driver to temporarily interrupt the power flow from the engine to the gear box and shift from one gear to the other

FREE AGRICULTURE K.C.S.E 2012 to 2015

ii) Gear- These are toothed wheels. They provide towards speed or reverse. The set of gears are

housed in the gear box.

iii) Differential- it is located between the wheel axial. it enables one wheel to move faster than the

other while negotiating a corner

iv) Driving axial- The final drive is brought about by driving axial which gets the power from the

differential.

When the axial rotates they rotate the wheels making the tractor to move either engorged gear

v. Wheels- Comprises of the tyres, tubes rims nuts and bolts. They must be inflated to the

movement of the tractor

19.

- Fuel system
- Electrical system
- Cooling
- Lubrication
- Transmission
- Ignition
- Hydraulic

20. Functions of clutch

- Connects or disconnects the drive shaft to or from the engine
- Facilitates smooth and gradual take off
- Provides power from the engine to the P.T.O (Power Take Off)

21. 2 adjustment on mould board plough

- Adjust the plough depth
- Front furrow depth
- Lowering/ raising ploughing pitch
- Front furrow width

22. a) Disc plough reject disc alone

b) X – Disc scrapper
Z – Disc

Y- Rear wheel/ furrow wheel

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- c)
- Replace broken discs
 - Clean plough after use reject wash plough
 - Lubricate hubs and furrow wheel bearing/ moving parts reject movable parts
 - Lighten loose nuts and bolts
 - Store in a cool dry place
 - Apply old engine oil to prevent rusting during long storage/ paint implement

23. a) Differences operational

Disc plough	Mould board plough
Can be used in fields with obstacles	Cannot be used in fields with obstacles
Ploughs/ cuts at varying depths	Ploughs/ cuts at constant depths or confirm depth
Requires less skills to operate	Requires more skills to operate
Works well in sticky soils	Does not work well in sticky soils
Rotates and not easily broken since rolls over obstacles	Easily broken by obstacles
Requires more harrowing	Requires fewer harrowing
Poor furrow slice inversion	Proper furrow slice inversion
Does not require constant replacement of parts	More power to pull

b)

- Poor communication network/ poor infrastructures
- Lack of cooling/ handling facilities/ processing facilities
- Competition with non- dairy products/ cheap imported dairy products
- Prevalence of Zoonotic diseases
- Inefficient/ poor management of marketing society/ dairy boards
- Late/ non- payment by marketing agents/ exploitation by marketing agents/ middle men
- Lack of capital to finance marketing activities
- Price fluctuation due to changes in supply
- Lack of market information

c) Reasons for culling livestock

- Old age
- Poor health
- Low libido/ infertile
- Physical deformities
- Hereditary defects
- To avoid inbreeding

24. a) Share-makes a horizontal cutting on the furrow slice

b) Mould board-completes the turning of the furrow slice

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c) Land side-stabilizes the plough by absorbing the side pressure ($\frac{1}{2} \times 3 = 1 \frac{1}{2}$ mks)

25. a) five maintenance practice of a mould board plough

- Lubricate the moving parts
- Sharpen blunt share
- Tighten bolts and nuts
- Clean the plough after use
- Coat the unpainted parts with old engine oil before any storage
- Replace worn out parts (1x5=5mks)

b) five structural and functional differences between the petrol and diesel engines

Petrol engine	Diesel engine
i)has a carburetor ii)fuel and air mixed in the carburetor iii)fuel ignited by an electric spark iv)produces little smoke v)is light in weight	i)Has an injector pump ii)Fuel and air mixed within the cylinder iii)fuel ignited by compression of air and fuel mixture in the cylinder iv)produces a lot of smoke v)relatively heavy

c) List five uses of farm fences

- keep of wild life ,predation and intruders
- demarcates boundaries
- separate crop field from pasture land
- divide pasture land into paddock
- control movement of animals and people within the farm and prevent formation of unnecessary pests
- control disease and parasites helps in isolate sick animals
- helps in controlling breeding
- provide security
- act as wind break (1x5=5mks)

26. Uses of ox-dram tine harrows

- Leveling of seed bed
- Breaking large soils clod
- Mixing up soil with organic matter
- Destroy weeds
- Cover seeds
- Collecting trash

27. Care and maintenance of tractor battery

- Maintain correct level of electrolyse by topping up with distilled water
- Scrap corroded terminals and smear with grease

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Fix battery tightly in box to avoid spillage and damage
- Charge regularly and periodically
- Under storage empty battery and keep it upside down
- Generator belt should always be functioned to charge

28. a) mouldboard

b) A – Shaire B- Mouldboard C- Disc coulter

c) E- Stabilize the plough

- absorb side thrust by pressing against furrow wall

F- Scrapes of mold from disk coulter

d) Care and maintenance

- Lubrication of moving parts
- Loose nuts and bolts should be tightened
- Clean after use/ remove trash and wet soil
- For long storage paint with old engine to prevent
- Replace/ repair worn out parts

29. Six uses of live fences

- Thorn species prevent wild animals and other invaders into the farm
- Tall varieties act as wind breakers
- Add aesthetic value to the homestead
- Roots hold soil firmly controlling soil erosion
- Species such as lantana canara can be used to feed livestock
- Provide shade to livestock and man
- Trimmed branches can be used as organic manure, wood fuel
- Some species have medicinal value

30. two possible causes of over heating in a tractor engine

- Slack fan belt
- Low oil level
- Low water level in radiator
- Deformed / broken yolk.

31. two events occur during induction stroke in a four stroke engine. (1mk)

- Piston moves down from TDC
- Exhaust valve is closed
- Inlet valve is open
- Air / fuel mixture get into combustion chamber
- Piston reaches BDC. ($4 \times \frac{1}{2} = 2\text{mks}$)

32. i) a) Spike tooth harrow ($1 \times 1 = 1\text{ m k}$)

b) two uses of the implement above.(2mks)

- Level seed bed
- Break soil clods
- Stir soil

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Destroy weeds
 - Incorporate fertilizer in the soil
 - Removing trash from the field. (2x1 = 2)
 - c) three maintenance practices carried out on the above implement.(3mk)
 - Replace worn out parts
 - Clean after work
 - Tighten loose bolts and nuts
 - Oil unpainted parts for storage. (3x1 = 3mks)
 - ii) a) Bucket pump /stir - up pump (1x1 = 1mk)
 - b) Spraying acaricide on livestock (1x1 =1mk)
 - c) W – Nozzle X – Trigger Y - Pail /bucket
 - d) For holding acaricide solution during spraying. (1x1 = 1)
33. a) the factors that influence the power output of farm animals (8mks)
- Training
 - Level of nutrition
 - Harnessing animals properly
 - Body weight
 - Age-mature ones produce more power than young
 - Handling of animals
- b) State the importance of farm fences (12mks)
- demarcates farm land from that of neighbours
 - keeps wild animals and other intruders from entering the farm
 - separates crop field from pastures facilitating mixed cropping
 - used to divide pastures into paddocks facilitating controlled grazing
 - controls movement of animals and people preventing formation of unnecessary paths in the farm
 - helps control spread of diseases and parasites in the farm by keeping wild animals away
 - helps isolate sick animals from the rest of the herd preventing the spread of diseases
 - enables farmer to control breeding rearing different animals in different paddocks
 - provide security to the homestead
 - they have aesthetic value
 - live fences act as animal feeds
 - live fences act as wind breakers (1x12=12mks)
34. (a) Methane
- (b) slurry
- (c) Fresh mixture of waste material
- (d) Initial capital investment is high hence very expensive
- Requires management skills that may not be available & it available expensive

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Requires large number of farm animals to produce animal waste.

- 35 . – Solar panels
- Petrol and diesel generators
- wind mills
- steam production from boilers using organic or inorganic
36. (a) Check engine oil, fuel, water level, electronic in the battery
-Tighten nuts and bolts
-Apply grease
-Remove large sediments from sediment bowl
-Check the tyre pressure and inflated or deflated appropriately
-Fan belt tension should be checked to ensure it defients between 1.9cm to 2.5cm when punched
-Grease the brake shaft and maintain brake fluid level
- (b) -Health of animal
-Level of feeding
-Animal slpeciesa
-Care and handling

FREE AGRICULTURE K.C.S.E 2012 to 2015

K.C.S.E PAST PAPERS

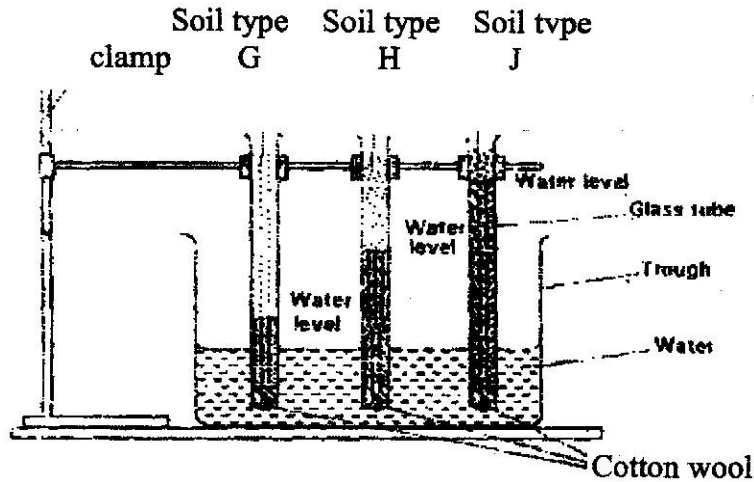
INTRODUCTION TO AGRICULTURE

1. What is plantation farming system?
2. State the main characteristics of shifting cultivation.
3. Differentiate between olericulture and pomoculture
4. State three characteristics of shifting cultivation.
5. State six reasons why agriculture is important in Kenya's economy
6. State two ways in which agriculture contributes directly to the development of industries.
7. Give the limitations of pastoral Farming.
8. State any two disadvantages of pastoral nomadism system of farming.

FACTORS INFLUENCING AGRICULTURE

K.C.S.E PAST PAPERS

1. List four environmental factors that affect crop distribution in Kenya.
2. State two ways by which wind affects the growth of crops.
3.
 - a) State two ways by which wind affects the growth of crops.
 - b) Give two roles of micro-organisms in the soil that are beneficial to crops
 - c) State three properties of soil that are influenced by its texture.
4. List down the four aspects of rainfall that affect agriculture.
5. Describe the environmental conditions that may lead to low crop yields.
6. Outline three effects of soil organisms which benefit crop growth.
7. State two causes of hard pan in a crop yield.
8.
 - a) Name two processes of rock weathering.
 - b) Differentiate between soil texture and soil structure.
 - c) State three benefits of good soil structure in crop production.
9.
 - a) List two aspects of light that influence crop growth.
 - b) The diagram below shows an experiment set up using soil types & H and J and observations made after 24 hours. Study the diagram and answer the questions that follow.



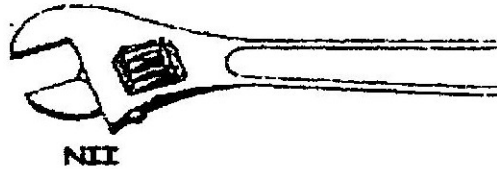
- i) What is the experiment represented above designed to study?
 - ii) Name the three soil types & H and J.
 - iii) What is the characteristic texture of soil types G and J?
 - iv) State how a farmer would improve the structure of soil type G.
10.
 - a) Give 3 reasons why soil is important to crops.
 - b) State two benefits of optimum soil temperature in crop production.
 - c) Give 3 factors of soil that influence soil productivity.
 11. Give three reasons why soil is important to crops.
 12.
 - a) Give four reasons why a well drained soil is suitable for crop production.
 - b) State two benefits of optimum soil temperature in crop production.
 - c) Give three soil factors that influence soil productivity.
 13. State three advantages of adding organic matter to sandy soil.
 14.
 - a) State two roles of good soil aeration in crop growth.
 - b) Give two roles of micro-organisms in the soil that are beneficial to crops.
 - c) State three properties of soil that are influenced by its texture.

FARM TOOLS AND MACHINERY

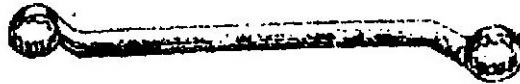
1. Give one use of each of the following hammers;
i) Ball pen hammer ii) Sledge hammer
State two maintenance practices that should be carried out on a wheelbarrow.
2. The diagram labeled NI, NII, NIII below represent some tools used in farms.



NI-----



NII



NIII

- i) What functional advantage does the tool labeled NII have over the tool labeled NIII?
 - ii) What is the function of tool labeled NI iii) Give the maintenance practice of NI
3. Study the photographs below carefully and answer the questions that follows:



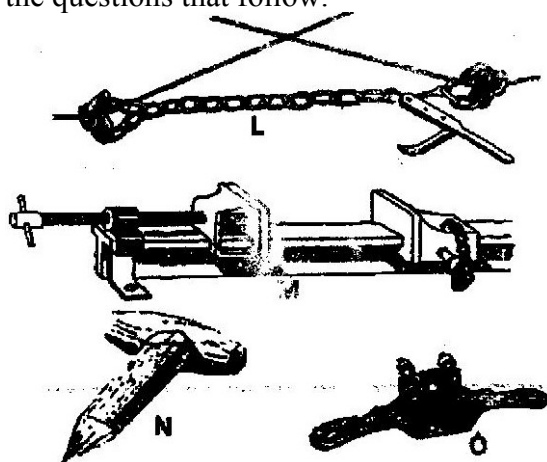
- a) i) Identify 1-----
2 -----
ii) Give the land preparation stage when 2 is used
- b) i) Which of the tools is suited for a field with roots, trash and

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other obstacles?

ii) Give a reason for your answer in b(i)

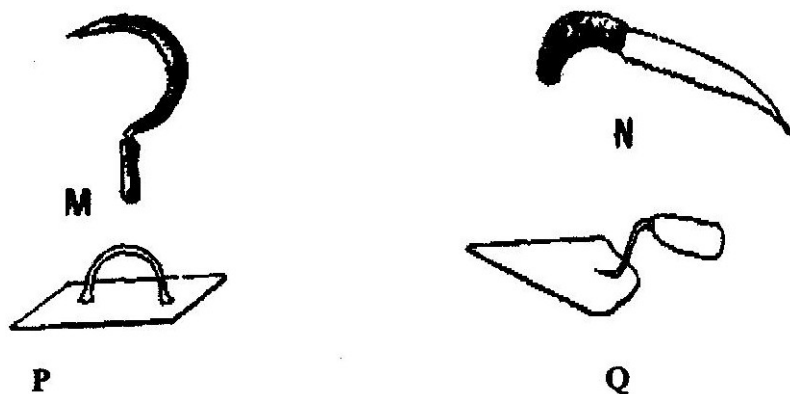
4. State one use of each of the following tools;
 - (i) Spoke shave
 - (ii) Plumb bob
5. Name two tools used for cutting galvanized iron pipes.
6. State four maintenance requirements of a jack plane.
7. List four tools used for laying concrete blocks when constructing a wall.
8.
 - a) What is the difference between a tenon saw and a crosscut saw?
 - b) What safety measure should be taken when using a crosscut saw?
 - c) Give three maintenance practices that should be carried out on crosscut saws.
9. Give the factor, that are considered when selecting a garden tool for primary cultivation.
- 10: (a) State one use of each of the following tools, (i) sickle
ii) Secateurs.
 - b) Give two reasons for proper maintenance of farm implements.
 - c) State three factors that should be considered when selecting gardener tools for primary cultivation.
12. L, M, N and O are diagrams of farm tools. Study them and answer the questions that follow.



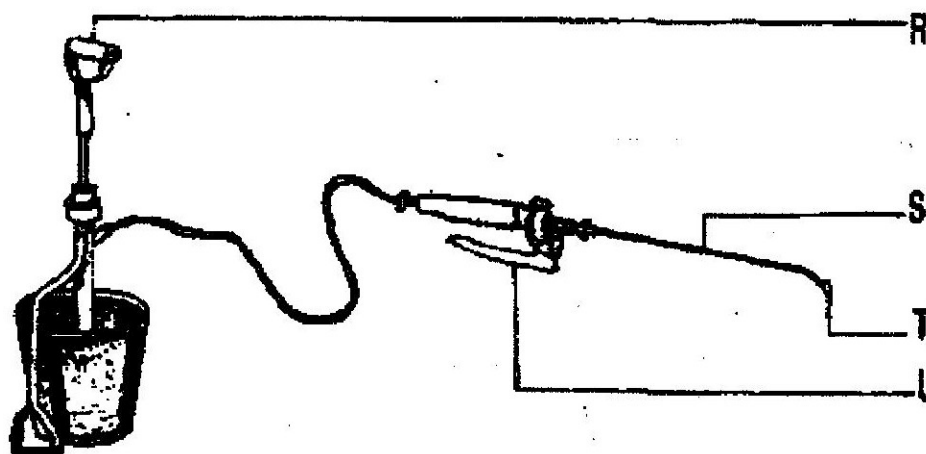
- i) Identify the farm tools; L, AA>N and O. (ii) State the use of each tool L, AA, N and O.
13. List the tool used for each of the following
 - i) Tightening barbed wires during fencing,
 - ii) Smoothing concrete floors during plastering.
 - iii) Administration of liquid medicine to livestock through the mouth.
 - iv) Processing butter-----
14. List five tools used when constructing a wooden fence.

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15. a) Diagrams M, N, P and Q represents some farm tools.



- Identify the tools; M,N,P and Q.
 - Give the use of each of the tools named above.
 - State two maintenance practices that should be carried out on tool M.
- (b) The diagram below shows a farm equipment. Study it and answer the questions that follow.



- Identify the equipment.
- Name the parts labeled R, S, T and U.

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CROP PRODUCTION – LAND PREPARATION

K.C.S.E PAST PAPERS

1. State 4 reasons for primary cultivation.
2. State four reasons for primary cultivation.
10. State four factors which determine the depth of Ploughing.
3.
 - (a) Give tow reasons why the use of fire should be discouraged in clearing land during seedbed preparation.
 - b) Give tow reasons for secondary cultivation.
 - c) State two benefits of minimum tillage in crop production.
4. List four implements used to carry out secondary cultivation
5. State two causes of hard pans in a crop field.
6. Give the maintenance practices of a disc plough.
7. State reasons why use of fire in clearing land is discouraged.
8. Outline the main reasons for secondary tillage.
9. State the main benefits of minimum tillage
10.
 - a) State one condition under which a farmer would prefer to use an ox-cart instead of tractor – drawn trailer.
 - b) Give three maintenance practices carried on an ox – plough.
11. Give four farming practices that may help in achieving minimum tillage.

WATER SUPPLY AND IRRIGATION

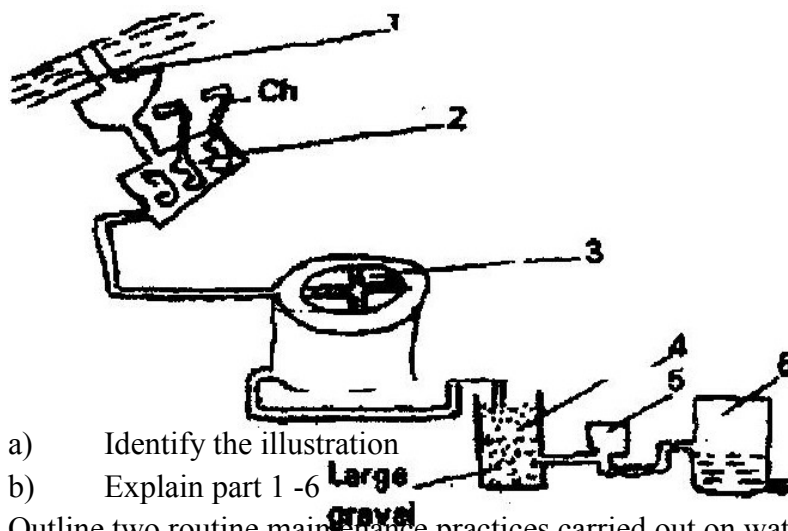
K.C.S.E PAST PAPERS

1. Give the main methods of conveying water from place to place.
2. State two methods of storing water on a farm.
3. State two ways of overcoming the problem of water logging in crop production.
4. State four methods of treating water for domestic use.
5.
 - (a) Give tow ways of conserving water for livestock use.
6. State two means by which water can be conveyed from the place of Storage to where it is needed on the farm.
7. Name four types of water pumps which can be used on the farm.
8. List two features of plastic pipes a farmer should consider before buying the pipes.
9.
 - a) State three factors to be considered before deciding on irrigation in crop production.
 - b) State three advantages of overhead irrigation compared to

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surface irrigation.

10. Give four features of plastic pipes a farmer should consider before buying the pipes.
11. Name four types of water pumps which can be used on a farm.
12. Which of these factors would you consider in deciding on irrigation in crop production.
13. In what way is water useful for agriculture activities?
14.
 - a) Explain the reasons for treating water on the farm.
 - b) State the uses of water in the farm.
 - c) Describe the process involved in water treatment using a chemical treatment system.
15. Give two factors that influence the quantity of water used in the farm.
16. State three farming activities which may cause pollution to water sources.
17. State three advantages of crop irrigation in a farm.
18. Study the illustration below carefully and answer the questions which follow.

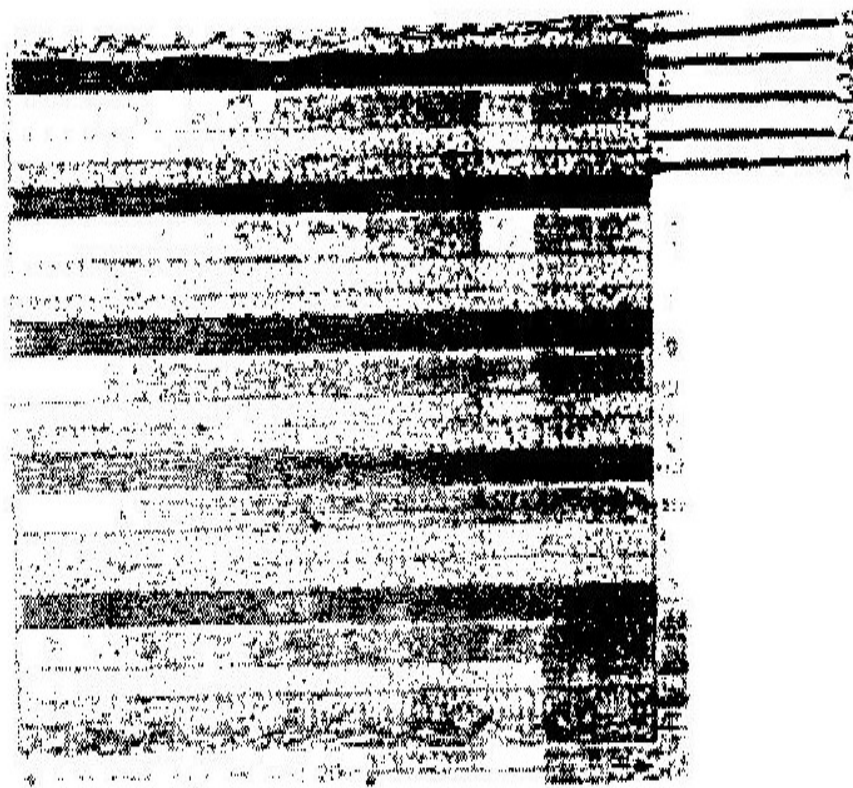


- a) Identify the illustration.
- b) Explain part 1 - 6.
19. Outline two routine maintenance practices carried out on water storage tanks.
20. List three types of surface irrigation in crop production.
21. List four methods used to drain farm land.
22.
 - a) List three surface water sources found in a farm.
 - b) Give two reasons for treating water before use in a farm.
23. Give the reasons why water treatment is important.

SOIL FERTILITY – ORGANIC MANURES

K.C.S.E PAST PAPERS

1. How is green maturing carried out on a farm?
 - (a) Give four reasons why it is advantageous to use farm yard manure instead of straight fertilizer.
 - (b) State four factors that determine the quality of farm yard manure.
2. State four ways by which plant nutrients may be lost from the soil.
3. Outline the various benefits of F.Y.M in the farm.
4.
 - a) Define soil fertility
 - b) List the major characteristics of a fertile soil.
5. The diagram below is a presentation of a cross section through a compost Heap. Study it and answer the questions which follow.



- a)
 - i) Name the parts labeled 1 –
Give the importance of 5, 4, 3
 - b) Why is it advisable that
 - i) A long sharp pointed stick driven into the heap at an angle.
 - ii) Compost pits be preferably alone in more drier area / weather.
9. Give four ways by which soil loses its fertility
 10. State four characteristics that make a crop suitable for green manuring.
 11. State three factors that determine the quality of compost manure.

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12. State four characteristics of a fertile soil.
13. Name three farming practices which may lead to soil erosion.
14. State three factors which should be considered when sitting a compost heap.
15. State three advantages of adding organic matter to sandy soil.

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LIVESTOCK PRODUCTION – (COMMON BREEDS)

K.C.S.E PAST PAPERS

2. i) Name the breed of camel that is used for provision of quicker mode of transport and is & better adapted for arid conditions.
- ii) Give two reasons why this species of camel is well adapted to North par of Kenya.
3. i) Give two reasons why jersey breeds is better suited for marginal areas than Friesians breed.
3. Name the major characteristics of indigenous cattle breeds.
4. Give the major features of exotic beef breeds
1. Name the exotic cattle with
 - i) Highest butter fat content
 - ii) Lowest butter fat content
5. State two characteristics of goats that make them adaptable to arid areas of Kenya.
6. Give three ways used to improve & production in indigenous cattle.

AGRICULTURE ECONOMICS – BASIC CONCEPTS AND FARM RECORDS

KCSE PAST PAPERS

1. List any four types of records a farmer should keep.
2. State four reasons for keeping health records in livestock production
3. State the conditions under which the opportunity cost is Zero in a farming enterprise.

SOIL FERTILITY II – INORGANIC FERTILIZERS

KCSE PAST PAPERS

1. State four characteristics of Nitrogenous fertilizers (2mks)
2. Calculate the amount of K_2O contained in 400kg of a compound fertilizer 25:10:5. (2mks)
3. State four functions of Potassium in plant growth.(2mks)
4. Give two symptoms of potassium deficiency in crops (2mks)
5. State four ways by which plant nutrients may be lost from the soil(2mks)
6. a) State three functions of nitrogen in crops (2mks)
 b) State two symptoms of nitrogen deficiency in a growing maize crop (2mks)

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7. State four effects of excessive application of Nitrogenous fertilizers on crop growth. (2mks)
8. Give two functions of sulphur in crops (2mks)
9. Give four deficiency symptoms of phosphorous in crops.(2mks)
10. List four ways of applying fertilizers in crops (2mks)
11.
 - a) Differentiate between macro – nutrients and mirco – nutrients (2mks)
 - b) State four functions of Calcium in plant growth and development (2mks)

CROP PRODUCTION II – PLANTING

KCSE PAST PAPERS.

1. State four benefits of using vegetative propagation in orange production.(2mks)
2.
 - (a) Give two advantages of growing cereal crops in rows instead of broadcasting.(2mks)
 - (b) Give two reasons for planting crops at correct spacing (2mks)
 - (c) Give two factors that determine the depth at which seeds should be planted. (2mks)
3. State any four factors that determine the spacing of a crop (2mks)
4. Give two reasons for sowing annual crops early in theplanting season.(2mks)
5. Why should legume seeds be inoculated before planting (1mks)
6. State two benefits a farmer would get by having the correct plant population in the production of annual crops. (2mks)
7. State four factors that influence spacing when planting a pure stand maize.(2mk)
8. Give two factors that would influence the time of planting beans (1mk)
9. Give four reasons for seed selection in crop production. (2mks)
10. State six factors that influence the spacing of an annual crop. (2mks)
11. Give four advantages of under sowing in pasture production. (2mks)
12.
 - (a) Give two harmful effects of high population density in a maize crop growth for grain production (2mks)
 - (b) Calculate the plant population per hectares of a maize crop planted at a spacing of 100cmx 50cm. Show your working (2mks)
13. Name two crop production practices carried after planting to achieve optimum plant population. (1mk)

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14. State five qualities of the mother plant which should be considered when selecting vegetative materials for planting. (2mks)
15. State four factors that determine the time of planting of a crop. (2mks)
16. Give four disadvantages of planting seed using broadcasting method. (2mks)
17. List four factors that determine (2mks)
18. Give six reasons for timely planting of annual crops. (3mks)

CROP PRODUCTION-NURSERY PRACTICES

KCSE PAST PAPERS

Give a reason for carrying out each of the following practices in a tomato nursery. (2mks)

- (i) Pricking out (ii) Hardening off.

Give two activities carried out during hardening off tomato seedlings (2mks)

Give two characteristics of a good root stock for grafting. (1mk)

CROP PRODUCTION (IV) – FIELD PRACTICES

KCSE PAST PAPERS

1. State four reasons for pruning fruit crops (2mks)
2. Give two reasons for drying grains before storage. (2mks)
3. (a) Why is training done in some crops. (1mk)
(b) Name two crops, which require training.
5. State two factors that determine the stage at which a grain crop is ready for harvesting (1mk)
6. State four benefits of using organic matter for mulching, sounds (2mks)
7. State two ways in which inorganic mulch helps to conserve moisture in the soil (1mk)
8. (a) What is crop rotation? (1mk)
(b) State three advantages of crop rotation. (1mk)
(c) State three factors considered when designing a crop rotation programme (2mks)

CROP PRODUCTION (V) – VEGETABLES

KCSE PAST PAPERS

- Give one cause of blossom end rot in tomatoes
 - State two methods of controlling blossom end rot in tomatoes (2mks)
- State four factors to consider when grading tomatoes for fresh market.

LIVESTOCK HEALTH – INTRODUCTION

KCSE PAST PAPERS

- State three advantages of keeping a herd of dairy cattle health. (3mks)
- State two reasons for maintaining livestock in good health. (2mks)
 - Name two noticeable diseases in cattle. (2mks)
- State two ways by which proper feeding contribute to disease control in livestock. (1mk)
- Explain measures used to control livestock diseases. (12 marks)
- Give four ways in which diseases can spread from one animal to the other within the farm.

LIVE STOCK HEALTH – PARASITES

KCSE PAST PAPERS

- Which livestock disease is transmitted by each of the following ticks? (2mks)
 - Blue tick (*Boophilous decoloratus*)
 - Brown ear tick (*Rhipicephalous appendicula tus*)
 - How many hosts does the red-legged tick (*Rhipicephalous averts*) require to complete its life cycle? (1mk)
- State four signs of infestation by external parasites in livestock? (2mks)
- State four signs of infestation by external parasites in livestock? (2mks)
 - Name the intermediate host for each of the following internal parasites. (2mks)
 - Tape worm (*Taenia solium*)
 - Liver fluke (*Fasciolahepatica*)
- Describe the life cycle of a three- host tick. (8mks)
- State four non-chemical methods of controlling ticks in cattle. (2mks)

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6. Give four measures that should be taken to control tapeworms on the farm.

(2mks)

LIVE STOCK PRODUCTION (II) – NUTRITION

KCSE PAST PAPERS

1. What is a production ration as used in animal nutrition? (1mk)
2. Name two groups into which vitamins are classified. (1 mk)
3. State 3 factors that influence the amount of water intake by a farm Animal. (1½ mks)
4. (a) Differentiate between a roughage and a concentrate feed in animal nutrition. (2mks)
(b) State three ways in which a production ration may be utilized by cattle. (3 mks)
5. Give four characteristics of a livestock roughage feedstuff.
6. Outline four functions of proteins in the body of an animal. (2mks)
7. Outline four factors that determine the nutritional requirements in Cattle (2mks)
8. Give 4 functions of calcium in dairy cow. (2mks)
9. State four factors that are considered when formulating a livestock ration (2 marks)
10. (a) Explain the term “production ration” as used in livestock productions. (1mk)
b) State four factor which determine the amount of feed an animal can consume. (4mks)
11. State three reasons for feeding livestock. (1½mks)

LIVE STOCK PRODUCTION (III)- SELECTION AND BREEDING

KCSE PAST PAPERS.

1. (a) How does crossbreeding improve livestock production.
2. State six signs that are likely to be observed when a cow is on heat
3. State four disadvantages of natural mating as a method of breeding in dairy cattle management.
4. Define the term out crossing in animal breeding.
5. The diagram below shows the reproductive system of a cow. Study it carefully and answer the questions that follow.

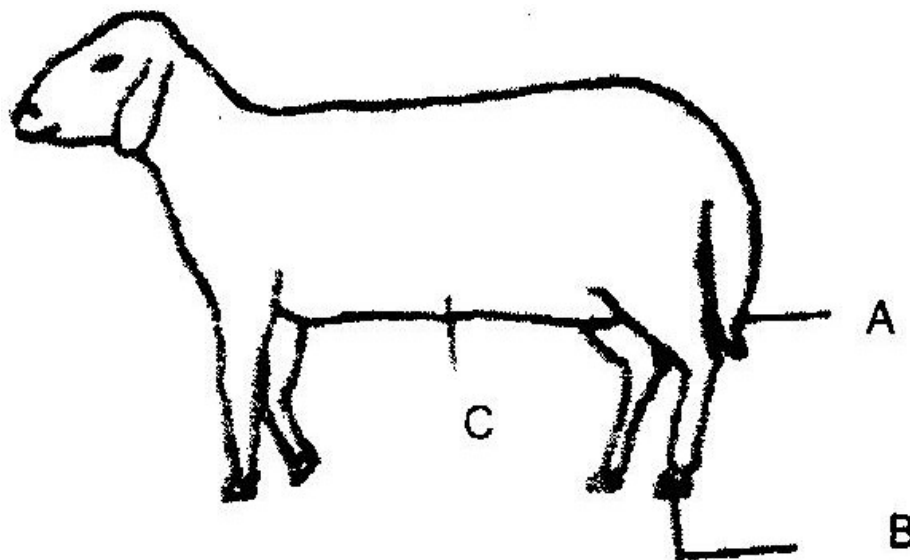


- a) i) Name the parts labeled A B and C.
- ii) State the function of each of the parts labeled a and b
- b) (i) Give two methods of mating in cattle
- (ii) How long is the oestrus cycle in cattle?
- 6. Explain the term hybrid vigour as used in livestock production.
- 7. Describe the factors a farmer should consider when selecting a young female pig (Gilt) for breeding.
- 8.
 - (a) Define the following terms as used in livestock breeding.
 - (i) Inbreeding
 - (ii) Out crossing
 - (b) Outline three disadvantages of artificial insemination in cattle management
 - (c) State three desirable characteristics to be considered when selecting a heifer for milk production.

LIVESTOCK PRODUCTION (IV) – REARING PRACTICES

KCSE PAST PAPERS

1. Name two kinds of livestock which can be castrated using a rubber ring.
2. Give four reasons why bees may swarm from a hive.
3. Below is a diagram of a sheep with some parts labeled A, B, and C. Study the diagram and answer the questions that follow.



- (i) What operation is usually carried out on the part labeled A during a sheep's early stages of life?
 - (ii) Why is it necessary to carry out the operation in (i) above?
 - (iii) At what stage of sheep should the operation in (i) above be carried out?
 - (iv) Give two methods of carrying out the operation in (i) above,
 - (v) Which operation is usually carried out on part labeled B
 - (vi) What problem would occur if the operation in (V) above is not carried out?
 - (vii) How should the sheep be held when shearing wool around part labeled C?
4. Why should smoke be used during harvesting of honey?
5. State four reasons for culling breeding sows.
6.
 - (a) Define the term colostrums.
 - (c) Explain three qualities that make colostrums suitable for newly born calves.
 - (d) Give three methods of feeding colostrums to a newly born calf.
6. Give six signs a cow would show just before parturition.
7. State four reasons for castrating male piglets.
8. Give two qualities of creep feed that makes it suitable for piglets.
9. State two reasons why it is necessary to place sugar syrup close to a beehive.
10. State four routine management practices that should be carried out on a

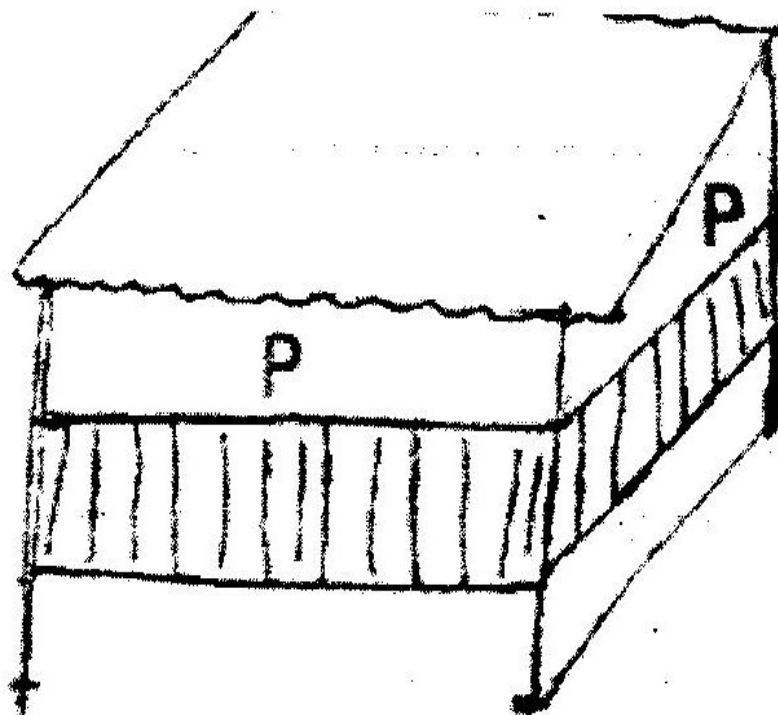
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lactating ewe.

11. State six management practices in fish rearing.
12. Give five signs, which indicate that a sow is about to farrow.
13. State four conditions which would make it necessary to feed bees.
14. State four management practices that should be carried out during the mating season in sheep.
15. Name four species of fresh water fish reared in Kenya.

FARM STRUCTURES.

1. The diagram below represents a calf pen. Study it to answer the questions that follow.



- (i) How high should the floor be above the ground level?
 - (ii) Why should the floor of the calf pen be raised?
 - (iii) Why should the parts of the pen marked p be open?
 - (iv) State three factors that should be considered in sitting a calf pen?
2. State two advantages of using wood in the construction of farm buildings
3. Describe the construction of a rabbit hutch under the following sub-headings.
4. Give two reasons for treating timber to the used in construction of farm buildings.
5. State one advantage and one disadvantage of using barbed wire instead of plain wire for fencing paddocks.
6. State two functions of ventilation in an animal house.
7.
 - a) One of the recommended ratio of mixing ingredients for making Concreteblock is 1:3:4, Name ingredients represented by the numbers 1,3and 4 in the mixture.
 - b) If stronger concrete blocks were to be made, name the ingredient that would be increased.
 - c) State three properties of concrete that make it suitable for constructing farm buildings.
 - d) In addition to concrete, name three other materials that would be

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required to construct the floor of milking shed.

8. State four features of a good maize granary.
9. State two reasons why maintenance of farm structures is important.
10.
 - a) State any four factors that would influence the siting of a calf pen.
 - b) State four factors to consider when selecting materials for constructing a calf pen.
 - c) Give four maintenance practices that should be carried out on a permanent calf pen.
11. Give two practices, other than use of preservatives, that can be carried out on wooden fencing posts to make them last long.
12. State six features of an ideal calf pen.
13. Outline any four maintenance practices that should be carried out in a deep litter poultry house.
14. Give two advantages of concrete blocks over timber as building materials.
15.
 - a) State the uses of fences in farms.
 - b) What factors would be considered when siting a farm structures.
16.
 - a) State four advantages of a hedge in a farm.
17. State four advantages of using a Kenya Top Bar Hive over log hive.
18. Outline two routine maintenance practices carried out on water tanks.
19.
 - a) Explain the uses of various hand tools in the construction a Kenya Top Bar Hive
 - b) Describe the procedure of erecting wooden posts for fencing.
20. State three disadvantages of using steel in construction of farm buildings.

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SOIL AND WATER CONSERVATION

KCSE PAST PAPERS

1. Give one way through which check dams control soil erosion.
2. State two ways by which trees help in soil conservation.
3. State two reasons for carrying out soil conservation in a farm.
4. State two ways by which grass cover help to conserve soil.
5. State two ways by which inorganic mulch help to conserve water in the Soil.
6. Define the terms;
 - a) Forestation
 - b) Re-a forestation
7. Outline three factors, which may influence soil erosion.

WEEDS AND WEED CONTROL

KCSE PAST PAPERS

1. State four reasons why timely weed control is advisable in crop production.
2. Give four ways of controlling weeds in a maize field. (2mks)
3. PP2: Diagram & and H show weeds.
 - i) Identify the weeds. (2mks)
 - ii) State the economic importance of the weed shown in diagram G (2mks)
 - iii) Why is it difficult to control weed in diagram G? (1mk)
4. State six disadvantages of weeds in crop production (3mks)
5. (i) State four factors that contribute to the competitive ability of weeds. (2mks)
6. The diagram below represents a weed.



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- i) Identify the weed (1/2 mrks)
- ii) Classify the weed according to its life span. (1/2 mark)
- iii) State one harmful effect of the weed to livestock. (1mk)

AGRICULTURE ECONOMICS (II)- LAND TENURE AND REFORMS

KCSE PAST PAPERS

- 1. State four disadvantages of communal land tenure system.
- 2. Give two ways in which land consolidation helps to improve farm management.
- 3. State four objectives of land settlement which have been undertaken in Kenya
- 4. Give two forms of collective land tenure system in

CROP PESTS AND DISEASES

KCSE PAST PAPERS

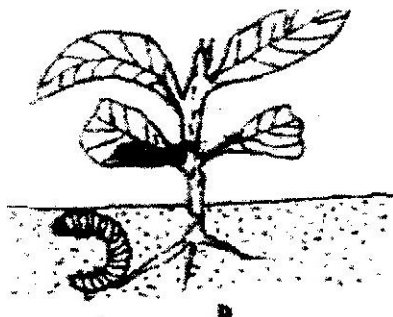
- 1. PP2
Below is a diagram of a bird labeled A. Which is a crop pest?



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- i) Identify the pest
- ii) State two ways by which the bird causes loss in crops.
- iii) State four methods, which are used to control the pests.

2. The diagram labeled D below shows a Kale crop invested by a pest

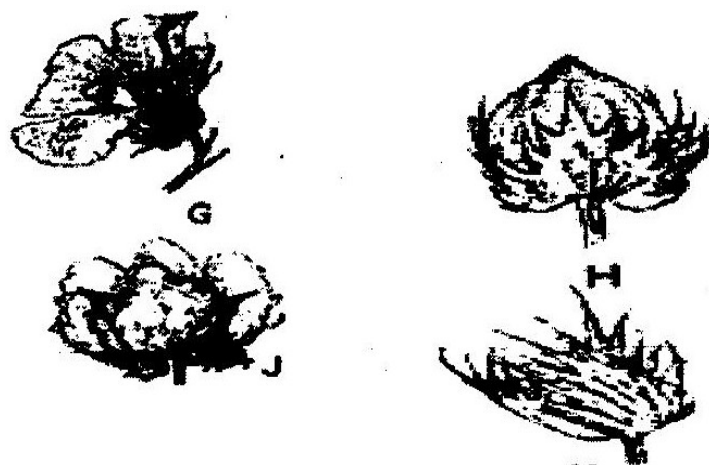


- i) Identify the pest.
 - ii) What damage does the pest cause the crop?
 - iii) State two methods of controlling the pest
3. Give two methods of controlling the pest?
4. State two cultural methods of controlling bollworms in a crop of cotton.
5. Give two ways of controlling bacteria blight in cotton.
6. State four factors that affect the effectiveness of a pesticide.
7. a) State two feeding habits of field insect pests.
b) State two cultural methods of pest control in stored grains.
8. Explain how various practices carried out in the field help to control crop diseases.
9. State three cultural ways of controlling nematodes in a field of bananas.
10. a) Define the term "Economic Injury Level" of a crop.
b) Give two ways by which pesticides kills crop pests.
c) State four disadvantages of chemical pest control in crop production.
11. Give three harmful effects of pests in crop production
12. State two cultural methods in controlling bollworms in a tomato crop.
13. Give two possible causes of swelling on the roots of bean plants.

CROP PRODUCTION (VI) FIELD PRACTICES (II)

KCSE PAST PAPERS.

1. Name two field pests and two diseases of millet.
2. Why is it advisable to apply a straight nitrogenous fertilizer to a crop of maize at a height of 30 – 45cm.
3. Which disease causes a mass of dark spores on the flowering parts of maize?
4. State any four non – chemical methods of controlling storage pests in a maize granary.
5. List four insect pests of maize in storage.
6. PP2 The diagram below labeled G,H,J and K shows different stages of cotton fruit.



- i) Rearrange the label G,H,J, and K to show the correct sequence in which the cotton fruit develops.
 - ii) What would be the effect of attack by cotton boll worms at the stage labeled K?
 - iii) State two conditions that should be observed when harvesting to ensure that cotton picked is of high quality.
 - iv) Name the two products which are obtained after processing cotton.
7. State four practices used to control maize streak in the field.
 8.
 - i) State two cultural methods of controlling pests in an established field of sorghum.
 - ii) List any four insect pests that attack maize in the field.
 9. Give four ways of controlling weeds in a field of maize.
 10. Give four control measures of maize streak virus.

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FORAGE CROPS

KCSE PAST PAPERS.

1. State two advantages of establishing a mixed grass legume pasture instead of planting a pure grass pasture.
2. Give two disadvantages of overstocking in cattle production.
3. State two roles of additives in silage making.
4.
 - a) Define the following term, (i) Under sowing (ii) Over sowing
 - b) State three methods of controlling weeds in a pure grass pasture.
 - c) Give three benefits of top dressing in the management of grass pastures.
5. State four ways by which a farmer can make efficient use of a pasture crop.
6. Give four factors that determined the nutrient content of hay.
7. State four advantages of under sowing in pasture production
8. Describe field production of Napier elephant grass under the following sub-headings.
 - i) Seedbed preparation
 - ii) Planting
 - iii) Fertilizer application
 - iv) Weed control
 - v) Utilization
9.
 - (a) List three pasture legumes grown in medium altitude zones.
 - (b) Give three advantages of rotational grazing.
 - (c) State three ways by which overheating can be prevented in the process of making silage.
10. State two advantages of proper stocking pasture management.
11. Explain the following terms as used in pasture establishment.
 - a) Seed Inoculation
 - b) Over sowing

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LIVE STOCK HEALTH (III) – LIVESTOCK DISEASES

KCSE PAST PAPERS.

1. List six routes through which pathogens can enter the body of an animal.
2. State two methods of controlling rinderpest disease in cattle.
3. Give four symptoms of Newcastle disease in poultry.
4. State four predisposing factors to the occurrence of mastitis in dairy cattle.
5. State any three symptoms of mastitis in dairy cattle.
6. Name two notifiable diseases in cattle.
7. State two measures that should be taken to prevent an outbreak of Newcastle disease in poultry.
8. **PP2:** The diagram below shows the head of a chicken having symptoms of a poultry disease.
 - i) Identify the disease
 - ii) Give two reasons why the disease is of economic importance to the farmer.
 - iii) State any two methods of controlling the disease.
9. State four symptoms of foot rot in sheep.
10. Name the causal agents for each of the following disease,
 - i) Coccidiosis
 - ii) Black quarter
11.
 - i) State two predisposing factors of foot rot in sheep.
 - ii) Give three symptoms of anaplasmosis disease.
12. Explain measures used to control livestock diseases. (12mks)
13. Give two signs that would indicate that a cow has died of anthrax.
14.
 - a) Name the causal organism of brucellosis in cows (1mks)
 - b) Give two symptoms of brucellosis in cows.
 - c) State four measures that should be taken to control brucellosis in cattle.
15. Give three methods of controlling rinderpest in cattle.
16. Mention four symptoms of East Coast Fever in cattle.
17.
 - a) State the cause of milk fever in dairy cows. (1mks)
 - b) Give four symptoms of milk fever in dairy cows.
 - c) State two methods of controlling milk fever.

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LIVESTOCK PRODUCTION – POULTRY

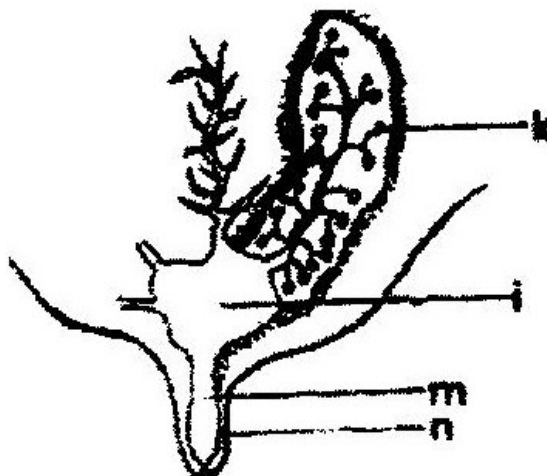
KCSE PAST PAPERS

1. Give three methods of controlling cannibalism in a flock of layers in deep litter system.
2. List four factors that should be considered when grading eggs for marketing.
3.
 - a) Describe the artificial rearing of layer chicks from day old up to the end of brooding.
 - b) Describe the characteristics of a poor layer, which should be considered during culling
4. Outline any four maintenance practices, which should be carried out in a deep litter poultry house. (2mks)
5. Give four conditions that reduce the quality of eggs for hatching. (2mks)
6. Give four measures that can control egg eating by hens in a deep litter system (2mks)
7.
 - a) State four observations on the behaviour of chicks which would indicate that the temperature of a brooder is too high. (4mks)
 - b) Give four advantages of deep litter system of poultry keeping. (4mks)
8. List six qualities of eggs suitable for incubation (3mks)
9. Describe the steps to be taken in maintaining hygiene in a deep litter poultry house. (5mks)

LIVESTOCK PRODUCTION (VI) – CATTLE

KCSE PAST PAPERS.

1. (a) The diagram below is a cross section of part of a cows adder



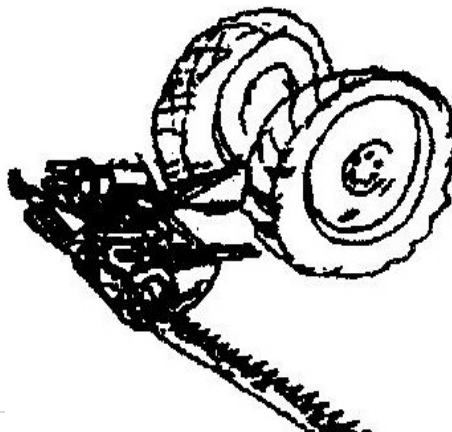
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- Label on the diagram the parts marked k, l, m and n. (2mks)
- (b) i) What is milk let down? (1mk)
ii) Which hormone stimulates milk let down. (1mk)
- (c) State three practices which are carried out to control mastitis in lactating cows.
2. Describe the management of a dairy heifer calf from birth until it is mature for first service. (20mks)
3. a) Name any two characteristics of good quality whole milk. (1mk)
b) State three advantages of artificial calf rearing. (3mks)
4. State four qualities of clean milk. (2mks)
a) Define the term colostrums (1mk)
b) Explain three qualities that make colostrums suitable for newly born calves. (3mks)
c) Give two methods of feeding colostrums to a newly born calf. (1mk)
5. State any six practices that would ensure clean milk production (3mks)
6. State six marketing problems affecting dairy farming in Kenya. (3mks)
7. Describe the management of a dairy calf using artificial rearing method from birth to weaning (20mks)

FARM POWER AND MACHINERY

KCSE PAST PAPERS

1. (a) States four advantages of farm mechanization (2mks)
(b) Give the functions of each of the following parts of a mould board plough. (4mks)
(i) Mould board (ii) Share
(iii) Frog (iv) Landslide.
(c) Give two daily maintenance practices that should be carried out on a mould board plough. (2mks)
2. The diagram below is a tractor drawn implement hitched at the rear of the tractor.



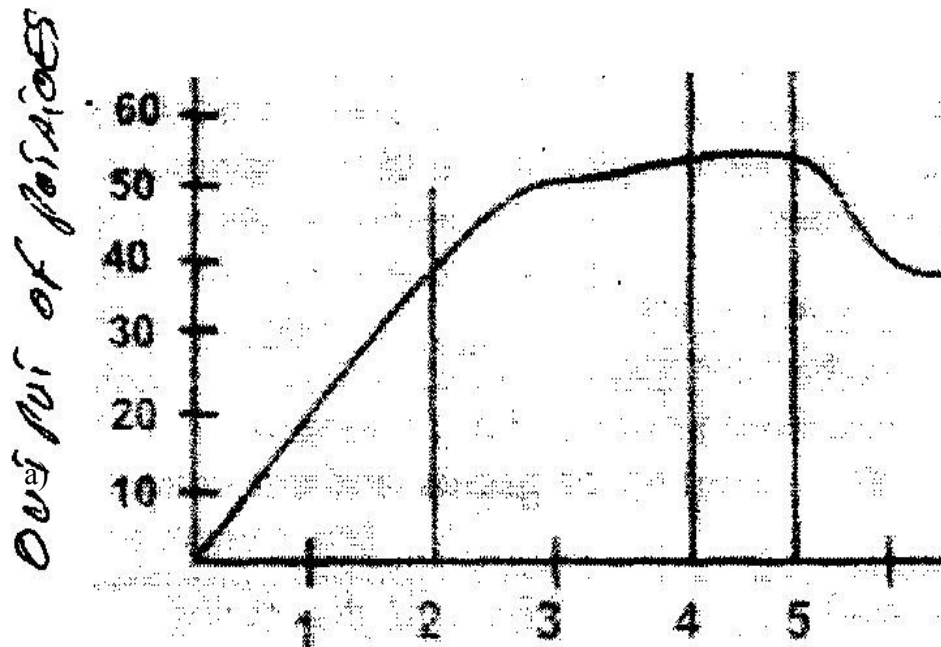
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- (i) Identify the implement (1mk)
- (ii) What is the method of power transmission for operating the implement? (1mk)
- (iii) State three maintenance practices that should be carried out on the implement. (3mks)
3. a) Compare the use of an ox-drawn mould board plough with that of a tractor-drawn mould board plough. (9mks)
- b) Describe the maintenance practices that should be carried out on an ox-drawn mould board plough (6mks)
- c) What are the advantages and disadvantages of using tractor hire service farming instead of owning and using your own tractors. (5mks)
4. a) State two reasons of applying oil and grease on a rotary mower. (2mks)
- b) State four maintenance practices required on a rotary mower besides oiling and greasing. (2mks)
- c) State four factors that a farmer should consider before buying a tractor for use as the source of power on the farm. (2mks)
5. a) Give two uses of ox –drawn harrow (2mks)
- b) Give two maintenance practices of a ox-drawn tine harrow. (2mks)
- c) State two advantages of an ox- drawn harrow over tractor-drawn harrow. (2mks)
6. Out line four maintenance practices of a disc harrow. (2mks)
7. a) State one method of increasing ploughing depth when using a disc plough. (1mk)
- b) State two reasons for maintaining a disc plough, (2mks)
- c) State three advantages of farm mechanization. (3mks)
8. a) State two functions of a coulter in a mould board plough. (2mks)
- b) Give three maintenance practices carried out on an 0x-drawn trailer. (3mks)
9. a) State two problems associated with tractor hire service that farmers encounter. (1mk)
- b) List four implements used to carry out secondary cultivation. (2mks)
10. a) State one condition under which a farmer would prefer to use an ox-cart instead of a tractor-drawn trailer.
- b) Give three maintenance practices carried out on an ox-plough. (3mks)

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AGRICULTURE ECONOMIC III- PRODUCTION ECONOMICS

1. (a) State four ways of improving the labour productivity of farm labour. (2mks)
(b) What is increasing returns in a production? (1mk)
(c) What are three classifications of farm credits according to the repayment periods? (1 ½ mks)
2. (a) Define the term opportunity cost as used in economics. (1mk)
b) What is working capital in a farming situation (1mk)
c) Define the term utility of a commodity as used in agriculture economics. (1mk)
3. (a) State any two sources of capital for farming
(b) Explain the advantages of budgeting in farm business. (5mks)
4. a) Explain the difference between fixed costs and variable costs in farming.
b) Give four variable costs in the production of coffee in an established field of coffee. (2mks)
c) Give 3 advantages of planning in a farm business. (3mks)
5. Give two uses of gross margin analysis in farm business.
6. a) Below is a graphical representation of The law of admonishing returns.



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Explain what happens in each of the Zones marked I,II and III in

relation to output. (3mks)

b) Which of the three is a rational zone of production? (1mks)

c) State any three precautions a potatoes farmer would take to minimize risks in the production of potatoes. (2mks)

7. a) State three ways of improving labour productivity in a farm.

b) Give two changes that would indicate improvement of labour efficiency in farm. (2mks)

c) State two ways of determining the rate of payment of casual labour in a farm. (3mks)

8. a) State four factors that influence the supply of casual labour in a farm. (2mks)

b) State six ways by which a farmer can risk and uncertainties.

c) State six reasons why agriculture is important in Kenya economy. (3mks)

9. State 2 reasons for choosing the right enterprise combination in farming business. (1mks)

10. a) Give four examples of joint products in livestock production.

b) List four variable inputs in poultry production.

c) A farmer can combine dairy meal and home made in

Dairy meal (kg)	Home made feed (kg)	Marginal rate of substitution
1	48	O
2	39	V
3	32	7
4	27	W
5	23	4
6	21	X
7	20	1
8	19	y

i) Given the above information, calculate the marginal rate of substitution and give values of V,W,X and X (4mks)

ii) Given that the price of dairy meal is Ksh.8.00 per kilogram and that of home made feeds is Ksh.2.00 per kilogram, calculate the least cost combination. (1mks)

11. a) Name three – product relationship in agriculture economics.

b) Explain the following terms as used in agriculture economics.

i) Production function

ii) Equi-marginal returns.

12. Name five sources of agriculture credit in Kenya (1 ½ mks)

13. Give two examples in each case of the following costs incurred in the

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production of milk.

- a) Variable costs (1mk)
 - b) Fixed costs (1mks)
14. a) Differentiate between partial budget and complete budget. (2mks)
- b) Explain how factors may adjust uncertain rules in farming business (2mks)

The cost of fertilizer is Kshs.1500 per unit and the price of maize in Ksh.1200 per bag.

- i) At what unit of fertilizer input should the farmer be advised to stop applying any more fertilizer to the maize.
- ii) Give two reasons for your answer in b (i) above.
- iii) Calculate the marginal return at the point of optimum production.

AGRICULTURAL ECONOMICS

(FARM ACCOUNTS)

- 1. a) List four types of financial books farmers should keep. (2mks)
- b) State two uses of a balance sheet. (2 mks)
- 2. Study the following information which was extracted from Mr. Rambo's farm record on 31 -12 – 95 and answer the question below.

	Kshs.
Loans payable to bank	300,000
Five milking cows	250,000
400 layers	80,000
20 goats	30,000
Debts payable to cooperative	20,000
Buildings and structures	600,000
Bonus payable to workers	19,000
Cattle feed in store	10,000
Animal drugs in store	4,000
Debts receivable	18,000
Breakages to repair	30,000
I cash at hand	20,000
I Cash in bank	30,000
Spray equipment	12,000

Prepare a balance sheet for Rambo's farm using the information above 7 mks

- 3. a) Explain the following terms as used in farm account.
- i) Cash account

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- ii) Ledger
- iii) Balance sheet
- iv) Purchase order.

Name two types of inventories used in farm accounts. (2mks)

4. On 5 – 1- 2001 Tamu farm purchased on credit the following items from a K. F. A shop.

- 20 bags of dairy meal, 70kg each @ sh. 1,100 per bag.
- 16 bags of bran, 70kg each @ sh. 700 per bag.
- 18 bags of D.S.P fertilizer, 50kg each @ sh. 1,500 per bag.
- 45 bags of seed maize, each 2kg @ Ksh.300 per bag.
- 8 shearing knives (medium size) @ sh. 300per knife.

- i) Prepare the purchase order that Tamu farm made to K.F.A. (6mks)
- ii) Calculate the value of each item purchased and the total value of the order.(3mks)

5. The following accounts information is from Mrs. Mbuta's farm for the year ended 31 – 12 – 2003.

Opening valuation	Ksh. 6,000/=
Paid wages	Ksh. 5000/=
Bought equipment worth	Ksh. 8,000/=
Bought pig feeds worth	Ksh. 4,000/=
Sold mature pigs worth	Ksh. 7,000/=
Bought drugs worth	Ksh. 3,200/=
Sold maize worth	Ksh.3,000/=
Closing valuation	Ksh. 4,000/=

- i) Using the information above, prepare a profit and loss account for Mrs. Mbuta's farm.
- ii) From the calculations in (i) above, state whether Mrs. Mbuta made a profit or a loss. (1mk)

6. a) What is opening valuation as used in farm account? (1mks)
- b) State the use of each of the following financial documents (3mks)
- i) Cash receipt
 - ii) Purchase order.
 - Mi) Delivery note.

AGRICULTURAL ECONOMICS

(AGRICULTURAL MARKETING AND ORGANIZATION)

1.
 - a) What is the minimum number of people required to form a co-operative society? (1mks)
 - b) State four factors that may influence the supply of a commodity in a market? (4mks)
 - c) State four problems that farmers are likely to face when marketing their produce. (4mks)
 - d) Name two marketing organizations for coffee in Kenya. (1mk)
2.
 - a) Differentiate between market and marketing? (2mks)
 - b) What is an imperfect market? (1mk)
 - c) How will the price of mangoes in the short run be affected if the quantity of mangoes supplied in a market is increased?
 - d) State any four problems a dairy farmer is likely to face in marketing milk. (2mks)
3.
 - a) State the law of demand. (1mk)
 - b) State four factors that determine the demand of a commodity in a free market economy. (4mks)
 - c) What is elasticity of demand for a commodity? (1mk)
 - d) Explain the functions of agricultural marketing boards. (15mks)
4.
 - a) Give four benefits a farmer would derive from being a member of a dairy co-operative society. (2mks)
5.
 - a) Explain the problems farmers face in marketing of agricultural produce. (9mks)
 - b) Describe the various agencies and institutions involved in marketing of a agricultural produce. (5mks)
6.
 - a) State six factors that influence demand for a commodity in a market. (3mks)
7.
 - a) Given that at a price of Ksh.100 per bag, 20 bags of maize are demanded, but when the price changes to Ksh.800 per bag, 22 bags of are demanded. Calculate the elasticity of demand. Show your working. (3mks)
8.
 - a) Describe the functions of agricultural marketing (10mks)
 - b) Explain the role of agricultural co-operatives in Kenya. (10mks)
9.
 - a) State six problems experienced by farmers in marketing agricultural produce. (3mks)

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MARKING SCHEMES

TOPICAL

INTRODUCTION TO AGRICULTURE.

1. This is a system of farming in which one crop is grown in large scale.
2. **Characteristics of shifting cultivation.**
 - When fertility of the soil goes down crops are not grown again until fertility is restored.
 - Plenty of land is available to the farming community
 - Practicable with annual crops not with the perennials.
 - Agricultural output from the whole system is low / subsistence production.
 - Input such as pesticides, fungicides, fertilizers are rarely used / build up of pests and diseases is avoided by periodic movements to the new lands.
 - Use of simple hand tools.
3.
 - It is a source of food for the population.
 - Earns foreign exchange for the country
 - Provides market industrial goods.
 - Farmers earn a lot of income
 - Provides employment both directly and indirectly.
4.
 - Providing market for industrial goods.
 - Providing raw materials that are used in industries.
5. **Limitations of pastoral farming**
 - Drought and aridity Diseases and parasites.
 - Diseases and parasites
 - Attack by wild animal
 - Soil erosion due to over stocking
 - Poor pastures species
 - Inadequate land/over population.
6.
 - Low production
 - Uncontrolled mating
 - Difficult to control parasites and diseases
 - Leads to overgrazing hence soil erosion.

FACTORS INFLUENCING AGRICULTURE.

1.

- Temperature/altitude
- Prevailing winds
- Soil types
- Rainfall

2.

- Cause physical damage / breakages to crops (accept any physical damage)
- Causes water stress / increases rate of transpiration in crops.
- Spread crop pests, diseases and weeds.
- Can cause stress to crops due to chilling caused by cold air.
- Cause soil erosion leading to loss of soil fertility.
- Encourage transpiration hence water and mineral uptake.

3.

- Rainfall intensity
- Rainfall reliability
- Rainfall distribution
- Rainfall amount

4.

- Poor soil fertility due to lack of manure and fertilizer application.
- Less rainfall/unreliable rainfall/too much rainfall.
- Poor soil type leading to water logging or excess leaching.
- Inappropriate pH/poor soil pH.
- Pest attack
- Poor weed control leading to competition.
- Too high or low temperature / inappropriate temperature.
- Excess wind that increase evapotranspiration/lodging of crop/floral abortion.
- Inappropriate humidity, either too high or too low.
- Extreme light intensity that may reduce photosynthesis / exhaustion of nutrients.
- Inappropriate topography that may limit crop growth.
- Hailstorm damage.

5.

- Decompose organic matter.
- Help to aerate the soil
- Atmospheric Nitrogen to nitrates.

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- Upon death and decay release plant nutrients.

6.

- Ploughing at the same depth season after season
- Use of heavy machinery on wet soil.

7.

a)

- Physical weathering
- Biological weathering
- Chemical weathering

b)

- Soil texture is the relative proportion of the different sized particles in the soil;
- Soil structure is the general appearance of the soil in relation to the arrangement of the individual soil particles.

c)

- Allows proper infiltration/drainage of water
- Has good aeration.
- It is not easily eroded.

8.

a)

- Light intensity.
- Light wavelength
- Light duration / photosynthesis

b)

- i) Capillarity in the three different soil samples.
- ii) G – Sandy soil
H – Loam soil
J – Clay soil
- iii) G – Rough and coarse texture
J – Fine textured
- iv) Addition of organic manure – Addition of lime

9.

a) **Why soil is important to crops.**

- Supports plant life anchorage.
- Provides nutrients and water
- Contains organic matter, food for micro – organism

b) **Benefits of optimum soil temperature.**

- Increase the rate of bio chemical reactions hence breakdown of materials to form organic matter i.e. for every 10°C rate doubles.
- Activates soil micro – organism especially the useful bacteria for nitrogen fixation,

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- Ensures maximum activities for bio chemical enzymatic reactions that bring about growth e.g. germination process.

c) Factors of soil which influence soil productivity.

- Good supply of plant nutrients and oxygen.
- Good depth
- Good drainage
- Abundance of useful soil and organism
- Adequate water retention.
- Freedom from plant pests and disease causing agents.

10.

- Provide anchorage to crop roots
- Provide nutrients to crops
- Provide water to crops
- Hold air/oxygen for crop growth.

11.

a)

- Good aeration is necessary for root respiration.
- Suitable for microbial activity
- Easy to work on.
- Raises soil temperature
- Removes toxic substances e.g. sodium, iron and magnesium salts.

b)

- Enhance seed germination
- Enhances plant growth.
- Enhance soil microbial activities
- Improves quality of crops e.g. Tea, pineapples.

c)

- Soil depth / drainage / aeration
- Water holding capacity
- Level of nutrients / cation exchange
- Soil pH/ Soil borne – pests and diseases.

12.

- Improves soil structure
- Reduces leaching.
- Improves water holding capacity
- Increase cation exchange capacity
- Improve nutrient status upon decomposition
- Moderate soil temperature. Buffer the soil pH.

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FARM TOOLS AND EQUIPMENT

1.
 - i) – Riveting / flattening metal sheets / driving cold chisel
 - ii) – Breaking / crushing hardcore / big stones
 - Driving wedges into wood.
2.
 - NI – pipe wrench
 - NII – Adjustable spanner
 - NIII – Ring spanner
 - i) Holds different sizes of nuts and bolts.
 - ii) NI is used for holding pipes during plumbing work.
 - iii) Maintenance of NI – Lubrication of adjustable screw.
3.
 - a)
 - i) 1 – Disc harrow
 - 2 – Disc plough
 - ii) Secondary cultivation / Harrowing
 - b)
 - i) 2 (Disc plough)
 - ii) It is able to ride over obstacles, bouncing on its springs.
4.
 - i) Smoothing rounded/curved edges of timber.
 - ii) To check verticalness of an upright object.
5. Hacksaw / pipe cutter.
6.
 - Cleaning it after use.
 - Lubricating the adjustment screw.
 - Sharpening the plane iron / blade
 - Tightening loose screws and nuts.
 - Proper storage
 - Replacing the broken parts.
7.
 - Spirit level/trowel/float.
 - Plumb bob/plumb line.
 - Mason's square/tape measure.
 - String/line/mason's hammer
 - Mason's chisel/bolster.
8. (a)
 - A tenon saw is shorter
 - A tenon saw has a rigid frame at the back
 - Unlike a cross cut saw.
 - A tenon saw has more teeth per unit length
 - Than a cross cut saw.

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- A tenonsaw is used in joinery work while the cross cut saw cuts wood across the grains.

b)

- Hold the work firmly.
- Ensure the saw is in good working condition / straight blade / sharp teeth / well set teeth/firm handle.

c)

- proper storage after use
- Oil the blade to avoid rusting
- Sharpen the teeth
- Set the teeth properly
- Tighten the handle screws if loose / replace broken handle.

9.

- Soil type/ hardness of the soil
- Vegetation covers of the field.
- Cost of the tool. Skills required
- Availability of the tool
- Availability of capital

10. **When to use a Jembe and not a disc plough.**

- When the land is very steep.
- Lack Technical skills in operation of disc plough
- When size of land is too small
- When capital is inadequate to acquire a disc plough.
- When it is cost effective to use a Jembe.
- When disc plough is not available.
- When inadequate time is available
- When land is rocky / stony.

11. a)

- i) – Pruning / cutting crops
 - Harvesting rice / grasses
- ii) Pruning
 - Cutting vegetative materials for planting / harvesting flowers.

b)

- To avoid injury to the user.
- To reduce repair / replacement cost
- To increase their durability / life span
- To make them more efficient.

c)

- Soil type/hardness of soil.
- Vegetation cover of the field

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- Cost of the tool
- Skill required / availability of the tool
- Availability of capital.

12. (i)

L – Monkey strainer / wire strainer.

M – Sash clamp

N – Dibber

O – Spoke

(ii)

L – Touting fencing wire during fencing.

M – Holding pieces of timber together when joining them together. i

N – Making holes for transplanting. J

O – Smoothing curved/round edges of timber.

13. **Tools used.**

- i. Monkey strainer/wire strainer.
- ii. Steel/wood float
- iii. Drenching gun.
- iv. Butter churner.

14.

- Hand saw/bow saw.
- Claw hammer/plumb bob
- Hole digger/soil anger/panga.
- Ramming rod/builders trowel.
- Soil scoop/spade/shovel.
- Tape measure/mallet/garden line.

15.

(a) M – Sickle

N – Pruning saw

P – Float

Q – Garden trowel.

(b) M – Harvesting rice/grasses

N – Pruning/cutting stems/branches in crops e.g. coffee citrus.

P – Spreading screed on concrete floors/wall

Q – Transplanting seedlings

LAND PREPARATION

1.
 - To kill the weeds
 - Bury crop residues/organic matter into the soil
 - Loosen up the soil/facilitate rainfall infiltration/improve aeration/easy penetration of roots.
 - Control soil borne pests/diseases by destroying their life cycles.
 - Make subsequent operations easier.
2. (ii)
 - Clean after use.
 - Hammer bent share
 - Replace worn out parts e.g. hooks, share.
 - Check, tighten loose nuts and belts during the day's work
 - Oil shiny parts e.g. mould board, if of use.
 - Paint metallic parts e.g. handle, beam and braces, to prevent rusting.
 - Store properly in a shed.
3.
 - To kill weeds
 - To bury organic matter/crop residual into the soil
 - To loosen up the soil/improve aeration/improve water infiltration/improve root penetration. To control soil borne pests/diseases.
 - To make subsequent operations easier.
4.
 - Type of crop/rooting system of crop to be grown.
 - Type of implement available.
 - Type of soil.
 - Soil moisture content at ploughing time
 - Presence of certain weeds e.g. couch grass.
 - Source of power.
5.
 - When the land steep
 - Lack of skill to operate the disc plough
 - When capital is inadequate to acquire disc plough
 - When it is cost effective to use a jembe
 - When there is adequate time.
 - Where the land is rocky/stony.
6. (a)
 - Kills soil organisms.

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- Leads to loss of nutrients.
- Destroys soil organic matter.
- Leads to accumulation of some nutrients to toxic levels e.g. potassium.

(b)

- Control weeds/control pests and diseases
- Incorporate organic matter in the soil
- Improve soil physical conditions/attain appropriate tilt
- To make seedbed appropriate for growing certain crops e.g. in ridging.

(c)

- Saves time/reduce cost of production.
- Maintain soil structure
- Minimizes soil erosion.
- Minimizes root/tubers disturbance.
- Less laborious/conserves soil moisture.

7.

- Use of heavy machinery/implements when soil is wet.
- Continuous shallow cultivation.

8. **Maintenance practices of a plough**

- Cleaning after work
- Storing under shed
- Tightening loose nuts and bolts
- Replacing worn out parts/ repair broken parts.
- Greasing moving / rotating parts
- Oiling / painting for long storage.

9. **Destroy soil organisms**

- Loss of soil nutrients / loss of soil fertility.
- Accumulation of some nutrients to toxic level e.g. potash.
- Destruction of organic matter by oxidation / burning.

10. **Reasons for secondary cultivation**

- Control weeds
- Control pests and diseases
- Incorporates organic matter in the soil
- Improves physical condition / form required tilth
- Make appropriate tilth for planting certain crops e.g. ridging, rolling, leveling.

11. **Benefits of minimum tillage**

- Saves time and costs of production.
- Maintains soil structure
- Minimizes soil erosion.
- Less laborious

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- Conserves moisture
 - Minimizes root damage.
12. i)
- Cost / expenses / when distance is short
 - Availability of skilled worker.
- 13.
- Disc harrow / rotavators
 - Spring tine harrow / rigid tine harrow / ox – tine harrow.
 - Spike toothed / peg – toothed harrow
 - Chain harrow / zigzag harrow
14. Application of herbicides.
- Use of mulch / timely cultivation
 - Use of cover crops
 - Uprooting / slashing / grazing to control weeds

WATER SUPPLY IRRIGATION

- 1.
- Transporting in containers.
 - Piping / use of channels.
2. Dams/ ponds, Tanks / containers
- 3.
- Draining the land / Any method of drainage.
 - Growing water tolerant crop e.g. rice
 - Ridging / making furrows.
- 4.
- Boiling water / filtration
 - Use of chemicals / chlorination / alum / soda ash
 - Aeration of water / sedimentation.
5. a) Dams / weirs tanks / containers
- b)
- Piping / use of pipes
 - Use of channels
 - Use of containers by vehicles , animals
- 6.
- Centrifugal / rotor dynamic pump.
 - Reciprocating / piston pump
 - Semi – rotary pump
 - Hydram pump / hydraulic ram.

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- Rotary pump / hydraulic ram.

7.

- Quantity of the material / durability.
- Size of the pipe / diameter / length
- Working pressure / thickness of the wall.
- Colour of the pipe of the pipe.

8. a)

- Type of soil
- Type of crop to be grown
- Source of water / quality of water
- Size of land to be irrigated
- Capital available / topography of land
- Profitability / viability of the enterprise.

b)

- Water is evenly distributed over the required area
- Less water is used / it more water economical.
- Causes less soil erosion in sloppy areas compared to surface irrigation
- Leveling of land is not necessary
- Help to clean the plant for ease of transpiration / photosynthesis.
- Soluble fertilizers can be applied with irrigation water more easily than in surface irrigation.

c)

- Use of sprinkler
- Use of hose pipes
- Use of watering cans.

9.

- Quality of material used for making the pipes / durability / resistance to rodent attack.
- Size of pipes i.e. diameter / length.
- Working pressure of the pipe / thickness of the pipe wall.
- Colour of the pipe.

10.

- Centrifugal motor dynamic pumps
- Piston / reciprocating pump
- Semi rotary/Hydrum
- Rotary pump.

11. **Uses of water**

- For diluting / mixing chemicals used to control pests, diseases, weeds for watering livestock e.g. drinking.
- Watering plants e.g. irrigation

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- In processing farm produce e.g. coffee, carrot hides.
- Domestic use e.g. drinking cooking
- For rearing fish.
- Mixing concrete in construction
- Recreation e.g. swimming pooling.
- Cooking and running machine engines water treatment plant and process.

12. a)

- To remove bad smell/taste from water thus makes it fit for human consumption.
- To kill disease causing organism that thrive in dirty water.
- To remove solid particles/impurities.
- To remove excess chemicals e.g. fluorine/that may cause hardness in water.

b)

- For diluting/mixing chemicals e.g. pesticides.
- Acaricides, herbicides, fungicides.
- For watering livestock.
- For irrigation/watering crops.
- For processing farm produce e.g. hides, coffee, carrots.
- For washing farm tools equipment and farm structures e.g. animal houses.
- For domestic use e.g. cooking, drinking, sewerage disposal.
- For rearing fish.
- Mixing concrete/mortar in construction.
- Recreation purposes e.g. swimming pools.
- For cooling/running machines

c)

Stage I: Filtration at intake: Water is made to pass through a series of sieves.
Sieves trap large solid impurities.

Stage II softening of water Soda ash (sodium hydrogen carbonate is added into the water).

Stage II coagulation and sedimentation Alum/aluminum sulphate is added into the water. Alum facilitates coagulation and sedimentation of solid.

Stage iv , filtration Water is passed through a filtration tank, that removes all solid impurities.

State v; chlorination Small amounts of chlorine solution is added to the water chlorine kills pathogens disease causing organisms.

Stage vi storage Treated water is stored in clean tanks before use/distribution.

13.

- Size of the farm
- Type of enterprises in the farm

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- Source of water
- Method of conveyance of water

15.

- It is water economical
- Can use water under low pressure
- Does not encourage fungal diseases
- Discourage growth of weeds between the crop rows

16.

- Repair/ replace worn out parts
- Regular cleaning
- Regular painting to prevent rusting

17. **The water treatment process**

Stage 1. Filtration of an intake

Water is made to pass through a series of sieves where large particles of impurities are trapped

Stage 2 Softening of water

Water is let to circulate in a tank (open) and mixed with soda ash to soften it

Stage 3 Coagulation and sedimentation

Alum is added to water to facilitate sedimentation and coagulation. Water stays here for 36 hours

Stage 4 Filtration

Water passes into infiltration tank where all the remaining solid particles are removed.

Stage 5 Chlorination

Filtered water enters chlorination tank where a small amount of chlorine is added to kill disease agents.

Stage 6. Storage

Treated water is stored before entering distribution channels for use.

18.

- Furrow irrigation/ corrugate irrigation
- Basin irrigation/ flood irrigation

19.

- Open ditches
- Underground perforated pipes
- French drains
- Cambered beds/ pumping water
- Breaking hard pans/ sub- soiling

20. (a) Dams/ weirs/ streams/ river ponds/ lakes

(b)

- Kill pathogens
- Remove sediments
- Remove bad smell/ taste
- Remove chemical impurities

21. Reasons for water treatment

- To remove bad smell and taste
- To kill harmful micro-organisms which thrive in dirty water e.g. bacteria.
- To remove solid particles e.g. soil, sand sticks
- To remove excess chemical impurities e.g. fluorides to soften water

SOIL FERTILITY

1. Benefits of farm yard manure

- Improves soil structure/ water holding capacity
- Supplies more than one plant nutrient
- Longer residual effect
- Promotes microbial activities in the soil
- Locally available
- Imparts a dark colour to the soil which help in temperature regulation
- Buffers soil PH/ improves cation exchange capacity

2.(a)

- Ability of a soil to produce and sustain high crop yields continuously by providing adequate moisture, nutrients, oxygen, space and freedom from pests and diseases damage.

(b)

- Adequate depth
- Enough aeration
- Adequate moisture
- Sufficient nutrients
- Freedom from pests and diseases

3. (a)

(i)

- Foundation materials e.g. maize stalks accept material that occur in abundance
- Kitchen refuse, leaves
- Farm yard manure/ any well rotten manure

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- Ash/ potassic fertilizer
- Top soil
- Leaves cover/ trash

(ii) Importance

- Provide source of micro- organism
- Enrich the manure with K and P.3
- Provide food for micro- organism that bring about decay

- (b) (i) Sharp pointed stick checks the temperature of the manure during its formation to avoid over heating
- (ii) Prevent entry of too much water causing water logging, poor decomposition and leaching of nutrients.

4. (a)

- (i) A- Pit method
B- Heap system

(ii)

- Drainage/ free draining soils on site
- Type of soil
- Topography/ slope should be gentle

5.

A crop is grown and then ploughed and incorporated into the soil while it is still green

7. (a)

- Farm yard manure (FYM) improves soil structure/ improves soils water holding capacity.
- Supplies a variety of plant nutrients
- Has a longer residual effect
- Promotes microbial activities in the soil
- It is locally/ easily available
- Moderates the soil pH/ increases cation exchange capacity

(b)

- Species of animal from which the waste is collected
- Food material eaten by the animal
- Material used as bedding
- Method of storage of the FYM
- Age of the animal which produces the waste matter
- Age of the farm yard manure.

(c)

- It is bulky hence difficult to apply/ laborious/ difficult to store/ handle
- It has less nutrients per given volume
- May spread weed seeds
- May spread crop diseases e.g. black scurf in potatoes
- Releases nutrients slowly

8.

- Leaching/ soil erosion
- Plant uptake
- Volatilization/ burning/ denitrification

9.

- Leaching/ soil erosion
- Change of soil pH
- Burning of land/ volatilization/ denitrification/ accumulation of salts
- Fixation of nutrients/ nitrogen lock up
- Uptake by plants/ weeds/ continuous cropping
- Soil capping/ formation of hard pan
- Presence of soil – borne pests/ monocropping

10.

- Rapid growth rate
- Production of abundant foliage
- Rich in plant nutrients/ leguminous/ rich in nitrogen
- Ability to decay quickly
- Adaptable to wide range of conditions/ hardy.

11.

- Method of storage
- Degree of decomposition
- Type of organic matter used
- Presence/ absence of non – biodegradable materials
- Amount and type of inorganic fertilizers added.

12.

- Appropriate depth
- Proper drainage/ good water infiltration
- Well aerated/ good water holding capacity

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- Adequate nutrients/ correct pH
- Free from excessive infestation of soil borne pests and diseases.

13.

- Cultivation along the slopes/ across contours/ along river banks
- Continuous cropping with annual crops
- Pulverization of soil due to over- cultivation
- Burning of vegetation/ overgrazing

14.

- Improves soil structure
- Reduces leaching
- Improves water holding capacity
- Increases microbial activities
- Increases cation exchange capacity
- Improves nutrient status upon decomposition
- Burning of vegetation/ overgrazing

LIVESTOCK PRODUCTION (COMMON BREEDS)

1.

(i) Dromedary *camelus dromedaries*

- Can stay for long periods without water
- Can withstand extremes of temperature
- Able to eat dry feeds
- It has thick hide and deep layer of subcutaneous which insulates in against effect of radiant heat
- Feet have large surface area which enable it walk on sand without sinking

(iii) **Characteristics of indigenous cattle**

- More resistant to tick borne diseases/ trop diseases
- Can withstand high temperatures
- Can survive on poor quality pastures
- Can walk long distance in search of pasture and water
- Have lower feed and water intake

(iv) **Features of exotic beef cattle**

- Blocky/ square/ rectangular in shape
- Low set/ short legs
- A fleshy body
- Thick neck

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- Smaller udders

3. (i) Jersey verses Friesian

- Jersey requires less food.
- Jersey are more tolerant to heat.
- Jersey can survive on poorer pastures.

ii. Characteristics of goats that adopt them rid areas

- Good foragers hence survive on poor pastures.
- Ability to eat dry feeds
- Heat tolerant tissues.

iii. Ways of improving indigenous cattle production.

- Cross breeding with exotic breeds with superior characteristics.
- Supplementary feeding.
- Proper control/prevention of livestock diseases.

- 1 (i) Jersey
- (ii) Friesian

4.

- Are more resistant to tick-borne diseases/tropical diseases.
- Can withstand high temperature.
- Can survive on poor quality pastures
- Can walk for long distances in search of pasture and water.
- HAVE lower feed and water requirement.
- Layers/Broilers/Dual purpose

AGRICULTURAL ECONOMICS

1. The four types of records that a farmer should keep

- Field operations records.
- Production records
- Consumable goods inventory.
- Permanent goods inventory.
- If there is no inventory records.
- Market records
- Labour records
- Breeding records.
- Breeding records
- Benefits of labour

2. Reasons for keeping health record

- For use in selection and culling of animals on health grounds.
- Help to detect theft on the farm
- Help in planning and making sound decision

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- Provide information for income tax assessment
- Help to determine how creditworthy is a farmer.
- Shows the assets and liabilities of the farmer
- Determine the value of the farms.
- Used in comparing performance between enterprises/seasons/different farms.
- Help to solve disputes on the farm.
- Provide a history of the farm
- Help to determine terminal help trace history of diseases for better approach in treatment.
- Show when to vaccinate, deworm.
- Show costs on health of the animals in assessing/no fat ability of animals.

3: Conditions under which opportunity cost is zero

Where there are no alternative enterprises to choose from/competing for resources available

This is the value of foregone best alternative/choice. When resources are free/unlimited.

9. If there is no choice. When resources are free/unlimited.

SOIL FERTILITY II- INORGANIC FERTILIZERS

1.

They are highly soluble water/ easily leached/ no residual
Have a scorching/ burning effect on crops
They are highly volatile

2.

5 kg of K_2O is contained in 100kg of 25: 10:5

Therefore 400kg of fertilizer contains $\frac{400 \times 5}{100} = 20\text{kg}$

3.

Help in Carbohydrate formation and translocation/plays role in plant metabolism
Assists in uptake of Nitrates from the soil
Imparts diseases resistance to crops
Strengthens straws of cereals
Formation of chlorophyll
Help neutralization of organic acids in plants.

4.

- Lodging in crops/ weak stems
- Scorching of leaf tips and margins.
- Chlorosis.
- Premature leaf fall.
- Mottling/brownish spots on leaves

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- Stunted growth.

5.

- Leaching
- Plant uptake.
- Volatilization/burning

6. (a)

- Needed in protein synthesis.
- Hastens vegetative growth in crops.
- A constituent of the chlorophyll molecule
- Increases the size of cereal grain
- Regulates the availability of phosphorous and potassium.
- Improves succulence in crops.

(b)

- Chlorosis of leaves
- Stunted growth.
- Premature ripening of crops.
- Premature leaf fall
- Formation of purple pigments.

7.

- Lodging? Weak stems? over succulence of stems
- Scorching of leaves
- Delayed maturity.
- Excessive foliage growth.
- Cause blossom end rot in tomatoes

8.

- Amino acids/protein synthesis
- Increase oil content and hormones
- Needed in formation of chlorophyll.
- Aid in Nitrogen fixation in legumes
- Needed in carbohydrates metabolism.

9.

Leave have purple pigmentation
Stunted growth
Poor root development
Lodging/weak stems in cereals
Poor seed and fruit formation

10.

- Broadcasting
- Foliar application

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- Side/row/basal application
- Fertigation
- Hole placement/drilling

11. a)

Macro nutrients are required by the plants in relatively large quantities, but macro nutrients are required in small quantities

b)

- Elongation of apices of roots and shoot,
- Strengthens plant cell wall
- Help in protein formation
- Help in formation of the middle lamella
- Used in cell division

CROP PRODUCTION- PLANTING

1.
 - Early maturity of the crop
 - Plant assumes desired shape and size.
 - Possible to obtain two or more varieties of oranges on one root stock
 - Highly yielding.
 - Maintains parental genetic characteristics.
 - Possible to propagate seedless e.g orange varieties.
2. (a)
 - Seed economical
 - Easy to achieve correct spacing
 - Make subsequent operations easyb)
 - To obtain optimum plant population/efficient land use.
 - Avoid competition for resources.
 - Control spread of pests and diseases.
 - Makes subsequent operations easy/easy to mechanize.
3.
 - Soil moisture content
 - Seed size.
 - Soil type.
 - Type of germination
 - Soil moisture content
 - Soil fertility.
 - Machinery to be used.
 - Intended use of the crop.
 - Growth habit
 - Prevalence of pests and diseases.
 - Cropping system used.
4.
 - To enable the crop to smoothens.
 - For maximum utilization of seasons rainfall.
 - Enable the crop evade serious attack by pests and diseases.
 - Enable maximum utilization of available nutrients before they are leached.
 - Reduced competition for labour.
 - To get good market price.
5.
 - To improve Nitrogen fixation in the roots of legumes
6.
 - Highly yield.

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- Eases control of weeds/pests/diseases.
- Aids in soil and water conservation.

7.

- Variety of maize
- Method of planting
- Purpose of the crop.
- Soil fertility
- Soil moisture content
- Number of seeds per hole.
- Use of machinery in subsequent operations

8.

- Rainfall pattern/reliability.
- Variety of beans
- Incidence of pests and diseases attack.
- Expected harvesting time.

9.

- To obtain high yield.
- Reduce pest/disease /weed attack
- To ensure high germination percentage,
- To obtain high quality produce.
- To obtain high quality produce.
- To obtain seeds suitable to the particular area.

10.

- Fertility status of the soil
- Use moisture content.
- Use to which the crop is to be put, machinery to be used.
- Growth habit of the crop/variety
- Number of seeds per hole.
- Prevalence of certain diseases, pests
- Crop stands either pure or mixed.

10.

- Intensive land use.
- Control soil erosion.
- Reduce cost of production
- Ensure early pasture establishment.

11. (a)

- Induces lodging
- Difficult to carry out field operations.
- Reduced yields.
- Low quality produced/ small cobs.

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(b)

$$1 \text{ ha} = 10,000\text{m}^2$$

$$1\text{m} \times 0.5\text{m}$$

$$= 20,000 \text{ plants/ha}$$

12.

- Thinning
- Gapping

13.

- Free from pest/ disease attack
- Resistant to diseases
- High yielding
- Has high quality produce
- High rooting ability
- Early maturing

15.

- Rainfall availability
- Incidence of pests and diseases
- Expected weather conditions during harvesting
- Market demand of the crop
- Type of soil

16.

- Uses higher seed rate
- Not easy to use machines
- Not easy to establish the right population
- Overcrowding in some areas
- Uneven germination
- Difficulty to carry out field practices

17.

- Soil type
- Soil moisture content
- Size of seeds
- Type of germination

18.

- Enable the crop to withstand competition from weeds
- Enable the crop to escape attack by pests and diseases
- To better utilization of nutrients in the soil
- For better utilization of available rainfall
- To get good market
- To reduce competition for labour

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- To time harvesting to occur during appropriate weather conditions

CROP PRODUCTION – NURSERY PRACTICES

1.

- (i) Reduce competition/ ensure healthy seedlings
- (ii) Prepare the seedlings for the harsh conditions expected in the field

2.

Removal of the shade
Reduce the frequency of watering

3.

- Free from pest/ diseases
- Resistant to diseases
- Adaptable to different soil pH
- Can withstand water loggings
- Compatible with several scions

4.

- Requires less skills to establish
- Easier to prune
- Higher yielding in the early years of bearing
- Does not allow accumulation of CBD due to frequent removal of old stems
- Comes to bearing earlier

CROP PRODUCTION – FIELD PRACTICES

1.
 - Allow adequate light penetration into the plant
 - Improve quality of the fruits
 - Reduce incidence of pest and disease attack
 - Enable effective use of chemical sprays
 - Facilities easy harvesting
2.
 - To prevent sprouting
 - Reduce pest attack
 - To prevent fungal diseases/ rotting
3. (a)
 - Facilitates field practices e.g spraying and harvesting
 - Improves the quality of crops by preventing soiling
 - Enable crop grow in the required direction
 - To improve yield
 - To control pests and diseases(b)

Passion fruits, tomatoes, cucumber, grapes, tea, pumpkin, water melons, Boungain vellia, garden peas, yams, gourds, indeterminate beans, roses, money plant, hops, okra
4.
 - Moisture content of grains
 - Colour of leaves and grains
 - Intended use of the crop
5.
 - Improves soil aeration upon decomposition
 - Reduce toxicity of plant poisons upon decomposition
 - Reduce soil erosion
 - On decomposition it improves soil structure
 - Modify the soil temperature
 - Add nutrients on decomposition
 - Improves water infiltration
 - Increases microbial activity
 - Control weeds
 - Reduces evaporation of water
 - Buffer soil pH upon decomposition
6.
 - Reduce run - off thus increase soil moisture
 - Reduce evaporation thus maintaining high moisture in the soil
7. (a) Growing of different types of crops on the same piece of land following

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orderly sequence.

(b)

- Ensure maximum utilization of nutrients
- Control soil – borne pests and diseases
- Control weed
- Add nitrates into the soil
- Control soil erosion
- Improve soil structure

(c)

- Deep rooted crops alternate with shallow rooted ones
- Crop easily weeded are alternated with those difficult to weed
- Crops of the same family should not succeed each other
- Heavy feeders should come first in the cycle
- Include a legume crop

CROP PRODUCTION V- VEGETABLES

1. (a)

- Irregular watering of the crop/ water stress
- Excess application of Nitrogen in early stages
- Deficiency of element calcium in young fruits

2. (a)

- Regular watering
- Addition of calcium into the soil
- Mulching

(b)

- Regular watering
- Addition of calcium into the soil
- Mulching
- Avoid excess Nitrogen in the soil

(c)

- Size of fruits
- Degree of ripeness
- Damage of tomatoes
- Shape of fruits

LIVESTOCK HEALTH – INTRODUCTION

3. (a)

- They have a longer productive life
- Produce high quality produce
- They are less expensive to keep

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- They are high yielding
- Do not spread diseases others/ man
- They breed regularly

2 (b)

- Foot and mouth diseases
- Anthrax
- Rinder pest
- Lumpy skirt disease
- Rabigs

3.

- Prevent deficiency disease
- Improves animals and ability to resist diseases

4.

- General farm hygiene/ cleanliness of houses. Feed/ water trough – W proper carcass
- Disposal; to destroy pathogens
- Isolation; prevents spread of the diseases
- Drenching; to control internal parasites
- Treat sick animals; prevent spread of the diseases
- Vaccination; develop resistance against diseases.
- Control vectors, prevent transmission of diseases
- Prophylaxis; avoids infection
- Slaughtering en- mass; prevent spread of diseases
- Proper breeding; control breeding diseases
- Quarantine; avoid spread of the diseases
- Hoof trimming; minimize occurrence of foot rot
- Proper housing; avoid predisposing causes of diseases

5.

- Through vectors/ external parasites
- Ingesting contaminated feed, water, milk
- Inhaling contaminated air
- Through surgical and instrument e.g. scalpels, needles, docking knife etc
- Mating
- Suckling young ones

LIVESTOCK HEALTH – PARASITES

2. (a) (i)

- Red water (Babesiasis)
- Gall sickness (Anaplasmosis)

(ii)

- East Coast Fever (ECF) Theileriosis
- Red water (Babesiasis)
- Nairobi sheep disease
- Texas fever

(b) Two hosts

2. (a)

- Anaemia
- Irritation/ scratching
- Loss of hair
- Wounds on skin
- Presence of parasites on the body

(b)

- Regular drenching
- Rotational grazing
- Drain swampy areas
- Use of latrines by farm workers
- Use appropriate chemicals on swampy areas/ sue CuSO_4 solution
- Burn heavily infested pastures
- Plough infested pastures
- Peripheral fencing
- Use of ducks on snail

(c)

(i) Pig

(ii) Fresh water snail

3.

- Larvae climbs on host
- Larva feed on 1st host
- Larva drop on the ground and moults into a nymph
- Nymph climbs – onto 2nd host
- Nymph feeds on 2nd host
- Nymph drops on the ground and moult into an adult
- Adult climb on 3rd host
- Adults feeds and mate on 3rd host
- Mated, engorged female drops and lay eggs on the ground
- Egg hatches into larvae

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4.

- Burning infested pastures
- Hand picking and killing of ticks
- Rotational grazing
- Double fencing of pastures
- Zero grazing
- Ploughing infested pastures

5.

- Proper meat inspection
- Proper disposal of infested carcass
- Routine deworming of livestock/ human beings pets
- Rotational grazing
- Proper use of latrines
- Proper cooking of meat
- Burning/ ploughing infested pastures

LIVESTOCK HEALTH – NUTRITION

1.

- Production ration is the feed given to an animal over and above maintenance level in order to produce a given product

2.

- Water soluble vitamins
- Fat soluble vitamins

3.

- Size of the animal
- Level of production
- Type of feed eaten
- Ambient temperature
- Physiological status of the animal
- Species of the animal

4.

(a) Roughage is a feed with high fibre content and low energy content; while concentrates is a feed with high protein/ energy content and low fibre content

(b)

- Synthesis/ formation of various products e.g. milk, meat
- Foetal development
- For work. Provide draft energy

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- Growth

4.

- Bulky
- Low digestibility
- Low in energy/ protein content
- Highly fibrous
- Plant origin

5.

- Raw materials for synthesis of livestock products e.g milk, eggs
- Growth of cells
- Production of energy
- Formation of enzymes. Hormones and antibodies
- Repair of worn out tissues

6.

- Body weight/ size
- Age of the animal
- Work done
- Level of production
- Physiological condition e.g. pregnancy
- Weather conditions ambient temperature

7.

- A component of milk
- Formation of the skeleton / teeth
- Blood clotting
- Nerve functioning/ control milk fever

8.

- Nutrient requirement of the animal
- Age of the animal
- Type of animal whether ruminant or non- ruminant
- Availability of feedstuffs
- Cost of the food stuffs

10. (a)

- Production ration is the feed given to an animal over and above maintenance level in order to produces a given product

(b)

- Body weight/ size
- Age of the animal
- Work done
- Level of induction

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- Physiological condition e.g. pregnancy
- Weather conditions/ ambient temperature

11.

- Provide energy/ maintenance of body temperature
- For growth and repair of body tissues
- For maintenance of good health
- Production of various products
- Enhance reproduction

LIVESTOCK PRODUCTION (III)–SELECTION & BREEDING

1.

By concentrating genes from the parents to the offspring's

2.

- Reddening and swelling of the vulva
- Clear mucus discharge from the vulva
- Frequent bellowing
- Cow mount others
- Cow stands still when mounted by others
- Cow becomes restless
- Slight rise in body temperature
- Slight drop in milk in lactating cows
- Loss of appetite urinating frequently

3.

- It is uneconomical to keep a bull
- May lead to uncontrolled mating
- High risk of transmission of breeding diseases
- Only a small number of cows can be served

4. Mating of two unrelated animals of the same breed

5. (a)

- (i) A- Oviduct/ fallopian tube
B- Ovary
C- Uterus
D – Vagina

(ii) A – passage of Ova from the ovary to the uterus
-Site of fertilization

B – Production of ova
Production of female sex hormones

(b)

(i) Artificial insemination

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(ii) Natural method

(c) 19 – 23 days/ 3 weeks

Increased ability and performance of the offsprings above the average of the parents

7.

- Should be of age/ mature S – 7 months old/ 90 – 100kg live weight
- Good mothering ability
- Able to grow fast
- Good conformation
- With no physical defects
- Healthy
- Has 12- 14 teats
- Highly prolific
- Able to withstand heat stress during mating

8.

(i) Mating closely related animals e.g brother and sister

(ii) Mating unrelated animals but within the same

(b)

(i) Harmful traits/ diseases can spread fast

(ii) Requires trained personnel/ skills

(iii) Laborious

(iv) Low chance of conception due to poor timing/ death of sperms,

(c)

- Good body conformation/ wedge shaped
- Docile/ mild temperament
- Her ancestors should have a record of high milk production
- Free from physical deformities
- Good health
- Good size/ weight for the breed

LIVESTOCK PRODUCTION (IV) – REARING PRACTICES

1.

- He goats
- Male calves
- Rams

2.

- Bad smell
- Overheating
- Lack of food/ water
- Sick/ infertile queen

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- Attack by predators/ destruction of brood
- Too much noise

3. J (i) Docking/ Tailing

(ii)

- To facilitate mating/ to ease mating
- To control blow fly attack
- To ensure uniform distribution of fat

(iii) 1-2 weeks after birth

(iv)

- Use of elastrators and rubber ring
- Use of sharp knife
- Use of a docking iron

(v) Hoof trimming

(vi) The animal would become lame

Foot rot may develop

(vii) In a sitting position facing away from the person shearing

4. To make the bees less aggressive

5.

- Old age
- Low fertility
- Difficult in furrowing
- Less milk production
- Passing undesirable characteristics to its young ones
- Poor hearth
- Poor mothering ability

6. (a) Thick yellow secreted by the mammary glands within the first week after giving birth

(b)

- Has a laxative value that helps to cleanse the bowels
- Rich in antibiotics that confers artificial immunity
- Highly nutritious is necessary for fast growth of the calf
- It is highly digestible, hence appropriate for the underdeveloped calf's digestive system

(c)

- Bucket feeding/ Artificial feeding
- Natural method/ calf sucks the dam

6.

- Restlessness
- Loss of appetite
- Swelling of vulva

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- Udder enlarges
- Mucus discharge from the vulva
- Presence of colostrums in the teats
- Relaxation of ligaments on either side of the pelvic bones
- Cow isolate herself

7.

- To make them docile
- To improve growth rate
- Control breeding diseases
- To prevent inbreeding
- To control breeding
- To improve meat quality

8.

- Highly palatable
- Has high digestibility
- Rich in nutrients

9.

- Prevent the bees from absconding
- To avail food during time of seed scarcity
- To attract bees into a new hive
- To encourage multiplication of bees
- To ease access of feed

10.

- Proper feeding
- Control internal parasites
- Control external parasites
- Vaccination
- Zugging/burlyng
- Hoof trimming
- Provision of adequate clean water
- Treat in case of infection

12.

- Routine feeding
- Application of manure/ fertilizer
- Cropping
- Maintenance of water flow/ changing the water regularly
- Control of predators

13.

- Restless
- Loss of appetite

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- Udder/ teats enlarge
- Present of milk in the teats 24hrs before farrowing
- Vulva enlarges
- Muscle on either side of the tail/ pelvic bones slackens
- Sow prepares a nest

14.

- During drought/ when there are no flowers
- Incase of a new colony
- When the colony is small/ to encourage fast multiplication

15.

- Deworming
- Flushing
- Crutching/ ringing/ cutting wool around reproductive organs
- Riddling
- Wiggling/ cutting wool around the face

16.

- Catfish
- Tilapia
- Trout
- Carps
- Bass
- Tench
- Blue gill
- Nile perch

FARM STRUCTURES

1.

- (i) 0.5m/50cm
- (ii) To prevent dampness
- (iii) To allow proper lighting
- (iv)
 - Nearness to the milking shed
 - Well drained area/topography.
 - Soil type
 - Security
 - Wind direction

2.

- It is cheap
- It is attractive
- It is easily available
- Easy to work with

3. Siting

- Should be near the homestead for security
- Should be in an accessibility place
- Sheltered from strong wind
- A well drained area
- On the leeward side of the farm house
- Where there is large space for possible expansion in future

Selection of building materials

- Use easily available materials
- Choose durable materials
- Consider the cost of the material
- Consider the skills needed to work with the materials
- Materials to choose from; iron sheets, wire mesh off cuts, timber, concrete

Requirement of the hutch

- Should be leak proof
- Should be well ventilate
- Should be free from draught
- Should be easy to clean
- Should have raised floor
- Floor should allow free drainage

3.

- To prevent warping/ bending / twisting
- To prevent rotting/ damage by fungi
- To prevent it from pest attack

FREE AGRICULTURE K.C.S.E 2012 to 2015

- To enable timber to achieve its maximum strength

4.

Advantage: It's more effective in preventing animals from forcing their way through the fence

Disadvantage: It's more expensive than plain wire
The bars may cause injury to the animal/ destroy wool.

5.

- Allows air circulation in the house
- Controls temperature in the house
- Prevent humid condition inside the house

7. 1 cement, 3 and, 4 ballast/ gravel

(b) 1/ cement

(c)

- It is durable
- It is fire proof
- It is easy to clean
- Can be moulded into various shapes

9.

- Should be raised from the ground
- Should be leak proof
- Should be easy to clean
- Should be well ventilated
- Should be rat proof/ vermin proof
- Should be easy to load/ offload
- Strong enough, spacious

10.

- Reduces incident of accident when using them
- Ensure efficient usage of the structure
- To increase its durability
- To give high re- sale value of the structure

10.

- Topography/ drainage
- Accessibility of the pen
- Wind direction
- Direction of the sun
- Security of the calf
- Location of existing, structure/ amenities
- Space for future expansion

(b)

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Strength/ durability of the material
- Cost of the material
- Availability of the material
- Availability of capital
- Workability/ skills required in using the material
- Type of pens (permanent/ temporary)
- Climate of the area
- Safety of the calf

(c)

- Repair/ replace worn out/ broken parts
- Clean the pen regularly
- Ensure drainage system is working
- White wash the walls

11.

- Partial burning/ charring of the posts
- Cut the top in a sloping manner
- Cover the top with a plastic/ metal sheet
- Reinforce with concrete

12.

- Well ventilated
- Easy to clean
- Leak proof
- Well drained floor
- Draught free
- Spacious
- Well lit

13.

- Repair/ replace broken parts
- Regular cleaning to remove dirt
- Dust/fumigate/ spray to control parasites and diseases
- Apply old engine oil on timber parts
- Ensure good drainage around the house
- Maintain a footbath at the entrance

14.

- It is more durable
- Its stronger
- Its fire resistant
- Its not attacked by vermin's

15. (a)

- Provide security from thieves, predators

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Enable paddocking/ rotational grazing/ mixed farming
- Control parasites and diseases by keeping away foreign animals
- Show boundaries between farms
- Hedges act as wind breaks
- Have an aesthetic value
- Hedges help to conserve soil and water
- Hedges may be source of fruits/ fodder/ firewood
- Provides privacy
- Enables isolation of animals for different purposes

(b)

- Wind direction
- Soil type
- Security
- Accessibility of the structure
- Location in relation to existing structure
- Topography/ drainage
- Government policy/ regulations
- Purpose of the structure
- Position of the sun
- Proximity to social amenities
- Space for future expansion
- Farmers tastes and preference

16. Adds beauty to the farm

- Source of firewood
- Source of fodder/ mulch material
- Serves as a wind break
- Control soil erosion
- Controls animal/ human movement
- Provides security/ privacy
- Mark farm boundaries

17.

- Top bars can easily be removed and replaced when inspecting the combs
- Easy to construct
- Ensure high quality honey
- Easy to harvest
- Help to avoid mass killing of bees
- Possible to exclude the queen from the honey combs

18.

- Paint metal tanks
- Regular washing of the tank

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Repair any leakage

19. (a)

- **Claw hammer;** for driving in and out nails from wood
- **Tape measure;** to measure the required sizes of timber
- **Tin snip;** for cutting iron sheets for roofing
- **Clamp/ sash camp;** for holding tightly together pieces of wood when cutting/ joining
- **Handsaw;** for putting timber to the required size
- **Wood chisel/ Brace/ hand drill;** for boring holes in wood
- **Mallet;** for hitting the chisel when boring holes
- **Pliers;** for cutting wires
- **Jack plane;** For smoothing timber surface
- **Marking/ mortise gauge;** for marking points for cutting / planning on timber
- **Ball pen hammer;** for straightening/ shaping metal sheet
- **Try square/ combination square;** for determining right angles on cutting points of timber

(b)

- Clear the site to be fenced
- Use a string to layout the fence line
- Determine the position of posts using a tape measure
- Dig the holes using a hole digger/ claw bar
- Use a ruler to determine the right hole depth
- Obtain the right length of the posts using a tape measure
- Obtain the posts to the required depth using a handsaw
- Put concrete at the bottom of the hole
- Place the posts in the holes
- Ensure posts are vertical/ right angles
- Fill up the hole with soil/ concrete
- Firm the soil/ concrete in the hole using a ramming rod
- Heap soil/ concrete at the base of post

20.

- It is more expensive
- It requires a lot of skills to work with
- Prone to rusting
- Its not easily available
- Its heavy and difficult to transport

SOIL AND WATER CONSERVATION

1.
 - Reduce the speed of water hence erosive ability
 - Trap soil being carried by moving water
 - Increase infiltration of water hence reduce surface run-off
2.
 - Reduce the volume of run-off due to increased infiltration
 - Acts as windbreakers
 - Reduce the impact of raindrops on the soil
 - Tree roots bind the soil particles together reducing its erodibility
 - Reduce the impact of raindrops on the soil
 - Reduce speed of run-off
 - Improve soil structures thus reducing the erodibility of soil
3.
 - To prevent loss of plant nutrients
 - To maintain good soil structures
4.
 - Reduce the speed of run-off thus lowering the erosive power of water
 - Reduce impact of raindrops thus reducing splash erosion
 - Cover the soil protecting it from wind erosion
 - Grass roots bind the soil particles together reducing the erodibility of soil
 - Reduce speed of run-off, thus reducing the erosive power of water
 - Organic matter from grass improves soil structure thus reducing erodibility of the soil
5.
 - Reduce evaporation
 - Reduce surface run-off
6.
 - (a) Afforestation is the practice of growing trees in areas where they had not existed,
 - (b) Re-afforestation is the practice of growing trees where they have been harvested.
6.
 - Topography of the area
 - Rainfall intensity/ amount
 - Type of soil
 - Soil depth
 - Vegetation cover

- Farming practices

WEEDS AND WEED CONTROL

1.

- Prevents weeds from establishing in the field
- Prevent allelopathic effects of weeds
- Reduce the cost of crop production
- Reduce multiplication and spread of the weeds
- Reduce spread of pests/ diseases for which weeds acts as alternate hosts
- Reduce competition between weeds and the crop
- Avoid contamination of crop with weed seeds
- Prevent injury to the farmer/ livestock

2.

- Uprooting
- Cultivation
- Mulching
- Application of herbicides
- Slashing

3. (i) G – Couch grass (*digitaria sealarum*)

H- Sodom Apple (*solanum incarium*)

(ii)

- Compete for resource with cultivated crops
- It increases the cost of crop production
- Lowers the quality of pastures

(iii) It has deep underground structures difficult to remove

4. 2001 Disadvantages of weeds

- Compete for resources with crops
- Increase the cost of crop production
- Some may contaminate the crop lowering the quality.
- Irritate the workers lowering labour efficiency.
- Some weeds are parasitic to crop pests and diseases
- Some weeds are allelopathic.

4.:

- Produce large quantities of seeds
- Seeds remain viable for along time
- They have effective mechanisms of dispersal
- Some weeds have the ability to propagate both by seeds and vegetatively.
- They have elaborate root system.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Some have underground structures difficult to control.
- Some are able to survive with limited nutrients.
- Some are able to compress their life cycle.
- Some weeds are allelopathic.

6.:

- (i) Thorn apple (*Datura stramonium*)
- (ii) Annual weed
- (ii) Poisonous if eaten

LAND TENURE AND REFORMS

1.

- Difficult to control pests and diseases/diseases and parasites spread faster.
- Land disputes are common
- No motivation to conserve land
- No motivation to make long term investments
- An individual cannot use land as security to acquire
- Difficult to control breeding in livestock.

2.

- Saves time and money
- Makes it easy to have a sound farm plan e.g rotation programme.
- Eases soil conservation.
- Eases supervision as all enterprises are at one place.
- Facilitate mechanization.

3.

- To settle the landless.
- To ease population pressure.
- To increase agricultural production.
- To improve people's standard of living.

4.

- Communal tenure
- Co-operative tenure

PESTS AND DISEASE CONTROL

1.
 - (i) Weaver bird
 - (ii)
 - By eating the grains
 - Opening the cab to water that leads to rotting of the grains,
 - (iii)
 - Use of scare crows
 - Trapping
 - Use of explosives
 - Use of resistant varieties
 2.
 - (i) Cutworm/Agrotis
 - (ii) Cuts the stem causing lodging.
 - Use of appropriate insecticide.
 - Removing it and killing it.
 3.
 - Its cheap
 - Saves on labour.
 - Does not cause environmental pollution.
 4.
 - Destroy the alternate host
 - Practice crop trapping.
 - Destroy the crop remains
 5.
 - Seed dressing using appropriate chemical.
 - Use of resistant varieties
 - Practice close season.
 - Practice field hygiene/destroy crop residual.
 6.
 - Concentration of the pesticide
 - Weather condition during and soon after application
 - Stage of development of the pest.
 - Persistence of the pesticide.
 - Formulation of the pesticide.
 - Mode of action of the pesticide.
- (a)
- Biting and chewing
 - Piercing and sucking
- (b)
- Store hygiene/ensure store is clean

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- Proper drying of grains.
- Ensure store has proper ventilation.
- Keep store airtight.
- Clear the bush around the granary.

8

- Proper plant nutrition, increase disease resistance/control deficiency diseases.
- Crop rotation breaks life cycle of pathogens
- Rogueing, prevent further spread of the disease.
- Use of disease free planting materials: prevents introduction of pathogens in the field.
- Close season breaks the life cycle of pathogens.
- Timely /early planting; help crop to establish early before attack.
- Proper spacing; creates unfavorable conditions for some pathogens.
- Weed control; eliminate weeds that could be alternate hosts for particular pathogens.
- Resistant varieties; ensure crop is not attacked by the pathogen.
- Use of clean equipment/tools, prevent spreading of the disease from one plant to the other.
- Quarantine; prevent spread of the pathogen from one farm to the other.
- Heat treatment; kills the pathogen.
- Pruning; creates unfavorable microclimate for some pathogens/prevent spread of the disease.
- Destroy crop residue; kills the pathogen.
- Control the vectors, prevent further spread of pathogens.

9.

- Plant resistant varieties
- Practice crop rotation.
- Practice field hygiene.
- Use of clean planting materials.
- Trim roots of suckers before planting.
- Plant Mexican marigold close to the banana plants.

10.

(a) Point at which damage on a crop by pest is beyond tolerance and has to be controlled.

(b)

- By contact
- Through ingestion
- By suffocation

(c)

- Pesticides are expensive.

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- Pesticides requires special skills to handle. They are poisonous/environmental hazard.
- Some are non-selective/they kill useful organisms.
- Some pest develop resistance to the pesticide.

11.

- Lower the quality of farm produce.
- Lower they yield.
- Increase cost of production
- Some pests are vetors of crop diseases.
- Some pest lower labour efficiency by irritation.

12.

- Early planting
- Rogueing
- Trap cropping

13.

- Attack by nematodes.
- Root nodules due to nitrogen fixing bacteria.

FIELD PRACTICES (II)

1.

- Field birds; *accept any appropriate example.*
- Shoot fly;
- Stem borers.
- Diseases; Rust, Downy mildew, Ergot, smut

2.

- The crop has well developed roots to absorb the nitrogen before it can be lost
- Maize is growing fast and requires a lot of nitrogen.

3. Head smut/smut

4.

- Puttingrat guards/ ensuresgranary is vermin proof.
- Ensure store is clean.
- Clear the vegetation around the granary.
- Harvest the grains on time to prevent attack in the field.
- Proper drying of grains.
- Use rat traps
- Use of cats.

5.

- Weevils,
- Flour beetles,
- Lesser grain borer;
- Greater grain borer;
- Khapra beetle;

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- Warehouse moth;
- Saw toothed grain beetle;
- Angoumois grain moth.

(i) K, G, H, J

(ii) Failure of boll to develop

(iii)

- Do not pick the lint when it is wet
- Pick on weekly bases
- Avoid dry twigs or leaves contaminating the cotton
- Do not use sisal bags to hold cotton as the sisal fibres may contaminate lint,

(iv)

- Cotton lint
- Cotton seed

6.

- Early/ timely planting
- Practice crop rotation
- Practice field hygiene
- Destroy crop residue
- Practice rogueing
- Use healthy / certified seeds
- Practice close season
- Use appropriate pesticide to control the vector
- Uproot volunteer crop

7. (i)

- Field hygiene/ destroy crop residue/ rogueing
- Use of trap crop
- Destroy alternate host/ weeds
- Timely harvesting

(ii)

- Stalk borers
- Armyworms
- Aphids
- Dusty brown beetles
- Weevils
- Termites
- Locusts
- Leaf hoppers/ grasshoppers
- Cystic beetle

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- Great/ lesser grain borer; cut worms
- American bollworms

9.

- Uprooting weeds cultivation
- Mulching; t application of herbicides
- Slashing/ cutting the weeds

10.

- Control the vector
- Use certified seeds
- Use of resistant varieties
- Practice field hygiene
- Early planting
- Rogueing
- Practice close season

FORAGE CROPS

1.
 - Mixed pasture is more nutritious
 - Economies on nitrogenous fertilizers by N- fixation
 - There's higher yield per unit area
 - There is security incase of failure of one crop
 - It provides a good soil cover hence control erosion
 - There is maximum utilization of soil nutrients
3.
 - Leads to exhaustion of pasture crop
 - May lead to high incidence of parasites and diseases
 - Leads to soil erosion
4.
 - To raise the carbohydrate level for proper fermentation
 - To increase the nutrient level of silage
 - To increase the palatability of silage
 - To restrict growth of undesirable microorganisms
4. (a)
 - (i) Under- sowing is the establishment of a pasture crop under a nurse crop
 - (ii) Over- sowing is the establishment of a high quality pasture crop on an existing grass pasture(b)
 - Slashing/ moving
 - Uprooting
 - Use of selective herbicides(c)
 - Replenish soil nutrients
 - Hasten growth/ increase herbage yield
 - Improve the nutritive value of the crop
 - Amend soil physical and chemical properties
 - Enhance decomposition of organic matter by micro- organism/ increase microbial activity
5.
 - Rotational grazing/ controlled grazing
 - Proper stocking rate
 - Conserve excess pasture
 - Timely defoliation
 - Practice zero grazing
 - Graze different classes/ species of animals
- 6.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Stage of growth at harvesting time
- Species of the forage crop used
- Duration of storage
- Soil fertility where the crop was grown
- Weather conditions during drying
- Length of drying period
- Pest/ disease attack on the crop
- Method of storage

8.

- There is intensive/ efficient use of land
- Reduces the cost of forage production
- Control soil erosion due to ground cover
- Ensure early pasture establishment

8. (i) Seedbed preparation

- Practice early seedbed preparation/ during the dry period
- Clear all the vegetation/ stumps
- Carry out primary tillage
- Dig deeply to remove all weeds/ perennial weeds
- Carry out secondary tillage
- Seedbed should have a medium tilth
- Prepare furrow/ holes for planting
- Spacing between furrows 90- 100cm for cuttings/ 90 – 100cm x 50cm for splits

(ii) Planting

- Plant at the onset of the rains/ early planting
- Select desirable Napier grass variety for the ecology of the area
- Use healthy planting materials
- Use cuttings/ canes or splits for planting
- Cutting/ canes should have 3-5 nodes
- Select cutting from mature canes/ stems
- Place planting materials in the furrows/ holes
- Cover the material with soil to the appropriate depth

(iii) Fertilizer application

- Apply phosphate fertilizer at planting
- Apply farm yard manure/ composite manure before planting
- Rate of organic manure should be 7 – 10 tons/ ha
- Apply organic manure after harvesting and dig it into the soil every year
- Top dress with Nitrogen and potassium 6-8 weeks after planting

(iv) Weed control

- Control weeds by; cultivation, uprooting, slashing, selective/ appropriate herbicides, intercropping with legumes that smother the weeds

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Practice timely weed control
- (v) Utilization**
- Cut and feed it to ruminants.
- Defoliate/ cut at the right stage of growth/ 3 - 5 months old/ when stems are 1 – 1.5 m high
- Cut the stems at 2.5 – 5 cm above the ground surface
- Use sharp panga for cutting
- Conserve excess as silage
- Chop Napier grass into small pieces before feeding
- Napier grass can be dried and used as mulch

FREE AGRICULTURE K.C.S.E 2012 to 2015

9. (a)
▪ Lucerne, silver/ green leaf, desmodium siratro, stylo
- (b)
▪ Reduce build- up of parasites and diseases
▪ Animal waste is evenly distributed in the fields
▪ Excess pastures can be conserved
▪ It is easy to carryout management practice
▪ Pasture is given time to regenerate
▪ Pasture is maximally utilized
- (c)
▪ Sprinkling some water
▪ Reduce compaction
▪ Fill the silo and seal it from air rapidly
10.
▪ Control soil erosion
▪ Ensure adequate pasture for animals
▪ Increases the useful life of the pasture crop
11. (a) Practice of coating legumeseeds with a nitro- culture/ rhizobium bacteria
(b) Growing alegume pastures over an existing grass pasture

LIVESTOCK HEALTH (III)

1. Oral (mouth), broken skin (wounds), nasal, ocular (eyes), anal, ears, reproductive organs, umbilical cord
2.
▪ Routine vaccination
▪ Slaughtering all infected animals
▪ Impose quarantine
▪ Treatment of wounds
3.
▪ Difficult in breathing/ respiratory problem
▪ Dullness/ birds stands with eyes closed
▪ Anorexia/ loss of appetite
▪ Nasal discharge
▪ Sneezing
▪ Droopy wings
▪ Yellow/ greenish watery diarrhoea
▪ Drop in production

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Soft- shelled eggs
- Torticollis/ bending of neck

4.

- Wounds on the udder/ teats
- High milk production
- Very old cows
- Unhygienic handling of the cow
- Poor udder placement/ pendulous udder
- Incomplete milking
- Genetic factors
- Early and late lactation period
- Poor milking technique

5.

- Blood/ pus in milk
- Painful udder/ teat/ animal kicks when teats are touched
- Swollen/ inflamed udder
- Clots/ thick milk
- Watery milk
- Blocked teat canals
- Drop in milk production
- Foyer

6.

- Anthrax
- Rinderpest
- Foot and mouth
- Lumpy skin disease

7.

- Routine vaccination
- Isolate infected birds/ kill all infected birds
- Disinfect the poultry house before introducing a new stock
- Obtain chicks from reliable sources
- Imposition of quarantine

8. (i) Fowl pox

- May lead to death of the birds
- Farmer incur extra expenses in its control

(ii)

- Kill and dispose off all infected birds
- Routine vaccination
- Control biting parasites
- Treat wounds

9.

- Wounds in infected hooves
- Foul smell from infected hooves
- Spend most of the time lying if both hind legs are infected
- Grazes when kneeling if fore legs are infected
- Loss of appetite
- Lameness/ limping
- Swelling of infected hooves
- Emaciation

10. (i) Protozoa/ coccidia/ Eimeria species

(ii) Bacterium/ clostridium spp/ clostridium chauvei

11. (i)

- Long hooves
- Damp environment/ poor hygiene
- Injury on hooves

(ii)

- Anaemia
- Fever
- Constipation
- Lack of appetite/ anorexia
- Inability to move
- Reduced milk production
- Animal becomes aggressive

12.

- General farm hygiene to kill the pathogens
- Isolate sick animals to prevent spread of the disease
- Deworm the animals to control endo-parasite
- Treat the sick animals to prevent spread of the disease
- Vaccinate the animals to develop resistance against disease attack
- Control vectors to prevent spread of the disease
- Routine administration of drugs/ prophylaxis to prevent infection
- Proper feeding to prevent deficiency / nutritional diseases
- Mass slaughter of infected animals to prevent spread of diseases
- Proper selection and breeding to control breeding and inheritable diseases
- Proper housing to avoid predisposing animal to diseases
- Foot trimming to minimize occurrence of foot rot disease
- Imposition of quarantine to prevent spread of diseases

13.

- Lack of stiffness of the carcass/ lack of rigor mortis
- Production of tar- like watery blood from all body openings
- Extensive bloating

14. (a) Bacterium/ *Brucella abortus*

(b)

- Spontaneous abortion/ premature birth
- Retention of after birth abortion
- Bareness
- Yellowish brown slimy odourless discharge from the vulva

(c)

- Use artificial insemination
- Cull infected animals
- Vaccinate all young animal
- Avoid direct contact with aborted foetus/ after birth
- Observe hygiene/ proper disposal of aborted foetus

15.

- Routine vaccination
- Improve quarantine
- Kill and dispose off infected animals
- Isolate sick animals
- Disinfect the animal house

16.

- Swollen lymph nodes
- High fever
- Profuse salivation
- Lachrymation/ production of tear
- Difficulty in breathing
- Bleeding in the vulva and mouth
- Coughing

17. (a) Deficiency of calcium in the wood of the cow

- Muscular twitching
- Staggering as the animal moves
- Animal become unconscious
- Loss of appetite
- Cessation of body functions
- Dullness

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Animal lies with the neck twisted for the head to lie on the shoulder

(c)

- Feed animal with diet rich in calcium
- Partial milking
- Intravenous injection with calcium salts/ calcium borogluconate

POULTRY PRODUCTION

1.

- Debeaking
- Making laying boxes/ nests dark
- Feeding on balanced diet
- Hanging green vegetation to keep birds busy scatter grains on the floor
- Isolating and treating cannibalized birds
- Control external parasites
- Keeping birds according to age/ avoid introducing new birds
- Provide adequate space

2.

- Size/ weight/ volume of egg
- Colour
- Cleanliness
- Shell quality e.g. broken, rough
- Shape of the egg

3. (a)

- Ensure brooder corners are rounded
- Provide enough brooding space accordingly. Clean and disinfect the brooder house/ equipment.
- Provide wood shavings/ proper litter on the floor
- Maintain appropriate temperature range according to the age of the chicks
- Temperature during the 1st one week should be 32 – 35⁰ c, then reduce accordingly
- Spread sheets of papers and sprinkle chick mash on them
- Provide fire guard around the heat source
- Maintain proper ventilation by adjusting the openings
- Provide fresh, adequate and qualify feed/ chick mash
- Provide brooder with reliable and appropriate lighting/ dim light
- Provide adequate and appropriate waters/ feeders according to age/ number of chicks
- Remove dead chicks from the brooder
- Control parasites by applying appropriate pesticides

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Control diseases appropriately
- Isolate and treat sick chicks immediately
- Keep proper records
- Gradually change the feed in the last one week in the brooder
- Debeak 8 – 10 days towards the end of brooding
- Provide adequate clean water all the time

(b)

- Combs and wattles are small, dry and cold/ combs have white scales
- The space between the pelvic bones is narrow 2-3 fingers cannot fit in the space between the pelvic bones
- Plumage is shiny, well preened/ sometimes moulting
- Yellowish pigmentation in the vent, shanks and beak
- Space between the keel bone and pelvic bone is small / 3-4 fingers cannot fit in the space
- Eyes are dull and yellow
- Abdomen is hard
- The layer is lazy and dull
- Hen becomes broody

4.

- Repair/ replace broken parts of the house
- Maintain a footbath at the door of the deep litter
- Clean and remove any dirt/ cob webs
- Fumigate against pests and diseases/ spraying/ dusting
- Paint some parts of the house/ apply old

5.

- Dirt
- Abnormal size/ undersize/ oversize
- Irregular/ broken/ soft shell
- Internal abnormalities e.g. double yolk, meat spots, poor candling qualities
- Poor storage/ long storage beyond 5 days

6.

- Ensure laying nets are dimly lit
- Provide adequate laying nests
- Debeak perpetual egg eaters
- Collect eggs regularly and frequently
- Ensure birds get a balanced diet
- Keep birds busy by hanging green vegetation in the house/ scatter grains on the liner.

7. (a)

- Chicks move away from heat source
- Parting/ opening beaks

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Opening the wings/ spreading wings
- Making abnormal noise
- Drinking water excessively
- Chicks may lie flat on their bellies

(b)

- Requires less space/ high stocking rate
- Less loss of eggs
- Easier collection of manure/ easier accumulation of manure
- Easier protection of birds from vermin's/ parasites/ diseases
- Less labour requirement/ easier to collect eggs/ easy handling
- Cheaper to set up than battery cage/ low initial cost

8.

- Should be fertilized
- Should be medium in size
- Should be oval in shape
- Should not be cracked
- Should be clean
- Should be free from abnormalities/ blood spots/ meat spots/ double yolk
- Should be 5 – 10 days old
- Should have smooth shell

9.

- Regularly wash and disinfect the feeders/ waterers/ perches
- Replace old wet litter/ turn litter regularly
- Control visitors into poultry house/ use of footbath before entering the house
- Avoid pouring water on the litter/ avoid dampness
- Isolate sick birds
- Treat sick birds
- Dispose of dead birds immediately

LIVESTOCK PRODUCTION (vi)

1. (a)

K- Alveolus
L- Gland cistern
M – Teat cistern
N – Teat

(b)

(i) Milk let down is the flow of milk from the upper/ alveolar region of the udder to the gland and teat cisterns

(ii) Oxytocin

(c)

- Practice farm hygiene/ milk infested cows last/ use a separate udder towel for each cow/ use disposable udder towel
- Immediate treatment of infected cows to avoid spread of the diseases/ treat any wounds on the teat/ udder
- Practice teat dips after milking
- Applying milk salve/ jelly to prevent drying and cracking of teats
- Practice good milking techniques

2.

- Ensure calf is breathing/ administer artificial respiration.
- Clean mucus from the calf/ ensure cow licks the calf dry
- Cut and disinfect the umbilical cord
- Ensure calf sucks colostrums from the mother within the first 8 hours
- Feed the calf on colostrums for the first 4 days
- Keep records on the performance of the calf, introduce whole milk or milk replacer after the 4th day.
- Feed the calf with warm milk at regular intervals
- Feed the calf 2-3 times per day for the first 1 – 4 weeks
- Feed the correct amount of milk up to weaning
- Observe strict hygiene in calf
- Protect the calf against adverse weather conditions by providing proper housing
- Provide adequate clean water from the 3rd week
- Introduce palatable dry fed e.g. concentrates and good quality grass/ mineral salts from the 3rd week
- Keep calf in individual pens until it is 3-4 months old
- Spray / dip calf against external parasites
- Drench deworm calf against internal parasites
- Vaccinate the calf against prevalent diseases
- Release the calf from the pen occasionally for exercises
- Wean calf at 8 weeks/ 16 weeks

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Deworm the calf using appropriate methods
- Graze calf on good quality pastures ahead of adult cows
- Separate heifer calves at puberty to avoid in breeding
- Weigh the calf regularly
- Treat sick calves
- Put appropriate identification
- Defeat if necessary
- Change in feed be done gradually
- Serve at the right age/ weight/ 15 – 20 months/ 250- 280 kg

3. (a)

- Should be clean/ free from physical contamination
- Has the right consistency/ no water added/ true to the breed
- No strange odours/ no foul smell
- Free from diseases causing organisms
- White in colour/ normal colour/ not tainted
- Normal taste/ flavour

(b)

- The calf can be reared artificially even if a mother dies during birth
- Many calves can be reared at a time
- The calf can be given correct amount of milk
- It is possible to keep clear records on milk
- Yield

4.

- Free from diseases causing organisms
- Free from dirt/ foreign materials
- Appropriate smell and flavour
- Chemical composition within the expected standards

5.

(a) The milk secreted by the mammary glands within the first week of lactation/ thick yellow milk secreted by the mammary glands within the first week after parturition.

(b)

- Has a laxative and helps to remove the faecal meconium/ first faecal matter/ opens up the alimentary canal/ cleanse the digestive system/ prevent constipation
- It is rich in antibodies that offers temporary immunity against diseases
- It is rich digestible proteins/ fats/ minerals/ vitamin/ highly nutritious
- It is highly digestible

FREE AGRICULTURE K.C.S.E 2012 to 2015

- (c) Bucket feeding/ bottle feeding/ artificial method. Dam suckling the calf/ calf sucks the dam/ natural method

6.

- The milk person should be clean
- Test for mastitis before milking
- Milk person should be healthy
- Ensure utensils/ equipment are clean
- Ensure milking parlour is clean
- Ensure milking herd is free from zoonotic disease e.g. TB
- Cows with mastitis should be milked last
- Clean the udder
- Sieve the milk
- Cover the milk
- Avoid feeds/ weeds that would taint the milk just before milking
- Proper storage of milk/ cool, dry place.

7.

- Clean mucus from calf soon after birth/ ensure cow licks dry
- Ensure calf is breathing / administer artificial respiration
- Cut and disinfect the umbilical cord. Ensure calf sucks the mother to get colostrums, within the first 8 hours
- Feed the calf on colostrums for the first 4 days. Introduce whole milk/ milk replacer after 4 days
- Feed the calf with milk at body temperature
- Weigh the calf regularly
- Provide adequate clean water
- Introduce palatable solid feed e.g. concentrates, good quality grass from 3rd week
- Put appropriate identification marks
- Treat calf if sick
- Castrate male calf
- Provide proper housing for calf
- Keep calf individually up to the weaning time
- Control external parasite with appropriate method/ spray/ dip
- Drench/ deworm to control internal parasites
- Observe hygiene
- Remove extra teats
- Dehorn/ disband using appropriate method
- Release calf occasionally for exercises
- Keep records on calf performance
- Wean calf at 8 weeks/ 16 weeks
- Reduce amount of milk gradually towards weaning

- Train calf to take milk from a bucket.

FARM POWER AND MACHINERY

1. (a)

- It makes farm operations timely/ faster
- Economizes on labour
- Work is done more efficiently
- Reduces drudgery/ can accomplish heavy task
- Cheaper per unit work done in large
- Operations

(b)

- Turns/ inverts the furrow slices thus covering surface vegetation
- Cuts the furrow slice horizontally
- Holds the frame on to the mould board land side and share
- Absorbs thrust exerted on the mould board to make the plough stable

(c)

- Always clean after use
- Check the nuts and bolts and tighten if loose
- Repair broken parts
- Replace worn out parts/ lost parts
- Proper storage

2.

- Reciprocating mower/ cutter bar mower
- Power take – off shaft
- Sharpen the cutting blades
- Tighten loose nuts and bolts
- Lubricate moving parts
- Repair broken parts
- Replace worn- out parts
- Clean after a days work
- Check the tension of the v- belt and adjust accordingly
- Coat with old engine oil for long storage

3. (a)

- Ox- drawn mould board is lighter hence does not compact the soil as much as the tractor – drawn mould board plough
- Ox – plough can be use for more farm operations e.g. weeding, ploughing harvesting roots crops than tractors mould board.
- Ox- plough requires less skills to operate compared to the tractor plough
- Tractor plough is faster than ox- plough hence can plough a large area with a short time

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Source of power for ox- plough is not as reliable as the source of power for tractor plough
- Ox- plough relatively shallow compared to tractor plough that plough deeper
- Ox- plough can be used in steeper lands where tractor plough cannot plough
- Ox- plough requires more people to operate than tractor plough
- Ox - plough is cheaper to buy than tractor plough
- Ox - plough is cheaper to maintain than tractor plough

(b)

- Grease the moving parts (i.e. movable parts)
- Paint frame/ oil before long storage
- Tighten loose nuts and bolts
- Clean it after work
- Store in a shed
- Repair damage parts
- Replace worn- out parts

(c) **Advantages**

- It is cheaper
- Farmer does not incur maintenance costs of the tractor
- There is no risk of losing the tractor
- Farm operations are carried out faster

Disadvantages

- Tractor may not be available when required
- Some operators may charge high fees for the service
- Some operators can carry out poor quality work

4. (a)

- To reduce wear and tear/ increase durability
- Avoid rusting

(b)

- Sharpen blades if blunt
- Replace worn – out parts
- Clean the mower after use
- Tighten loose nuts and bolts
- Repair worn- out guard
- Paint when necessary
- Proper storage in shed

(c)

- Size of land
- Cost of the tractor
- Availability of services facilities/ availability of skilled labour e.g drivers, mechanics, spare parts

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Topography of the land
- Time available for the operation
- Availability of capital
- Number/ type of enterprises on the farm
- Availability of tractor hire services/ availability of other sources of power

5. (a)

- Incorporating manure into the soil/ stirring the soil
- Breaking soil clods
- Leveling the seed bed
- Covering broadcasted seeds
- Gathering/ removing trash

(b)

- Clean after use
- Store in a shed
- Repair/ replace any broken tine/ part
- Apply oil for long storage

(c)

- Ox- drawn harrow is cheaper than a tractor- drawn harrow
- Ox – drawn harrow can be used where tractor drawn harrows cannot be used

6.

- Clean after work storing in a shed
- Tighten loose nuts
- Replacing worn- out parts
- Greasing moving parts
- Oiling/ painting for long storage

7. (a)

- Decrease the angle of cut
- Use of hydraulic/ draught control lever
- Adding weights on the plough beam
- Raising the land wheel

(b)

- To avoid injury
- To make disc plough last longer
- To make disc plough work efficiently
- To reduce replacement/ maintenance cost

(c)

- Saves time
- Improves efficiency
- Saves on labour cost
- Reduces drudgery/ less tedious

FREE AGRICULTURE K.C.S.E 2012 to 2015

8. (a)

- To make a vertical cut into the soil that separate the furrow slice from the unploughed land
- Cut trash which would prevent the share from penetrating deep into the soil

(b)

- Lubricate moving parts
- Adjust the tyre pressure
- Replace worn-out tyres
- Tighten loose nuts and bolts
- Proper storage/ in shed
- Clean after use
- Paint the trailer
- Repair broken parts

9.

- Poor quality work unless under strict supervision/ poor skill of the operator
- Not readily available leading to late land
- Preparation
- Disc harrow
- Spring tine harrow/ rigid tine harrow/ ox- tine harrow
- Spike toothed harrow/ peg toothed harrow
- Chain harrow
- Rotavator
- Zigzag harrows

10. (a)

- If a farmer has inadequate capital
- If a farmer has little load to carry
- If the area is too steep to use a tractor

(b)

- Tighten loose nuts and bolts
- Straighten bent/ repair worn out/ broken parts
- Store in a dry place
- Apply oil on exposed metal parts when plough is not in use
- Repaint the appropriate parts when necessary

AGRICULTURE ECONOMIC III

1. (a)

- Training the labour force
- Giving incentives to employees
- Efficient supervision of labour
- Assigning specific tasks to workers
- Proper remuneration of a worker
- Provide efficient tools
- Mechanization of some operations
- Provide transport within the farm

(b)

This is the production in which each addition unit of input results to a larger increase in output than the proceeding unit of input

(c)

- Short term credit
- Medium term credit
- Long term credit

2. (a)

Opportunity cost is the value of foregone best alternative/ revenue foregone because of choosing the best alternative

(b)

Refers to the raw materials used up in the process of production, e.g. seeds, fuel fertilizer, feeds

(c)

Utility is the satisfaction one gets by using a commodity

3. (a)

- Borrowing from financial institutions/ individuals co-operative/ credit
- Person savings from ones income
- Inheritance/ gifts/ donations

(b)

- Assists the farmer in estimation of the required production resources
- Assists the framer when negotiating for farm credit
- Assist the farmer in making management decisions when comparing alternative projects
- Help to reduce uncertainty in farming process
- Encouraging farmers to efficient as to meet the projected targets
- Show progress/ lack of progress in the farm business/ FREE profit or foresee losses

4. (a)

Fixed costs are those that do not change with the level of production while variables costs are those that change with level of production

(b)

- Casual labour costs

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Fertilizer/ manure costs
- Costs of chemicals
- Cost of fuel
- Costs of repair of machinery
- Cost of hiring machinery

(c)

- Planning helps in selection of the best enterprise/ production option to undertake
- Helps in setting production targets/ goals
- Helps in allocation of resources to various enterprises
- Helps to identify the weakness and strength of farm operations
- Helps in timely and careful considerations in decision making
- Assists in negotiation of credit
- Maximize use of resources

5.

- To compare the performance of one farm with another
- To compare the performance of the farm between one season and another
- To compare the contribution of one enterprise and another on the same farm
- Acts as a measure of profit in a farm

6. (a)

- Zone I. For each addition unit of fertilizer applied, the output of potatoes increases at an increasing rate because resources are under utilized
- Zone II. For each additional unit of fertilizer applied, the out put of potatoes increases at a decreasing rate as the resources are utilized to the maximum
- Zone III. For each additional unit of fertilizer applied the out put of potatoes decreases since the resource is excessively applied.

(b) Zone II

(c)

- Flexibility in production.
- Produce under contract
- Input rationing/appropriate allocation of resource input.
- Insurance of the crop.
- Use of modern technology e.g. disease resistant varieties,
- Use of pesticides and fungicides, use of fertilizers.

6. (a)

- Training/ educating labour force
- Mechanizing farm operations/proper working tools.
- Giving incentive/proper housing/transport bonuses.
- Supervision of labour

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Assign specific tasks to workers.
- Proper remuneration
- Assigning tasks according to skills/ability/interest.

(b)

- Improvement in quality of work/quality of produce.
- Reduced time of performing a given work.
- Increase in returns per unit of labour.

(c)

- Amount of work/ task performed e.g. kg of tea leaves picked per head.
- Duration of work e.g. 30/= per hour/day.
- Labour regulations/market rates
- Nature of work
- Quality of work/skill of labour

8.

- Amount of money to be paid to them as wages.
- Number of people in the labour market
- Health /fitness of the work force
- Ability/skills of the labour force
- Working conditions/incentives
- Nature of work

(b)

- Adopting modern methods of production
- Flexibility in production methods
- Input rationing
- Taking insurance cover
- Selecting more reliable enterprises
- Diversification.

(c)

- Provides employment
- Source of food
- Earns the country foreign exchange
- Source-of raw materials for industries
- Provide market for industrial goods
- Source of income for farmers
- Improves infrastructure relationship.

9.

- Maximize profit
- To maximize cost of production
- To spread/ reduce risks

10. (a)

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Milk and butter
- Beef and hide
- Honey and wax
- Mutton and wool
- Pork/bacon and bristles
- Rabbit meat and skin/pelts
- Mutton and skin

(b)

- Feeds
- Pesticide
- Replacement stock
- Veterinary services
- Drugs
- Casual labour
- Packing materials e.g trays and carton boxes

(c)

$$(i) V = \frac{48-39}{2-1} = 9$$

$$W = \frac{32-27}{4-3} = 5$$

$$X = \frac{23-21}{6-5} = 2$$

$$Y = \frac{20-19}{8-7} = 1$$

L.C.C occurs where MRS =

$$L.C.C. = \frac{X_1}{PX_2}$$

Where Δ = change

P = price

X_1 = dairy meal

X_2 = Home made feed

$$\frac{8}{2} = 4$$

L.C.C. is where MRS = 4 I.E where 5 units of dairy meals are mixed with 23 units of home made feed.

FREE AGRICULTURE K.C.S.E 2012 to 2015

11. (a)

- Joint products
- Competitive products
- Supplementary products
- Complimentary products

- b) (i) Production function is the physical relationship between inputs and output (products). The quantity of product expected from a certain-combination of in put.
- (ii) Equi-marginal returns states that limited amounts of resources should be allocated in such away that the marginal returns those resources is the same in all alternative to which they are put.

12.

- Crop boards/marketing board/statutory boards.
- Commercial banks
- Cooperative societies
- Agricultural finance corporation (A.F.C)
- Settlement fund trustees
- Private money lenders/Non –Governmental Organizations (NGOS) insurance companies/ Hire purchase companies.

13 a)

- Costs of feeds
- Wages casual labour
- Cost of pesticide / chemicals/drugs
- Cost of insemination services

b)

- depreciation of machinery /buildings
- Land rent
- Salaries of regular/permanent labour
- Interest on borrowed capital

14 (a)

- Partial budget is prepared when minor changes are to be made in an enterprise
- Complete budget is spread when major changes are to be made in an enterprise/ starting arrow enterprise.

b)

- Diversification of enterprises to avoid total loss
- Insurance against tosses to maintain high liquidity/for compensation /to access money easily for any eventuality.
- Strategic farming/keeping crops produce and selling when prices are high.
- Flexible enterprises-engage in enterprises that can be stopped and changed
- Rationing of inputs use of insufficient inputs such that incase of failure losses are not too high.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Contracting for marketing-making arrangements with marketing agencies in advance.
- Selection of more certain enterprises/ selection of enterprises that can do well in an area/section of enterprises with ready market and less price fluctuation to reduce degree of risk.

AGRICULTURE ECONOMICS- FARM INPUTS

1. (a)

- Journal
- Cash book
- Ledger
- Inventory

b)

- Shows the assets and liabilities of the farm business,
- Shows farm net worth/net capital/ owners worth/ can be used to negotiate for credit/correct income tax assessment,
- Used in decision making;
- Show profit or loss.

AGRICULTURAL ECONOMICS (MARKETING &ORGANIZATION)

1. -Ten members

(b)

- Number of sellers
- Price of the commodity
- Availability/ seasonally of the commodity / weather.
- Technique of production
- Market information
- Transportation
- Price expectations of the commodity.
- Government policy/taxation.

(c)

- Price fluctuations/ low prices
- Lack of transportation.
- perish ability of some products
- poor storage facilities.
- Competition with substitute products.
- Delayed payments
- Some government policy

(d)

- Kenya Planters Co-operative Union
- Coffee Board of Kenya.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- a) Marketing is the performed of business activities that direct the flow of goods and services from producers to consumers.
- b) An imperfect market is g situation in which some buyers, some setters or both have limited knowledge of goods and services offered for sale at various prices.
- c) The price of mangoes will go down.
- (d)
 - Milk is highly perishable we hence needs cool storage which the farmer may not afford.
 - Lack of vehicles/poor roads/high transportation costs
 - Containers for handling milk are expensive.
 - Lack of market information
 - Price fluctuation.
 - Delayed payments.

2.

- a) The quantity of the product demanded varies inversely with the price 7 as the price declines the corresponding quantity demanded rises and as the price increases the corresponding quantity demanded falls. (Mark as a whole)
 - Advertisements/ sales promotions
 - Price of related goods, price of the goods,
 - Level of income
 - Price expectations
 - Tastes and preferences.
 - Tastes and preferences
 - Population,
 - Religious beliefs / taboos.
- (c)
 - Elasticity of demand of a commodity is the percentage in quantity demanded of a commodity resulting from a percentage change in existing price.
 - The degree of responsiveness of quantity demanded, to a- percentage change in existing price.
 - Buy farmers produce / delegates buying to an approved agent
 - Arrange for supply of inputs.
 - Fix prices of farm produce in consultation with the government.
 - Collect farm produce from areas of production to the stores/factories.
 - Inspect the production process to ensure and maintain quality of the produce.
 - Provide storage facilities for farmers produce.
 - Provide credits to farmers,
 - Provide technical advice on production/extension services where applicable
 - Process farm produce e.g. K.T.D.A.
 - Undertake research services on techniques of production

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Sell farm produce for farmers
- Regulate production to prevent under supply of the produce.
- Pack/package the farm produce.
- Invest accrues profits.
- Advertise/promote sales of the produce.

3.

- Members buy inputs at lower prices.
- Has easy access to credit facilities from the society.
- Gets advice/education from the society.
- Society share overhead costs with the member.
- Society share overhead costs with the member.
- Provide special services e.g. A.I, Banking
- Bar society bargains for better milk price on behalf of the member.

5.

(i) Most agricultural produce are perishable hence, farmers incur extra costs in transportation-processing/storage/incur losses due to spoilage.

- Most of them are bulky, occupy large space/expensive to transport.
- Poor transport network/lack of vehicles lead to loss due to spoilage.
- Most of them are seasonal; hence create storage problems/over supply at times leading to price fluctuation.
- Due to bulkiness they are expensive to store/difficult to store.
- Due to changes in market demand, there is time e.g. between decision to produce and actual availability of the product making it difficult to respond immediately to market demand.
- Change in supply due to under/over production/competition from cheap imports cause price fluctuation.
- Lack of perfect market information makes selling difficult/many farmers are ignorant on the prevailing prices of their produce in other parts of the country.
- Delayed payments lead to lack of capital for farm operations.

(b)

- Itinerant traders/ middlemen buy and resell produce from farmers.
- Processors / Manufacturers buy and process produce from farmers.
- Wholesalers- Buy produce in bulk from farmers and resell/ process.
- Brokers / Commission agents- acts on behalf of other business people for a fee commission.
- Co-operative societies/ unions- buy farmers produce locally.
- Marketing boards- promote production and marketing of agricultural produce/buy produce from farmers.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Retailers- buy from wholesalers and resell to consumers.

6.

- Price of related commodity.
- Price of the commodity.
- Size of population of consumers,
- Tastes and preference of consumers.
- Income of consumers.
- Advertisements/sales promotion
- Government policy
- Price expectations.
- Religious beliefs/religious taboos/ cultural beliefs.

7.

Elasticity of demand = $\frac{\% \Delta \text{ in quantity}}{\% \Delta \text{ in price}}$

$\Delta \text{ in price quantity} = 22 - 20 = \text{bags}$

$\% \Delta \text{ in price} = \frac{22 - 20}{20} = 10\%$

$\Delta \text{ in price} = 100 - 800 = 200/=$

$\% \Delta \text{ in price} = \frac{200 \times 100}{1000} = 20\%$

Elasticity of demand = $\frac{10\%}{20\%} = 0.5$

7. (a)

- Carrying out advertisement of the farm to increase demand
 - Finance agricultural activities.
 - Transportation of farm produce to areas of consumption.
 - Storage of farm produce to minimize losses/ as a marketing strategy.
 - Selling the produce on behalf of the farmers.
 - Packing/package farm produce to ease transport/ reduce storage space.
 - Grade farm produce to provide uniform standards/ cater for various consumers.
 - Process farm produce from areas of production for bulking/ transportation
 - Protect the farm produce from damage by use of chemicals/ insurance/ bear risks.
 - Buy from produce from the producers.
 - Gather, analyze and interpret market information to determine appropriate market and price.
- b)
- Co-operators pool their resources together to buy expensive machinery e.g tractor for use by farmers.

FREE AGRICULTURE K.C.S.E 2012 to 2015

- Provide education/technical information to members.
- Provide credits to members inform of inputs and cash.
- Negotiate for higher produce prices for members.
- Reduce overhead costs e.g transportation, storage and use of machinery.
- Bargain with suppliers to give discount on seed fertilizer and other farm inputs /provide inputs at lower prices.
- Provide employment for their members and other people.
- Benefits farmers from lower taxes charged
- Market farmers produce.
- Provide strong bargaining power for members on policy issues.
- Invests and pay dividends to members.
- Help to negotiate loans for their members without security.
- Provides banking services to it's members.

8.

- Perishability of the produce.
- Inadequate supply to spread supply over a long period.
- Drastic changes in supply/seasonality.
- Poor infrastructure e.g. poor roads no vehicles/piped water/ telephone / electricity,
- Bulkiness
- Lack of market information,
- Delayed payments.

FREE AGRICULTURE K.C.S.E 2012 to 2015

Mr. Rambo's farm balance sheet as at 31-12-95

ASSETS			LIABILITIES		
	Kshs	Cts		Kshs	Cts
Fixed			Long term		
Asset			Liabilities		
Buildings and structures	60000		Loan payable Dank	300,000	-
Five cows	250,000	=			=
400 layer	80,000	=			=
20 goats	30,000	=			=
Spray equipment	12,000	=			=
TOTAL	972,000				
Current			Current		
Assets			Liabilities		
Cattle feeds in store	10,000		Debts to co-p	20,000	=
Animal drugs in store	4,000	=	Bonus payable to workers	19,000	
Debts receivable	18,000	=	Breakages and repair	30,000	-
Cash at hand	20,000	=			
Cash at bank	30,000	-			
Total	82,000	=	Total	69,000	=
Total Assets	1,054,000		Total liabilities	369,000	
			Capital	685,000	
TOTAL	1054,000		TOTAL	1054,000	
ASSETS			LIABILITIES		
	Kshs	Cts		Kshs	Cts
Fixed			Long term		
Asset			Liabilities		
Buildings and structures	600000		Loan payable dank	300,000	-
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20 goats	30,000	=			=
Spray equipment	12,000	=			=
TOTAL	972,000				
Current			Current		
Assets			Liabilities		
Cattle feeds in store	10,000		Debts to co-p	20,000	=
Animal drugs in store	4,000	=	Bonus payable to	19,000	

FREE AGRICULTURE K.C.S.E 2012 to 2015

			workers		
Debts receivable	18,000	=	Breakages and repair	30,000	-
Cash at hand	20,000	=			
Cash at bank	30,000	-			
Total	82,000	=	Total	69,000	=
Total Assets	1,054,000		Total liabilities	369,000	
			Capital	685,000	
TOTAL	1,054,000		TOTAL	1,054,000	

FREE AGRICULTURE K.C.S.E 2012 to 2015

3.

- (a) **Cash account;** is a record that shows all cash receipts and payments,
Ledger: This is a financial book that shows all financial transactions in the farm business in a summarized form,
Balance sheet: This is a financial statement that shows the value of assets and liabilities of a business at the end of an accounting period.
Purchase order. This is a financial document drawn by a buyer to a supplier requested goods on credit.

(b)

- Permanent goods inventor
- Consumable goods inventory.

5.

PROFIT AND LOSS ACCOUNT FOR MRS. MBUTA'S FARM FOR THE YEAR ENDED 31-12-03			
PURCHASES AND EXPENSES	Shs. Cts	SALES AND RECEIPTS (CREDITS)	Kshs. Ct
Opening valuation	6,000.00	Pig sales	7,000.00
Wages	5,000.00	Piglet sales	4,000.00
Equipment	8,000.00	Maize sales	3,000.00
Pig feeds	4,000.00	Closing valuation	4,000.00
Drugs	3,200.00		
		Total	18,000.00
		Loss	8,200.00
TOTAL	26,200.00		2,600.00

Marks allocation

- (i) Correct columns (sales& receipts) ½ mk
(ii) Purchases & expenses ½ mk
(iii) Correct entries – Sales & receipts – ½ mk
Purchases & expenses – ½ mk
(iv) Correct totals – sales & receipts – ½ mk
(v) Purchases & expenses – ½ mk
(vi) Correct balance / profit/ loss ½ mk
3 ½ mk
(vii) Mrs. Mbuta made loss
6. (a) it is an entry in a financial statement showing the worth of all assets of an enterprise at the beginning of an accounting period
(b) (i)
- Issued to the buyer as evidence of cash payments for goods or services rendered

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- Used for requisition for the supply of goods/ services on credit
- Given to the buyer by the seller as evidence of goods supplied

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K.C.S.E AGRICULTURE PAPER 1

SECTION A (30 marks)

Answer all the questions in this section in the spaces provided

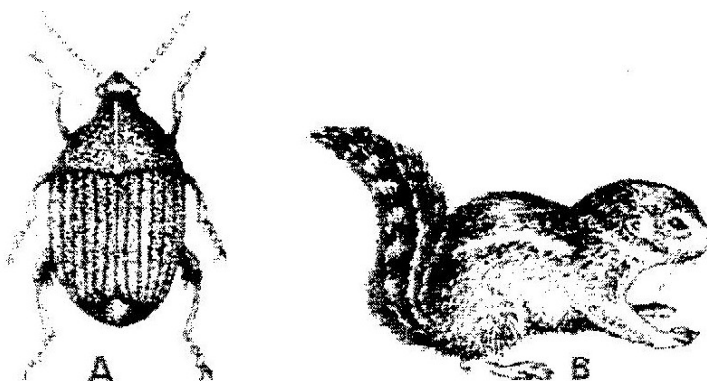
1. Differentiate between Olericulture and pomoculture as used in crop production
(1 mk)
2. State three ways by which biological agents can enhance the process of soil formation
(1 ½ mk)
3. State four advantages of drip irrigation
(2 mks)
4. State four advantages of adding organic manure to a sandy soil
(2 mks)
5. State two factors that would determine the amount of fertilizer to be top dressed to a crop in the field
(1 mk)
6. State four advantages of applying lime as a measure of improving soil condition
(2 mks)
7. Give four reasons for using certified seeds for planting
(2 mks)
8. Give four reasons for planting crops at the correct spacing
(2 mks)
9. State three effects of soil erosion
(2 mks)
10. Name four methods used to control weeds in pastures
(2 mks)
11. State two benefits of conserving forage crops
(2 mks)
12. Mention four practices that should be carried out to maintain grass pasture
(1 ½ mks)
13. Define the following terms as used in agriculture economics
 - (a) Gross domestic product (GDP)
(1 ½ mks)
 - (b) Per capita income
(½ mks)
14. What is profit maximization in agriculture economics?
(½ marks)
15. State four benefits of budgeting to a farm manager
(2 mks)
16. Give two reasons why farmers keep farm accounts
17. State activities carried out by young farmers club in Kenya
(2 mks)
18. State four ways by which afforestation helps in land reclamation
(2 mks)
19. State three advantages of multiple stem pruning over single stem pruning in coffee
(1 ½ mks)

SECTION B (20 mks)

Answer ALL the questions in this section in the spaces provided

20. Two maize pests are shown in the diagram below. Study them and answer the questions that follow,

s



FREE AGRICULTURE K.C.S.E 2012 to 2015

- (a) Identify the pests in the diagram labeled A and B (1 mk)
(b) at what stage of maize production does each damage the crop?
(c) Give one way of controlling each of the pests in the field
- 21 (a) state the law of diminishing returns in a production process
(b) Use the information on the table below to answer the questions that follow

Fertilizer input (units)	Maize yield (bags)	Marginal productions (bags)
0	50	12
1	62	12
2	66	4
3	68	2
4	69	1
5	69	0

The cost of fertilizer is Kshs 1500 per unit and the price of maize is Kshs 1200 per bag.

- (i) At what unit of fertilizer input should the farmer be advised to stop applying any more fertilizer to the maize? (1mk)
(ii) Give a reason for your answer in (b) above
(iii) Calculate the marginal return at the point of optimum production (1mk)
22. (a) Describe the procedure which should be followed in spraying a crop in tomatoes using a fungicide in powder form, water and a knapsack sprayer. (3mks)
- (b) Name one fungal disease of tomatoes that can be controlled using the above procedure. (1mks)
(c) State four safety measures that should be taken while spraying the crop with the fungicide. (2mks)

23. The diagram below shows a weed



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- a) Identify the weed (1mk)
- b) State two reasons for controlling the weed. (2mks)
- c) Name two herbicides that can be used to control the weed in a field of maize (1mk)
- d) At what stage of growth of maize should the weed be controlled using a post emergence herbicide?

SECTION C (40 MARKS)

Answer any TWO questions in this section in the spaces provided at the end of the section.

- 24. Describe the establishment of kales under the following sub – headings:
 - a) Nursery preparation
 - b) Establishment in the nursery
 - c) Management of seedlings in the nursery.
 - d) Transplanting of seedlings.
- 25. a) Outline the factors necessary for proper functioning of farmers' co-operative societies in Kenya. (5mks)
 - b) Explain how farmers overcome risks and uncertainties in a farming business.
 - c) Describe the steps farmers should follow when planning a farm business
- 26. a) List various methods of harvesting water in a farm
 - b) Outline farming activities which may encourage soil erosion.
 - c) Explain how various farming practices would help to conserve soil in a farm.

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K.C.S.E. PAPER 2 SECTION A (30 MARKS)

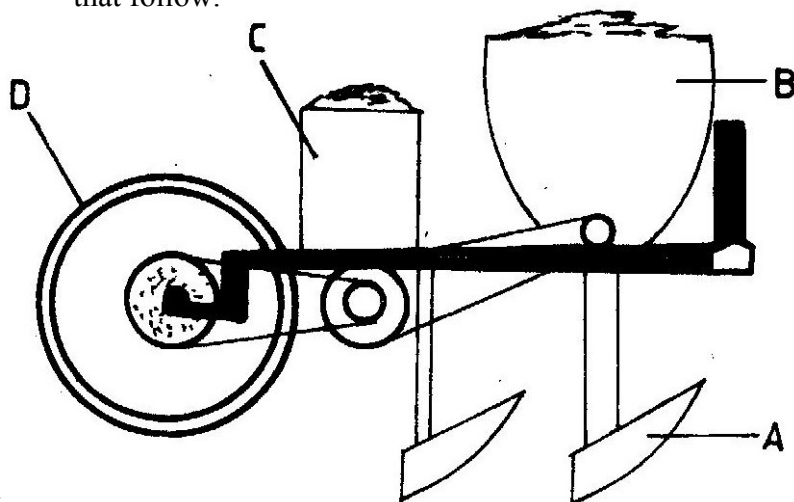
Answer ALL the questions in this section in the spaces provided.

1. Name a breed of sheep with a Lambing percentage of above 125 and whose fleece may be inferior due to black fibres. (1mk)
2. List two appropriate hand tools needed to finish off the handle of a fork-jembe. (1mk)
3. What is “cropping” in fish farming? (1mk)
4. State four functions of lubrication system in a tractor. (2mks)
5. Give four maintenance practices carried out on the water cooling system of a tractor. (2mks)
6. State reasons why a farmer would choose to use a disc plough rather than a mould board plough. (2mks)
7. State four construction features necessary in a fish pond. (2mks)
8. Give four ways in which disease causing organisms can gain access into a newly born calf (2mks)
9. State four ways of controlling tsetse flies. (2mks)
10. Give two predisposing factors of foot-rot in sheep. (1mk)
11. State four factors which should be considered when selecting dairy goats for breeding. (2mks)
12. Give four reasons why camels are suited to living in arid areas. (2mks)
13. Name two functions of the crop in the digestive system of chicken. (1mk)
14. State four methods of dehorning (2mks)
15. Mention six causes of stress to a flock of layers. (3mks)
16. State four functions of the worker bees in a bee colony. (2mks)
17. State four features of a good pig house. (2mks)

SECTION B (20 MARKS)

Answer ALL the questions in this section in the spaces provided.

18. (a) A diagram of a planter is shown below. Study it and answer the questions that follow.



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- (i) Identify the parts labelled A, B, C, and D, (2mks)

A _____
B _____
C _____
D _____

- (ii) State two maintenance practices carried out on the planter. (2mks)

- b) Study the diagrams of workshop tools shown below

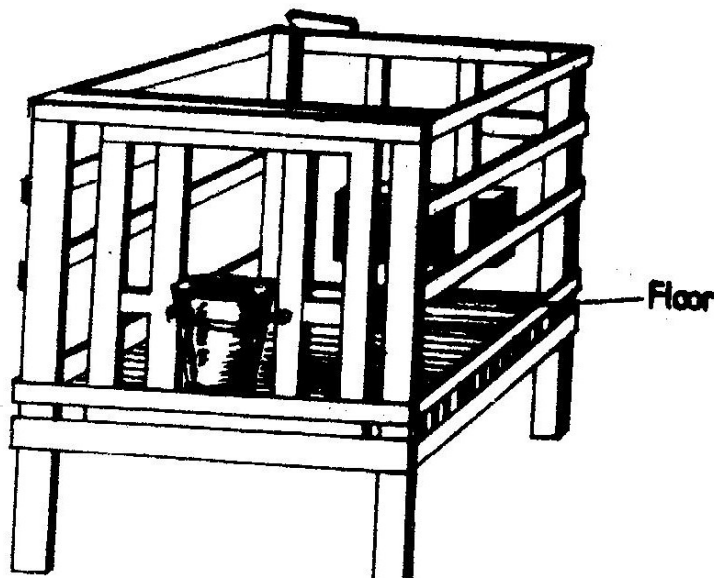


- (i) Identify the tools labeled E and F (1mk)

E _____
F _____

- (ii) What functional advantage does tool E have over tool F? (1mk)

19. The diagram below represents a calf pen. Study the diagram and answer the questions that follow.



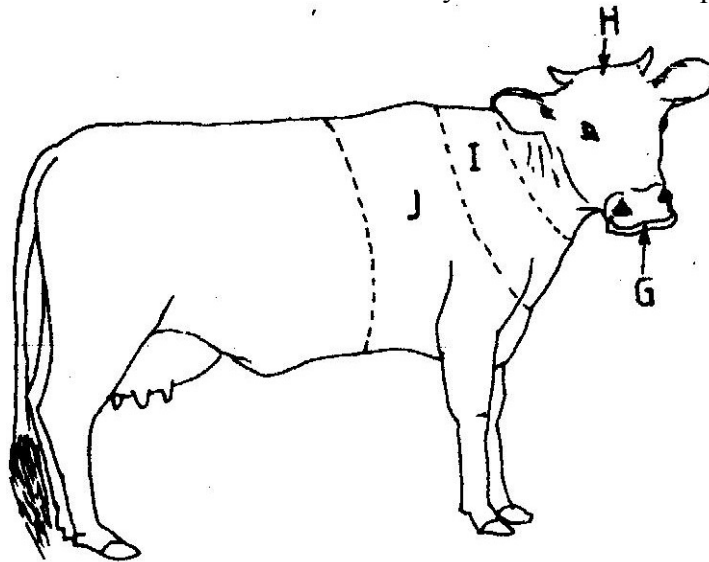
- (a) (i) Identify the type of floor. (½ mk)
(ii) How high should the floor be raised above the ground level? (1mk)

- (b) (i) Give one reason for having the floor of the calf pen raised. (1mk)
(ii) State three factors that should be considered in sitting the calf pen. (3mks)

20. (a) Define the term digestible Crude Protein (DCP) (½ mk)

- (b) A farmer wanted to prepare a 200kg of calf rearing ration containing 20% DCP. Using the Pears Square Method, calculate the amount of Maize containing 10% DCP and Sunflower containing 35% DCP the farmer would need to prepare the ration. (Show your work) (4mks)

21. A diagram of a cow is shown below. Study it and answer the questions that follow.



- (a) Name the parts labeled G, H, I and J.

G _____
H _____
I _____
J _____

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- (b) Name four parts of the animal preferred by a two host tick. (2mks)

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SECTION C (40 MARKS)

Answer any TWO questions in this section in the spaces provided at the end of the section.

22. a) Outline the procedure followed when hand spraying cattle to ensure effective use of acaricides to control ticks. (10mks)
- b) Discuss Foot and Mouth disease under the following headings:
- (i) Casual organisms. (1mk)
 - (ii) Livestock species attacked. (2mks)
 - (iii) Symptoms of attack. (4mks)
 - (iv) Control measures. (3mks)
23. a) Describe the management practices that a farmer should carry out to improve milk production in a low yielding herd of dairy cattle.(15mks)
- b) Describe the management practices that would ensure maximum yield of fish in a fish pond. (5mks)
24. a) What are the advantages of farm mechanization? (6mks)
- b) Explain the differences between a two stroke and a four stroke cycle engine. (6mks)
- c) Outline the daily maintenance practices that should be carried out on a farm tractor (8mks)

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K.C.S.E AGRICULTURE PAPER 1 MARKING SCHEME

SECTION A

1. Olericulture is growing of vegetables while pomocuhure is growing of fruits
(Mark as whole) 1mk
2.
 - Movements of animals in large numbers
 - Decomposition of plants and animals remains by soil micro-organism
 - Physical breaking of rocks by roots of higher plants
 - Man's activities e.g. cultivation, mining and road construction
 - Mixing up of soil burrowing animals e.g. earth worms and termitesAny 3 x 1 ½ = (1 ½ mks)
3.
 - Little amount of water is used/economics water use
 - Reduces incidences of certain leaf diseases/ Fugal discs
 - Can be used in sloppy areas because there is no risk of surface run off/no risk of soil erosion
 - Water under low pressure can be used
 - Some fertilizers and pesticides can be applied with irrigation water.
 - Minimizes growth of weeds (Any 4 x ½ = (2mks)
4.
 - Adds nutrients.
 - Increases microbial activity in the soil
 - Improves water holding capacity/reduces leaching/ improves capillarity
 - Buffers soil PH
 - Moderates soil PH
 - Moderates soil temperatures
5.
 - Type of cop-soil nutrient status
 - Stage of growth of crop
 - Expected yield
6.
 - Lowers soil acidity raises soil ph(modifies ph
 - Increases the calcium content of organic matter
 - Improves soil structure through flocculation of soil particles/improves drainage.
 - Facilities the availability and absorption of Nitrogen and prosperous
 - Improves legume nodulation and N fixation
 - Encourages multiplication of micro-organization in the soil
7.
 - Free from foreign materials e.g. weeds
 - Gives rise to vigorously growing plants

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- Have high germination percentage
 - Free from pest and diseases attack/healthy
 - True to type not contaminated
- (Any 4 x $\frac{1}{2}$ = 2mks)

8.

- Easy to determine plant population in a given area
- Ensure high quality produce.
- Ensures high production
- Facilities optimum use of nutrient moisture and light
- Permits use of machines when carrying on subsequent farm operations
- Facilities control of pests and e.g. ground

9.

- Pollution of the environment
 - Loss of plant nutrients and soil microorganisms
 - Siltation of dams and rivers
 - Reduction of soil depth
 - Destruction of farm structures
- (Any 4 x $\frac{1}{2}$ = 2 mks)

10.

- Burning pasture during grazing season, cultural method
- Moving/physical method
- Use of biological agents/ Biological methods

11.

- Provide feed during period of scarcity/ distribute available forage for livestock through the year
- Ensure better and proper utilization of land
- Can be sold for money

12.

- Top dress with N fertilizers occasionally/ apply manure
 - Control weed
 - Practice controlled grazing to avoid denudation
 - Cut back dry and unpalatable stems to encourage fresh re-growth/ topping
 - Re- seeding when necessary
 - Irrigation when necessary
 - Control of pest
- (Any 4 x $\frac{1}{2}$ = 2 mks)

13 (a)

GDP- is the sum total of goods and services produced by a country within a period of one year.

(1 x $\frac{1}{2}$ = $\frac{1}{2}$ mk)

16.

- To keep check on income and expenditure / profit and loss
- To know which activities are financially viable/ weakness and strength of the business

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- To obtain knowledge of the total value of the farm/ the value of assets and liabilities for farm planning
- To assess credit worthiness
- To provide information for tax purposes
- Organizing agriculture field days for the local community
- Participating in agricultural exchange programs both locally and internationally

(Any 4 x $\frac{1}{2}$ = 1 $\frac{1}{2}$ mk)

17.

- Organizing and participating in annual YFC rallies and camps
- Participating and completing in ASK show activities e.g livestock judging
- Planting trees/ carrying out agricultural project in schools
- Organizing agricultural field days for the local community.
- Participating in agriculture exchange programs both locally and internationally.

(Any 4 x $\frac{1}{2}$ = 2 mks)

18.

- Adds- organisms matters
- Recycles soil nutrients
- Helps to control soil erosion
- Improves drainage of swampy areas
- Plays an important part in the hydrological

B- Trapping and killing

Use of scare crows/ scaring

Poisoning/ rodenticide usage

1x1 = (1 mk)

21.

- (a) The law state that “if successive units of one input are added to fixed units of other inputs, a point is eventually reached where additional output per additional unit of input will decline” (mark as a whole)
- (i) At the end of the third unit of fertilizers application
- (ii) This is the least profitable unit of fertilizer application beyond which there would be a loss
- (iii) Marginal returns (MR) at the point of optimum production
 $MR = Kshs\ 1200 \times 2 = 2400/=$

22. (a)

- Read the label/ the manufactures instruction
- Measure the requirement amount of fungicide
- Place it into a container and mix thoroughly
- Powder has dissolved completed/ has formed slurry
- Pour the mixture into the knapsack sprayer though the sieve
- Spray the mixture onto the crop

(b) Blight (late or early) powdery mixture

(1 mk)

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(c)

- Spray following the direction of the wind
- Wear protective clothing
- Avoiding eating or smoking while handling fungicides
- Avoid spillage of the fungicide/ avoid contaminating the environment
- Do not suck/ blow a blocked nozzle (Any 4 x Vi = (2 mks)

23.

(a) blackjack/ *Bidens pilosa*

(b)

- To avoid competition for nutrients, moisture and light
- Black jack seeds may contaminate some crops/ farm practice
- Blackjack may be an alternate host to some pest e.g aphids which may attack crops like beans
- Black jack seed prick and irritate workers (Any 2 x 1 = 2 mks)

(c) - MCPA

- 2, 4-D

(d) At what stage of growth of maize should the weed be controlled using a pest?

- 10 to 15 cm high

- 2 to 4 weeks after emergence

1 x 1 = 1 mk

SECTION C

24.

- Clear the place, if bushy
- Dig/prepare the site to a desirable tilt/ Fine with
- Remove roots and stones from the site
- Prepare nursery beds 1- 1.54 wide by any convenient length
- Prepare raised or sunken nursery bed depending on moisture content available
- Level the Nursery bed (Any 4 x 1 = mks)

(b)

- Make shallow furrow drills/ about 10cm apart
- Apply phosphates fertilizers in the furrows/ Drill and mix with the soil
- Sow seeds by drilling
- Cover the seed lightly with soil
- Apply some mulch after sowing seeds
- Water the nursery thoroughly (Any 3 x 1 = 3 mks)

(c)

- Remove the mulch as soon as seedling emerges
- Water the nursery at least twice a day, preferably morning and late evenings
- Remove weeds as they come up

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- Thin young seedlings if over crowded/ prick seedlings
- Control diseases
- Harden off the seedling/ remove shade gradually and reduce frequency of watering
(Any 5 x 1 = 5 mks)

(d)

- Water nursery thoroughly before transplanting
- Dig the planting holes at appropriate depth
- Select healthy seedlings
- Uproot seedlings carefully with as much as possible to avoid root damage/ use a garden trowel
- Transport seedling carefully to the end field using appropriate means
- Transport on a cloudy day or late in the afternoon
- Place insecticide in the hole to control soil borne pests
- Place the seedling in the planting holes at the same depth they were in the nursery bed
- Fill the hotels with soil and firm around the seedlings
- Apply mulch or erect a shade
- Water the seedling thoroughly
(Any 5 x 1 = 5 mks)

25. (a)

- Availability of adequate funds or capital/ inputs
- Training of personnel or availability of advisory services on managerial skills
- Loyalty on the part of all farmers, co-operators and officials to support their organization
- Proper and accurate record keeping and accountability for all operations
- Efficiency with which produce from farm are marketed
- Honest on the part of personnel with regard to the handling of co-operative finances
- Timely payment of farmers dues

(b)

- Diversification/ growing a variety of crop or having various enterprises so that if one fails has something to rely on.
- Insurance against losses/ taking insurance policy for farming activities so that in case of failure the enterprises are covered.
- Inventory marketing/ strategic farming keeping farm product and selling at when prices are favorable
- Flexible enterprises engaging in enterprises that can be stopped or started early as condition change.
- Rationing of inputs using just sufficient inputs such that in case of losses the cost are not too high
- Using more certain husbandry practices using practices that the farmer is sure of and has used in the pas.

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- Hedging/ contract marketing making arrangements with marketing agencies in advance so that changes in price after the arrangement do not change the price of the farmer's produce.
- Selecting more certain enterprises selection of enterprises that the done well in the area/ tried though research (Any 7 x 1 = 7mks)

C.

- Determination of the farmer's objectives and preference in order to eliminate those production possibilities that are unsuccessful
- Determination of available resources to the farmer in order to establish his/her abilities and limitations.
- Determination of possible productive enterprises
- Determination of tentative budget/ translation of physical plan into a financial
- Determination of yield f various enterprises
- Development of financial flow in order to establish the capital requirements
- Examination of the plan to ensure that is is consistence, workable and desirable
- Determination of government policies and regulation to make the plan realistic.

(Any 8 x 1 = 8 mks)

26. (a)

- Ponds/ water pumps
- Dams/ weirs
- Roof catchments
- Rock catchments
- Retention ditches/ level terraces

(b)

- Continuous cropping without giving the land a rest
- Burning
- Ploughing along the slopes/ farming on step land
- Deforestation
- Ploughing along river banks
- Cultivating when the soil is too dry or wet
- Overgrazing/ overstocking
- Flooding/ application of a large amount of water at high rate
- Over cultivating the land to fine tilth/ pulverizing the soil

(c)

- Mulching by reducing the speed of run- off and reducing the impact of raindrops
- Contour farming by reducing the speed run off
- Terracing effective length of the slope and consequently slowing down speed of running off
- Planting trees/ holding soil particles together hence reducing effects of wind erosion and reducing the impact of rain drops

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- Establishing and maintaining vegetated water; by reducing the impact of livestock on the soil erosion
- Establishing trash lines/ sones lines by reducing speed of run- off an effects of wind erosion

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K.C.S.E AGRICULTURE PAPER 2 MARKING SCHEME

SECTION A

1. Hampshiredown (1 mark)
2. Cross cut saw/ Tenon saw/ Back saw/ spokes have/ circular plane $2 \times \frac{1}{2} = 1 \text{ mk}$
3. Removal/ harvesting of marketable size fish from the pond (1 mk)
4. Prevents metal engine parts from rusting
 - Promotes free movement of engine parts by reducing friction
 - Traps foreign materials e.g. soot, dirt and dust
 - Lowers engine temperature by conducting away excess heat
 - Helps in sealing compression between the piston and cylinder (4 x $\frac{1}{2}$ = 2mks)
5.
 - Keeps radiator fins free of rubbish and dirt. Water pump lubricated regulated/ weekly
 - Ensure that the fan belt is tightly fitted/ proper tension/ lock bolts and nuts should be tightened
 - All pipes should be fitted tightly to avoid leakage
 - To top up the level of water in the radiator before using the tractor (4 x $\frac{1}{2}$ = 2mks)
6.
 - Disc ploughs work better in dry/ sticky and hard soils than mould board plough
 - There is less hindrance to operations chances of breakages because the discs roll/ ride over obstacles
 - The maintenance costs of disc plough are lower than the moldboard.
 - Disc plough require less tractor- power to pull than moldboard
7.
 - An outlet to drain off excess water
 - An inlet for fresh water supply
 - A spill way channel to take away excess water/ overflow water
 - A screen to prevent escaping of fish/ entry of unwanted objects/ fish
 - A fence to keep away predators/security
 - Dikes walls embankment/ levees (4 x $\frac{1}{2}$ = 2 mks)
8.
 - Through the mouth/ natural openings
 - Through umbilical cord
 - Through respiratory track
 - Through injury/ wounds on the body
 - Through bites by disease vectors (4 x $\frac{1}{2}$ = 2 mks)
9.
 - Spraying insecticides the breeding places
 - Clearing the vegetation
 - Use of appropriate insecticides to spray cattle

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- Sterilization of the male tsetse flies (4 x ½ = 2mks)
- 10.
 - Overgrown hooves
 - Wet and muddy conditions
 - Physical foot injuries (2 x ½ = 1 mk)
- 11.
 - High milk yields
 - Good health
 - Fast growth/ early maturity
 - High growth/ maturity
 - Good mothering ability
 - Good body conformation (4 x ½ = 2 mks)
- 12.
 - They can browse and survive on poor vegetation
 - They have hooves with hard pads which enable them to traverse large area sandy ground/ flat hooves
 - They can tolerate high temperature/ have thick skins
 - They can travel long distances for several days with very little water
 - Store fats in humps/fats can be metabolized to metabolic
 - Long eye lashes to prevent entry of sand/ have nose flaps
- 13.
 - Softening moistening of the food
 - Storage of food (2 x ½ = 1 mks)
- 14.
 - Using of caustic potash stick
 - Use of disbudding iron/ dehorning
 - Use of dehorning saw or wire
 - Use of rubber ring and elastrator
 - Use of dehorning collar (4 x ½ = 2 mks)
- 15.
 - Overcrowding
 - Pest infestation/ pest diseases
 - Noise/ strangers
 - Lack of food and water
 - Sudden change in routine/ management
 - Unbalanced diet
 - Fluctuation in temperature
 - Introducing new bird in the flock
- 16.
 - Feeding the queen / the broods

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- Protecting the hive from intruders
- Collecting nectar, pollen, gums and water/ Foraging
- Cleaning the hive
- Building combs and sealing cracks
- Making honey bee wax
- Scouting (4 x $\frac{1}{2}$ mk= 2 mks)

17.

- Should be rain- proof/ leak proof
- Should be well ventilated
- Should be easy to clean
- Should be well lit
- Should have adequate space
- Drought free
- Good drainage (4 x $\frac{1}{2}$ = 2 mks)

SECTION B

18. (i) A- Furrow opener
B- Fertilizer hopper
C- seed hopper
D- Press wheel (4 x $\frac{1}{2}$ = 2 mks)

Clean hopyjers/ tuirow openers after use

- Lubricate/ grease moving parts
 - Replace worn out lost bolts and nuts
- (ii)
- Check tension of chains/ drive sprockets before use
 - Tighten loose bolts and nuts (Any 2 x 1 = 2mks)

- (b) (i) E - adjustable spanner
F - Ring spanner (2 x $\frac{1}{2}$ = 2mks)

(iii) Tool E can be used for tightening or loosening more than two sizes of nuts and belts (Rejects one is adjustable (1 mk)

19

- (a)
(i) Slatted floor 1 x1 = 1 mk

(ii) (H 40 – 60 cm high

(b) (i)

- To allow urine and dung to pass through
- To keep the floor dry (Any 1x 1 = 1mk)

(ii)

- Prevailing direction of the wind
- Safety/ security
- Proximity to the dairy shed/ accessibility of the dairy shed

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- Drainage
- Topography (Any 3 x 1 = 3 mks)

20. (a) Term used to express that amount of the crude protein absorbed by an animal's body from a feed (1 x ½ = 1mk)
- (b) Pearson's square method

Maize 10% DCP35-



20 = 15 parts of maize

Sunflower 35%DCP

Sunflower

Amount of maize $15/25 \times 200 = 120\text{kg}$

Amount of sunflower $10/25 \times 200 = 80\text{ kg}$

Mark as shown in the diagram

(4 x 1 = 4 mks)

- 21 (a) G - Muzzle
 H - Poll
 I - Shoulder
 J - Heart girth

(4 x ½ = 2 marks)

- (b) Ear lobes/ deep in the ear

Anus

Vulva

Under tail

(4 x ½ = 2mks)

22. (a)

- Spray the entire backline from my shoulder to the tail head
- Spray the sides in a zigzag motion to trap me retain the wash from the backline
- Spray the belly with me nozzle facing upward
- Spray the scrotum/ udder and the hind flanks carefully
- Spray both hind legs up to and including the heels
- Spray under the tail head and the area around the anus and vulva
- Hold the tail switch on to the rump and spray it thoroughly to ensure complete wetting
- Spray the neck and the foreleg; from the flanks to the heels
- Spray the head and face making sure that bases of the horns are thoroughly wetted,
- Spray the inside of the ears (10 x 1 = 10 mks)
- (b) (i) Causal organisms – Virus/ virus types O, A, C/ south African types SAT1, SAT2, SAT3, / Asian type 1 (1 x 1 = 1mk)

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(ii)

- Cattle
- Pigs
- Goats
- Sheep
- Profuse salivation (Any 2 x 1=2 mks)
- Blisters which are painful around the mouth and hooves of the feet leading to lameness
- Drop in milk production in lactating cows
- Sharp rise in temperature/ high fever
- Emaciation
- Complete loss of appetite
- Diarrhoea (Any 4 x 4 = 4 mks)

(iv) Quarantine

- (a) Vaccination every 6 months
- (b) Slaughter and destruction of carcass
- (c) Regulations of livestock movement by issue of movement permits
- (d) Burn/ bury dead animals

23 (a)

- Select good animals on the basis of high yielding cows
- Select animal with good health
- Select animals having high fertility
- Select animal having good dairy conformation
- Cull poor producers
- Use superior bulls/ semen from superior bulls to service the cows
- Mate heifers when fully mature considering weight/ age
- Breed cows 60- 90 days after calving to maintain after calving interval of one year
- Keep animals healthy by routine vaccination
- Control internal parasites by routine drenching using appropriate drugs
- Treat sick animals
- Avoid physical injuries to the animals/ predisposing disease factors
- Improve sanitation/ cleanliness in the farm
- Feed the cattle on a balanced diet
- Give adequate feeds
- Give clean and uncontaminated feed
- Provide plenty of clean water
- Provide minerals/ vitamins

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- Provide housing/ avoid overcrowding/ provide shelter that is leak proof
- Use proper milking techniques
- Milk at regular intervals (Any 15 x 1 = 15mks)

(b)

- Control stocking rate
- Control of water pollution
- Supply adequate feed regularly
- Provide appropriate feed
- Aerate the water by ensuring constant inflow and outflow of water
- Control predators
- Harvest fish at the correct maturity stage
- Maintain appropriate water level in the fish pond always
- Add manure or fertilizer in pond to encourage growth of planktons (Any 5 x 1 = 5 mks)

24. (a)

- Farm operations can be achieved on time
- Large area can be covered within a short time
- Reduce drudgery/ makes work easy and enjoyable
- Better job is done mechanically than human labor/ increased efficiency
- High yields are obtained because farm operations are carried out on time
- Pest and disease outbreak can be controlled relatively in a shorter time
- Tends to encourage farmers to consolidate their land
- Farmers benefit from economies of scale
- Use less labor (Any 6 x 1 = 6 mks)

(b) TWO STROKE CYCLE ENGINE

- Cheap to buy and easy to maintain
- Produce less power/ do less heavy
- Mainly air cooled
- Inefficient in fuel and oil utilization
- Easy to transport to different areas of the farm land e.g hilly areas\
- Require two complete upward and downwards movements of piston to be position, and one revolution of crankshaft
- There is no provision of oil in the sump, during induction, to lubricate the crankshaft
- Simple in construction with no valves
- Has 2 openings exhaust

(c) FOUR STROKE CYCLE ENGINE

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- Expensive to buy and maintain
 - Produce more power/ do heavy work
 - Efficient in fuel and oil utilization
 - Mainly water cooled
 - Difficult to transport easily due to weight
 - Require 4 complete upwards and downwards
 - 2 revolutions of 1 he crankshaft
 - Engine have oil in the sump to lubricate the crankshaft bearings
 - Complex in constructions with two valves (inlet and outlet)
 - Has no parts and inductors ports (Any 6x 1 = 6 mks)
-
- Using a dip – stick to check the level of oil in the sump
 - Check the fuel tank to ensure there is adequate fuel for the day's job
 - Check the level of the electrolyte in the battery and adjust accordingly.
 - Grease/oil moving parts
 - Check-fan belt. Tension' and condition and adjust accordingly
 - Check level of water in radiator and top up if necessary
 - Check air cleaner to ensues that there is no dirt/check level of oil
 - Check tyre pressure before work and adjust accordingly
 - Tighten bolts, nuts and pins
 - Open and remove the dirt from sediments bowels (Any 8x1=8marks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

K.C.S.E AGRICULTURE PAPER 1

SECTION A [30 MARKS]

Answer ALL the questions in this section in the spaces provided.

1. Give **four** conditions of the land which may make it necessary to carry out reclamation practices. [2marks]
2. List **three** physical weathering agents in the soil formation process [1½]
3. State **two** mechanical methods of separating soil particles according to size during soil analysis [1marks]
4. Give **two** benefits of possessing a land Title Deed to a farmer. [1mark]
5. Give four advantages of crop rotation [2 marks]
6. State four factors that should be considered when classifying crop pest
7. State **three** functions of boron in crop development. [1½]
8. Outline **four** observable indicators of economic development of a nation [2marks]
9. Give three factors that may influence the price of an agricultural commodity. [1½]
10. Name three examples of leguminous fodder crops. [1½]
11. Give two factors that may determine the size of a pit for silage making [1mark]
12. Give three reasons for controlling weeds in pastures. 1½
13. State six characteristics of a productive soil. (3 mks)
14. State any five qualities that should be considered when selecting seeds for planting (2 ½ mk)
15. (a) State four practices which encourage soil erosion (2 mks)
(b) Name two forms of gully erosion (1 mk)
16. (a) State four advantages of land consolidation (2 mks)
(b) Give two advantages of leasehold tenure system in farming (1 mk)

SECTION B (20 MARKS)

Answer all the questions in this section in the spaces provided

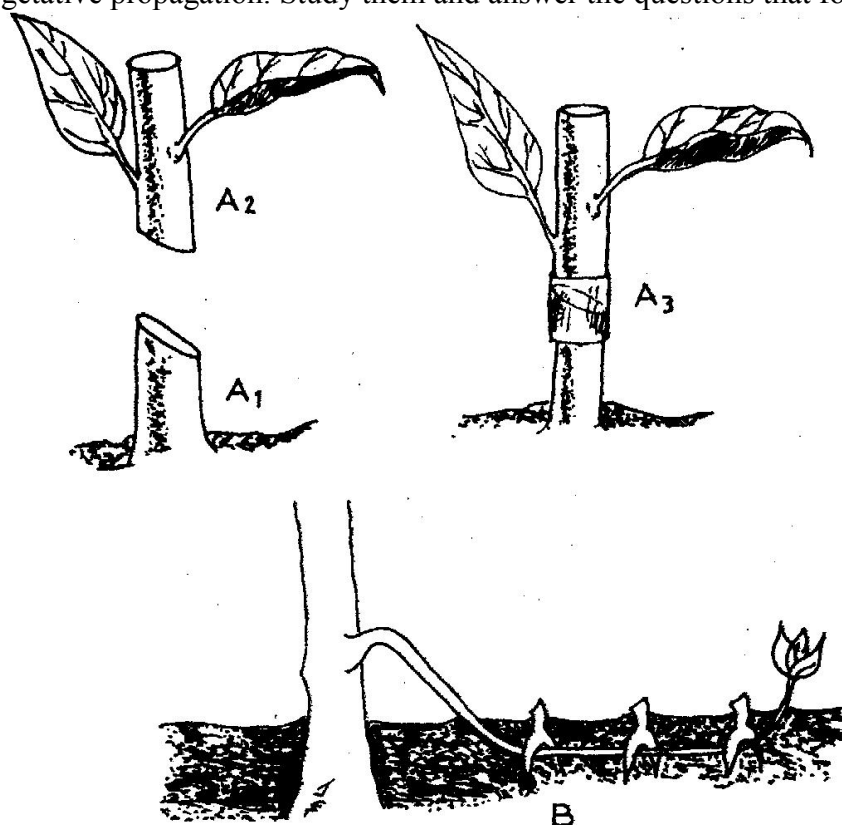
17. The table below shows the demand and supply of potatoes at UKULIMA market.

Price (Kshs)	Quantity demanded (in bags)	Quantity supplied (in bags)
1200	50	250
1000	90	200
800	150	150
600	225	70
400	335	0

- (a) Using suitable scales, draw and label a graph showing the relationship between the demand and supply of the potatoes at UKULIMA market. (5mks)

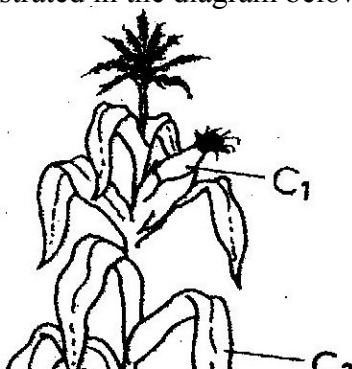
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- (b) What is the equilibrium price of the potatoes? (1 mk)
- (c) From the graph determine:
- (i) The number of bags of potatoes that would be bought if the price per bag is Kshs 900/= (1 mk)
- (ii) The price of a bag of potatoes if 180 bags are supplied (1 mk)
18. The diagrams labeled A₁, A₂, A₃, and B below illustrate materials and methods of vegetative propagation. Study them and answer the questions that follow.



- (a) Name the parts labeled A₁, and A₂ (2 mks)
- A₁
- A₂
- (b) Name the methods of propagation illustrated in diagrams A₃ and B (2 mks)
- A₃
- B

19. Study the crop illustrated in the diagram below and answer the questions that follow



- (a) Name one insect pest which attacks the part labeled C_1 and one disease which attacks the part labeled C_2 (2 mks)

C_1

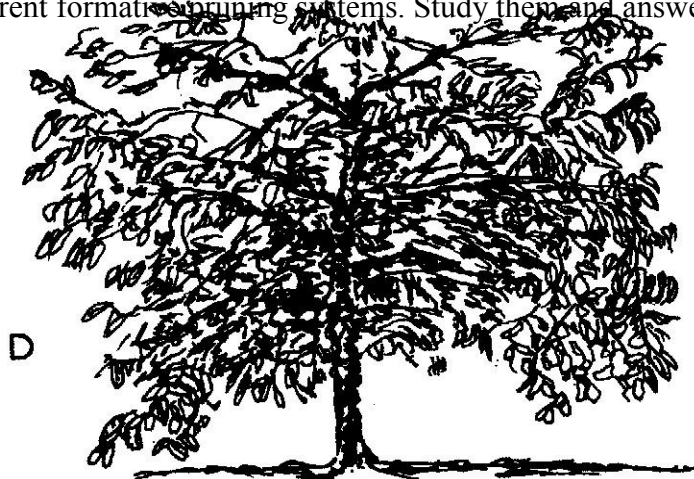
C_2

20. A member of young farmers club was advised to apply a complete fertilizer 30: 20:10 in a tomato plot measuring 10m long by 5m wide at the rate of 300kg per hectare

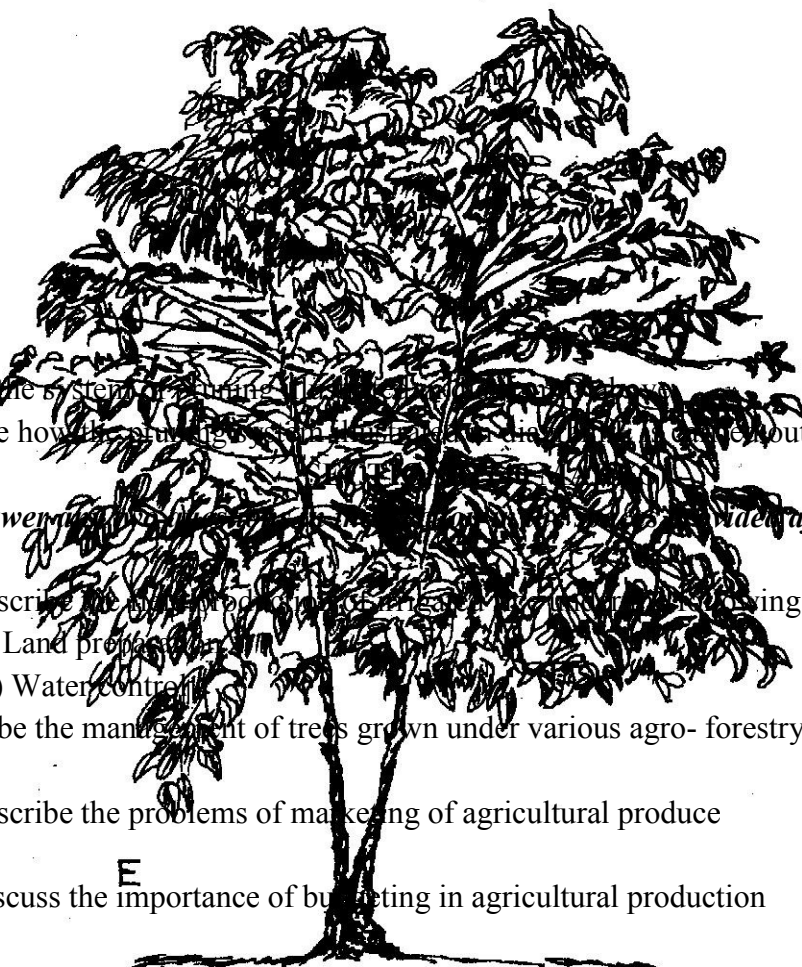
- (a) State the percentage of P_2O_5 in the complete fertilizer (1 mk)
(b) Calculate the amount of fertilizer the member would require for the plot
(2 mks) (Show your working)

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21. The diagrams labeled D and E below are illustrations of coffee established using two different formation/pruning systems. Study them and answer the questions that follow.



D



E

- (a) Name the system of pruning illustrated in diagram D (1mk)
 (b) Outline how the pruning system in diagram E can be carried out (2mks)

Answers must be given after questions 24

22. (a) Describe the management of a farm using sub-headings
 (i) Land preparation (7 mks)
 (ii) Water control (6 mks)
 (b) Describe the management of trees grown under various agro- forestry systems (7 mks)
23. (a) Describe the problems of marketing of agricultural produce (10 mks)
- (b) Discuss the importance of budgeting in agricultural production (10 mks)
24. (a) Discuss the importance of irrigation in farming (12 mks)
- (b) Explain the factors that influence the type of irrigation to be used in a farm

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(8 mks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

K.C.S.E AGRICULTURE PAPER 2

SECTION A (30 marks)

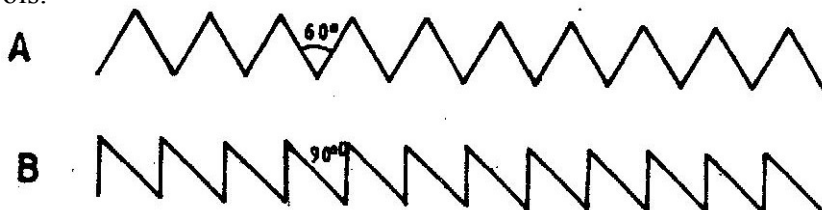
Answer ALL the questions in this section in the spaces provided.

1. Give two reasons for using litter in a poultry house. (1mk)
2. Name two diseases of poultry that are controlled by vaccination. (1mk)
3. State two factors that could lead to failure to conceive in sows after service. (1mk)
4. Give two causes of scouring in calves. (1mk)
5. State three factors that would determine the amount of concentrate fed to dairy cattle. (1 ½ marks)
6. Give three ways of stimulating milk let-down in a dairy cow. (1 ½ marks)
7. State two reasons for dehorning cattle. (1mk)
8. List two equipment used in handling cattle during an agricultural exhibition. (1mk)
9. State three signs of anthrax infection disease observed in the carcass of cattle. (1 ½ mks)
10. Give three effects of external parasites that are harmful to livestock. (1 ½ mks)
11. State four factors to consider when siting a fish pond. (2mks)
12. State three adjustments that should be carried out on a tractor – mounted mouldboard plough in preparation for ploughing. (1 ½ mks)
13. a) Name four breeds of dairy goats. (2mks)
b) Mention two distinguishing characteristics of the Bactrian camel breed. (1mk)
14. State five methods of maintaining good health in livestock. (2 ½ mks)
15. List four sources of farm power which are environmental friendly. (2mks)
16. State three maintenance practices that should be carried out on a feed trough. (1 ½ mks)
17. Name four systems of a tractor engine. (2mks)
18. List three types of calf pens. (1 ½ mks)
19. State four conditions that would encourage hens to eat eggs in poultry production (2mks)

SECTION B (20 MKS)

Answer ALL the questions in this section in the spaces provided.

20. The diagrams labeled A and B below show the teeth arrangements in hand workshop tools.



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- a) Identify the tools represented with by the teeth arrangements A and B. (1mk)

A

B

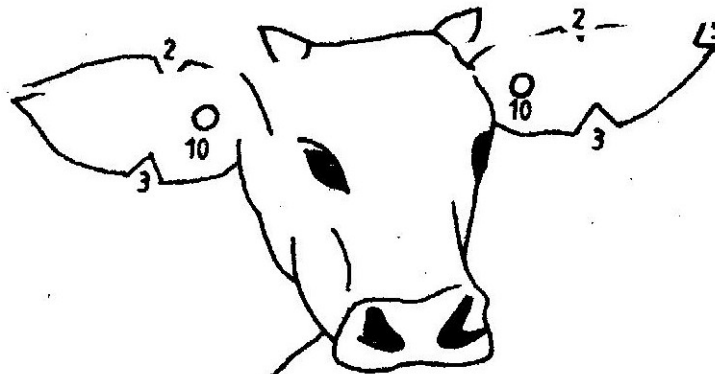
- b) State one functional difference between tools represented by the teeth arrangements A and B.

A

B

- c) Give two maintenance practices for the tools represented by the teeth arrangement shown above. (2mks)

21. a) The diagram below illustrates a method of identification in livestock production. Study the diagram and answer the Questions that follow.



- i) Name the type of identification illustrated above. (1mks)
- ii) Give the identification number of the animal illustrated in the diagram above. (1mk)
- iii) Using diagrams illustrate how you can identify animals Nos 24 and 36 using the above method. (2mks)
- Animal No. 24
- Animal No. 36
- (b) If a sow was successfully served on 27th September, 2006, state the date she is likely to have furrowed. (1mks)

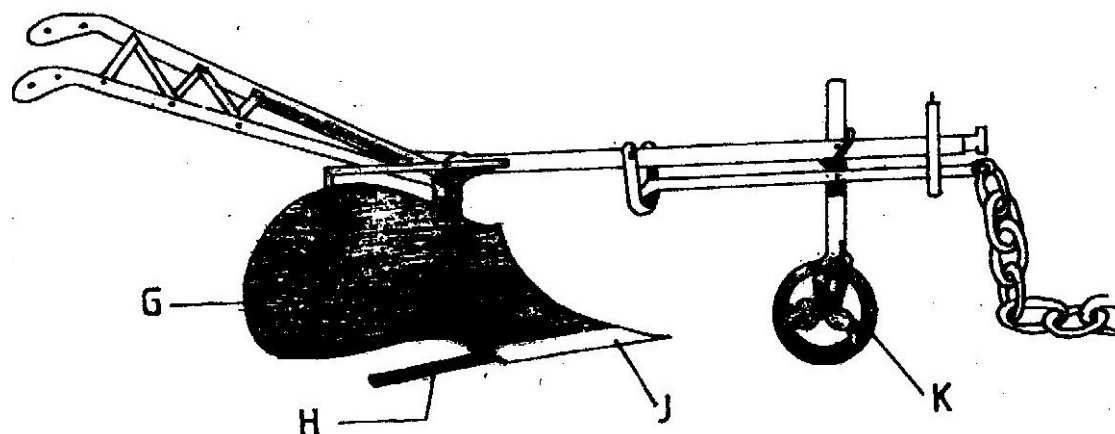
22. The diagram below shows a type of a farm gate. Study the diagram and answer the questions that follow.



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- a) Identify the type of gate shown (1/2 mk)
- b) Name the parts labeled C,D and E. (1 ½ mks)
- C
- D
- E
- c) i) State one function of the part labeled F. (1mk)
- F
- ii) State two functions of the gate illustrated above. (2mks)

23. The diagram below shows a farm implement. Study it and answer the questions that follow.



- a) Identify the farm implement illustrated above. (1mk)
- b) Name the parts labeled G,H,J and K.
- G
- H
- J
- K
- c) State four functions of the farm implement illustrated above. (2mks)

SECTION C (40 marks)

Answer any TWO questions in this section in the spaces provided after question 26.

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24. a) Describe the advantages of the battery system of rearing layers. (10mks)
b) Outline the factors to consider when selection livestock for breeding.
25. a) Name the strokes in a four stroke engine and describe how each operates. (12mks)
b) Describe the functions of the gear box in a tractor. (8mks)
26. a) Name and describe the features of an ideal calf pen. (9mks)
b) Discuss pneumonia in calves under the following sub – headings:
i) Predisposing factors (3mks)
ii) Symptoms (5mks)
iii) Control measures (3mks)

FREE AGRICULTURE K.C.S.E 2012 to 2015

KCSE AGRICULTURE MARKING SCHEMES PAPER 1

1.
 - Very steep land
 - Water logging / marshy area.
 - Forested / Bushy area.
 - Rocky / Aridity/Tsetse fly infested areas.
2.
 - Wind / Moving water
 - Temperature changes
 - Moving ice/ Glacier
3.
 - Using a sieve / sieve analysis.
 - Sedimentation method
4.
 - Can be used as a security for credit.
 - Encourage long term investments
 - Reduces land disputes
 - Motivates the farmer to conserve soil water.
5.
 - Improves soil structure
 - Controls soil borne pests and diseases.
 - Ensure maximum utilization of farm labour.
 - Aids in weed control
 - Improves soil erosion.
 - Security incase of failure of one crop.
 - Add nitrogen through N – fixation by Rhizobium bacterial when legumes are included.
6.
 - Crop attacked / mode of feeding .
 - Whether field or storage pest.
 - Crop parts attacked.
 - Stage of crop growth attacked.
 - Scientific classification e.g. insects, mite, rodents.
7.
 - Important in calcium utilization.
 - Necessary in sugar translocation
 - Needed in water absorption.
 - Aids in translocation of sugar nitrogen and phosphorous.
 - Aids in fruit development.
- 8.

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- Development of infrastructure.
- Housing status of the citizens.
- Increase in recreation facilities.
- Ratio of teachers to students.
- Improvement in the level of technology/ more industrialization.

9

- Price of substitutes.
- Price expectations in future.
- Quality of the commodity
- Tastes and preference of the commodity.

10

- Medicago sativa/Lucerne
- Leucaena leucocephala/calliondra.
- Artemisia annual/Artemisia.
- Calliandra calothyrsus/calliandra
- Desmodium species
- Kenya white clove/ Infolium sepiosum

11

- Quantity of forage available for ensiling.
- Number of animal to cater for.
- Length of the period of forage scarcity.
- Bulkiness of the material.

12

- To avoid poisoning of livestock.
- Minimize diseases spread.
- To ensure the forage is of high palatability.
- Minimize competition for nutrients, space light.
- To increase the life span of the pasture.

13

- Has appropriate depth
- The right PH/ Good soil structure.
- Good water logging capacity.
- Well aerated/good drainage.
- Free from soil borne pests and diseases.
- Rich in nutrients in the right proportions.

14

- Should be of high purity.
- Should be free from pest and disease attack.
- Should be appropriate size

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- Should be mature.
- Should be free from any physical damage.
- Should be of high percentage of germination.
- Should be suitable to the ecology of the area.

15 (a)

- Over –cultivation, overstocking/overgrazing.
- Deforestation/planting annual crops on steep slopes.
- Burning of the vegetation.
- Ploughing up and down the slope.

(b) V- shaped gullies U-shaped gullies.

16. (a)

- There is proper supervision of the farm.
- Reduces costs on traveling
- Easy to get extension services.
- Allows good farm planning.
- It enhances proper pests, diseases and weed control.
- Encourages long term investments.

(b)

- Landlord can earn income from the land.
- People who have no land are able to access to farming.
- Idle land is put into agricultural use.
- Tenant is able to increase/decrease the size of land leased depending on profitability.

17. (a) Shs. 800

(b) (i) 120 bags ii) 900

18. a) A₁- root stock

A₂- Grafting

b)A₃ Grafting

b- Trench layering

19. a) C₁ – Maize stalk borer, maize weevil, Aphids

C₂ – Maize streak, white leaf blight.

20. a) p₂ o₅ = 20%

b) 1 ha = 10,000m² requires 300kg of fertilizer.

5mx 10m₂= 50m₂ requires x of the fertilizer

10,000 x = 300 x 50

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$$X = \frac{300 \times 50}{10,000} = \frac{3}{2}$$

21. a) Single stem pruning.
b) The main stem is capped at 38cm above the ground to encourage more suckers to grow. Select two strong and healthy suckers and remove the others. The selected suckers should form a U-shaped to avoid splitting.
22. (a)
- Clear the land
 - Divide the land into plots of 0.4 ha
 - Construct /repair bunds /dykes.
 - Construct/ repair inlet and outlet channels
 - Flood the field to a height of 7.5 – 10cm above the soil surface.
 - Carry out primary tillage
 - Puddle the soil to a fine mud.
 - Uprooted weeds should be heaped on the bunds.
 - Level the plots by dragging a wooden board/ jembe.
- (ii)
- Flood the plots to a depth of 7.5 – 10 cm.
 - Leave the field flooded for 4 days.
 - During transplanting, drain the field to a depth of 5cm,
 - Introduce water gradually as the crop establishes.
 - Maintain the water level at 1/3 the height of the crop
 - Change water every 2-3 weeks or when it is cold.
 - Water should allow to flow slowly through the field
 - Drain the field 2-3 weeks before harvesting.
- (b)
- Irrigation during the dry season.
 - Timely pest control.
 - Timely weed control
 - Pruning, Coppicing/pollarding/capping.
 - Thinning/selective harvesting.
 - Protection against damage by animals.
 - Grafting/budding.
 - Fertilizer/manure application
 - Construction of micro-catchments
 - Structures around the trees
 - Provision of shade/mulch to reduce evaporation.
23. (a)
- Competition from cheap/synthetic / products, causing loss.
 - Change in supply of the produce; leading to price fluctuation

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- Change in market demand; leading to price fluctuation.
- Lack of market information; leading to exploitation by middle .
- Inadequate capital; hence poor financing of various marketing functions.
- Poor quality of produce; leads to price fluctuation.
- Seasonally of produce; leads to price fluctuation.
- Bulkiness of most agricultural produce; making it expensive and difficult to transport.
- High perishability; this leads to low quality of produce
- Poor storage structure; leading to heavy losses of the produce.
- Lack of knowledge in marketing leading to heavy losses.
- Government interference through its agents leading to price fluctuation
- Acts as a record for future reference.
- Helps in deciding the viability of the enterprise
- Assist in securing credit.
- Helps to predict the profitability of the enterprise.
- Aids in detecting problems easily hence correction is done in good time.
- Aids in making management decisions especially when comparing between enterprises.
- Helps in making changes in the farm.
- Ensures periodic analysis of the farm business.
- Encourage the farmer to be efficient so as to meet the target.

24 (a)

- Enables one to grow crops during the dry seasons.
- It's a method of land reclamation/ allows crop production in arid and semi-arid areas.
- Makes it possible to grow crops in special structures e.g. green house.
- Enables one too grow crops that require high amount of water e.g. paddy rice.
- It supplements rainfall in case it inadequate in crop produce.

(b)

- Topography,
- Soil type
- Type of crop to be irrigated.
- Amount of water available.
- Technology available.
- Distance of the source of water to the field.
- Capital available, skills available
- Climate factors of the area.

FREE AGRICULTURE K.C.S.E 2012 to 2015

K.C.S.E PAPER 2 MARKING SCHEMES

1.
 - To keep the house warm.
 - To absorb moisture from poultry droppings.
 - Keeps birds busy scratching, thus reducing cannibalism.
2.
 - Marks's disease, avian spirochaetosis.
 - Fowl typhoid, Gumboro/ infectious bursa disease.
 - New castle, fowl pox, infectious bronchitis.
 - Chronic respiratory disease.
 - Infectious coryza of chicken.
3.
 - If the sow is barren.
 - Poor nutrition if the calf cold milk.
 - Poor timing services
4.
 - Overfeeding/ giving the calf cold milk.
 - Lack of colostrums.
 - Irregular feeding of calf.
 - Feeding milk at wrong temperature.
 - Feeding milk in dirty containers/ feeding contaminated milk.
5.
 - Level of milk production
 - Quality of roughages.
 - Availability of the concentrates.
 - Economic factors/cost of concentrates.
 - Physiological status.
6.
 - Washing the udder with warm water.
 - Allow the calf to suck for a while
 - Feeding the cow during milking.
 - Regular milking time
 - Sound associated with milking.
 - Massaging the udder when washing it.
7.
 - To make the animal docile
 - Reduce, incidence of animals injuring each other/attendant.
 - Reduce incidence of animals damaging farm structures.
 - Increase feeding, watering transportation space.
 - Add aesthetic value to the animal.

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8.

- Halters, Nose bull ring and leading stick. Rope.

9

- Carcass lacks rigor mortis.
- Excess bloating
- Water tar-like blood oozes from body openings
- Oozing blood clot.
- Rapid purification.

10

- Introduce toxins that are harmful to the animal.
- Cause anaemia/transmit diseases.
- Cause wounds that allows secondary infection.
- Cause irritation which leads to scratching/destroy wool.

11.

- Source of water/Type of soil
- Topography.
- Closeness to homestead/accessibility.
- Closeness to the market/consumers.
- Far away natural sources of fish.

12.

- Adjust the depth of ploughing
- Adjust furrow width of ploughing
- Front furrow depth.
- Lowering /raising ploughing pitch.

13.

(a)

-Saanen, anglo-Nubian, Toggenburg. British alpins, Jamnapari.

14

- proper feeding. prophylaxis, quarantine.
- Proper housing, control of parasite.
- Practice farm hygiene.
- Routine vaccination.
- Use of healthy breeding stock.
- Timely treatment of the sick livestock.
- Control of vectors, dipping, spraying.

15

- Wind power. Water power, animal power solar energy. Human power, Biogas
Geothermal.

16.

- Painting metallic parts

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- Regular washing.
- Repair broken parts/cracks.
- Replace lost parts.

17.

- Fuel systems, 1
- Lubrication system.
- Electrical system.
- Ignition system,
- Cooling system,
- Hydraulic system
- Power transmission system

18

- Permanent calf pen.
- Movable calf pen.
- Concrete floor calf pen.
- Slatted floor calf pen

19.

- Calcium deficiency in the birds body.
- Blight light in the laying nests
- Birds laying on the floor.
- Presence of broken, soft shelled eggs.
- Prolonged stay of eggs in the laying boxes.
- Idleness of birds.
- Inadequate feeding.

20

(a) A- cross-cut saw B- rip saw

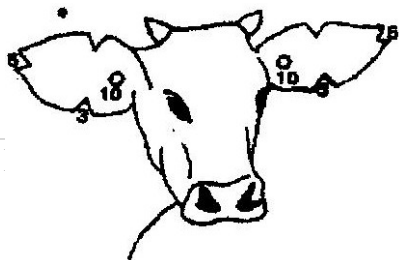
(b) A- cutting across the grain B- cutting along the grains

(c)

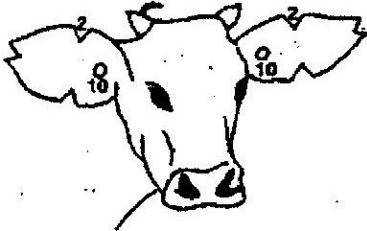
- Wipe blade with an oily rug.
- Regular sharpening of the teeth.
- Ensure the handle is firm,
- Teeth setting.
- Straighten the blade if bent.
- Proper storage of the foods.

21(a) (i) ear notching (ii) Number 40 (forty)

(ii) AC. $10+5+5+2+2$ ($17+17$)



Animal No 36



Animal no 34

- (b) Between 18-1-2007 and 20-1-2007
22. (a) Barbet wire gate
(b) C – gate post/King post/strainer
D – Wire loop
E – Dropper.
- (c) (i)
- Support the gate post
 - To ensure the barbet remains taut.
- (ii)
- Prevent movement of farm animals outside
 - Keep away livestock from outside.
 - Used as entrance into/exit from the farm.
23. (a) Animal/ ox-drawn plough.
(b) G – Mould board.
J – Share
H – Land slide
K – Land wheel
- (c)
- Plough/ridging
 - Harvesting root crops e.g. groundnuts.
 - Weeding row planted crop.

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- Opening furrows for planting.

24. (a) **Advantages of battery system.**

- Higher egg produce due to less energy wastage.
- Easy to keep individual production records.
- Control cannibalism and egg eating.
- No contamination of water and feed.
- Birds are not exposed to predators, parasites and diseases.
- Facilitates culling and handling.
- Easy to collect eggs
- Egg losses are reduced.
- Many birds are kept in a given/high stocking rate.
- Eliminates broodiness.
- Birds still have tender meat at culling due to confinement.
- Facilitates mechanization.
- Keeps eggs clean.

(b) **Factors considered in selecting livestock k for breeding.**

- Body confirmation.
- Fertility/breeding ability.
- Adaptability of the breed to the arts/hardiness.
- Mothering ability in case of females.
- Production potential/yielding capacity.
- Temperament/behaving e.g. cannibalism egg eating.
- Deformities/abnormalities e.g. one eye lameness.
- Offspring performance
- Age of animal.
- Growth rate, quality produce
- Disease resistance, prolificacy.
- Lifespan/reproductive life.

25. (a) **Operation of a four stroke engine.**

(i) **Induction stroke/sunction.**

- The piston moves down the cylinder, causing the inlet valve to open and draw in fresh supply of petrol vapour and air into the cylinder, exhaust valve closed

(ii) **Compression stroke.**

- The inlet valve closes and the piston moves up the cylinder. This compresses the fresh fuel mixture into the combustion chamber, exhaust valve to close.

(iii) **The power stroke.**

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- Fully compresses the fresh fuel mixture and as a result a spark is produced at the spark plug. This causes the fuel mixture to ignite and expand resulting in pressure that forces the piston down the cylinder. Inlet valve closed exhaust valve closed.

(b) Functions of gearbox.

- Helps the driver to select any forward or reverse gear.
- Adjust speed of the driver from the engine crankshaft to the driver shaft.
- Helps to alter the speed ratio.
- Enables the power from the engines to be more easily applied to the work done by the tractor.
- Enables the driver to stop the tractor movement without stopping the engine or without foot oppressing on the clutch all the time.

26. **(a) Features of an ideal calf pen.**

- **Concrete/raised stated floor** – Easy to maintain cleanliness.
- **Dry litter/bedding** – Maintain warmth.
- **Proper lighting** – Should have good supply of natural light/sunlight.
- **Proper drainage** – facilitate free flow of urine and water to avoid dampness.
- **Draught free** – The structure should stop strong winds from blowing into the calf pen.
- **Proper ventilation** – Structure should allow for fresh air circulation.
- **Security** – Should be strong enough to keep away intruders/wild animals.

(b) Pneumonia in calves.

(i) Predisposing factors

- Overcrowding of calves in the pen.
- Dampness/chilliness in the pen.
- Poor ventilation.
- Age/younger calves are more prone to pneumonia than older calves.
- Effects of diarrhea and other illness.

(ii) Symptoms.

- Rough hair coats/ruffled hair.
- Loss of appetite.
- Abnormal lungs sounds e.g. whizzing.
- Emaciation, frequent coughing.
- Nasal discharge.
- Fluctuating body temperature.
- Dull and reluctant to move.

(iii) Control measures.

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- Treating the sick calve with antibiotics.
- Providing warmth in pens.
- Maintaining good sanitation in pens.
- Isolating sick calves to avoid spread of the disease.